Introduction to Stochastic Calculus in Valuation of Options

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Abstract

In this report, we would delve into stochastic calculus and its application in valuation of options. The main purpose is to develop and implement advanced stochastic models for option pricing, especially Merton jump-diffusion process. We give an overview of theorems related to stochastic process, such as Brownian motion, Itô integral, Poisson process and stochastic integral for jump processes. Other mathematical equations in estimating the value of derivatives in finance market, like Black-Scholes-Merton equation, would also be introduced and employed. This report provides two methods to derive the Black-Scholes-Merton equation: partial differential equation method and risk-neutral measure. The Lévy measure and Lévy-Khintchine formula would be interpreted to derive the characteristic components and cumulants of Lévy processes. We leverage cumulants to calculate variance, skewness, kurtosis and conduct sensitivity analysis for stochastic processes. The basic ideas of change of measure would be demonstrated when introducing jump processes. Later, we leverage the methodologies into an example of valuation of options with Python. In this report, it is noteworthy that we present an innovative method to derive Merton's jump diffusion model. During the calculation, we focus on the inclusive compound Poisson process, along with its mean jump size, which would provide a completely different understanding compared to Merton (1976). This is the first version of the report.

Keywords: Brownian Motion, Itó Integral, Poisson Process, Black-Scholes-Merton Equation, Merton's Jump Diffusion Model, Lévy Measure, Lévy-Khintchine Formula, Cumulants, Valuation of Options, Sensitive Analysis

Contents

1 Introduction

This report is meant for undergraduates, postgraduates and anyone who yearns for delving into stochastic process and valuation of options. We would particularly discuss the properties and theories regarding stochastic calculus and how it could be utilized in asset pricing. In the report, we mainly focus on computing methods rather than analysis. However, some basic propositions in measure theory and functional analysis would be employed. This report is my final year project for the award of the degree of BSc Applied Mathematics.

This report is suitable for training of professions, namely, people could have a sound grasp of handling stochastic differential equations and utilizing stochastic models in asset pricing from this report. Besides, after reading the report, people could develop a strong proficiency of stochastic modeling techniques and gain an experience with Python for valuation of options using numerical analysis methods.

In the first part, we begin with Brownian motion (or Wiener process), the starting point of stochastic process.

We would introduce the techniques that using mathematical models to describe random walks. Bernoulli random walk and its properties would be involved in this part. And symmetric random walk is a particular case of Bernoulli random walk, essential to the construction of Brownian motion. Readers would have a solid concept that Brownian motion is a nowhere differential but continuous function. That is why we employ stochastic calculus to analyze the behavior of stochastic process. This chapter contains martingale, quadratic variation which would be frequently utilized in this report.

The second part, based on Brownian motion, we demonstrate Itô integral and Itô-Doeblin formula.

Itô integral, though not quite adequate, could estimate the change in asset prices over time. As said, since Brownian motion is a nowhere differentiable function, Itô integral is defined in a space of square-integrable stochastic processes. We would firstly analyze Itô integral for simple integrands and then generate into Itô integral for general integrands. We would give a proof for general integrands using function convergence in a metric space. Although it can be verified by real analysis easily, but reader could create intuitive feelings on the reason for function convergence using the proof in this chapter. And then it follows Itô-Doeblin formula, which could be used to deal with functions depending on Brownian motion. The formula is crucial in learning stochastic process and it has many variants. Anytime people want to figure out the integral form or differential form of a stochastic process, it may be of help.

The next part deals with Black-Scholes-Merton equation.

We would introduce geometric Brownian motion, which is an improved stochastic model in describing stock prices. Black-Scholes-Merton partial differential equation is deduced from geometric Brownian motion to estimate portfolio value over time. Then we would derive the equation in terms of boundary conditions using partial differential equation method. The basic idea of deriving the equation comes from Wilmott, Howison and Dewynne (1995). Firstly, we introduce one dimensional heat equation, the principles of delta function would be employed to examine the general solution of one dimensional heat equation. Then we introduce quarter plane problems to further discuss heat equations. Secondly, we transfer the Black-Scholes-Merton partial differential equation into one dimensional heat equation by change of variables. Lastly, we would solve the heat equation and replace the variables with original ones. The change of measure required for transferring the partial differential equation is greatly intricate, prompting thoughts of partial differential equation methods. Subsequently, replacing variables in the computing results with original ones requires complicated calculations.

In the next part, we interpret the methods for sensitive analysis on stochastic processes. The principles of variance, skewness and kurtosis would be introduced, and these parameters are crucial for sensitivity analysis for stochastoc processes. Then we would discuss the methods of sensitive analysis for Lévy's processes. The definition of Lévy's process and infinite divisibility would be presented. Next, we would leverage Lévy-Khintchine formula for calculating the characteristic components of Lévy's processes, and during the calculation, the Lévy measure would be introduced. Subsequently, we would interpret the definition of cumulants. For Lévy processes, cumulants could be derived from the characteristic components of Lévy processes. And the variance, skewness and kurtosis is relevant to the first four cumulants. Finally, we would illustrate the way to employ these concepts to analyze the sensitivity for Black-Scholes-Merton equation. Since Black-Scholes-Merton equation is not a Lévy's process, we would calculate the characteristic components and cumulants for the geometric Brownian motion, which is related to Black-Scholes-Merton equation. Then the variance, skewness and kurtosis, which is called sensitive analysis methods, could be determined.

The next part interprets risk-neutral measure in solving Black-Scholes-Merton equation. After a revision of probability space, we introduce the basic principles of change of measure. Then the Radon-Nikodým derivative process would be introduced and the adapted process is defined under Radon-Nikodým derivative process. Then we give an example of change of measure on Brownian motion and notice that the drift of Brownian motion varies under new probability measure. Next, we would discuss risk-neutral measure on stock prices by defining the random discounted rate and discounted stock price process, the discounted stock price process could be converted into a martingale under new probability measure, entitling us to deriving the formula for stock prices. Then we create hedges using the portfolio. Subsequently, we employ the theorems of hedging processes to derive the Black-Scholes-Merton equation.

The next part discusses the properties and theorems of jump process. Jump process is a stochastic process that changes its state suddenly at random moments of time. We begin with Poisson distribution, and then derive some common jump processes, such as Poisson process and compensated Poisson process. Then we extend Itô integral to the stochastic integral for processes that depend on jump processes. Then we introduce two significant properties: Itô product rule and Doleans-Dade exponential, which would be leveraged in calculations in change of measure. Then we introduce change of measure that can vary the intensity that describes the frequency of jump happening. We start from the adapted process and new probability measure and we provide an example of change of measure on discrete compound Poisson process and non-discrete compound Poisson process.

The next part introduce the Merton's jump diffusion model. This report presents a completely innovative method to explain Merton's jump diffusion model, which is an extension of the approaches from Shreve (2004) and the principles of partial differential equations. We would firstly discuss the probability distributions of log stock price jump size and log-return density function, which is crucial for sensitive analysis in the later subsection, this part refers to Matsuda (2004). To derive the Merton's jump diffusion model, we consider the rate of return for call prices, and we calculate the differential term of the call price. After that, we consider a portfolio consisting of three components to construct a hedging process, then we could apply some methods to derive the Merton integral partial differential equation with boundary conditions. To solve the equation, we compute the derivatives of Black-Scholes-Merton equation, then we could verify the solution for Merton integral partial differential equation. The Merton's jump diffusion model is built on Merton integral partial differential equation. During the derivation, we focus on the inclusive compound Poisson process, along with its jump sizes, in the return of call prices. Due to this computing method, there would be one more term, the radio of the call price before and after the jump. However, it would provide a clearer understanding of the underlying compensated Poisson process, implying its mean rate of return and why it is non-arbitrage. The method is innovative and different from Merton (1976), while the result is the same.

In the next part, we conduct two examples about option pricing with Python. In this part, we compare the efficacy of Black-Scholes-Merton equation and Merton's jump diffusion model in valuation of options. The formulas for these models are shown and the least square error is applied as the error analysis method. Then we interpret the optimization problem to determine the parameters of Black-Scholes-Merton equation and Merton's jump diffusion model. To figure out the optimal solution, we introduce pure Newton's method, DFP algorithm and BFGS algorithm. In the first example, we choose the example from Dow Jones Industrial Average from Sep. 28, 2022 to Feb. 28, 2023. We leverage the train set to determine the parameters and the test set to examine the accuracy. We also analyze the convergent rate and complexity of these two models. Next, we would choose an example from MKS instruments with a long duration to analyze the effects of different time to maturity on valuation of options. Lastly, we conduct sensitive analysis by calculating variance, skewness and kurtosis. We would fix the other optimal parameters and change one parameter to determine its sensitivity.

This report is inspired by *'Stochastic Process for Finance'* written by Shreve (2004), and when I write the report, I was deliberating how to throw a light on every property and every theorem in this report. Eventually, I could say that this report provides detailed and systematic explanations on many theorems and proofs.

The first advantage is that this report contains more details of calculation. Actually, reader can notice this in every piece of proof that needs calculation. For instance, I modify the proof for Itô-Doeblin formula to enhance rationality. Also, in the proof that geometric Poisson process has mean rate of return α, we calculate the formula of $X(t)$ using the definition of Riemann integral, and then find that $X(t)$ is a generalized Brownian motion before calculating the expected value; while in Shreve's book, it is purposed without proof. Another one which is impressive should be the property of Doleans-Dade exponential. I amply describe how to utilize Itô-Doeblin formula on an exponential function. By reading my report, readers can generate meaningful ideas about employing known theorems and deriving differential equation, which significantly improve their confidence in handling stochastic problems.

This report also tries to use some preliminaries in later chapters, such as Jensen's inequality, supermartingale, submartingale and bracket process for cross variation. This report also provides several principles in functional analysis and optimization theory. In function analysis, we introduce Banach space, Dual space, inner product space and Hilbert space. In optimization theory, we explain the convex sets, convex functions, Farkas's theorem and Gordan's theorem. The principles from the preliminaries are quite useful in some problems which we would encounter later. For instance, the cross variance is defined under inner product space, then we could leverage the theorems of inner product space; the space of infinite divisible random variables is a closed vector space, which leads to the introduction of Lévy measure and Lévy-Khintchine formula.

This report employs partial differential equation method and risk-neutral measure techniques to solve the Black-Scholes-Merton equation, and then provides a personal interpretation on the derivation of Merton's jump diffusion model. For solving the Black-Scholes-Merton equation with partial differential equations, please refer to Wilmott, Howison and Dewynne (1995); for Merton's jump diffusion model, please refer to Merton (1976).

Another advantage is that this report discusses the principles of sensitivity analysis for stochastic processes by transferring them into corresponding Lévy's processes. We further calculate the characteristic components, cumulants and sensitivity analysis methods such as variance, skewness, kurtosis of Lévy's process.

Narrating financial meaning of the stochastic equations is significantly important in the report. It is inadvisable to learn the formula without understanding what it can be used for. So this report would introduce many financial examples, such as a finance meaning of geometric Brownian motion.

This report leverages traditional methods in metric spaces to verify that stochastic functions could be approximated by step functions; this report uses dynamic systems to calculate the mean rate of return; this report defines the Black-Scholes-Merton log-return density function and employs the Lévy-Khintchine formula to determine the variance, skewness, kurtosis for the designed function; this report leverages the principle of delta function to find the general solution for one-dimensional heat equation; this report presents an innovative method to derive Merton's jump diffusion model which focuses on the compensated Poison process during the calculation; this report compares the accuracy of the Black-Scholes-Merton equation and Merton's jump diffusion model using the call options in a tech company. This report aims to provide detailed explanation of stochastic calculus and partial differential equation methods for undergraduates and beginners by including complete calculations and intuitive explanations.

I sincerely hope that everyone can enjoy reading and benefit from the report.

2 Literature Review

We first discuss the literature review for stochastic process.

Brownian motion was first discovered by Robert Brown, a Scottish biologist, while studying plant materials in 1827. He found that pollen grains would move randomly in the water without external forces. Initially, Brown thought that these tiny grains are living creatures. But after some experiments, he inferred that it is due to some physical phenomenon. In 1900, Louis Bachelier, a pioneer of stochastic process, wrote an article 'The Theory of Speculation', referring to Bachelier (1900), and it is the first time of using mathematics models to describe finance markets. Then Einstein (1905) published an article 'On the Movement of Small Particles Suspended in a Stationary Liquid Demanded by the Molecular-Kinetic Theory of Heat' that offered a theoretical explanation for Brownian motion in 1905.

The breakthrough happened in 1923, Norbert Wiener's work employed stochastic models in describing Brownian motion and triggered the development of stochastic process. Please refer to the article 'The Homogeneous Chaos' by Wiener (1923), and 'Differential-Space' Wiener (1923). In order to calculate the stochastic integral related to Brownian motion, Kiyosi Itô delved into this field and established some principles in solving stochastic integral. And Itō (1944) published the article 'Stochastic Integral', utilizing step functions in solving stochastic integral, which is called Itô integral later. Then another article was written by Itō (1947): 'On Stochastic Differential Equations'. This article explored stochastic differential equations and introduced Itô formula in finding the differential form of stochastic processes. Then Samuelson discovered a modified stochastic process, which is called geometric Brownian motion to describe asset prices.

Black and Scholes (1973) employed stochastic models to value options from their article 'The Pricing of Options and Corporate Liabilities'. And then Merton (1973) improved Black-Scholes pricing by considering credit risk from his article 'Theory of Rational Option Pricing'. Black-Scholes model is quite significant in finance since it can be used to estimate the options with fair value assuming no transaction costs and constant volatility. Then Merton (1976), on the article 'Option Pricing and the Valuation of Corporate Debt', he argued that the volatility and fluctuation of the underlying assets could affect the debts in corporations. Kou (2002) purposed double exponential jump-diffusion model in his article 'A Jump-Diffusion Model for Option Pricing', and he believed that humans feelings should be different when stock prices rise or drop, which is demonstrated in the double exponential model, entitling Kou to becoming a leading light in the field of quantitative finance.

Steven Kou graduates in Columbia university is 1995 with a Ph.D. degree in statistics. After his graduate, Kou taught at Rutgers University (1995-1996), University of Michigan (1996-1998), Columbia University (1998-2014), National University of Singapore (2013-2018), and Boston University (2018-). Kou's expertise in quantitative finance and stochastic modeling motivated him to publish over 30 papers on academic journals such as Management Science, Mathematical Finance and Operations Research. And one of his prominent contributions should be Kou's double exponential jump-diffusion model. His contributions helped him win numerous awards, such as Erlang Prize from INFORMS Applied Probability Society and Shahdadpuri Faculty Research Award from Boston University.

Then we would talk about the literature review for Poisson process.

Let us start with Poisson distribution which named after French mathematician Siméon Denis Poisson. In his book 'Research on the Probability of Judgments in Criminal and Civil Matters' referring to Poisson (1837), Poisson distribution is employed to describe the probability of winning a range of games for a gambler. And for a large number of trials, the number of wins would approximate the expected value. Then Erlang (1909) found that the arrival of phone calls follows a Poisson distribution, as purposed in his book 'The Theory of Probabilities and Telephone Conversations'. In the meantime, Physicist Rutherford worked as a doctorial tutor in Manchester university. He guided his student Geigar to implement a famous experiment which is called Geiger–Marsden experiment. In the experiment, they utilized α-rays to strike the thick gold foil. They observed that the majority of particles could pass through the gold foil, while a few particles could not. The results of the experiment disobeys J.J.Thomson's Plum pudding model. And Rutherford & Geiger (1910), the book 'The probability variations in the distribution of α particles' demonstrates that the count particles passing through the foil would follow a Poisson distribution. Then French mathematician Paul Lévy made great distribution to Poisson process and he purposed the concept of compensated Poisson process. His works is accessible from 'Théorie de l'addition des variables aléatoires' by Levy (1937) and 'Processus Stochastiques et Mouvement Brownien' Levy (1948).

3 Preliminaries

3.1 Metric Space

3.1.1 Normed Space

Definition 3.1 (vector space)**.** *We say X is a vector space if it satisfies the addition property and multiplication property. So suppose $x, y \in X$, then we have*

$$x + y \in X, \tag{3.1}$$

and

$$\alpha x \in X,\ \forall \alpha \in R. \tag{3.2}$$

Definition 3.2 (subspace)**.** *Suppose that X is a vector space, then $Y \subset X$ is called a subspace of X if and only if*

$$\alpha x + \beta y \in Y,\ \forall x, y \in Y,\ \alpha \in R. \tag{3.3}$$

Definition 3.3 (l^p space)**.** *For $1 \leqslant l < \infty$, the l^p space is defined as*

$$l^p = \left\{ (a_n)_n \in R : \sum_k |a_k|^2 < \infty \right\}. \tag{3.4}$$

Definition 3.4 (l^∞ space)**.** *The l^∞ vector space is defined as*

$$l^\infty = \left\{ (a_n)_n \in R : (a_n)_n \text{ is bounded} \right\}. \tag{3.5}$$

Definition 3.5 (normed space)**.** *Consider a vector space X, a norm in X has the following properties:*
i) $\|x\| \geqslant 0$, $\forall x \in X$ and the equality holds only when $x = 0$;
ii) $\|\alpha x\| = |\alpha| \|x\|$, $\forall x \in X$ and $\alpha \in R$;
iii) $\|x + y\| \leqslant \|x\| + \|y\|$, $\forall x, y \in X$.

Then the normed space operated on $\|\cdot\|$ should be $(X, \|\cdot\|)$.

3.2 Functional Analysis

3.2.1 Young's inequality, Hölder's inequality, Minkowskin's inequality

Theorem 3.6 (Cauchy-Schwards inequatlity)**.** *The Cauchy-Schwards inequality is described as*

$$\sum_{k=1}^{n} |x_k y_k| \leqslant \left(\sum_{k=1}^{n} |x_k|^2 \right)^{\frac{1}{2}} \left(\sum_{k=1}^{n} |y_k|^2 \right)^{\frac{1}{2}}. \tag{3.6}$$

Proof.

$$\begin{aligned} 0 &\leqslant \sum_{k=1}^{n} |x_k + \lambda y_k|^2 \\ &= \sum_{k=1}^{n} x_k^2 + 2\lambda \sum_{k=1}^{n} x_k y_k + \lambda^2 \sum_{k=1}^{n} y_k^2. \end{aligned} \tag{3.7}$$

Therefore, we have

$$\sum_{k=1}^{n} |x_k y_k| \leqslant \left(\sum_{k=1}^{n} |x_k|^2 \right)^{\frac{1}{2}} \left(\sum_{k=1}^{n} |y_k|^2 \right)^{\frac{1}{2}}. \tag{3.8}$$

□

Theorem 3.7 (Young's inequality). *The Young's inequality denotes that, $\forall p, q \in R$ with $\frac{1}{p} + \frac{1}{q} = 1$, it gives that*

$$ab \leqslant \frac{1}{p}a^p + \frac{1}{q}b^q. \tag{3.9}$$

Theorem 3.8 (Hölder's inequality). *Consider the normed space $\|\cdot\|_p$ and $\|\cdot\|_q$, suppose that $\frac{1}{p} + \frac{1}{q} = 1$ and $p > 1$, then it follows that*

$$\sum_{k=1}^{n} x_k y_k \leqslant \|x\|_p \|y\|_q. \tag{3.10}$$

Especially, when $p = q = \frac{1}{2}$, the inequality becomes Cauchy-Schwards inequality.

Proof. We first consider the term $\frac{x_k}{\|x\|_p}\frac{y_k}{\|y\|_p}$, then according to Young's inequality, we have

$$\begin{aligned} \frac{x_k}{\|x\|_p}\frac{y_k}{\|y\|_p} &\leqslant \frac{|x_k|}{\|x\|_p}\frac{|y_k|}{\|y\|_q} \\ &\leqslant \frac{|x_k|^p}{p\|x\|_p^p}\frac{|y_k|^q}{q\|y\|_q^q} \end{aligned} \tag{3.11}$$

And then

$$\sum_{k=1}^{n} \frac{|x_k|}{\|x\|_p}\frac{|y_k|}{\|y\|_q} = \frac{1}{p} + \frac{1}{q} = 1, \tag{3.12}$$

which implies that

$$\sum_{k=1}^{n} x_k y_k \leqslant \|x\|_p \|y\|_q. \tag{3.13}$$

□

Theorem 3.9 (Minkowskin's inequality). *The Minkowskin's inequality shows that*

$$\|x + y\|_p \leqslant \|x\|_p + \|y\|_p. \tag{3.14}$$

Proof.

$$\begin{aligned}
\|x+y\|_p^p &= \sum_{k=1}^{n}|x_k+y_k|^p \\
&= \sum_{k=1}^{n}|x_k+y_k||x_k+y_k|^{p-1} \\
&\leqslant \sum_{k=1}^{n}|x_k||x_k+y_k|^{p-1} + \sum_{k=1}^{n}|y_k||x_k+y_k|^{p-1} \\
&\leqslant \left(\sum_{k=1}^{n}|x_k|^p\right)^{\frac{1}{p}}\left(\sum_{k=1}^{n}|x_k+y_k|^{(p-1)q}\right)^{\frac{1}{q}} + \left(\sum_{k=1}^{n}|y_k|^p\right)^{\frac{1}{p}}\left(\sum_{k=1}^{n}|x_k+y_k|^{(p-1)q}\right)^{\frac{1}{q}} \\
&= \left[\left(\sum_{k=1}^{n}|x_k|^p\right)^{\frac{1}{p}} + \left(\sum_{k=1}^{n}|y_k|^p\right)^{\frac{1}{p}}\right]\left(\sum_{k=1}^{n}|x_k+y_k|^p\right)^{\frac{p-1}{p}} \\
&= (\|x\|_p+\|y\|_p)\|x+y\|_p^{p-1}.
\end{aligned} \tag{3.15}$$

Thus, by dividing $\|x+y\|_p^{p-1}$ on both side of the inequality, we can derive that

$$\|x+y\|_p \leqslant \|x\|_p + \|y\|_p. \tag{3.16}$$

□

3.2.2 Banach space

Definition 3.10 (Banach space)**.**
A normed space X is said to be Banach if it is complete.

Properties 3.11. *$(l^p, ||\cdot||_p)$ is Banach for $1 \leqslant p < \infty$.*

Proof. Suppose $(X_n)_n$ is a Cauchy sequence, then

$$\forall \epsilon > 0, \exists N \in \mathbb{N},\ s.t.\ \forall n, m \geqslant N,\ ||X^n - X^m||_p < \epsilon.$$

Therefore,

$$||X^n - X^m||_p = \left(\sum_{k=1}^{r}|X_k^n - X_k^m|^p\right)^{\frac{1}{p}} < \epsilon.$$

Thus, $|X_k^n - X_k^m| < \epsilon$, which implies that X_k^n is a Cauchy sequence in complete set R. We suppose it converges to X_k^*. From the equation,

$$\left(\sum_{k=1}^{r}|X_k^n - X_k^m|^p\right)^{\frac{1}{p}} < \epsilon,$$

let $m \to \infty$, we have

$$\left(\sum_{k=1}^{r} |X_k^n - X_k^*|^p \right)^{\frac{1}{p}} < \epsilon,$$

which implies that X^n converges to X^*. And we can verify that $X^* \in l^p$ since

$$\sum_{k=1}^{r} |X_k^*|^p \leqslant \sum_{k=1}^{r} |X_k^n|^p + \sum_{k=1}^{r} |X_k^n - X_k^*|^p < \epsilon_0^p + \epsilon^p \ for \ some \ \epsilon_0.$$

□

Theorem 3.12 (subspace of a Banach space).
Suppose Y is a subspace of X and X is a Banach space, then the following two properties are equivalent:
i) $(Y, \|\cdot\|)$ is Banach;
ii) $(Y, \|\cdot\|)$ is closed.

Sketch of proof.

- Suppose property i) is true, then for a sequence $(x_n)_n \in Y$, it converges to a point $x \in Y$. Therefore, Y is closed.

- For a Cauchy sequence $(x_n)_n \in Y \subseteq X$, it converges to a point $x \in X$ since X is Banach. And Y is closed, so $x \in Y$.

□

Theorem 3.13 (infinite series).
The series $(S_n)_n$ is the sum of X_n in normed space X, say

$$S_n = \sum_{k=1}^{n} X_n. \tag{3.17}$$

Then if $\sum_{k=1}^{n} \|X_k\|$ exists, we have $\sum_{k=1}^{n} X_k$ converges and X is Banach.

Sketch of proof.

- To prove that $(S_n)_n = \sum_{k=1}^{n} X_k$ converges, we only need to verify that $(S_n)_n$ is Cauchy.

- To prove that X is Banach, we need to verify that Cauchy sequence $(X_n)_n \in X$ is convergent. Define the sequence X_k^n with $\|X_k^n - X^n\| \to \epsilon_k$ and $\epsilon_k = \frac{1}{2^k}$, we can set

$$S'_n = X_1^n + \sum_{k=2}^{n} (X_k^n - X_{k-1}^n), \tag{3.18}$$

which is definitely convergent since it is absolutely convergent. Also, note that $S'_n = X_n^n$ converges, then it is proved.

□

Definition 3.14 (Schauder basis)**.**
If any $x \in X$, we can find unique $(a_n)n$ such that

$$\|x - \sum_{k=1}^{n} a_k e_k\| \to 0, \tag{3.19}$$

then $(e_k)_k$ is defined to be Schauder basis of X.

Theorem 3.15 (completion)**.**
For a normed space X, there is a subspace $\hat{X}$ such that $\exists Y \subseteq X$, which is dense in $\hat{X}$, such that Y is isometric to X.

Definition 3.16 (proper subspace)**.**
We say Y is a proper subspace of X, which means that Y is a subspace of X and $Y \neq X$. Recall subspace must satisfy addition property and scalar multiplication property.

Lemma 3.17 (Riesz's Lemma)**.**
Suppose that Z is a normed space, and Y is a proper subspace of Z. Then there exists $z \in Z \backslash Y$ such that $\|z\| = 1$ and for any $\epsilon \in (0, 1)$, we have

$$\|z - y\| > \epsilon, \ \forall y \in Y. \tag{3.20}$$

Proof. For $v \in Z \backslash Y$, suppose that

$$d = \inf_{y \in Y} \|v - y\|. \tag{3.21}$$

Then $\exists\ y' \in Y$ such that

$$d \leqslant \|v - y'\| < \frac{d}{\epsilon} \tag{3.22}$$

Assume that $z = \frac{v - y'}{\|v - y'\|}$, then $\|z\| = 1$. Therefore, for any $u \in Y$, we have

$$\|z - u\| = \frac{1}{\|v - y'\|} \|v - (y' + u\|v - y'\|)\| \tag{3.23}$$

Note that $y' + u\|v - y'\| \in Y$(linear combination of y' and u), which implies that

$$\begin{aligned} \|z - u\| &= \frac{1}{\|v - y'\|} \|v - (y' + u\|v - y'\|)\| \\ &\geqslant \frac{d}{\|x - y\|} \geqslant \frac{d}{d/\epsilon} = \epsilon \end{aligned} \tag{3.24}$$

□

Theorem 3.18 (finite dimension)**.**
A normed space X is finite dimensional if and only if the closed ball $\bar{B}_x(0, 1)$ is compact.

Sketch of proof.

- Apply Heine-Borel Theorem, in finite dimension, closed and bounded set is compact.
- According to Riesz's lemma, we can construct a sequence $(e_n)_n$ such that $\|e_i\| = 1$ and $\|e_i - e_j\| > \frac{1}{2}$.

□

3.2.3 Linear Operator

Definition 3.19 (linear operator).
$T : X \to Y$ is a linear operator if it satisfies the following property:

$$T(\alpha x + \beta y) = \alpha Tx + \beta Ty. \tag{3.25}$$

Lemma 3.20 (linear combinations). *For linear independent vectors $\{x_1, x_2, \cdots, x_n\}$, $\exists c > 0$ such that*

$$\|\sum_{k=1}^{n} \alpha_k x_k\| \geqslant c \sum_{k=1}^{n} |\alpha_k|. \tag{3.26}$$

Definition 3.21 (boundedness of linear operators). *We say a linear operator $T : X \to Y$ is a bounded linear operator if there exists $c > 0$, we have the property that*

$$\|Tx\|_Y \leqslant c\|x\|_X \ \forall x \in X. \tag{3.27}$$

Theorem 3.22.
The linear operator $T : X \to Y$ is bounded if X is finite dimensional.

Proof. We would use the definition of bounded linear operators to verify the theorem, and we can simply use an inequality to prove it. Note that $x \in X$ could be denoted by a linear combination since X is finite. Then we have

$$x = \sum_{k=1}^{n} x_k e_k. \tag{3.28}$$

Therefore,

$$\begin{aligned} \|Tx\|_Y &= \|T \sum_{k=1}^{n} x_k e_k\|_Y \\ &\leqslant \sum_{k=1}^{n} |x_k| \|Te_k\|_Y \\ &\leqslant \max_k \|Te_k\|_Y \sum_{k=1}^{n} |x_k| \\ &\leqslant \left(\max_k \|Te_k\|_Y\right) \|x\|_X \leqslant c\|x\|_X \end{aligned} \tag{3.29}$$

□

Theorem 3.23 (continuity and boundedness). *For the linear operator $T : X \to Y$, we have the properties that*
i) If T is bounded, then it is continuous;
ii) if T is continuous, then it is bounded;
iii) if T is continuous at one point, then it is continuous.

Proof. i) Since T is bounded, for some $x, x_0 \in X$, the linear combination $x - x_0 \in X$, then it follows that

$$\|Tx - Tx_0\|_Y = \|T(x - x_0)\|_Y \leqslant c\|x - x_0\|_X. \tag{3.30}$$

So when $\|x - x_0\|_X < \frac{1}{c}\epsilon$,

$$\|Tx - Tx_0\|_Y < \epsilon, \tag{3.31}$$

which implies that T is continuous.

ii) The continuity shows that $\forall \epsilon > 0$, $\exists \delta > 0$, such that $\|x - x_0\|_X < \frac{1}{c}\epsilon$, if $\|Tx - Tx_0\|_Y < \epsilon$. Then we can define

$$x - x_0 = \frac{\delta}{\|y\|_X} y. \tag{3.32}$$

Then it follows that

$$\|Tx - Tx_0\|_Y = \|T\left(\frac{\delta}{\|y\|} y\right)\|_Y = \frac{\delta}{\|y\|_X}\|Yy\|_Y < \epsilon \tag{3.33}$$

iii) The continuity of T at a point illustrates that it is bounded, and the boundedness implies the continuity of T. □

Then we would introduce a norm corresponding to bounded linear operation and its properties, and this is quite significant to the establishment of Dual space.

Definition 3.24 (the norm of bounded linear functions)**.** *Consider linear and bounded operator T, since we have the property that*

$$\|Tx\| \leqslant c\|x\|, \tag{3.34}$$

then we can define

$$\|T\|_{\mathscr{B}} = \sup_{x \neq 0} \frac{\|Tx\|_Y}{\|x\|_X}. \tag{3.35}$$

Properties 3.25. *For the norm of bounded linear functions, it suggests that*

$$\|T\|_{\mathscr{B}} = \sup_{\|x\|_X = 1} \|Tx\|_Y \tag{3.36}$$

The equation above is a simplified form, but it is commonly employed in several proofs.

Definition 3.26. *We use the notation $\mathscr{B}(X, Y)$ to denote the set of all the linear and bounded operators $T : X \to Y$, and it satisfies the formula that*

$$\|T\|_{\mathscr{B}} = \sup_{x \neq 0} \frac{\|Tx\|_Y}{\|x\|_X} = \sup_{\|x\|_X = 1} \|Tx\|_Y. \tag{3.37}$$

Properties 3.27. *$\mathscr{B}(X, Y)$ is a Banach space if Y is a Banach space.*

Proof. In order to prove that $\mathscr{B}(X,Y)$ is a Banach space, we need to show that it is complete, namely, every Cauchy sequence would converges to a point in $\mathscr{B}(X,Y)$.

So, consider a Cauchy sequence $(T_n)_n$, then $\forall \epsilon > 0$, $\exists N > 0$, s.t. $\forall n, m > N$,

$$\|T_n - T_m\|_{\mathscr{B}} < \epsilon. \tag{3.38}$$

Since

$$\|T_n - T_m\|_{\mathscr{B}} = \sup_{x \neq 0} \frac{\|(T_n - T_m)x\|_Y}{\|x\|_X}, \tag{3.39}$$

then we have

$$\|(T_n - T_m)x\|_Y < \epsilon \|x\|_X. \tag{3.40}$$

It is noteworthy that $\epsilon\|x\|_X$ is constant, so the sequence $(T_n x)_n$ is also a Cauchy sequence. Since $T_n x = y_n \in Y$ and Y is Banach, then $T_n x$ converges to some point $Tx = y$. It follows that

$$T_n \to T \ as \ n \to \infty. \tag{3.41}$$

Then we still need to prove that $T \in \mathscr{B}$, which is equivalent to show that T is linear and bounded.

For $x, y \in X$, we have

$$\begin{aligned} T(\alpha x + y) &= \lim_n T_n(\alpha x + y) \\ &= \alpha \lim_n T_n x + \lim_n T_n y \\ &= \alpha T x + T y, \end{aligned} \tag{3.42}$$

which implies that T is linear operator.

From the equation that

$$\|(T_n - T_m)x\|_Y < \epsilon \|x\|_X, \tag{3.43}$$

by taking $m \to \infty$, it suggests that

$$\|T_n x - Tx\| < \epsilon \|x\|_X, \tag{3.44}$$

which implies that $T_n - T$ is bounded, and then $T = T_n - (T_n - T)$ should also be bounded.

The linear and boundedness shows that $T \in \mathscr{B}$. And since the arbitrary Cauchy sequence $(T_n)_n$ converges to $T \in \mathscr{B}$, we verified that $\mathscr{B}(X,Y)$ is a Banach space for Banach space Y. □

3.2.4 Dual Space

Definition 3.28 (dual space)**.** *A dual space is defined to be*

$$X^* = \mathscr{B}(X, R), \tag{3.45}$$

where X^* is called the dual space of X.

Properties 3.29. *The dual space of R^n should be R^n.*

Proof. Note that for a linear operator $T \in (R^n)^*$, it can be represented as

$$Tx = \sum_{k=1}^{n} x_k T e_k. \tag{3.46}$$

Therefore, according to Cauchy-Schwarz inequality, it could be derived that

$$\|Tx\|_1 \leqslant \sum_{k=1}^{n} |x_k T e_k| \leqslant \left(\sum_{k=1}^{n} x_k^2\right)^{\frac{1}{2}} \left(\sum_{k=1}^{n} (Te_k)^2\right)^{\frac{1}{2}} = \|x\|_2 \left(\sum_{k=1}^{n} (Te_k)^2\right)^{\frac{1}{2}}. \tag{3.47}$$

Since the norm of bounded and linear operator is defined as

$$\|T\|_{\mathscr{B}} = \sup_{\|x\|_2=1} \frac{\|Tx\|_1}{\|x\|_2} \tag{3.48}$$

And actually, note that

$$\|T\|_{\mathscr{B}} = \inf\{M > 0 : \|Tx\|_1 \leqslant M\|x\|_2\}, \tag{3.49}$$

which implies that

$$\|T\|_{\mathscr{B}} = \left(\sum_{k=1}^{n} (Te_k)^2\right)^{\frac{1}{2}}. \tag{3.50}$$

We have verified that $\mathscr{B}(R^n, R)$ is the Euclidean norm in R^n. Due to the equivalence of norms in finite dimensional spaces, we have $(R^n)^* = R^n$. $\square$

Properties 3.30. *The property of dual space says that*

$$(l^1)^* = l^\infty. \tag{3.51}$$

Proof. Consider a linear operator $T \in (l^1)^*$, then for some $x \in l^1$ with $\|x\| = \sum_{k=1}^{\infty} |x_k| < \infty$, it follows that

$$Tx = \sum_{k=1}^{\infty} x_k T e_k. \tag{3.52}$$

Then according to the definition of dual space, we have

$$|Tx| = |\sum_{k=1}^{\infty} x_k T e_k| \leqslant \max\{Te_k\} \sum_{k=1}^{\infty} |x_k| = \|x\|_1 \|Te\|_\infty. \tag{3.53}$$

And it is noteworthy that, with respect to any k,

$$\|T\|_{\mathscr{B}(l^1,R)} = \sup_{\|e\|_1=1} \|Te\|_\infty = \sup \|Te\|_\infty \|e\|_1 \geqslant |Te_k|, \tag{3.54}$$

which implies that $\|Te\|_\infty < \infty$, thus the boundedness of Tx could be verified and $T \in l^\infty$. $\square$

3.2.5 Inner Product Spaces

Definition 3.31 (inner products)**.** *Normally, the inner product on X is defined to be $\langle x, y\rangle$ for $x, y \in X$. For $x, y, z \in X$ and scalars α, the inner product must satisfies the following properties:*
i) Bilinearity property:

$$\langle \alpha x + y, z\rangle = \alpha\langle x, z\rangle + \langle y, z\rangle; \tag{3.55}$$

ii) Symmetry property:

$$\langle x, y\rangle = \langle y, x\rangle; \tag{3.56}$$

iii)Positive definiteness:

$$\langle z, z\rangle \geqslant 0, \tag{3.57}$$

where the equality holds only when $z = 0$.

Definition 3.32 (inner product space)**.** *The space $(X, \langle\cdot,\cdot\rangle)$ is called an inner product space. A norm in the space could be defined as*

$$\|x\| = \sqrt{\langle x, x\rangle}. \tag{3.58}$$

And then we have the property corresponding to the norm defined in the inner product space,

$$\|y - z\| = \sqrt{\langle y - z, y - z\rangle}. \tag{3.59}$$

Properties 3.33 (parallelogram equality)**.** *Consider an inner product space, there is a crucial equation in terms of the norm, and the equality is derived as*

$$\|x + y\|^2 + \|x - y\|^2 = 2(\|x\|^2 + \|y\|^2). \tag{3.60}$$

Proof. We can use the norm to verify the parallelogram equality, and it follows that

$$\begin{aligned} \|x + y\|^2 + \|x - y\|^2 &= \langle x + y, x + y\rangle + \langle x - y, x - y\rangle \\ &= \|x\|^2 + \|y\|^2 + 2\langle x, y\rangle + \|x\|^2 + \|y\|^2 - 2\langle x, y\rangle \\ &= 2(\|x\|^2 + \|y\|^2). \end{aligned} \tag{3.61}$$

□

Properties 3.34. *Two elements in the inner product space satisfies the property that*

$$\langle x, y\rangle = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2). \tag{3.62}$$

Definition 3.35 (orthogonal)**.** *Consider an inner product space $(X, \langle\cdot,\cdot\rangle)$, $x, y \in X$ are said to be orthogonal if their inner product is zero, namely, $\langle x, y\rangle = 0$.*

Theorem 3.36. *In the space $L^2[a, b]$, the inner product could be defined as*

$$\langle f, g\rangle = \int_a^b f(s)g(s)ds, \tag{3.63}$$

and $f, g \in L^2[a, b]$ are real functions.

Theorem 3.37. *In the space l^2, the inner product could be defined as*

$$\langle x, y\rangle = \sum_{i=1}^{\infty} x_n. \tag{3.64}$$

We would then introduce some inequalities to verify the validity of the norm.

Theorem 3.38 (Schwarz inequality)**.** *The inner product satisfies the Schwarz inequality that*

$$|\langle x, y\rangle| \leqslant \|x\|\|y\|. \tag{3.65}$$

Proof. For $\alpha \in R$, we could calculate the term $\langle x - \alpha y, x - \alpha y\rangle$, and it follows that

$$0 \leqslant \langle x - \alpha y, x - \alpha y\rangle = \|x\|^2 + \alpha^2\|y\|^2 - 2\alpha\langle x, y\rangle. \tag{3.66}$$

Suppose that α is a random variable, then the equality above is always true if and only if

$$4\alpha^2|\langle x, y\rangle|^2 \leqslant 4\|x\|^2\|y\|^2, \tag{3.67}$$

which can then be simplified into the Schwarz inequality that we would like to verify. □

Theorem 3.39 (triangle inequality of the norm under the inner product)**.** *The inner product satisfies the following property:*

$$\|x + y\| \leqslant \|x\| + \|y\|. \tag{3.68}$$

The inequality above is suggested to be the triangle inequality.

Proof. What we need to do is to rewrite the term $\|x + y\|$ into the form $\langle\cdot, \cdot\rangle$, it gives that

$$\begin{aligned} \|x + y\|^2 &= \langle x + y, x + y\rangle \\ &= \|x\|^2 + \|y\|^2 + 2\langle x, y\rangle, \end{aligned} \tag{3.69}$$

and then the Schwarz inequality could be applied to verify that $\|x + y\|^2 \leqslant (\|x\| + \|y\|)^2$. □

Theorem 3.40. *With the introduction of Schwarz inequality and triangle inequality, then we can show the validity of the norm that is defined under the inner product space.*

Proof. We have verified the triangle inequality, which is the one of the property that a norm have to satisfy, and we still need to prove the other two properties of a norm. For the first property, we have

$$\|x\| \geqslant 0, \tag{3.70}$$

since the norm is defined as the square root of the inner product. And when $\|x\| = 0$, the inner product $\langle x, x\rangle = 0$, which is equivalent to say that $x = 0 \in R^n$. Then for the second property, for $\alpha \in R$, the norm can be calculated as

$$\|\alpha x\|^2 = \langle \alpha x, \alpha x\rangle = \alpha^2\langle x, x\rangle = \alpha^2\|x\|^2. \tag{3.71}$$

With this properties, we can define the norm under the inner product. □

Theorem 3.41 (continuity of inner product)**.** *Consider an inner product space $(X, \langle\cdot,\cdot\rangle)$ and the sequences $(x_n)_n, (y_n)_n \in X$. Suppose $x_n \to x$ and $y_n \to y$, it follows that $\langle x_n, y_n\rangle \to \langle x, y\rangle$.*

Proof. Note that

$$\begin{aligned}|\langle x_n, y_n\rangle - \langle x, y\rangle| &= |\langle x_n, y_n\rangle - \langle x_n, y\rangle + \langle x_n, y\rangle - \langle x, y\rangle| \\ &\leqslant |\langle x_n, y_n - y\rangle| + |\langle x_n - x, y\rangle| \\ &\leqslant \|x_n\|\|y_n - y\| + \|x_n - x\|\|y\| \to 0.\end{aligned} \tag{3.72}$$

□

3.2.6 Orthogonal Projection and Separable Normed Space

Definition 3.42 (orthogonal complement)**.** *Let us consider a Hilbert space H, and a subspace $X \subseteq H$, then we can define the orthogonal complement with respect to the subspace X, denoted as $X^\perp$. Then we have*

$$X^\perp = \{p \in H : \langle p, x\rangle = 0, \forall x \in X\}. \tag{3.73}$$

Definition 3.43 (orthogonal projection)**.** *Then we define the projection from the Hilbert space H to the subspace X. Suppose the projection is denoted by P_Y, then it follows that*

$$\exists p = P_Y x, \forall x \in H. \tag{3.74}$$

Theorem 3.44. *Consider the subspace $X \subseteq H$ where H is a Hilbert space, the orthogonal complement $X^\perp$ is said to be a closed and linear space in the Hilbert space H.*

Proof. Note that we have introduced the orthogonal complement, that is

$$X^\perp = \{p \in H : \langle p, x\rangle = 0, \forall x \in X\}. \tag{3.75}$$

To prove that $X^\perp$ is linear, consider $p_1, p_2 \in X^\perp$, and this suggests that

$$\langle p_1, x\rangle = \langle p_2, x\rangle = 0, \forall x \in X. \tag{3.76}$$

For the vector $p' = \alpha p_1 + p_2$, the inner product satisfies the property that

$$\langle p', x\rangle = \alpha\langle p_1, x\rangle + \langle p_2, x\rangle = 0, \forall x \in X, \tag{3.77}$$

which implies that $p' \in X^\perp$, so $X^\perp$ is linear. To verify that the subspace is closed, define the sequence $(p_n)_n \in X^\perp$ such that $\langle p_n, x\rangle = 0$. Suppose that $p_n \to p$, and it implies that

$$\|p - p_n\|^2 = \langle p - p_n, p - p_n\rangle = 0. \tag{3.78}$$

Then, according to the linearity of inner product, we have $\langle p_n, x\rangle \to \langle p, x\rangle$, which shows that

$$\langle p, x\rangle = 0. \tag{3.79}$$

Thus, we verify that p is in the space $X^\perp$ and $X^\perp$ is closed. □

Definition 3.45 (Seperable).
We say a normed space X is seperable if it has a dense countable set.

Theorem 3.46. *Suppose that X is a normed space, then the following statement is equivalent:*
i) X is seperable;
ii) $\partial B(0,1)$ is seperable;
iii) X has a countable set $\{x_n\}_n$ such that

$$X = span\{x_1, x_2, \cdots, x_n, \cdots\}. \tag{3.80}$$

Theorem 3.47. *l^p is seperable for $1 \leqslant p < \infty$.*

Proof. It is equivalent to show that l^p has a countable set that spans the whole space. Take $(e^k)_k$ to be the set of unit vector such that the k-th element of $(e^k)_k$ is one and the other elements are zero, say

$$e^k = (0, \cdots, 0, 1, 0 \cdots). \tag{3.81}$$

Then we need to show that for any $x \in l^p$, it could be denoted by the linear combination of $(e^k)_k$. Suppose that $x = (x_1, x_2, \cdots) \in l^p$, define

$$x^n = \sum_{k=1}^{n} x_k e^k = (x_1, x_2, \cdots, x_n, 0, 0, \cdots). \tag{3.82}$$

Therefore, we have

$$\|x - x^n\|_p^p = \sum_{k=n+1}^{\infty} |x_k|^p. \tag{3.83}$$

Since $x \in l^p$, then $\|x\|_p^p = \sum_{k=1}^{\infty} |x_k|^p < \infty$. By taking $n \to \infty$, it follows that

$$\|x - x^n\|_p^p = \sum_{k=n+1}^{\infty} |x_k|^p \to 0. \tag{3.84}$$

Thus, x could be indicated by the linear combination of $(e^k)_k$ and l^p is seperable. □

Theorem 3.48. *It is notable that l^∞ is not seperable.*

Proof. To verify that l^∞ is not seperable, it is equivalent to show that any dense set $A \in l^\infty$, it is not countable. We can define an uncountable set B such that

$$B = \{x = (x_1, x_2, \cdots, x_n, \cdots) : x_k \in \{0, 1\}\}. \tag{3.85}$$

Then we need to show that there exists a surjection from B to A such that $\|x - x'\| < \epsilon$ for $x \in B$ and $x' \in A$. And note that we can verify that for any two points $x', y' \in A$, they are not the same. □

Definition 3.49 (orthogonal sets). *Suppose $Y \subseteq X$ where X is defined as an inner product space. The sequence $(x_n)_n \in X$ is called an orthogonal set of X if and only if for any x_i, x_j, it follows that*

$$\langle x_i, x_j \rangle = \begin{cases} 0, & i \neq j, \\ \alpha_j, & i = j. \end{cases} \tag{3.86}$$

And the set is called an orthonormal set when $\alpha_j = 1$ for any j.

Theorem 3.50. *For a finite orthogonal set, since we could not find a linear combination of one element with respect to the other elements, we say the elements in the orthogonal set should be linear independent.*

Theorem 3.51. *The sequence $\{e_k\}_{k=1}^{k=n}$ is an orthonormal set in X and suppose that $X = span\{e_1, \cdots, e_n\}$. Then for any elements in $x \in X$, it can be rewritten as a linear combination such that*

$$x = \sum_{i=1}^{n} \langle x, e_i \rangle e_i. \tag{3.87}$$

Proof. The property of span implies that $x \in X$ can be rewritten as

$$x = \sum_{i=1}^{n} \alpha_i e_i. \tag{3.88}$$

Then it can be calculated that $\langle x, e_i \rangle = \alpha_i$, which verifies the theorem. □

Theorem 3.52 (Bessel inequality). *Suppose that the inner product space X contains an orthonormal set $\{e_k\}_{k=1}^{k=n}$, then the Bessel inequality is derived as*

$$\sum_{k=1}^{n} |\langle x, e_k \rangle|^2 \leqslant \|x\|^2. \tag{3.89}$$

Proof. We would firstly calculate the square of the norm of x and the validity of the norm has been interpreted. Therefore,

$$\begin{aligned} \|x\|^2 &= \left\langle \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k, \sum_{k=1}^{\infty} \langle x, e_k \rangle e_k \right\rangle \\ &= \sum_{k=1}^{\infty} \sum_{j=1}^{\infty} \langle x, e_k \rangle \langle x, e_j \rangle e_k e_j \\ &= \sum_{k=1}^{\infty} \langle x, e_k \rangle^2 \geqslant \sum_{k=1}^{n} |\langle x, e_k \rangle|^2. \end{aligned} \tag{3.90}$$

Thus, we have verified the validity of Bessel inequality. □

3.2.7 Hilbert Space

Definition 3.53 (Hilbert Space).
Suppose that H is an inner product space, and H is complete, then H is a Hilbert space.

3.2.8 Introduction to Convex Analysis

Definition 3.54 (convex function).
The function $f : X \to R$ is convex, suppose that consider any $x_1, x_2 \in X$ and $t \in [0, 1]$,

$$tf(x_1) + (1 - t)f(x_2) \geqslant f(tx_1 + (1 - t)x_2). \tag{3.91}$$

Theorem 3.55 (Jensen's inequality).
Random variables λ_i satisfies $\sum_i \lambda_i = 1$. For convex function $f(x)$, we have

$$f\left(\sum_i \lambda_i x_i\right) \leqslant \sum_i \lambda_i f(x_i). \tag{3.92}$$

Theorem 3.56 (probability form of Jensen's inequality).
For convex function $f(x)$, we have

$$f(\mathbb{E}(x)) \leqslant \mathbb{E}(f(x)). \tag{3.93}$$

3.3 Optimization Theory

3.3.1 Unconstrained Optima

Definition 3.57 (local minimum and local maximum). *Consider the function $f : X \to R$ where $X \subset R^n$, and $y \in X$. Then we say*
i) y is a strict local minimum if $\exists \epsilon > 0$ such that

$$f(y) \leqslant f(x),\ \forall x \in X \cup B(y, \epsilon); \tag{3.94}$$

ii) y is a strict local maximum if $\exists \epsilon > 0$ such that

$$f(y) \geqslant f(x),\ \forall x \in X \cup B(y, \epsilon); \tag{3.95}$$

iii) y is an unconstrained local minimum if $\exists \epsilon > 0$ s.t. $B(y, \epsilon) \subseteq X$,

$$f(y) \leqslant f(x),\ \forall x \in B(y, \epsilon); \tag{3.96}$$

iv) y is an unconstrained local maximum if $\exists \epsilon > 0$ s.t. $B(y, \epsilon) \subseteq X$,

$$f(y) \geqslant f(x),\ \forall x \in B(y, \epsilon); \tag{3.97}$$

Definition 3.58 (continuity). *For the function $f : X \to R^m$ where $X \subset R^n$, we say the function f is differential at the point $x \in X$ if and only if $\forall \epsilon > 0$, $\exists \delta > 0$ s.t.*

$$\|f(y) - f(x) - A(y - x)\| < \epsilon \|y - x\|,\ \forall y \in B(x, \delta) \tag{3.98}$$

Here, note that $A \in R^{m \times n}$, $x \in R^n$ and $y \in R^m$.

Theorem 3.59. *According to the definition of differentiation, the $m \times n$ matrix A should be equal to the transpose of gradient of $f(x)$, that is*

$$A = \nabla f(x)^T. \tag{3.99}$$

And suppose $x = \{x_1, x_2, \cdots, x_n\} \in R^n$, then we have

$$A = \nabla f(x)^T = \left\{ \frac{\partial}{\partial x_1} f(x), \frac{\partial}{\partial x_2} f(x), \cdots, \frac{\partial}{\partial x_n} f(x) \right\}. \tag{3.100}$$

Theorem 3.60 (directional derivative)**.** *For the function $f : X \to R^m$ where $X \subset R^n$, the directional derivative of function f at the direction v is denotes by $f'(x, v)$, and the following formula holds that*

$$f'(x, v) = \nabla f(x)^T v \tag{3.101}$$

Proof. From the definition of differentiation, for some $y \in B(x, \delta)$, it follows that

$$\|f(y) - f(x) - \nabla f(x)^T (y - x)\| < \epsilon \|y - x\|, \tag{3.102}$$

Then we can set $y = x + tv$, therefore,

$$\|f(x + tv) - f(x) - \nabla f(x)^T tv\| < \epsilon t \|v\| \tag{3.103}$$

By dividing t on both sides of the equation and let $t \to 0$, we have

$$\begin{aligned} \left| f'(t, v) - \nabla f(x)^T v \right| &= \left| \lim_{t \to 0} \frac{\|f(x + tv) - f(x)\|}{t} - \nabla f(x)^T v \right| \\ &\leqslant \lim_{t \to 0} \frac{1}{t} \|f(x + tv) - f(x) - \nabla f(x)^T tv\| < \epsilon \|v\| \end{aligned} \tag{3.104}$$

Then let $\epsilon \to 0$, it gives that

$$f'(x, v) = \nabla f(x)^T v \tag{3.105}$$

□

Theorem 3.61. *Suppose $f : X \to R^m$ and $X \subset R^n$, for the point $x \in \int X$, it gives that $\nabla f(x) = 0$ if x is a local minimum or local maximum.*

Proof.

- Assume that x is a local minimum and $\nabla f(x) \neq 0$, taking $v = -\nabla f(x)$, then

$$f'(x, v) = \lim_{t \to 0} \frac{f(x + tv) - f(x)}{t} = \nabla f(x)^T v = -\|v\|^2 < 0, \tag{3.106}$$

  which implies that $f(x + tv) < f(x)$, contradictory to the assumption that x is a local minimum.

- Assume that x is a local maximum and $\nabla f(x) \neq 0$, taking $v = \nabla f(x)$, then

$$f'(x, v) = \lim_{t \to 0} \frac{f(x + tv) - f(x)}{t} = \nabla f(x)^T v = \|v\|^2 > 0, \tag{3.107}$$

  which implies that $f(x + tv) > f(x)$, contradictory to the assumption that x is a local maximum.

□

Definition 3.62 (Hessian matrix)**.** *The function $f : X \to R^m$ and $X \subset R^n$, so the Hessian matrix describes the second derivative of f, that is*

$$H(x) = \begin{bmatrix} f_{x_1x_1} & f_{x_1x_2} & \cdots & f_{x_1x_n} \\ f_{x_2x_1} & f_{x_2x_2} & \cdots & f_{x_2x_n} \\ \cdots & \cdots & \cdots & \cdots \\ f_{x_nx_1} & f_{x_nx_2} & \cdots & f_{x_nx_n} \end{bmatrix}. \tag{3.108}$$

Definition 3.63 (positive and negative semidefinte)**.** *A $n \times n$ matrix H is said to be*
i) positive semidefinite if and only if

$$v^T H v \geqslant 0; \tag{3.109}$$

ii) negative semidefinite if and only if

$$v^T H v \leqslant 0; \tag{3.110}$$

Theorem 3.64. *Suppose $f : X \subset R^2 \to R^n$ and $f \in C^2$, then we have the following properties*
i) If x is an local minimum, then $\nabla f(x) = 0$ and $H(x)$ is positive semidefinite;
ii) If x is an local maximum, then $\nabla f(x) = 0$ and $H(x)$ is negative semidefinite.

Proof. The first property would be verified. The Taylor expansion of $f(y)$ shows that

$$f(y) = f(x) + \nabla f(x)^T (y - x) + \frac{1}{2}(y - x)^T H(x)(y - x) + \epsilon\|y - x\|^2. \tag{3.111}$$

By taking $y = x + tv$ where $t \to 0$ and $\epsilon \to 0$, we could show that

$$\frac{1}{2}(y - x)^T H(x)(y - x) + \epsilon\|y - x\|^2 \geqslant 0. \tag{3.112}$$

Therefore, $v^T H(x)v \geqslant 0$ and $H(x)$ is positive semidefinite.

Similarly, we can prove the second property. □

3.3.2 Convex Sets

We mainly talk about the convex analysis in R^n.

Definition 3.65 (line segment).
Consider two points $x_1, x_2 \in R^n$, the line segment is a convex combination, say

$$[x_1, x_2] = \{\alpha x_1 + (1-\alpha)x_2 | \alpha \in [0,1]\}. \tag{3.113}$$

So the set $[x_1, x_2]$ is called the line segment.

Definition 3.66 (convex).
The set $S \subset R^n$ is convex if $\forall x_1, x_2 \in S$, we have

$$[x_1, x_2] \subset S. \tag{3.114}$$

Properties 3.67. *For convex sets S_1 and S_2 in R^n, the set*

$$S_+ = \{x_1 + x_2 : x_1 \in S_1, x_2 \in S_2\} \tag{3.115}$$

is a convex set.

Proof. Suppose $x_1, y_1 \in S_1$ and $x_2, y_2 \in S_2$, then $x_1 + x_2, y_1 + y_2 \in s_+$, the linear combination

$$\alpha(x_1 + x_2) + (1-\alpha)(y_1 + y_2) = \alpha x_1 + (1-\alpha)y_1 + \alpha x_2 + (1-\alpha)y_2 \tag{3.116}$$

Notice that $\alpha x_1 + (1-\alpha)y_1 \in S_1$ and $\alpha x_2 + (1-\alpha)y_2 \in S_2$, which implies that $\alpha(x_1 + x_2) + (1-\alpha)(y_1 + y_2) \in S_+$. □

Then we would introduce convex hull, which can be generated from any sets in R^n. A convex hull is convex.

Definition 3.68 (convex hull).
We suppose $S \subset R^n$ and $S \neq \emptyset$, the convex hull $conv(s)$ satisfies the property that

$$conv(s) = \{x : x = \sum_{j=1}^{k} \alpha_j x_j, \sum_{j=1}^{k} \alpha_j = 1, \alpha_j > 0\}. \tag{3.117}$$

Theorem 3.69 (Carathéodory Theorem).
Suppose $S \subset R^n$ and $x \in conv(s)$, then $\exists x_1, x_2, \cdots, x_{n+1}$ such that

$$x \in conv(x_1, x_2, \cdots, x_{n+1}). \tag{3.118}$$

Sketch of proof. From the definition of convex hull, we can write

$$x = \sum_{j=1}^{k} \alpha_j x_j. \tag{3.119}$$

And we only need to consider the case that $k > n+1$. However, more than k vectors must be linear dependent in R^n space, that is our starting point. □

Definition 3.70 (interior and closure point).
i) The vector $x \in X$ is a interior point of normed space $\|\cdot\|$ if and only if $\exists\ \epsilon > 0$ s.t.

$$B(x, \epsilon) = \{y : \|y - x\| < \epsilon\} \in X; \tag{3.120}$$

ii) The vector $x \in X$ is a closure point of normed space $\|\cdot\|$ if and only if $\forall\ \epsilon > 0$ s.t.

$$B(x, \epsilon) = \{y : \|y - x\| < \epsilon\} \cap X \neq \emptyset; \tag{3.121}$$

Definition 3.71 (interior, closure, boundary sets).
i) A collection of interior points of X should be the interior set, denoted by $int\ X$;
ii) A collection of closure points of X should be the closure set, denoted by $cl\ X$;
iii) The boundary set is the intersection of interior set and closure set, say $\partial X = int\ X \cup cl\ X$.

Properties 3.72. *We have the property to represent the relationship between interior set and closure set, that is*

$$int\ X \subseteq X \subseteq cl\ X = int\ X \cup\ \partial X \tag{3.122}$$

Properties 3.73. *If X is a convex set in R^n, then $cl\ X$ is also convex.*

Proof. Suppose $x, y \in\ cl\ X$, then according to the property of closure set, and take $\epsilon = \frac{1}{k}$

$$B(x, \frac{1}{k}) \cap X \neq \emptyset,\ B(y, \frac{1}{k}) \cap X \neq \emptyset \tag{3.123}$$

Then we set

$$x_k \in B(x, \frac{1}{k}) \cap X,\ y_k \in B(y, \frac{1}{k}) \cap X \tag{3.124}$$

The convexity of X says that $\alpha x_k + (1 - \alpha)y_k \in X$, and when $k \to \infty$, we have

$$\alpha x + (1 - \alpha)y \in X, \tag{3.125}$$

which verifies that $cl\ X$ is a convex set. □

Properties 3.74. *If X is a convex set in R^n, then $int\ X$ is also convex.*

Sketch of proof. To prove the property, it is sufficient to prove a more generalized property: set $x_1 \in int\ X$, $x_2 \in cl\ S$, then we have $(x_1, x_2) \subseteq int\ X$. To prove the property, we suppose $x_\alpha = \alpha x_1 + (1 - \alpha)x_2$ where $\alpha \in (0, 1)$. And it is equivalent to prove that $B(x_\alpha, \epsilon) \subseteq X$ for some $\epsilon > 0$. Fix $y_\alpha \in B(x_\alpha, \epsilon)$, then we need to show that y_α can be written as

$$y_\alpha = \alpha y_1 + (1 - \alpha)y_2,\ for\ some\ y_1, y_2 \in X. \tag{3.126}$$

Therefore, we can fix y_2 in the proximity of x_2, and then we only need to prove that $y_1 = \frac{y_\alpha - (1-\alpha)y_2}{\alpha}$ is in the proximity of x_1. So, since x_1 is an interior point, $\exists \epsilon_1$ such that $B(x_1, \epsilon_1) \subseteq X$. Then we can verify that

$$\|y_1 - x_1\| < \epsilon_1. \tag{3.127}$$

□

However, pay attention to the following statement, and it is prone to get confused. If $cl\ X$ is convex, $int\ X$ may not be convex; and if If $int\ X$ is convex, $cl\ X$ may not be convex. For the first statement, consider the set $X = (0, 1) \cup (1, 2) \in R$; for the second statement, consider the set $X = \{1\} \in R$.

3.3.3 Separation

Definition 3.75 (distance function). *Suppose C is a closed convex set, and $x \notin C$, then the distance function $d_C(x)$ is defined to be*

$$d_C(x) = \inf\{\|c - x\| : c \in C\} \tag{3.128}$$

Properties 3.76. *For closed and convex set C, we can find exactly one point c_x such that*

$$\|x - c_x\| = d_C(x), \tag{3.129}$$

and c_x is called the projection of x on C.

Theorem 3.77. *Suppose closed convex set $C \neq \emptyset$, then c_x is the projection of x on C if and only if*

$$(x - c_x)^T(c - c_x) \leqslant 0, \ \forall c \in C. \tag{3.130}$$

Proof.

- Suppose that c_x is the projection point, set $c' = c_x + \alpha(c - c_x)$, then according to the definition of projection, we have

$$\|x - c_x\| \leqslant \|x - c'\|. \tag{3.131}$$

  Then this suggests that

$$\|x - c_x\|^2 \leqslant \|x - c_x\|^2 - 2\alpha(x - c_x)^T(c - c_x) + \alpha^2\|c - c_x\|^2. \tag{3.132}$$

  By taking $\alpha \to 0$, it follows that

$$(x - c_x)^T(c - c_x) \leqslant 0, \ \forall c \in C \tag{3.133}$$

- Suppose $c \in C$, then we have

$$\|x - c\|^2 = \|x - c_x\|^2 + (x - c_x)^T(c - c_x) + \|c_x - c\|^2 \geqslant \|x - c_x\|^2, \tag{3.134}$$

  which implies that c_x is the projection.

□

Definition 3.78 (separation). *We say the hyperplane*

$$H_\alpha = \{y : p^T y = \alpha\} \tag{3.135}$$

separates the sets C_1 and C_2 if we have

$$p^T c_1 \leqslant \alpha \leqslant p^T c_2, \ \forall c_1 \in C_1, c_2 \in C_2. \tag{3.136}$$

Meanwhile, the two sets are said to be strictly separated if

$$p^T c_1 < \alpha < p^T c_2, \ \forall c_1 \in C_1, c_2 \in C_2; \tag{3.137}$$

and the two sets are said to be strongly separated if for some $\epsilon > 0$,

$$p^T c_1 \leqslant \alpha < \alpha + \epsilon \leqslant p^T c_2, \ \forall c_1 \in C_1, c_2 \in C_2. \tag{3.138}$$

3.3.4 Farkas's Theorem and Gordan's Theorem

Theorem 3.79 (Farkas's Theorem).
Suppose that $A \in R^{m\times n}$ and $c \in R^n$, then exactly one of the following systems has a solution:
***System 1**: $\exists x \in R^n$ such that $Ax \leqslant 0$ and $c^T x > 0$.*
***System 2**: $\exists y \in R^m$ such that $A^T y = c$ and $y \geqslant 0$.*

Theorem 3.80 (Gordan's Theorem).
Suppose that $A \in R^{m\times n}$, then exactly one of the following systems has a solution:
***System 1**: $\exists x \in R^n$ such that $Ax < 0$.*
***System 2**: $\exists y \in R^m$ such that $A^T y = 0$ with $y \geqslant 0$ and $y \neq 0$.*

3.3.5 Convex Cones and Extreme Direction

Definition 3.81 (convex cones).
The set $X \in R^n$ is defined to be a convex cone suppose that consider $x \in X$, we have $\lambda x \in X$, $\forall \lambda \geqslant 0$.

Definition 3.82 (polar cones).
X^* is defined to be the polar cone of X if

$$X^* = \{p \in R^n : p^T x \leqslant 0, \forall x \in X\}. \tag{3.139}$$

Properties 3.83. *For a nonempty set X, it satisfies the following properties that*
i) X^* is a convex cone;
ii) $X \subseteq X^{**}$;
iii) $X_1 \subseteq X_2$ implies that $X_2^* \subseteq X_1^*$.

Proof. i) Since $X^* = \{p \in R^n : p^T x \leqslant 0, \forall x \in X\}$, suppose that $p \in X^*$, then $p^T x \leqslant 0$. It follows that

$$\lambda p^T x \leqslant 0, \forall \lambda \geqslant 0. \tag{3.140}$$

So $\lambda x \in X^*$, which implies that X^* is a convex cone. Consider a sequence $(p_n)_n \in X^*$ convergent and $p_n \to p$, then we have

$$|p^T - p_n^T| < \epsilon, \forall \epsilon > 0. \tag{3.141}$$

Therefore, when $\epsilon \to 0$, $p^T x < p_n^T x + \epsilon x \leqslant \epsilon x$ and $p^T x \leqslant 0$.
ii) Note that

$$X^{**} = \{q \in R^n : q^T p \leqslant 0, \forall p \in S^*\}. \tag{3.142}$$

If $x \in X$, $p^T x \leqslant 0$, then $x^T p \leqslant 0$. Therefore, $x \in X^{**}$.
iii) If $p \in X_2^*$, we have $p^T x_2 \leqslant 0$ for all $x_2 \in X_2$. Since $X_1 \subseteq X_2$, then $p^T x_1 \leqslant 0$ for all $x_1 \in X_1$, then we finish our proof. □

Properties 3.84. *Suppose that $C \in R^n$ is a closed convex cone, then the property holds that $C = C^{**}$.*

Sketch of proof. We need to prove that $C^{**} \subseteq C$. We firstly suppose $\exists x \in C^{**}$ but $x \notin C$, then

$$p^T c \leqslant \alpha < p^T x, \forall c \in C. \tag{3.143}$$

Then $p^T c \leqslant \alpha$ and $p^T \lambda c \leqslant \alpha$, which shows the contradiction. □

Definition 3.85 (polyhedral sets)**.** *A set is a polyhedral set if it can be denoted as the intersection of closed half-spaces. Every half space has the form $\{x : p^T x \leqslant 0\}$. Normally, a polyhedral set is defined as*

$$S = \{x \in R^n : Ax \leqslant b\}. \tag{3.144}$$

Definition 3.86 (extreme point)**.** *If $x \in X$ and $x = \alpha x_1 + (1-\alpha)x_2$ where $\alpha \in (0,1)$ and $x_1, x_2 \in X$ shows that $x = x_1 = x_2$, then we say x is an extreme point.*

Definition 3.87. *For a closed and convex set $X \subseteq R^n$ and $d \in R^n$, if $x \in X$ shows that $x + \lambda d \in X$, $\forall \lambda \geqslant 0$, then d is called a direction.*

Properties 3.88. *For a polyhedral set $S = \{x \in R^n : Ax = b, x \geqslant 0\}$, $A \in R^{m\times n}$ with rank m. We have x is an extreme point if and only if A could be denoted as $[B, N]$ and $x = [x_B, x_N]^T = [B^{-1}b, 0]^T \geqslant 0$. And x is said to be a basic feasible solution of S.*

Sketch of proof.

- For an extreme point x, suppose

  $$x = (x_1, x_2, \cdots, x_k, 0, 0, \cdots, 0)^T. \tag{3.145}$$

  We can verify that the first K columns of A, say $A_1, A_2, \cdots, A_k$ are independent. If not, $\exists \lambda = (\lambda_1, \lambda_2, \cdots, \lambda_k, 0, \cdots, 0)$ such that $A\lambda = 0$. We take

  $$x_1 = x + \alpha\lambda,\ x_2 = x - \alpha\lambda. \tag{3.146}$$

  We can show that $x_1, x_2 \in S$ and $x = \frac{1}{2}(x_1 + x_2)$, which contradicts that x is an extreme point. Since A has rank m, so the set $\{A_1, \cdots, A_k, A_{k+1}, \cdots, A_m\}$ should be an independent set of A. Then we have

  $$B^{-1}b = (x_1, x_2, \cdots, x_k, 0, \cdots, 0)^T. \tag{3.147}$$

  Then we can verify the property.

- Suppose that A could be written as $[B, N]$ and $x = [x_B, x_N]^T = [B^{-1}b, 0]^T$. If we assume

  $$x = \alpha x_1 + (1-\alpha)x_2,\ \forall \alpha \in (0,1). \tag{3.148}$$

  We first shows the second terms of x_1 and x_2 are zero, then use the criterion that $x_1, x_2 \in S$. It implies that $x = x_1 = x_2 = [B^{-1}b, 0]^T$.

□

Theorem 3.89. *Suppose that $S = \{x \in R^n : Ax = b, x \geqslant 0\}$ is a polyhedral set. Then S has finite number of extreme points.*

Proof. According to Properties 3.88, the matrix A can be decomposed into $[B, N]$, if $A \in R^{m\times n}$ with rank m, then we can choose n columns from m columns, which is finite. □

Theorem 3.90. *Consider $S = \{x \in R^n : Ax = b, x \geqslant 0\} \neq \emptyset$, then it follows that S has at least one extreme point.*

Sketch of proof. Let us define $x = (x_1, x_2, \cdots, x_k, 0, 0, \cdots, 0)^T$ and $\exists\lambda = (\lambda_1, \lambda_2, \cdots, \lambda_k, 0, \cdots, 0)$. Then we take

$$\alpha = \min\left\{\frac{x_j}{\lambda_j} : \lambda_j > 0\right\}. \tag{3.149}$$

Then define

$$x'_j = x_j - \alpha\lambda_j, \tag{3.150}$$

and we can verify that x'_j is an extreme point. □

Properties 3.91. *Consider $S = \{x \in R^n : Ax = b, x \geqslant 0\} \neq \emptyset$, if d is a direction of S, then it follows that*

$$Ad = 0. \tag{3.151}$$

This can be proved by using the definition of direction.

Properties 3.92. *Let $S = \{x \in R^n : Ax = b, x \geqslant 0\} \neq \emptyset$, then d' is an extreme direction of S if and only if $A = [B, N]$ and $B^{-1}N_j \leqslant 0$ where N_j is the j-th column of N, then*

$$d' = \lambda d = \lambda[-B^{-1}N_j, e_j]^T. \tag{3.152}$$

Sketch of proof.

- Since d' is an extreme direction, we can suppose that

$$d' = (d'_1, \cdots, d'_k, 0, \cdots, d'_j, 0, \cdots, 0). \tag{3.153}$$

  To prove that $A_1, A_2, \cdots, A_k$ are independent.We can prove by contradiction. Define

$$\lambda = (\lambda_1, \cdots, \lambda_k, 0, \cdots, 0). \tag{3.154}$$

  And for $d_1 = d' + \alpha\lambda$ and $d_2 = d' - \alpha\lambda$. We can show that d_1 and d_2 are directions, $d' = \frac{1}{2}(d_1 + d_2)$, which is contradict. Then for $B = [A_1, \cdots, A_m]$, we have

$$Ad' = BD_m + d'_j e_j = 0. \tag{3.155}$$

  Then we write $d' = d'_j[-B^{-1}N_j, e_j]^T$.

- Suppose that $A = [B, N]$ and $B^{-1}N_j \leqslant 0$ where N_j is the j-th column of N, then
$$d' = \lambda d = \lambda[-B^{-1}N_j, e_j]^T. \tag{3.156}$$
Then we have
$$Ad = [B, N][-B^{-1}N_j, e_j]^T = 0, \tag{3.157}$$
so d is a direction. Then we need to show that d is an extreme direction. Suppose that $d = \lambda_1 d_1 + \lambda_2 d_2$, where
$$d_1 = [d_{11}, \alpha_1 e_j]^T, \tag{3.158}$$
and similar to d_2. Then it can be calculated that
$$d_1 = \alpha_1[-B^{-1}N_j, e_j]^T,\ d_2 = \alpha_2[-B^{-1}N_j, e_j]^T \tag{3.159}$$
This implies that d is extreme direction since $d_1 = \lambda d_2$.

□

Properties 3.93. *For $S = \{x \in R^n : Ax = b, x \geqslant 0\}$, we say S has finite number of extreme directions.*

Proof. We can choose n from m to determine the number of possible $-B^{-1}N_j$ and N_j has $n - m$ choices. □

3.3.6 Convex Functions

Definition 3.94 (convex functions)**.** *Suppose that C is a convex set and $f : C \to R$. The function f is convex on C if and only if*
$$f(\lambda x_1 + (1-\lambda)x_2) \leqslant \lambda f(x_1) + (1-\lambda) f(x_2),\ \forall x_1, x_2 \in C, \forall \lambda \in (0,1). \tag{3.160}$$

Definition 3.95 (convex functions)**.** *Suppose that C is a convex set and $f : C \to R$. The function f is said to be strictly convex on C if and only if*
$$f(\lambda x_1 + (1-\lambda)x_2) < \lambda f(x_1) + (1-\lambda) f(x_2),\ \forall x_1, x_2 \in C, \forall \lambda \in (0,1). \tag{3.161}$$

Definition 3.96. *The function $f : C \to R$ is concave if $-f$ is convex on C.*

Definition 3.97. *The function $f : C \to R$ is affine if it is both convex and concave.*

Properties 3.98. *Suppose that C is a convex set and $f : C \to R$, then the following properties holds that:*
i) We have
$$f(\alpha_1 x_1 + \cdots + \alpha_k x_k) \leqslant \alpha_1 f(x_1) + \cdots + \alpha_k f(x_k) \text{ with } \alpha_j \geqslant 0, \sum_{j=1}^{k} \alpha_j = 1; \tag{3.162}$$
ii) Suppose that g is a non-decreasing convex function, then $g(f(x))$ is convex;
iii) Suppose that g is an affine function, then $f(g(x))$ is convex;
iv) Suppose that $g \neq 0$ is a concave function, then $\frac{1}{g(x)}$ is a convex function.

Definition 3.99 (level set)**.** *The level set is defined as*

$$S_\alpha = \{x \in C : f(x) \leqslant \alpha\}, \tag{3.163}$$

or

$$S_\alpha = \{x \in C : f(x) \geqslant \alpha\}. \tag{3.164}$$

Properties 3.100. *Suppose $f : C \to R$ is a convex function, then each level set is a convex set.*

Proof. Consider two points $x_1, x_2 \in S_\alpha$, then we can derive that

$$f(\lambda x_1 + (1-\lambda)x_2) \leqslant \lambda f(x_1) + (1-\lambda) f(x_2) \leqslant \lambda\alpha + (1-\lambda)\alpha = \alpha, \tag{3.165}$$

which implies that $\lambda x_1 + (1-\lambda)x_2$ is a point in S_α. □

Properties 3.101. *Suppose that $f : C \to R$ is convex and C is convex and finite. Then we can verify that f is a continuous function on int C.*

Proof. Say $x' \in C$ and $x \in R^n$, then the increment could be calculated by

$$x - x' = \sum_{i=1}^{n} (x_i - x_i')e_i, \tag{3.166}$$

then assume

$$\|x - x'\|_1 \leqslant \delta = \frac{\delta'}{n} \min\left\{1, \frac{\epsilon}{\theta+1}\right\}, \tag{3.167}$$

where

$$\theta = \max\{\max\{f(x' + \delta' e_i) - f(x'), f(x' - \delta' e_i) - f(x')\} : i = 1, \cdots, n\}. \tag{3.168}$$

Therefore,

$$\begin{aligned} |x - x'| \leqslant \sum_{i=1}^{n} |(x_i - x_i')e_i| &= \sum_{i=1}^{n} \frac{|x_i - x_i'|}{\delta'} |sgn(x_i - x_i')\delta' e_i| \\ &= \sum_{i=1}^{n} \alpha_i |sgn(x_i - x_i')\delta' e_i|, \end{aligned} \tag{3.169}$$

where $\alpha_i = \frac{|x_i - x_i'|}{\delta'}$. And we note that

$$\|x - x'\| = \delta' \sqrt{\sum_{i=1}^{n} \alpha_i^2} \leqslant \delta = \frac{\delta'}{n} \min\left\{1, \frac{\epsilon}{\theta+1}\right\}, \tag{3.170}$$

which shows that

$$\alpha_i \leqslant \sqrt{\sum_{i=1}^{n} \alpha_i^2} \leqslant \frac{1}{n} \min\left\{1, \frac{\epsilon}{\theta+1}\right\}. \tag{3.171}$$

so it follows that

$$|x - x'| \leqslant n\frac{1}{n}\min\left\{1, \frac{\epsilon}{\theta+1}\right\}\delta' = n\delta. \tag{3.172}$$

Then it follows that

$$\begin{aligned} f(x) &= f\left(x' + \sum_{i=1}^{n}\alpha_i sgn(x_i - x_i')\delta' e_i\right) \\ &= f\left(\sum_{i=1}^{n}\frac{1}{n}(x' + n\alpha_i sgn(x_i - x_i')\delta' e_i)\right) \leqslant \frac{1}{n}\sum_{i=1}^{n} f(x' + n\alpha_i sgn(x_i - x_i')\delta' e_i) \\ &\leqslant \frac{1}{n}\sum_{i=1}^{n}[(1 - n\alpha_i)f(x') + n\alpha_i f(x' + sgn(x_i - x_i')\delta' e_i)]. \end{aligned} \tag{3.173}$$

Therefore,

$$f(x) - f(x') \leqslant \theta\sum_{i=1}^{n}\alpha_i \leqslant \epsilon. \tag{3.174}$$

We can also verify that $f(x) - f(x') \geqslant -\epsilon$ by setting $y = 2x' - x$. □

Properties 3.102 (Another statement of Properties 3.101)**.**
Suppose $f : C \to R$ is a convex function and C is convex. Then f is continuous when C is open; f is continuous on interior points when C is closed.

Proof. We prove the property when C is open, consider a sequence $(x_n)_n \to x \in C$, then $\forall \epsilon > 0$, there $\exists K \in N$ such that $|x_k - x| < \epsilon$ for any $k \geqslant K$. Suppose that $x_k = \alpha x + (1-\alpha)y$ for some $y \in C$. According to the convexity of C, we have

$$f(x_k) = f(\alpha x + (1-\alpha)y) \leqslant \alpha f(x) + (1-\alpha)f(y). \tag{3.175}$$

Then we let $k \to \infty$ and $\alpha \to 1$, it follows that

$$\lim_{k\to\infty} f(x_k) \leqslant f(x). \tag{3.176}$$

On the other hand, set $x = \beta x_k + (1-\beta)z$, then

$$f(x) = f(\beta x_k + (1-\beta)z) \leqslant \beta f(x_k) + (1-\beta)f(z). \tag{3.177}$$

Again, let $k \to \infty$ and $\beta \to 1$, we have

$$f(x) \leqslant \lim_{k\to\infty} f(x_k). \tag{3.178}$$

Then $\lim_{k\to\infty} f(x_k) = f(x)$, which implies that $f(x)$ is continuous. □

Theorem 3.103. *Suppose f is convex defined on a convex set $C \subseteq R$, thus, for $x_1 < x_2 < x_3$, it follows that*

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} < \frac{f(x_3) - f(x_1)}{x_3 - x_1} < \frac{f(x_3) - f(x_2)}{x_3 - x_2}. \tag{3.179}$$

Proof. To leverage the properties of convexity, we need to rewrite x_2. Suppose that $x_2 = x_1 + \lambda_1$ and $x_3 = x_1 + \lambda_2$ where $\lambda_2 > \lambda_1 > 0$, then

$$x_2 = x_1 + \lambda_1 = \frac{\lambda_1}{\lambda_2}(x_1 + \lambda_2) + (1 - \frac{\lambda_1}{\lambda_2})x_1 = \frac{\lambda_1}{\lambda_2}x_3 + (1 - \frac{\lambda_1}{\lambda_2})x_1. \tag{3.180}$$

According to the convexity, we can derive that

$$f(x_2) = f(\frac{\lambda_1}{\lambda_2}x_3 + (1 - \frac{\lambda_1}{\lambda_2})x_1) \leqslant \frac{\lambda_1}{\lambda_2}f(x_3) + (1 - \frac{\lambda_1}{\lambda_2})f(x_1) \tag{3.181}$$

It is equivalent to say that

$$\frac{f(x_2) - f(x_1)}{\lambda_1} < \frac{f(x_3) - f(x_1)}{\lambda_2}. \tag{3.182}$$

We can apply the similar ideas to verify other inequalities. □

Theorem 3.104. *Let f be a convex and finite function, then the directional derivative $f'(x, d)$ exists where d is a direction of f.*

Sketch of proof. Suppose $\lambda_2 > \lambda_1 > 0$. We could firstly leverage the equation that

$$x + \lambda_1 d = \frac{\lambda_1}{\lambda_2}(x + \lambda_2 d) + \left(1 - \frac{\lambda_1}{\lambda_2}\right)x, \tag{3.183}$$

to verify the inequality

$$\frac{f(x + \lambda_1 d) - f(x)}{\lambda_1} \leqslant \frac{f(x + \lambda_2 d) - f(x)}{\lambda_2}. \tag{3.184}$$

This implies that $\frac{f(x+\lambda d)-f(x)}{\lambda}$ should be decreasing when $\lambda \to 0$. Then we apply the formula

$$x = \frac{\lambda}{1+\lambda}(x - d) + \frac{1}{1+\lambda}(x + \lambda d), \tag{3.185}$$

which shows

$$f(x) - f(x - d) \leqslant \frac{f(x + \lambda d) - f(x)}{\lambda}. \tag{3.186}$$

Hence, $\frac{f(x+\lambda d)-f(x)}{\lambda}$ is bounded below, which implies that

$$f'(x, d) = \lim_{\lambda \to 0^+} \frac{f(x + \lambda d) - f(x)}{\lambda} = \inf \left\{ \frac{f(x + \lambda d) - f(x)}{\lambda} \right\}. \tag{3.187}$$

□

3.3.7 Subgradients, Supporting Hyperplanes to Epigraphs and Hypographs

Definition 3.105 (graph)**.** *The graph of a function is defined as*

$$G_f = \{(x, f(x)) : f : C \subseteq R^n \to R\}. \tag{3.188}$$

Definition 3.106 (epigraph and hypograph)**.** *Consider the function $f : C \to R$, then the epigraph is defined as*

$$epi\ f = \{(x, y) : f(x) \leqslant y\}, \tag{3.189}$$

and similarly, the hypograph is denoted as

$$hyp\ f = \{(x, y) : f(x) \geqslant y\}, \tag{3.190}$$

It is noteworthy that the epigraph is the upper area of function f, while hypograph is the lower area of function f.

Theorem 3.107. *Suppose that $f : C \to R$ and C is convex. We say f is a convex function when $epi\ f$ is convex and the inverse statement is also true.*

Proof.

- Consider two points $(x_1, y_1) \in epi\ f$ and $(x_2, y_2) \in epi\ f$, then $f(x_1) \leqslant y_1$ and $f(x_2) \leqslant y_2$. The convexity of f verifies that

  $$f(\alpha x_1 + (1-\alpha)x_2) \leqslant \alpha f(x_1) + (1-\alpha) f(x_2) \leqslant \alpha y_1 + (1-\alpha) y_2. \tag{3.191}$$

  Therefore,

  $$\alpha(x_1, y_1) + (1-\alpha)(x_2, y_2) = (\alpha x_1 + (1-\alpha)x_2, \alpha y_1 + (1-\alpha)y_2) \in epi\ f. \tag{3.192}$$

- Let $epi\ f$ be convex, so consider two points $(x_1, f(x_1)) \in epi\ f$ and $(x_2, f(x_2)) \in epi\ f$. We have

  $$(\alpha x_1 + (1-\alpha)x_2, \alpha f(x_1) + (1-\alpha) f(x_2)) = \alpha(x_1, f(x_1)) + (1-\alpha)(x_2, f(x_2)) \in epi\ f. \tag{3.193}$$

  By the definition of epigraphs, we derive that

  $$f(\alpha x_1 + (1-\alpha)x_2) \leqslant \alpha f(x_1) + (1-\alpha) f(x_2). \tag{3.194}$$

□

Theorem 3.108. *A function f is concave if and only if $hyp\ f$ is convex.*

Definition 3.109 (subgradient)**.** *Consider the convex function f defined on a convex set C, we say ξ is a subgradient of f at x' if and only if*

$$f(x) - f(x') \geqslant \xi^T (x - x'), \forall x \in C. \tag{3.195}$$

As for concave functions, the formula becomes

$$f(x) - f(x') \leqslant \xi^T (x - x'), \forall x \in C. \tag{3.196}$$

Definition 3.110 (subdifferential)**.** *The set of subgradients is denoted by $\partial f(x)$, which is called the subdifferential. We write*

$$\partial f(x) = \left\{ \xi : f(x) - f(x') \geqslant \xi^T(x - x'), \forall x \in C \right\}. \tag{3.197}$$

Properties 3.111. *The function $f : C \to R$ is convex where C is convex. Suppose that $x' \in int\ C$, then there exists at least one $\xi \in R^n$ that is the subgradient of f at x'.*

Proof. For a point $(x', f(x'))$ on the boundary of *epi* f, there must exists a hyperplane such that

$$(\xi_0^T, \mu)[(x, y) - (x', f(x'))] \leqslant 0, \forall (x, y) \in epi\ f, \tag{3.198}$$

where $(\xi_0, \mu) \neq 0$. We can verify that $\mu < 0$, otherwise ξ_0 would also be zero which is contradictory. Assume $\xi = \xi_0/|\mu|$, then we have

$$\xi^T(x - x') + f(x') - y \leqslant 0. \tag{3.199}$$

There exists ξ s.t.

$$f(x) \geqslant f(x') + \xi^T(x - x'). \tag{3.200}$$

□

Theorem 3.112. *Assume $f : C \to R$ is convex where the set C is convex. Suppose that f is differentiable at x', then there exists unique ξ such that*

$$f(x) \leqslant f(x') + \xi(x - x'). \tag{3.201}$$

Proof. According to Properties 3.111, there must exists ξ such that

$$f(x) \geqslant f(x') + \xi(x - x'). \tag{3.202}$$

Take $x = x' + \lambda d$, then

$$f(x' + \lambda d) = f(x') + \lambda \nabla f(x)^T d + \|x - x'\| \alpha(x, x') \geqslant f(x') + \lambda \xi^T d \tag{3.203}$$

Then it implies that

$$(\nabla f(x)^T - \xi^T)d + \|d\| \alpha(x, x') \geqslant 0. \tag{3.204}$$

Suppose $d = \nabla f(x) - \xi$, then $\xi = \nabla f(x)$. □

Theorem 3.113. *It suggests that the differentiable function f is convex if and only if*

$$f(x) \geqslant f(x') + \nabla f(x')^T(x - x'). \tag{3.205}$$

Properties 3.114. *Assume $f : C \to R$ is convex where the set C is convex. The the following statements should be equivalent:*
i) the function f is convex;
ii) for any $x_1, x_2 \in C$, we have

$$\left[\nabla f(x_1) - \nabla f(x_2)\right]^T (x_1 - x_2) \geqslant 0. \tag{3.206}$$

Sketch of proof.

- Substituting x_1 and x_2 into the formula for differentiable functions.
- By mean value theorem, $\exists x = \lambda x_1 + (1-\lambda)x_2$ for $\lambda \in (0,1)$, then
$$f(x_2) - f(x_1) = \nabla f(x)^T(x_2 - x_1). \tag{3.207}$$
Then
$$\nabla f(x)^T(x_2 - x_1) \geqslant \nabla f(x_1)^T(x_2 - x_1). \tag{3.208}$$

□

3.3.8 Optima of Convex Functions

Theorem 3.115. *Assume f is a convex function on a convex set C, then the minimization problem of the convex function could be denoted by*

$$\min\{f(x) : x \in C\}. \tag{3.209}$$

The the following properties holds true that
i) A local minimum solution is equivalent to a global minimum solution.
ii) A strict local minimum solution should be the unique minimum solution.

Sketch of proof. i) We can use contradiction to verify the property. By the definition of local minimum x', it suggests that $f(x') \leqslant f(x), \forall x \in C$. Now suppose $\exists y \in C$ s.t.

$$f(y) \leqslant f(x'). \tag{3.210}$$

Then let us consider the points between y and x', say $\lambda y + (1-\lambda)x'$, we have

$$f(\lambda y + (1-\lambda)x') \leqslant \lambda f(y) + (1-\lambda)x' \leqslant f(x'). \tag{3.211}$$

Take $\lambda \to 0^+$, it would contradict to the definition of locla minimum.
ii) Again, suppose $f(y) = f(x')$ for some $y \in C$. Consider the middle point, and the convexity of C would give a contradiction. □

Theorem 3.116. *The function f is a convex function on a convex set C. Then x' is a minimum solution for* $\min\{f(x) : x \in C\}$ *if and only if*

$$\xi^T(x - x') \geqslant 0, \tag{3.212}$$

where ξ is the subgradient.

3.4 Probability and Statistics

3.4.1 Martingale, Stopping Time and Dood's Optional Stopping Theorem

Definition 3.117 (martingale).
Endow $(\Omega, \mathscr{F}, \mathbb{P})$ with the meaning of probability space, and $\mathscr{F}_n$ epitomizes the series of σ-algebra satisfies $\mathscr{F}_n \subset \mathscr{F}_{n+1}$. For a stochastic process X_n, we say X_n should be a martingale if and conversely, X_n satisfies
i) $\mathbb{E}|X_n| < \infty$;
ii) X_n is $\mathscr{F}_n$ measurable, or we say adapted;
iii) $\mathbb{E}[X_{n+1}|\mathscr{F}_n] = X_n$.

Definition 3.118 (supermartingale and submartingale).
i) X_n constitutes a supermartingale when

$$\mathbb{E}[X_{n+1}|\mathscr{F}_n] \leqslant X_n. \tag{3.213}$$

ii) X_n constitutes a submartingale when

$$\mathbb{E}[X_{n+1}|\mathscr{F}_n] \geqslant X_n. \tag{3.214}$$

Theorem 3.119. *i) Suppose X_n is a martingale, then for $m > n$, $\mathbb{E}[X_m|\mathscr{F}_n] = X_n$;*
ii) Suppose X_n is a supermartingale, then for $m > n$, $\mathbb{E}[X_m|\mathscr{F}_n] \leqslant X_n$;
iii) Suppose X_n is a submartingale, then for $m > n$, $\mathbb{E}[X_m|\mathscr{F}_n] \geqslant X_n$.

Sketch of proof.
i) The theorem holds true for $m = n+1$. And for $m > n+1$, we have the property that

$$\mathbb{E}[X_m|\mathscr{F}_n] = \mathbb{E}\left[\mathbb{E}[X_m|\mathscr{F}_{m-1}]|\mathscr{F}_n\right] = \mathbb{E}[X_{m-1}|X_n]. \tag{3.215}$$

Then we can use induction to prove the theorem, similar to ii) and iii). □

Theorem 3.120. *$\{X_n|\mathscr{F}_n\}$ is a martingale, and convex function $f(x)$ satisfies $\mathbb{E}|f(x)| =$ constant, then we have $\{f(X_n)|\mathscr{F}_n\}$ is a submartingale.*

Proof. According to Jensen's inequality, we have

$$\mathbb{E}[f(X_{n+1}|\mathscr{F}_n)] \geqslant f(\mathbb{E}[X_{n+1}|\mathscr{F}_n]) = f(X_n). \tag{3.216}$$

It implies that $\{f(X_n)|\mathscr{F}_n\}$ is a submartingale. □

Definition 3.121 (stopping time).
Suppose τ belongs to the closed set of T, namely, $\tau \in \overline{T}$. We say τ is a stopping time if and only if for any $t \in T$, we have

$$\{\tau \leqslant t\} \in \mathscr{F}_t. \tag{3.217}$$

The condition means that for any time t, what happened before and at time t is known and $\mathscr{F}_t$ measurable.

Actually, if T is a finite number, we can write $\{\tau = t\} \in \mathscr{F}_t$. The crucial thing is that we know what happens immediately after time t if we have acquired all the information before time t.

Properties 3.122 (properties of stopping time)**.**
Suppose τ *,* $u \in \overline{T}$ *are stopping time, then*
i) $\tau + u$ *is stopping time;*
ii) if $A \in \mathscr{F}_\tau$*, then* $A \cap \{\tau \leqslant u\} \in \mathscr{F}_u$*;*
iii) if $\tau \leqslant u$*, then* $\mathscr{F}_\tau \subset \mathscr{F}_u$*.*

Proof. i) $\{\tau + u = T\} = \sum_{i \leqslant T} \{\tau = i\}\{u = T - i\} \in \mathscr{F}_T$.
ii) $A \in \mathscr{F}_\tau \subset \mathscr{F}_u$, so $A \in \mathscr{F}_u$. And $\{\tau \leqslant u\} \in \mathscr{F}_u$. Thus, $A \cap \{\tau \leqslant u\} \in \mathscr{F}_u$.
iii) From ii), for $\forall A \in F_\tau$, we have

$$A \subset A \cap \{\tau \leqslant u\} \in \mathscr{F}_u. \tag{3.218}$$

So, $A \cap \{\tau \leqslant u\} \in \mathscr{F}_u$. □

3.4.2 Cross Variation and Bracket Process

Theorem 3.123 (cross variation)**.** *Consider the stochastic functions* X *and* Y*, then the cross variation is specified as*

$$[X, Y](T) = \lim_{\|\Pi\| \to 0} \sum_{j=0}^{n-1} \left(X(t_{j+1}) - X(t_j)\right)\left(Y(t_{j+1}) - Y(t_j)\right), \tag{3.219}$$

where Π *denotes a partition of* $[a, b]$*.*

Theorem 3.124. *For the set defined by*

$$\mathcal{N} = \{f : R \to R \mid f \textit{ is nowhere differentiable or } f = 0\}, \tag{3.220}$$

it could be verified that the set is a vector space.

Proof. Consider the functions f, g in the set, when $f = g = 0$, $\alpha f + g = 0$; when f is nowhere differentiable and $g = 0$, $\alpha f + g = \alpha f$ is nowhere differentiable; when $f = 0$ and g is nowhere differentiable, $\alpha f + g = g$ is nowhere differentiable; when f, g are both nowhere differentiable, $\alpha f + g$ should also be nowhere differentiable. Thus, we have verified the validity of the set as a vector space. □

Properties 3.125. *The cross variation of two stochastic functions in the subspace* $\mathcal{N} = \{f : f \textit{ is nowhere differentiable or } f = 0\}$ *is an inner product space. For* $\alpha \in R$ *and* X, Y, Z *are stochastic processes, we have following properties.*
i) $[\alpha X + Y, Z] = \alpha[X, Z] + [Y, Z]$*;*
ii) $[X, Y] = [Y, X]$*;*
iii)$[X, X] \geqslant 0$ *and* $[X, X] = 0$ *if and only if* $X = 0$*.*

Proof. i) Presume the partition $\|\Pi\| \to 0$, hence,

$$
\begin{aligned}
[\alpha X+Y, Z] &= \sum_{j=0}^{n-1}\left(\alpha X(t_{j+1})+Y(t_{j+1})-\alpha X(t_j)-Y(t_j)\right)\left(Z(t_{j+1})-Z(t_j)\right) \\
&= \alpha \sum_{j=0}^{n-1}\left(X(t_{j+1})-X(t_j)\right)\left(Z(t_{j+1})-Z(t_j)\right) \\
&\quad + \sum_{j=0}^{n-1}\left(Y(t_{j+1})-Y(t_j)\right)\left(Z(t_{j+1})-Z(t_j)\right) \\
&= \alpha[X, Z]+[Y, Z].
\end{aligned} \tag{3.221}
$$

ii) Change in the order of the stochastic functions would not affect the value of cross variation.
iii) $[X, X]$ is the sum of quadratic values, which should be greater than zero; when $[X, X] = 0$, it follows that $X(t_{j+1}) = X(t_j)$ for all j, implying that X is a constant, and the only constant function in the set we have defined should be $X = 0$. □

Theorem 3.126 (bracket process)**.**
For $X, Y \in \mathcal{N}$, we can calculate cross variation between X and Y by calculating quadratic variation of X and Y respectively. And the equation holds that

$$
[X, Y](t) = \frac{1}{4}\Big\{[X+Y, X+Y](t)-[X-Y, X-Y](t)\Big\}. \tag{3.222}
$$

Proof. Since $(\mathcal{N}, [\cdot,\cdot])$ is an inner product space, according to Properties 3.34, it emerges that

$$
\begin{aligned}
[X, Y](t) &= \frac{1}{4}\Big\{\|x+y\|^2-\|x-y\|^2\Big\} \\
&= \frac{1}{4}\Big\{[X+Y, X+Y](t)-[X-Y, X-Y](t)\Big\}.
\end{aligned} \tag{3.223}
$$

□

Theorem 3.127 (Kunita-Watanabe inequity)**.**
For $M, N \in \mathcal{N}$ and X, Y are stochastic processes,

$$
\int_0^t [X, X](s)\cdot[Y, Y](s) d[M, N](s) \leqslant \left(\int_0^t X^2(s) d[M, M](s)\right)^{\frac{1}{2}}\left(\int_0^t Y^2(s) d[N, N](s)\right)^{\frac{1}{2}}. \tag{3.224}
$$

Theorem 3.128. *If X, Y are two independent processes, then*

$$
[X, Y](t) = 0. \tag{3.225}
$$

3.4.3 Quadratic Variation

Definition 3.129. *Quadratic variation is a special case of cross variation, such that for $X \in \mathcal{N}$,*

$$[X, X](T) = \lim_{\|\Pi\| \to 0} \sum_{j=0}^{n-1} \left(X(t_{j+1}) - X(t_j)\right)^2 . \tag{3.226}$$

4 Brownian Motion

4.1 Random Walk

4.1.1 Bernoulli Random Walk

We would begin with Bernoulli random walk and analyze its properties.

Definition 4.1 (Bernoulli random walk)**.**
Bernoulli random walk is an unrestricted random walk. It is defined as a sequence S_n with

$$S_0 = 0, \tag{4.1}$$

$$S_n = \sum_{i=0}^{n} X_i. \tag{4.2}$$

X_i is a random variable and $X_i \in \{-1, 1\}$, X_i takes the value $+1$ or -1 with fixed probability, say

$$\begin{cases} \mathbb{P}(X_i = +1) = p, \\ \mathbb{P}(X_i = -1) = q, \end{cases} \quad for\ p + q = 1. \tag{4.3}$$

Suppose we use H to denote the case that $X_i = +1$ with probability p, while T denotes the case that $X_i = -1$ with probability q. And w_i is the state of X_i. Then we write

$$X_i = \begin{cases} 1, & if\ w_i = H, \\ -1, & if\ w_i = T. \end{cases} \tag{4.4}$$

With the definition, we can then research into its mean, variance and the distribution corresponding to the path.

Properties 4.2.
The mean and variance of S_n can be calculated

$$\mathbb{E}[S_n] = n(p - q), \tag{4.5}$$

$$Var[S_n] = 4npq. \tag{4.6}$$

Proof. We could firstly calculate the mean and variance of X_i, say

$$\mathbb{E}[X_i] = 1 \cdot P + (-1) \cdot q = p - q, \tag{4.7}$$

$$\begin{aligned} Var[X_i] = \mathbb{E}[X_i^2] - (\mathbb{E}[X_i])^2 &= 1 - (p-q)^2 \\ &= 1 - (p+q)^2 + 4pq = 4pq. \end{aligned} \tag{4.8}$$

Then

$$\mathbb{E}[S_n] = \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i] = n(p-q), \tag{4.9}$$

$$Var[S_i] = \sum_{i=1}^{n} Var[X_i] = 4npq. \tag{4.10}$$

□

Properties 4.3. *It is interesting to note that S_n is even if and only if n is even; the negative proposition is also true. It follows that*

$$\mathbb{P}(S_{2n} = 2k+1) = \mathbb{P}(S_{2n+1} = 2k) = 0. \tag{4.11}$$

Properties 4.4. *It is prone to find that the probability distribution of S_{2n} and S_{2n+1} follows a Bernoulli distribution.*

$$\mathbb{P}(S_{2n} = 2k) = \begin{pmatrix} 2n \\ n+k \end{pmatrix} p^{n+k} q^{n-k}, \tag{4.12}$$

$$\mathbb{P}(S_{2n+1} = 2k+1) = \begin{pmatrix} 2n+1 \\ n+1+k \end{pmatrix} p^{n+k+1} q^{n-k}. \tag{4.13}$$

4.1.2 Symmetric Random Walk

Then we introduce symmetric random walk, which is a particular case of Bernoulli random walk, and it is significant to the establishment of Brownian motion.

Definition 4.5 (Symmetric random walk)**.**
Similarly to Bernoulli random walk (Definition 4.1), for symmetric random walk, the only difference is that we set the probability $p = q = \frac{1}{2}$.
The notation of symmetric random walk is M_n, which satisfies

$$M_n = \sum_{i=1}^{n} X_i, \tag{4.14}$$

where

$$\begin{cases} P(X_i = 1) = \frac{1}{2}, \\ P(X_i = -1) = \frac{1}{2}. \end{cases} \tag{4.15}$$

Properties 4.6. *After replacing p, q into Properties 4.2 , we have*

$$\mathbb{E}[M_n] = 0 \text{ and } Var[M_n] = n. \tag{4.16}$$

Definition 4.7 (increments of symmetric random walk)**.**
Assume $0 < n_0 < n_1 < \cdots < n_k$, we define the increments to be

$$M_{n_{i+1}} - M_{n_i} = \sum_{j=n_i+1}^{j=n_{i+1}} X_j \text{ for } i = 1, 2, \ldots, k. \tag{4.17}$$

It follows that

$$\mathbb{E}[M_{n_{i+1}} - M_{n_i}] = \mathbb{E}[M_{n_{i+1}}] - \mathbb{E}[M_{n_i}] = 0, \tag{4.18}$$

$$Var[M_{n_{i+1}} - M_{n_i}] = Var[M_{n_{i+1}}] - Var[M_{n_i}] = n_{i+1} - n_i. \tag{4.19}$$

Theorem 4.8. *Symmetric random walk is a martingale.*

Proof. For $k < l$, it follows that

$$\begin{aligned} \mathbb{E}[M_l|\mathscr{F}_k] &= \mathbb{E}[M_l - M_k + M_k|\mathscr{F}_k] \\ &= \mathbb{E}[M_l - M_k|\mathscr{F}_k] + M_k \\ &= \mathbb{E}[M_l - M_k] + M_k = M_k. \end{aligned} \tag{4.20}$$

Then M_n is a martingale from the definition. □

Theorem 4.9. *The quadratic variation of symmetric random walk satisfies*

$$[M, M]_n = \sum_{i=1}^{n} (M_i - M_{i-1})^2 = n. \tag{4.21}$$

4.2 Introduction to Brownian motion

The limitations of random walk is conspicuous, as the frequency of change in direction occurs per time unit. That is unacceptable, because in real life, if we would like to portray the stock price, change would happen in each tiny time interval. And That is why we introduce Brownian motion, an approach to deeply delving into stochastic process.

Definition 4.10 (scaled symmetric random walk)**.**
For fixed $n \in N$, the scaled symmetric random walk is defined to be

$$W^{(n)}(t) = \frac{1}{\sqrt{n}} M_{nt}. \tag{4.22}$$

Here, we can easily find the value of $W^{(n)}(\tau)$ if $n\tau$ constitutes an integer. Nevertheless, when $n\tau$ is a fractional number, we simply delay the point $(\tau, W^{(n)}(\tau))$ on the straight line between two consecutive points where nt is an integral.

Sometimes, the formula for scaled symmetric random walk seems confusing, as it looks like a function of two variables. Actually, n should be regarded as a fixed value since n is used to control the size of time intervals. Or we say $W^{(n)} : [0, T] \to \mathbb{R}$. Especially, when $n = 1$, the formula becomes symmetric random walks.

Let us illustrate how the formula works when $n \neq 1$. Suppose we use changing points to describe the time that the function have to decide whether to go up or go down. Then for changing points $0 = t_0, t_1, t_2, \ldots$, we must have $nt_i = i$, or $t_i = \frac{i}{n}$. And at each t_i, the function should go up $\frac{1}{\sqrt{n}}$ units or go down $\frac{1}{\sqrt{n}}$ units.

If $n > 1$, then between two changing points, the walk length would vary faster than time. And the slope of $W^{(n)}(t)$ should be $\frac{1/\sqrt{n}}{1/n} = \sqrt{n}$.

Example 4.11. Now we consider an example of scaled random walks.
By taking $n = 400$ and time interval $[0, 2]$. Then there should be $2 \cdot n = 800$ changing points, and each time, it would step up or down with size $\frac{1}{20}$.

Properties 4.12. *Now we consider the mean and variance of scaled symmetric random walk.*

$$\mathbb{E}[W^{(n)}(t) - W^{(n)}(s)] = 0, \tag{4.23}$$

$$Var[W^{(n)}(t) - W^{(n)}(s)] = t - s. \tag{4.24}$$

Proof. Let $s = t_0 < t_1 < \cdots < t_k = t$ be changing points between time s and time t. Note that $k = n(t - s)$. Then

$$\mathbb{E}[W^{(n)}(t) - W^{(n)}(s)] = \sum_{i=1}^{k} \mathbb{E}[W^{(n)}(t_i) - W^{(n)}(t_{i-1})] = 0. \tag{4.25}$$

$$\begin{aligned} Var[W^{(n)}(t) - W^{(n)}(s)] &= \sum_{i=1}^{k} Var[W^{(n)}(t_i) - W^{(n)}(t_{i-1})] \\ &= \sum_{i=1}^{k} \frac{1}{n} Var(M_i - M_{i-1}) \\ &= \sum_{i=1}^{k} \frac{1}{n} \cdot 1 = t - s. \end{aligned} \tag{4.26}$$

□

Properties 4.13. *$W^{(n)}(t)$ is a martingale.*

Proof. We can simply use the definition of martingale to prove the property.

$$\mathbb{E}[W^{(n)}(t) | \mathscr{F}(s)] = \mathbb{E}[W^{(n)}(t) - W^{(n)}(s)] + W^{(n)}(s) = W^{(n)}(s). \tag{4.27}$$

□

Properties 4.14. *The quadratic variation of scaled random walk satisfies*

$$[W_{(n)}, W_{(n)}](t) = t. \tag{4.28}$$

Proof. At time t, there should be nt changing points, denoted by $t_0, t_1, \ldots, t_{nt}$. It follows that

$$\begin{aligned} [W_{(n)}, W_{(n)}](t) &= \sum_{i=1}^{nt} (W_{(n)}(t_i) - W_{(n)}(t_{i-1}))^2 \\ &= \sum_{i=1}^{nt} (\frac{1}{\sqrt{n}}(M_i - M_{i-1}))^2 \\ &= \sum_{i=1}^{nt} \frac{1}{n} \cdot 1 = t. \end{aligned} \tag{4.29}$$

Then we clarify the property. □

Definition 4.15 (Brownian motion).
Consider the scaled symmetric random walk $W^{(n)}(t) = \frac{1}{\sqrt{n}} M_{nt}$*, suppose that* $n \to \infty$*, then we would obtain the formula for Brownian motion, denoted by* $W(t)$*.*

$$W(t) = \lim_{n\to\infty} W^{(n)}(t) \tag{4.30}$$

Properties 4.16 (properties of Brownian motion).
Generally speaking, the properties of Brownian motion should be the same with scaled random walk. This suggests that:
i) $\mathbb{E}[W(t) - W(s)] = 0$;
ii) $Var[W(t) - W(s)] = t - s$;
iii) $W(t)$ is a martingale;
iv) $[W, W](t) = 1$;
v) $\mathbb{E}[(W(t_{i+1}) - W(t_i))^2] = t_{i+1} - t_i$ as $t_{i+1} - t_i \to 0$.

Definition 4.17 (generalized Brownian motion).
We define $X(t)$ *where* $t \in [0, \infty]$ *to be a generalized Brownian motion corresponding to* (μ, σ) *if* $X(t)$ *satisfies following conditions,*
i) for $0 < t_0 < t_1 < \cdots < t_n$,

$$(X(t_1) - X(t_0)), (X(t_2) - X(t_1)), \ldots, (X(t_n) - X(t_{n-1})) \tag{4.31}$$

are mutually independent;
ii) $X(t+s) - X(t) \sim N(\mu s, \sigma^2 s)$,
iii) $X(t)$ is a continuous function.

Suppose $X(t)$ satisfies above conditions, then $X(t) \sim B(\mu, \sigma)$ is generalized Brownian motion with drift μ and volatility σ.

Especially, $B(0, 1)$ is standard Brownian motion, then $X(t) = W(t)$ as defined.

Theorem 4.18 (generalized Brownian motion, differentiable form).
Postulate that $x(t) \sim B(\mu, \sigma)$, then

$$X(t) = X(0) + \mu t + \sigma W(t). \tag{4.32}$$

Or we say

$$X(t) \sim N(X(0) + \mu t, \sigma^2 t). \tag{4.33}$$

5 Stochastic Integral

Itô integral is a extension of Brownian motion, which has been widely applied to forecast the portfolio asset in different situations. Itô integral adopts the configuration of

$$I(t) = \int_0^t \Delta(u) dw(u). \tag{5.1}$$

Here, $\Delta(t)$ describes the position in the asset at time t and w(t) is the Brownian motion. Then we would particularly explain the derivatives and properties of Itô integral.

5.1 Itô Integral for Simple Process

5.1.1 Definition of Itô Integral

To simplify, we first introduce Ito process under simple process and then extend to general integral. For simple process, we assume that integrant $\Delta(t)$ is a piecewise constant function. Specifically, we divide the interval $[0, T]$ into n subintervals ,say $[t_0, t_1), [t_1, t_2), \dots [t_{n-1}, t_n)$ where the first $n - 1$ interval shares the same length. And $\Pi = \{t_0, t_1, \dots, t_n\}$ constitutes the division of $[0, T]$. Then for $t \in [t_i, t_{i+1})$, we have $\Delta(t) = C_i$ where C_i are constants for $i = 0, 1, \dots, n - 1$.To rewrite $\Delta(t)$ in one equation, we introduce an indicator:

$$1_{A_i}(x) = \begin{cases} 1, & x \in A_i \\ 0, & x \notin A_i \end{cases} \quad where \quad A_i = [t_i, t_{i+1}), \tag{5.2}$$

$$\Delta(t) := \sum_{i=0}^{n-1} C_i \cdot 1_{[t_i, t_{i+1}]}(x). \tag{5.3}$$

Then we need to verify that the formula

$$I(t) = \int_0^t \Delta(u) du, \tag{5.4}$$

has some financial meanings.

Proof. We would prove that Itô integral can be used to describe fluctuations in the asset price at time t, or we say the expected gain from investing at time t. In each interval $[t_i, t_{i+1})$, the gain is the difference in Brownian motion multiple the position in asset, namely, $\Delta(t_i)[W(t_{i+1}) - W(t_i)] = C_i[W(t_{i+1}) - W(t_i)]$.

Then we have

$$\begin{aligned} I(t) &= \Delta(0)[W(t_1) - W(0)] + \Delta(t_1)[W(t_2) - W(t_1)] + \dots \\ &= \sum_{i=0}^{n-1} C_i \cdot [W(t_{i+1} - W(t_i))]. \end{aligned} \tag{5.5}$$

According to Lebesgue measure, we could infer $I(t) = \int_0^t \Delta(u)dW(u)$. □

5.1.2 Properties for Itô Integral

Properties 5.1. *Itô integral is a martingale.*

Proof. We are acquired to prove that

$$\mathbb{E}[I(t)|\mathscr{F}_s] = I(s), \tag{5.6}$$

where $0 \leqslant s < t = t_n \leqslant T$ and $\mathscr{F}_s$ is a collection of information at time $s > 0$ measured by σ-algebra.

Assume $s \in [t_n, t_{n+1}]$ and $t \in [t_m, t_{m+1}]$. We extend the Itô integral into basic forms which are shown below

$$\begin{aligned} I(t) = \sum_{i=0}^{i=n-1} \Delta(t_i)[w(t_{i+1}) - w(t_i)] + \Delta(t_n)[w(s) - w(t_n)] \\ +\Delta(t_n)[w(t_{n+1}) - w(s)] + \sum_{i=n}^{i=m-1} \Delta(t_i)[w(t_{i+1}) - w(t_i)]. \end{aligned} \tag{5.7}$$

Note that the leading pair of components situated on sinistral side of the equation should be $I(s)$, and $\mathbb{E}[I(s)|\mathscr{F}_s] = I(s)$. For the other terms situated on sinistral side

$$\begin{aligned} \mathbb{E}[\Delta(t_n)[W(t_{n+1}) - W(s)]|\mathscr{F}_s] &= \Delta(t_n)[\mathbb{E}[W(t_{n+1})|\mathscr{F}_s] - W(s)] \\ &= \Delta(t_n)[W(s) - W(s)]|\mathscr{F}_s] = 0, \\ \mathbb{E}[\Delta(t_i)[W(t_{i+1}) - W(t_i)]|\mathscr{F}_s] &= \mathbb{E}[\mathbb{E}[\Delta(t_i)[W(t_{i+1}) - W(t_i)]|\mathscr{F}(t_i)]|\mathscr{F}_s] \\ &= \mathbb{E}[\Delta(t_i)[\mathbb{E}[W(t_{i+1})|W(t_i)] - W(t_i)]|\mathscr{F}_s] \\ &= \mathbb{E}[\Delta(t_i)[W(t_i) - W(t_i)]|\mathscr{F}_s] = 0. \end{aligned} \tag{5.8}$$

Then we have prove that $\mathbb{E}[I(t)|\mathscr{F}_s] = I(s)$ by adding the equations together, so $I(t)$ is a martingale and $\mathscr{F}(t)$-measurable. □

Since $I(t)$ embodies a martingale with the initial value $I(0) = 0$, then

$$\begin{aligned} \mathbb{E}[I(t)] &= 0, \\ Var(I(t)) &= \mathbb{E}[I^2(t)] - \mathbb{E}^2[I(t)] = \mathbb{E}[I^2(t)]. \end{aligned} \tag{5.9}$$

Theorem 5.2 (Itô isometry).

$$\mathbb{E}[I^2(t)] = \mathbb{E}\int_0^t \Delta^2(u)du \tag{5.10}$$

Sketch of Proof.
i) Rewrite the Itô integral into basic form and extend square of I(t).
ii) During the calculation, we use the fact that

$$\begin{aligned} &\mathbb{E}\Big[A(t_i,t_j)(W(t_{i+1})-W(t_i))\cdot(W(t_{j+1})-W(t_j))\Big] \\ =& \mathbb{E}\Big[A(t_i,t_j)(W(t_{i+1})-W(t_i))\Big]\cdot\mathbb{E}\Big[(W(t_{j+1})-W(t_j))\Big]=0, \end{aligned} \tag{5.11}$$

And

$$\mathbb{E}\Big[(W(t_{i+1})-W(t_i))^2\Big] = t_{i+1}-t_i. \tag{5.12}$$

□

Theorem 5.3 (Itô Integral, quadratic variation).

$$[I,I](t) = \int_0^t \Delta^2(u)du \tag{5.13}$$

Proof. Consider quadratic variation on an interval $[t_i, t_{i+1}]$, $t_{i+1}-t_i=\tau$. We split the interval into $m=n\tau$ equal parts with length $\frac{t_{i+1}-t_i}{n\tau}=\frac{1}{n}$, say $t_i=s_0<s_1<s_2<\cdots<s_m=t_{i+1}$, and let $m\to\infty$. Then

$$[I,I](t) = \sum_{i=0}^{n\tau-1}\Delta^2(t_i)[W(s_{i+1})-W(s_i)]^2. \tag{5.14}$$

Note that

$$\begin{aligned} \sum_{i=0}^{n\tau-1}[W(s_{i+1})-W(s_i)]^2 &= \sum_{i=0}^{n\tau-1}[\frac{1}{\sqrt{(n)}}(M_{i+1}-M_i)]^2 \\ &= \sum_{i=0}^{n\tau-1}\frac{1}{n}\cdot 1 \\ &= n\tau\frac{1}{n} \\ &= \tau. \end{aligned} \tag{5.15}$$

Hence

$$[I,I](t) = \Delta(t_i)\tau = \Delta(t_i)(t_{i+1}-t_i),\ for\ t\in[t_i,t_{i+1}]. \tag{5.16}$$

Then for quadratic variation accumulated to time t, we have

$$[I,I](t) = \sum_{i=0}^{n-1}\Delta^2(t_i)(t_{i+1}-t_i) = \int_0^t \Delta^2(u)du. \tag{5.17}$$

We can rewrite it into differential form, that is

$$dI(t)dI(t) = \sum_{i=0}^{n-1} \Delta^2(t_i)(t_{i+1} - t_i) = \int_0^t \Delta^2(u)du. \tag{5.18}$$

□

5.2 Itô Integral for General Integrands

Generally, for $\Delta(t)$ in L_2, adapted, $\int_0^T \Delta(t)dW(t)$ can be defined. Here, we would like to introduce a method in a metric space to verify function convergence, which would give an intuitive sense that general integrands $\Delta(t)$ can be approached by a step function.

5.2.1 An Assumption for General Case

Now we would consider the case that the integrand $\Delta(t)$ is a piecewise continuous function on the domain $[0, T]$.

Recall that for simple process, we can extend Itô integral using definition of Lebesgue integral because the integrand $\Delta(t)$ is a piecewise constant function, giving $\int_{t_i}^{t_{i+1}} \Delta(u)dW(u) = C_i \int_{t_i}^{t_{i+1}} dW(u) = C_i[W(t_{i+1}) - W(t_i)]$.

However, for general case, we could not use the method above to calculate Itô integral. So we would utilize a sequence of simple integrands on $[0, T]$ to approach $\Delta(t)$. Say

$$\Delta_n(t) = \begin{cases} \Delta(0), & 0 \leqslant t < \frac{T}{n} \\ \Delta(\frac{T}{n}), & \frac{T}{n} \leqslant t < \frac{2T}{n} \\ \dots \\ \Delta(\frac{(n-1)T}{n}), & \frac{(n-1)T}{n} \leqslant t < T. \end{cases} \tag{5.19}$$

We can also say that $\Delta(t)$ is a step function. Suppose we can verify that $\Delta_n(t)$ approaches $\Delta(t)$ when $n \to \infty$, then we could also prove that

$$\int_0^T \Delta(t)dW(t) = \lim_{n\to\infty} \int_0^T \Delta_n(t)dW(t). \tag{5.20}$$

The formula help us analyze the property of Itô integral for general integrands.

5.2.2 Function Convergence of Integrand $\Delta(t)$ in L_p metric

In the first place, we need to ascertain that $\Delta_n(t) \to \Delta(t)$. It denotes one tricky problem since proving the convergence under any conditions is quite hard. And since both $\Delta(t)$ and $\Delta_n(t)$ may not be differentiable, we have to use Lebesgue measure in computing the integral. So, in this subsection, we would introduce a method in metric space to prove function convergence in L_p metric, which is sufficient for later calculation. To begin with, we verify function convergence in L_1 metric.

Suppose $\Delta(t)$ has jumps at $t = a_1, a_2, \dots, a_{l-1}$ and $a_0 = 0$, $a_l = T$. Here, the integral must be defined on a closed interval $[0, T]$ and there are finite jumps. Otherwise, the assumption may not be true.

Theorem 5.4.
i) We can find $M > 0$, thus, $-M \leqslant \Delta(t) \leqslant M$;
ii) $\exists k > 0$ such that $\forall x \in [a_i, a_{i+1})$, $-k(x - \Delta(x)) \leqslant \Delta(t) \leqslant k(x - \Delta(x))$ for $t \in [x, a_{i+1}]$.

Proof. i) From analysis, a continuous function $f : [a, b] \to \mathbb{R}$ should be bounded.
For $\Delta : [a_i, a_{i+1}) \to R$, we define the extended function

$$\begin{aligned} &\Delta^* : [a_i, a_{i+1}] \to \mathbb{R}, \\ where \quad &\Delta^*(t) = \Delta(t) \quad for \quad t \in [a_i, a_{i+1}), \\ and \quad &\Delta^*(a_{i+1}) = \lim_{t \to a_{i+1}} \Delta(t). \end{aligned} \tag{5.21}$$

Since $\Delta^*(t)$ is continuous on the closed interval $[a_i, a_{i+1}]$, then it is bounded. Say $\exists M_i \in R$, s.t. $|\Delta^*(t)| \leqslant M_i$ for $\forall t \in [a_i, a_{i+1}]$.

Since $\Delta(t) \subseteq \Delta^*(t)$, we have $|\Delta(t)| \leqslant M_i$ for $\forall t \in [a_i, a_{i+1})$. Since $[0, T] = \cup_{i=0}^{i=n-1}[a_i, a_{i+1}) \cup T$ and Δ is bounded on each $[a_i, a_{i+1})$. So we take $M = max\{M_1, M_2, \dots, M_{l-1}, \Delta(T)\}$, $-M \leqslant \Delta(t) \leqslant M$.

ii) Since Δ is bounded, assume $k_i = max|\frac{d}{dt}\Delta(t)|$ on $[a_i, a_{i+1})$ and $k = max\{k_1, k_2, \dots, k_{l-1}\}$. Then for $t \in [x, a_{i+1})$, the value of $\Delta(t)$ should be between the two lines passing through $(x, \Delta(t))$ with slope k and -k. □

Properties 5.5. *With the aim of demonstrating that $\Delta_n(t) \to \Delta(t)$, we need to show that for $\forall \epsilon > 0$, $\exists N \in N^*$ s.t. for $\forall n \geqslant N$, we have $d_1(\Delta_n(t), \Delta(t)) \leqslant \epsilon$.*

Proof.

$$d_1(\Delta_n(t), \Delta(t)) = \int_0^T |\Delta_n(t), \Delta(t)| dx = \sum_{i=0}^{n-1} \int_{\frac{iT}{n}}^{\frac{(i+1)T}{n}} |\Delta(t) - \Delta(\frac{iT}{n})| dx \tag{5.22}$$

Since we do not know the exist position of jumps a_i in $\Delta(t)$ and jumps $t_i = \frac{iT}{n}$ in $\Delta_n(t)$, we need two consider two possible cases.
Case A) if $t_i \in [a_j, a_{j+1})$, for some $t \in [t_i, t_{i+1})$, we have $t \in [a_j, a_{j+1})$
Case B) if $t_i \in [a_j, a_{j+1})$, for some $t \in [t_i, t_{i+1})$, we have $t \notin [a_j, a_{j+1})$

Case A describes normal conditions, while Case B describes the condition that jump happens. For later calculation, we define

$$S_i = \begin{cases} t_{i+1}, & t_{i+1} < a_{j+1}, \\ a_{j+1}, & t_{i+1} > a_{j+1}. \end{cases} \tag{5.23}$$

For $t \in A$, if $\frac{iT}{n} \leqslant t \leqslant \frac{(i+1)T}{n}$, $-k(x-\Delta(x)) \leqslant \Delta(t) \leqslant k(x-\Delta(x))$. And then $-k(x-\Delta(x)) - \Delta(\frac{iT}{n}) \leqslant \Delta(t) - \Delta(\frac{iT}{n}) \leqslant k(x-\Delta(x)) - \Delta(\frac{iT}{n})$.
Hence

$$\begin{aligned} S_A &= \sum_{i=0}^{n-1} \int_{\frac{iT}{n}}^{S_i} |\Delta(t) - \Delta(\frac{iT}{n})| dx \\ &\leqslant \sum_{i=0}^{n-1} \int_{\frac{iT}{n}}^{S_i} \max\{|k(x-\Delta(x)) - \Delta(\frac{iT}{n})|, |k(x-\Delta(x)) + \Delta(\frac{iT}{n})|\} dx \\ &\leqslant \sum_{i=0}^{n-1} \frac{T}{n} \cdot k\frac{T}{n} \\ &= n \cdot \frac{kT^2}{n^2} \\ &= \frac{kT^2}{n}. \end{aligned} \tag{5.24}$$

For $t \in B$, we have

$$\begin{aligned} S_B &= \sum_{i=0}^{n-1} \int_{S_i}^{\frac{(i+1)T}{n}} |\Delta(t) - \Delta(\frac{iT}{n})| dx \\ &\leqslant (l-1) \cdot \frac{T}{n} \cdot [\inf \Delta(t) - \sup \Delta(t)] \\ &\leqslant (l-1) \cdot \frac{T}{n} \cdot 2M \\ &= \frac{2MT(l-1)}{n}. \end{aligned} \tag{5.25}$$

The formula above is because $\Delta(t)$ only have $(l-1)$ jumps.

Then it emerges that

$$\begin{aligned} d_1(\Delta_n(t), \Delta(t)) &= \sum_{i=0}^{n-1} \int_{\frac{iT}{n}}^{\frac{(i+1)T}{n}} |\Delta(t) - \Delta(\frac{iT}{n})| dx \\ &= \sum_{i=0}^{n-1} \int_{\frac{iT}{n}}^{S_i} |\Delta(t) - \Delta(\frac{iT}{n})| dx + \sum_{i=0}^{n-1} \int_{S_i}^{\frac{(i+1)T}{n}} |\Delta(t) - \Delta(\frac{iT}{n})| dx \\ &= S_A + S_B \\ &\leqslant \frac{kT^2}{n} + \frac{2MT(l-1)}{n}. \end{aligned} \tag{5.26}$$

So, for $\forall \epsilon > 0$, we choose N $= \frac{kT^2 + 2MT(l-1)}{\epsilon}$. Then for $\forall n \geqslant N$, $d_1(\Delta_n(t), \Delta(t)) \leqslant \epsilon$. We have finished our proof that $\Delta_n(t) \to \Delta(t)$ in L_1 measure. $\square$

For L_m metric, we employ induction techniques to prove function convergence.

Theorem 5.6. *In L_m metric, for $\forall \epsilon > 0$, we choose $N = \frac{(2MT)^{m-1}[kT^2+2MT(l-1)]}{\epsilon}$. Then for $\forall n \geqslant N$, $d_m(\Delta_n(t), \Delta(t)) \leqslant \epsilon$.*

Proof. *i) It holds for $m = 1$.*
ii) Suppose it holds for $m = r$, then we have for $\forall n \geqslant N$,

$$d_r(\Delta_n(t), \Delta(t)) \leqslant \frac{(2MT)^{r-1}[kT^2 + 2MT(l-1)]}{n}. \tag{5.27}$$

Then for d_{r+1},

$$\begin{aligned} d_{r+1} &= \int_0^T |\Delta_n(t) - \Delta(t)|^{r+1} dt \\ &\leqslant \max |\Delta_n(t) - \Delta(t)| \cdot T \cdot \int_0^T |\Delta_n(t) - \Delta(t)|^r dt \\ &\leqslant 2MT \cdot \int_0^T |\Delta_n(t) - \Delta(t)|^r dt \\ &= 2MT \cdot d_r(\Delta_n(t), \Delta(t)) \\ &\leqslant 2MT \cdot \frac{(2MT)^{r-1}[kT^2 + 2MT(l-1)]}{n} \\ &= \frac{(2MT)^r[kT^2 + 2MT(l-1)]}{n}. \end{aligned} \tag{5.28}$$

So in L_{r+1} metric, for any $\epsilon > 0$, we can choose $N = \frac{(2MT)^r[kT^2+2MT(l-1)]}{\epsilon}$. Then for $\forall n \geqslant N$, $d_{r+1}(\Delta_n(t), \Delta(t)) \leqslant \epsilon$. □

From then, we can definitely say that the integrand $\Delta(t)$ could be approached by the sequence of functions $\Delta_n(t)$. The only thing we have not verified is that $\Delta_n(t)$ converges to $\Delta(t)$ in L_∞ metric, but we would not prove this since it is useless in this chapter.

5.2.3 Function Convergence for Itô Integral

Recall that we have verified that step function $\Delta_n(t)$ converges to $\Delta(t)$. Suppose $I_n(t) = \int_0^t \Delta_n(t)dW(t)$. And we still need to prove that $I_n(t)$ converges to some $I(t)$.

Theorem 5.7. *The Itô integral for general integrands $I(t) = \int_0^t \Delta(t)dW(t)$ could be approximated using the sequence $(I_n(t))_n$:*

$$I(t) = \lim_{n\to\infty} I_n(t), \tag{5.29}$$

which is equivalent to say that

$$\int_0^t \Delta(u)dw(u) = \lim_{n\to\infty} \int_0^t \Delta_n(u)dw(u). \tag{5.30}$$

Proof.
Note that for $\forall \epsilon > 0$, $\exists N_1 > 0$ s.t. for $\forall n \geqslant N_1$, $d_2(\Delta_n(t), \Delta(t)) \leqslant \epsilon$. Similarly, for $\forall \epsilon > 0$, $\exists N_2 > 0$ s.t. for $\forall n \geqslant N_2$, $d_2(\Delta_n(t), \Delta(t)) \leqslant \epsilon$.

Then take $N = \max\{N_1, N_2\}$, for $\forall n, m > N$, we have

$$d_2(\Delta_n(t), \Delta_m(t)) \leqslant \Big| d_2(\Delta_m(t), \Delta(t)) - d_2(\Delta_n(t), \Delta(t)) \Big| \leqslant \epsilon. \tag{5.31}$$

And recall d_2 satisfies

$$d_2(\Delta_n(t), \Delta_m(t)) = \int_0^t \Big| \Delta_n(u) - \Delta_m(u) \Big|^2 du. \tag{5.32}$$

Due to Itô isometry, for $n, m > N$, it follows that

$$\mathbb{E}(I_n(t) - I_m(t))^2 = \mathbb{E} \int_0^t \Big| \Delta_n(u) - \Delta_m(u) \Big|^2 du \leqslant \epsilon. \tag{5.33}$$

This implies that $I_n(t)$ is a Cauchy sequence in L_2 metric. Using Riesz-Fisher theorem, Cauchy sequence in L_2 metric is complete and has limits. So

$$\exists \lim_{n\to\infty} I_n(t) \ or \ \exists \lim_{n\to\infty} \int_0^t \Delta_n(u) dw(u). \tag{5.34}$$

Then we can define that

$$I(t) = \lim_{n\to\infty} I_n(t). \tag{5.35}$$

□

Since Itô integral for general integrand can be approximated using the step function, then we can analyze the properties of Itô integral.

Properties 5.8. *Itô integral for general integrand $I(t) = \int_0^t \Delta(t) dw(t)$, $I(t)$ has following properties:*
i) Itô integral is a martingale.

$$\mathbb{E}[I(t)|F(s)] = I(s). \tag{5.36}$$

ii) The Itô isometry has the form

$$\mathbb{E}[I^2(t)] = \mathbb{E} \int_0^t \Delta^2(u) du. \tag{5.37}$$

iii) The quadratic variation satisfies

$$[I, I](t) = \int_0^t \Delta^2(u) du. \tag{5.38}$$

Or we may scribe

$$d[I, I](t) = dI(t) dI(t) = \Delta^2(t) dt. \tag{5.39}$$

5.3 Itô-Doeblin Formula

In this section, we would explore how to differentiate the expressions $f(W(t))$ where $W(t)$ denotes Brownian motion and f constitutes a differentiable function. Since $W(t)$ does not have derivatives and contains nonzero quadratic variation. So we could not use the chain rule for ordinary calculus directly. Actually, it follows that

$$df(W(t)) = f'(W(t))dW(t) + \frac{1}{2}f''(W(t))dt. \tag{5.40}$$

After introducing Itô-Doeblin formula in this section, we can prove the formula and understand the reason for the formula.

Theorem 5.9 (Brownian motion, Itô-Doeblin formula).
For differentiable function $f(t,x)$ where $\frac{\partial f}{\partial t}$ and $\frac{\partial f}{\partial x}$ are well-defined. Endow $W(t)$ with the meaning of Brownian motion. We can conclude that

$$\begin{aligned} f(T, W(T)) =& f(0, W(0)) + \int_0^T f_t(t, W(t))dt \\ &+ \int_0^T f_x(t, W(t))dW(t) + \frac{1}{2}\int_0^T f_{xx}(t, W(t))dt. \end{aligned} \tag{5.41}$$

Sketch of proof.
Let $\Pi = \{t_0, t_1, \dots, t_n\}$ constitute the division of $[0, T]$, and presume $\|\Pi\|_\infty = t_{i+1} - t_i \to 0$. Then we can employ Taylor's expansion in the time interval $[t_i, t_{i+1}]$. This suggests that

$$\begin{aligned} &f(t_{i+1}, W(t_{i+1})) - f(t_i, W(t_i)) = \\ &\qquad \int_{t_i}^{t_{i+1}} f_t(t, W(t))dt + \int_{t_i}^{t_{i+1}} f_x(t, W(t))dW(t) + \frac{1}{2}\int_{t_i}^{t_{i+1}} f_{xx}(t, W(t))dW(t)dW(t) \\ &\qquad + \int_{t_i}^{t_{i+1}} f_{xt}(t, W(t))dW(t)dt + \frac{1}{2}\int_{t_i}^{t_{i+1}} f_{tt}(t, W(t))dtdt \\ &\qquad = f_t(t_i, W(t_i))(t_{i+1} - t_i) + f_x(t_i, W(t_i))(W(t_{i+1}) - W(t_i)) \\ &\qquad + \frac{1}{2}f_{xx}(t_i, W(t_i))(W(t_{i+1}) - W(t_i))^2 + f_{xt}(t_i, W(t_i))(W(t_{i+1}) - W(t_i))(t_{i+1} - t_i) \\ &\qquad + \frac{1}{2}f_{tt}(t_i, W(t_i))(t_{i+1} - t_i)^2. \end{aligned} \tag{5.42}$$

Then we have

$$\begin{aligned} f(T, W(T)) - f(0, W(0)) =& \sum_{i=0}^{n-1} f_t(t_i, W(t_i))(t_{i+1} - t_i) + \sum_{i=0}^{n-1} f_x(t_i, W(t_i))(W(t_{i+1}) - W(t_i)) \\ &+ \sum_{i=0}^{n-1} \frac{1}{2}f_{xx}(t_i, W(t_i))(W(t_{i+1}) - W(t_i))^2 + \sum_{i=0}^{n-1} f_{xt}(t_i, W(t_i))(W(t_{i+1}) - W(t_i))(t_{i+1} - t_i) \\ &+ \sum_{i=0}^{n-1} \frac{1}{2}f_{tt}(t_i, W(t_i))(t_{i+1} - t_i)^2. \end{aligned} \tag{5.43}$$

Recall that for Brownian motion $w(t)$, it emerges

$$(W(t_{i+1}) - W(t_i))^2 = t_{i+1} - t_i, \tag{5.44}$$

as $\|\Pi\|_\infty \to 0$. Then we say $t_{i+1} - t_i$ converges to zero faster than $W(t_{i+1}) - W(t_i)$ when time increment $\|\Pi\|_\infty \to 0$. It follows that

$$\begin{aligned}
&\sum_{i=0}^{n-1} f_{xt}(t_i, W(t_i))(W(t_{i+1}) - W(t_i))(t_{i+1} - t_i) \\
&\qquad\qquad \leqslant \max\{|t_{i+1} - t_i|\} \cdot \sum_{i=0}^{n-1} f_{xt}(t_i, W(t_i))(W(t_{i+1}) - W(t_i)) \\
&\qquad\qquad \leqslant 0 \cdot \int_0^T f_{xt}(t, W(t))dW(t) = 0.
\end{aligned} \tag{5.45}$$

The idea is that $\|\Pi\|_\infty = t_{i+1} - t_i \to 0$ after taking limits. We remain the term $W(t_{i+1}) - W(t_i)$ since it tends to zero slower than the time increments.
Similarly,

$$\begin{aligned}
&\sum_{i=0}^{n-1} f_{tt}(t_i, W(t_i))(t_{i+1} - t_i)^2 \\
&\qquad\qquad \leqslant Max\{|t_{i+1} - t_i|\} \cdot \sum_{i=0}^{n-1} f_{xt}(t_i, W(t_i))(t_{i+1} - t_i) \\
&\qquad\qquad \leqslant 0 \cdot \int_0^T f_{tt}(t, W(t))dt = 0.
\end{aligned} \tag{5.46}$$

Since $(W(t_{i+1}) - W(t_i))^2 = t_{i+1} - t_i$ as $\|\Pi\|_\infty \to 0$, we have

$$\begin{aligned}
\sum_{i=1}^{n-1} f_{xx}(t_i, W(t_i))(W(t_{i+1}) - W(t_i))^2 &= \sum_{i=1}^{n-1} f_{xx}(t_i, W(t_i))(t_{i+1} - t_i) \\
&= \int_0^T f_{xx}(t, W(t))dt.
\end{aligned} \tag{5.47}$$

And then (5.43) can be rewritten as

$$\begin{aligned}
f(T, W(T)) =& f(0, W(0)) + \int_0^T f_t(t, W(t))dt \\
&+ \int_0^T f_x(t, W(t))dW(t) + \frac{1}{2}\int_0^T f_{xx}(t, W(t))dt.
\end{aligned} \tag{5.48}$$

We have verify the Itô-Doeblin formula for Brownian motion. □

From Theorem 5.9, we have proved that as $\|\Pi\|_\infty \to 0$,

$$(t_{i+1} - t_i)(W(t_{i+1}) - W(t_i)) = (t_{i+1} - t_i)^2 = 0, \tag{5.49}$$

which implies that

$$dW(t)dt = dtdW(t) = dtdt = 0. \tag{5.50}$$

On the other hand,

$$dW(t)dW(t) = dt. \tag{5.51}$$

6 Black-Scholes-Merton Equation

6.1 Geometric Brownian Motion

Note that Brownian motion is definitely not appropriate for describing portfolio value or exchange rate since Brownian motion can take negative values. Thus, geometric Brownian motion is released to fit the finance market.

Recall that Itô process is specified by the formula:

$$X(t) = X(0) + \int_0^t \Delta(u)\ dW(u) + \int_0^t \Theta(u)\ du. \tag{6.1}$$

In geometric Brownian motion, we can assume $X(0) = 0$, $\Delta(u) = \sigma$ and $\Theta(u) = \alpha - \frac{1}{2}\sigma^2$, where σ and α are both constant. Then the Itô can be rewritten to be:

$$X(t) = \int_0^t \sigma\ dW(u) + \int_0^t \alpha - \frac{1}{2}\sigma^2\ du. \tag{6.2}$$

In finance market, we describe the asset price by

$$S(t) = f(X(t)) = S(0)e^{X(t)} = S(0)e^{\int_0^t \sigma dW(u) + \int_0^t \alpha - \frac{1}{2}\sigma^2 du}. \tag{6.3}$$

Formula (6.3) is called geometric Brownian motion.

Proposition 6.1. *The function $S(t)$ is random and we cannot find the relevant discrete model based on $S(t)$. However, the expected value of $S(t)$ is nonrandom and the graph of $\mathbb{E}[S(t)]$ can be portrayed after we acquire initial value $S(0)$ and rate of return α. Besides, contrary to Brownian motion, $S(t)$ is not a martingale and it has the tendency to increase.*

Proof.

$$X(t+\tau) - X(t) = \int_t^{t+\tau} \sigma\ dW(u) + \int_t^{t+\tau} \alpha - \frac{1}{2}\sigma^2\ du. \tag{6.4}$$

As $\tau \to 0$, applying ordinary integral,

$$X(t+\tau) - X(t) = \sigma[W(t+\tau) - W(t)] + (\alpha - \frac{1}{2}\sigma^2)\tau. \tag{6.5}$$

If we divide the time interval T into n equal parts, say $\Pi = t_0, t_1, \ldots, t_n$ where $t_0 = 0, t_n = T$ constitute the division of time interval T. Let $\tau = \frac{T}{n}$ As $n \to \infty$, then

$$X(T) - X(0) = \sum_{i=0}^{n} X((i+1)\tau) - X(i\tau) = \sigma[W(T) - W(0)] + (\alpha - \frac{1}{2}\sigma^2)T, \tag{6.6}$$

$$X(t) = \sigma W(t) + (\alpha - \frac{1}{2}\sigma^2)t. \tag{6.7}$$

So $X(t) \sim N((\alpha - \frac{1}{2}\sigma^2)t, \sigma^2 t)$,

$$\mathbb{E}(S(t)) = S(0)E(e^{X(t)}) = S(0)e^{(\alpha - \frac{1}{2}\sigma^2)t + \frac{1}{2}\sigma^2 t} = S(0)e^{\alpha t}. \tag{6.8}$$

Formula $\mathbb{E}[S(t)] = \mathbb{E}[S(0)]e^{\alpha t}$ implies the expect value of $s(t)$ is nonrandom. Then according to the Lemma 6.2 which is proposed below, we would find that the rate of return is α. $\square$

Lemma 6.2. *Presume that $S(t)$ constitutes the stochastic process, then we should validate that $S(t)$ conforms with the equation $S(t) = S(0)e^{\alpha t}$ if and only if the rate of return is α.*

Proof. i) Prove the necessity. Since $S(t)$ is the asset price with return rate α, then we can calculate $S(t)$ by dividing the time period t into n parts. So

$$S(t) = \lim_{n\to\infty} \left(1 + \alpha\frac{t}{n}\right)^n S(0) = S(0)e^{\alpha t}. \tag{6.9}$$

ii) Prove the efficiency. We have known the formula $S(t) = S(0)e^{\alpha t}$, and we want to compute the interest rate. The measure is to change the continuous model into the discrete model. So if $S(t) = S(0)e^{\alpha t}$ and $t = n\tau$ where $n \to \infty$, then

$$S(k\tau) = S(0)e^{\alpha k\tau} = S(0)(e^{\alpha\tau})^k \approx S(0)(1 + \alpha\tau)^k, \quad k \in 0, 1, \ldots n \tag{6.10}$$

The last term uses Taylor expansion $e^{\alpha\tau} = 1 + \alpha\tau + \frac{1}{2}\alpha\tau^2 + \cdots \approx 1 + \alpha\tau$. And from this equation we can see that $S((k+1)\tau) \approx S(k\tau)(1 + \alpha\tau)$, implying that the return rate

$$r_\tau = \lim_{\tau\to 0} \frac{S((k+1)\tau) - S(k\tau)}{S(k\tau)\tau} = \alpha. \tag{6.11}$$

Then the interest rate is α as computed. $\square$

Example 6.3 (A finance meaning of geometric Brownian motion)**.**
Assume we would like to measure the stock prices at the end of each day, then we need to describe the discrete form of geometric Brownian motion with time $t_0 = 0, t_1 = 1, \ldots, t_n = n$.

We let

$$L_i = \frac{S(t_i)}{S(t_{i-1})}. \tag{6.12}$$

Notice that $L_i - 1$ should be the return rate at the end of each day. And then

$$\ln\left(\frac{S(t_i)}{S(t_{i-1})}\right) = X(t_i) - X(t_{i-1}) = X(t_i) - X(t_{i-1}).$$

Since $X(t) = (\alpha - \frac{1}{2}\sigma^2)t + \sigma W(t)$, then the Brownian motion can be designated by $X(t)$ with drift $\mu = \alpha - \frac{1}{2}\sigma^2$ and volatility σ, say $X(t) \sim B(\mu, \sigma)$. Then $X(t+s) - X(t) \sim N(\mu s, \sigma^2 s)$.

In this situation, since $t_i - t_{i-1} = 1$, we have

$$X(t_i) - X(t_{i-1}) = X(t) - X(t-1) \sim N(\mu, \sigma^2). \tag{6.13}$$

This implies that $\ln(L_i)$ is a normal distribution with mean μ and standard deviation σ. Or

$$L_i \sim LN(\mu, \sigma). \tag{6.14}$$

Then we can see that

$$\begin{aligned} E[L_i] &= \exp\{\mu + \frac{1}{2}\sigma^2\} \\ &= \exp\{\alpha - \frac{1}{2}\sigma^2 + \frac{1}{2}\sigma^2\} \\ &= e^{\alpha}. \end{aligned} \tag{6.15}$$

It is equivalent to say that $E[S(t_i)] = E[S(t_{i-1})]e^{\alpha}$, which implies that mean rate of return for the discrete model is α.

If we rewrite $S(t)$ using L_i, we would get

$$S(t) = S(0) \cdot L_1 \cdot L_2 \cdot \cdots \cdot L_n \tag{6.16}$$

Suppose we use discrete Brownian motion to represent stock price, it seems that the asset would show an oscillating growth trend. However, in finance market, we should not ignore the effect of inflation. Actually,

$$actual\ interest\ rate \ = \ nominal\ interest\ rate \ - \ inflation\ rate \tag{6.17}$$

So if we let the rate of return to be exactly the same with inflation rate, then actual interest rate should be zero and stock prices are non-arbitrage.

In summary, when utilizing this model to describe the stock price, it takes non-negative values at any time and has no tendency to offer arbitrage. So it is a feasible model in finance market.

Definition 6.4 (generalized geometric Brownian motion)**.**
The generalized geometric Brownian motion is defined to be

$$S(t) = S(0)e^{X(t)}, \tag{6.18}$$

where

$$X(t) = \int_0^t \sigma(s)dW(s) + \int_0^t (\alpha(s) - \frac{1}{2}\sigma^2(s))ds. \tag{6.19}$$

The difference between generalized geometric Brownian motion and normal one is that the adapted process α and σ becomes a function of t.

Despite the difference, we could still apply Itô-Doeblin formula and find that

$$dS(t) = \alpha(t)S(t)dt + \sigma(t)S(t)dW(t) \tag{6.20}$$

Then at time t, the asset price $S(t)$ has mean rate of return $\alpha(t)$ and volatility $\sigma(t)$. Note that the mean and volatility might change over time.

Then we introduce a theorem where the mean rate of return for generalized geometric Brownian motion would be utilized cunningly. In the proof of the example, we would demonstrate how to construct the form of geometric Brownian motion in some problems. This proof provides us some valuable ideas in solving models on finance market.

Theorem 6.5 (Itô integral of a deterministic integrand)**.**
Recall that the Itô integral is defined to be $I(t) = \int_0^t \Delta(s)dw(s)$. We suggest that $I(t)$ follows normal distribution with mean zero and variance $\int_0^t \Delta^2(s)ds$.

Proof. Actually calculating mean and variance is not a difficult thing. Because $I(t)$ is a martingale, we have $\mathbb{E}[I(t)] = I(0) = 0$. Correspondingly,

$$Var[I(t)] = \mathbb{E}[I^2(t)] - (\mathbb{E}[I(t)])^2 = \int_0^t \Delta^2(s)ds. \tag{6.21}$$

Nevertheless, it is a tricky thing to prove that $I(t)$ follows a normal distribution. The key point here is to illustrate that the moment-generating function of $I(t)$ is the same with a random variable follows $N\left(0, \int_0^t \Delta^2(s)ds\right)$.
That is, we need to prove that

$$\mathbb{E}[e^{uI(t)}] = \exp\left\{\frac{1}{2}u^2 \int_0^t \Delta^2(s)ds\right\} \; for \; \forall u \in \mathbb{R}. \tag{6.22}$$

Or the equation can be written as

$$\mathbb{E}[\exp\left\{uI(t) - \frac{1}{2}u^2 \int_0^t \Delta^2(s)ds\right\}] = 1. \tag{6.23}$$

Then it is equivalent to prove that

$$\mathbb{E}[\exp\left\{\int_0^t u\Delta(s)dW(s) - \frac{1}{2}\int_0^t (u\Delta)^2(s)ds\right\}] = 1. \tag{6.24}$$

Note that $\exp\left\{\int_0^t u\Delta(s)dW(s) - \frac{1}{2}\int_0^t (u\Delta)^2(s)ds\right\}$ is the generalized geometric Brownian motion with mean rate of return $\alpha = 0$. So it is martingale and expected value should be equal to the value at $t = 0$, which is equal to 1. Then we have proved that $I(t)$ follows a normal distribution. $\square$

6.2 Black-Scholes-Merton Partial Differential Equation

In this section we are going to introduce one of the most exciting and meaningful model both in mathematics and in finance, Black-Scholes-Merton model. Black-Scholes-Merton model is deduced from geometric Brownian motion, and it can be used to describe non-dividend stock derivatives. The general idea is to construct a portfolio of options and underlying risk-free trades, and the expected rate of return should be the same with risk free rate so that the model would be no arbitrage.

6.2.1 Description of Portfolio Value

Suppose an investor processes a portfolio with value $X(t)$ at time t. And the portfolio consists of risk-free trades, for instance, money market account with rate of return r and a stock portrayed by geometric Brownian motion. So the stock price $S(t)$ satisfies

$$\frac{dS(t)}{S(t)} = \alpha dt + \sigma dW(t) \tag{6.25}$$

Assume that the investor holds $\Delta(t)$ shares of stock at time t. Then the amount of asset invested in derivatives should be

$$\Delta(t)S(t). \tag{6.26}$$

While the amount of asset invested in underlying risk-free trades should be the remaining, that is

$$X(t) - \Delta(t)S(t). \tag{6.27}$$

From the definition, we can then determine the change in the portfolio value during a very small time interval. For, $\tau \to 0$, it follows that

$$\begin{aligned} dX(t) \approx X(t+\tau) - X(t) &= \Delta(t)(S(t+\tau) - S(t)) + \Big(X(t) - \Delta(t)S(t)\Big)(1 + r\tau - 1) \\ &= \Delta(t)dS(t) + r\Big(X(t) - \Delta(t)S(t)\Big)dt. \end{aligned} \tag{6.28}$$

By replacing $dS(t)$ with $\alpha S(t)dt + \sigma S(t)dW(t)$, we have

$$dX(t) = rX(t)dt + \Delta(t)(\alpha - r)S(t)dt + \Delta(t)\sigma S(t)dW(t). \tag{6.29}$$

As for why equation (6.28) is same with equation (6.29), we can give a finance explanation. Equation (6.28) describes the change in portfolio by adding change in stock (derivatives) and change in money market account (underlying risk-free assets) separately.

While in equation (6.29), $rX(t)dt$ implies the expected rate of return in portfolio; $\Delta(t)(\alpha - r)S(t)dt$ means increase or decrease rate in stock price and $\Delta(t)\sigma S(t)dw(t)$ determines the volatility of stock price.

Definition 6.6 (discrete form of portfolio value)**.**
Although it is not easy to figure out the expression of portfolio value $X(t)$, we may consider the discrete form, denoted by $\{X_n\}$.

Equation (6.28) suggests that

$$X_{n+1} - X_n = \Delta_n(S_{n+1} - S_n) + r\Bigg(X_n - \Delta_n S_n\Bigg). \tag{6.30}$$

Then it gives that

$$X_{n+1} = \Delta S_{n+1} + (r+1)\Bigg(X_n - \Delta_n S_n\Bigg). \tag{6.31}$$

Properties 6.7 (considering discounted rate)**.**
In finance market, we should consider the discounted rate of stock price and portfolio value, denoted by $e^{-rt}S(t)$ and $e^{-rt}X(t)$ respectively. Then we can calculate the differential form.

$$d(e^{-rt}S(t)) = (\alpha - r)e^{-rt}S(t)dt + \sigma e^{-rt}S(t)dW(t), \tag{6.32}$$

$$d(e^{-rt}X(t)) = \Delta(t)d(e^{-rt}S(t)). \tag{6.33}$$

Proof. Assume that $f(t,x) = e^{-rt}x$. By applying Itô-Doeblin formula, we have

$$\begin{aligned} d(e^{-rt}S(t)) &= df(t,S(t)) \\ &= f_t(t,S(t))dt + f_x(t,S(t))dt + f_{xx}(t,S(t))dS(t)dS(t) \\ &= -re^{-rt}dt + e^{-rt}dS(t) \\ &= (\alpha - r)e^{-rt}S(t)dt + \sigma e^{-rt}S(t)dW(t). \end{aligned} \tag{6.34}$$

Similarly,

$$\begin{aligned} d(e^{-rt}X(t)) &= d(f(t,X(t))) \\ &= f_t(t,X(t))dt + f_x(t,X(t))dX(t) + \frac{1}{2}f_{xx}(t,X(t))dX(t)dX(t) \\ &= \Delta(\alpha - r)e^{-rt}S(t)dt + \Delta(t)\sigma e^{-rt}S(t)dW(t) \\ &= \Delta(t)d(e^{-rt}S(t)). \end{aligned} \tag{6.35}$$

□

Note that equation 6.34 can be rewritten as

$$\frac{d(e^{-rt}S(t))}{e^{-rt}S(t)} = (\alpha - r)dt + \sigma dW(t). \tag{6.36}$$

This formula tells us that the mean rate of return for discounted portfolio value is $\alpha - r$. And money market account can not offer arbitrage for the investor. Especially, if $\alpha - r > 0$, the investor is expected to earn profit from the investment.

6.2.2 Description of Option Value

Now we consider a European call option, the investor has the right to purchase shares of stock on a certain date. Then the profit should be $(S(T)-K)^+$ at time T, where $S(T)$ is the stock price and K is the strike price on the contract. $(S(T)-K)^+$ is also called the value of the call. Let $c(t,x)$ represent the value of the call at time t, where $x=S(t)$. And we hope that $c(t,x)$ can denote the value of European call in the future.

Again, we determine the change in the value of the call from Itô-Doeblin formula, say

$$\begin{aligned} dc(t,S(t)) &= c_t(t,S(t))dt + c_x(t,S(t)dS(t)) + \frac{1}{2}c_{xx}(t,S(t))dS(t)dS(t) \\ &= c_t(t,S(t))dt + c_x(t,S(t))(\alpha S(t)dt + \sigma S(t)dW(t)) + \frac{1}{2}c_{xx}(t,S(t))\sigma^2 S^2(t)dt \\ &= \left[c_t(t,S(t)) + \alpha S(t)c_x(t,S(t)) + \frac{1}{2}\sigma^2 S^2(t)c_{xx}(t,S(t))\right] dt + \sigma S(t)c_x(t,S(t))dW(t) \end{aligned} \tag{6.37}$$

Properties 6.8 (considering the discounted rate)**.**
The discounted option price for European call becomes $e^{-rt}c(t,S(t))$. If we assume $f(t,x) = e^{-rt}x$, then it follows that

$$\begin{aligned} d(e^{-rt}c(t,S(t))) &= df(t,c(t,S(t))) \\ &= f_t(t,c(t,S(t)))dt + f_x(t,c(t,S(t)))dc(t,S(t)) \\ &= e^{-rt}\left[-rc(t,S(t)) + c_k(t,S(t)) + \alpha S(t)c_x(t,S(t)) + \frac{1}{2}\sigma^2 S^2(t)c_{xx}(t,S(t))\right] dt \\ &\quad + e^{-rt}\sigma S(t)c_x(t,S(t))dW(t). \end{aligned} \tag{6.38}$$

6.2.3 Deriving the Equation

Note that the value of European call $c(t,S(t))$ is unknown to us, and here, we simply extend our conclusion in a broader situation, say $c(t,S(t))$ represents the value of option. To price the option, the general idea is to construct non-arbitrage pricing and self-financing dynamic hedging. We suppose that the portfolio value $X(t)$ consisting of risk-free trades and underlying assets can simulate payoff of the option such that the option has no arbitrage. It follows that

$$c(t,S(t)) = X(t) \ for \ \forall t \in [0,T]. \tag{6.39}$$

Actually, we only need to ensure the following two equations:

$$\begin{aligned} c(0,S(0)) &= X(0) \ and \\ dc(t,S(t)) &= dX(t) \ for \ t \in [0,T). \end{aligned} \tag{6.40}$$

Recall from Property 6.7 and Property 6.8, we can conclude that

$$\begin{aligned}
&\Delta(t)(\alpha - r)S(t)dt + \Delta(t)\sigma S(t)dW(t) \\
&= \left[-rc(t, S(t)) + c_k(t, S(t)) + \alpha S(t)c_x(t, S(t)) + \frac{1}{2}\sigma^2 S^2(t)c_{xx}(t, S(t))\right] dt \qquad (6.41) \\
&\quad + \sigma S(t)c_x(t, S(t))dW(t).
\end{aligned}$$

From the equality, we can equate $dW(t)$ term and dt term and obtain two significant equations, which would be introduced below.

Definition 6.9 (delta-hedging rule)**.**
By matching $dW(t)$ term from equation (6.41), we can see that

$$\Delta(t) = c_x(t, S(t)). \qquad (6.42)$$

Here, equation (6.42) is delta-hedging rule and $c_x(t, S(t))$ is called delta of the option. We would like to emphasize that delta of the option not necessarily satisfies delta-hedging rule, and it only satisfies the rule when hedging the non-arbitrage option. In real life trading, using PnL to denote profit and loss, then it follows that

$$option\ PnL = delta\ PnL\ +\ gamma\ PnL\ +\ vega\ PnL\ +\ higer\ order\ terms. \qquad (6.43)$$

We are not going to discuss how the factors delta, gamma and vega affect option PnL. But anyway, let us see an example about delta hedging.

Example 6.10 (an example about delta hedging)**.**
Suppose an investor has 100 shares of out-of-the-money call with a delta of -0.6 in short position. Then according to equation (6.42), if the underlying stock's price increases 1 dollar, the portfolio value is expected to increase $0.6 \times 100 = 60$ dollars.

Coming back to equation (6.41), we would equate dt term on both sides.

Definition 6.11 (Black-Scholes-Merton partial differential equation)**.**
Equation (6.41) suggests that

$$c_t(t, x) + rxc_x(t, x) + \frac{1}{2}\sigma^2 x^2 c_{xx}(t, x) = rc(t, x), \qquad (6.44)$$

where x represents $S(t)$ and $t \in [0, T)$ as defined. The formula is known as Black-Scholes-Merton partial differential equation.

In order to find the general solution for the PDE, we would have to find a particular solution. Note that the initial condition gives that $c(0, S(0)) = X(0)$, but $X(0)$ is an uncertain value, it could not be regarded as a particular solution. Actually, here we define the terminal condition to be

$$c(T, x) = (x - K)^+ \qquad (6.45)$$

The deduction here may seems a bit weird, so we would roughly sort out the ideas. Our purpose is to figure out a portfolio that can hedge the short position option. We firstly assume the payoff at the last day of the contract, and then we can solve the Black-Scholes-Merton partial differential equation with the terminal condition. After obtaining the general solution of $c(t,x)$, the value of option at time t, we can determine the portfolio value that can hedge the option. Specifically speaking, we need to ensure the initial value of portfolio satisfies that $X(0) = c(0, S(0))$ and the position in the portfolio satisfies that $\Delta(t) = c_x(t,x)$. Then we can say the option is non-arbitrage, which fits our assumption.

6.3 Solving Black-Scholes-Merton Partial Differential Equation

Recall that we have derived the Black-Scholes-Merton partial differential equation in last section, and it has the form that

$$c_t(t,x) + rxc_x(t,x) + \frac{1}{2}\sigma^2 x^2 c_{xx}(t,x) = rc(t,x), \tag{6.46}$$

with the terminal solution

$$c(T,x) = (x-K)^+. \tag{6.47}$$

Then, in this section, we would discuss how to find the general solution of second order partial differential equation. It is noteworthy that the equation is similar to heat equation if you rearrange it. So, we would firstly interpret heat equation, then introduce a method to transfer equation (6.46) into heat equation and find its general solution.

We start from heat equation.

6.3.1 One dimensional Heat Equation

In real life, a common phenomenon demonstrates that heat can transmit through different materials and it usually flows from hot areas to cold areas. In mathematics, a formula interprets this phenomenon, which is heat equation. The generalized heat equation has the form that

$$u_t = k\widetilde{\Delta}u, \tag{6.48}$$

where $\widetilde{\Delta}$ is a Laplace operator. But in this section, we are only interested in one-dimension heat equation.

Definition 6.12 (one dimensional heat equation)**.**
From equation (6.48), we can note that one dimensional heat equation should be

$$u_t = ku_{xx}. \tag{6.49}$$

In this formula, heat only transmits along x-axis, t denotes the time and k is the transmission rate proportional to temperature gradient.

Properties 6.13 (general solution of one dimensional heat equation)**.**
For many years, mathematicians put a great deal of effort to derive the general solution of equation (6.49). One of particular solutions should be

$$\phi(x,t) = \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}}. \tag{6.50}$$

And then the general solution for one dimensional heat equation has the form

$$\begin{aligned} u(x,t) &= \int_{-\infty}^{\infty} \phi(x-y,t) g(y) dy \\ &= \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} e^{-\frac{(x-y)^2}{4kt}} g(y) dy. \end{aligned} \tag{6.51}$$

However, it is hard to figure out the relationship between $u(x,t)$ and $g(x)$. So we would introduce a method using delta function.

Definition 6.14 (delta measure, point-mass measure)**.**
The delta measure $\delta(x)$ satisfies the properties that

$$\delta(x) \to \begin{cases} \infty & , x = 0, \\ 0 & , otherwise. \end{cases} \tag{6.52}$$

And on the other hand,

$$\int_{-\infty}^{\infty} \delta(x) = 1. \tag{6.53}$$

Properties 6.15 (Strict definition of delta measure)**.**
The delta measure can be strictly defined employing a convergent sequence of functions, define

$$\delta_\epsilon(x) = \begin{cases} \frac{1}{\epsilon}, x \in [-\frac{\epsilon}{2}, \frac{\epsilon}{2}]; \\ 0, otherwise. \end{cases} \tag{6.54}$$

When $\epsilon \to 0$, the delta measure would be approached.

Proof. Note that $\int_{-\infty}^{\infty} \delta_\epsilon = \frac{1}{\epsilon} \cdot (\frac{\epsilon}{2} + \frac{\epsilon}{2}) = 1$. When $\epsilon \to 0$, $\delta_\epsilon \to \infty$. □

Properties 6.16. *Suppose that $\delta(x)$ is the delta function and $g(x)$ is a Lipschitz continuous function, then we have*

$$\int_{-\infty}^{\infty} \delta(x-y) g(y) dy = g(x). \tag{6.55}$$

Proof. From the definition of delta measure, $\delta(x-y)$ is non-zero only when $x=y$. Then,

$$\begin{aligned}\int_{-\infty}^{\infty}\delta(x-y)g(y)dy &= \lim_{\epsilon\to 0}\int_{-\infty}^{\infty}\delta_\epsilon(x-y)g(y)dy \\ &= \lim_{\epsilon\to 0}\frac{1}{\epsilon}\int_{x-\frac{\epsilon}{2}}^{x+\frac{\epsilon}{2}} g(y)dy \\ &= \lim_{\epsilon\to 0}\frac{1}{\epsilon}\int_{x-\frac{\epsilon}{2}}^{x+\frac{\epsilon}{2}} \left(g(x)+F(x,y)\right)dy \\ &= g(x)+\lim_{\epsilon\to 0}\frac{1}{\epsilon}\int_{x-\frac{\epsilon}{2}}^{x+\frac{\epsilon}{2}} F(x,y)dy.\end{aligned} \tag{6.56}$$

Since g is Lipschitz continuous function, $\exists M>0$ such that

$$|g(x)-g(y)|\leqslant M|x-y|\leqslant M\epsilon. \tag{6.57}$$

Therefore, $|F(x,y)|\leqslant M\epsilon$, which implies that

$$\lim_{\epsilon\to 0}\frac{1}{\epsilon}\int_{x-\frac{\epsilon}{2}}^{x+\frac{\epsilon}{2}} F(x,y)dy=\lim_{\epsilon\to 0}\epsilon=0. \tag{6.58}$$

It follows that $\int_{-\infty}^{\infty}\delta(x-y)g(y)dy=g(x)$. □

Properties 6.17. *We can check that $\phi(x,0)$ in equation (6.50) has the same property with the delta measure, or*

$$\phi(x,0)\equiv\delta(x). \tag{6.59}$$

Proof. To prove the property, note that

$$\lim_{t\to 0}\frac{1}{\sqrt{4\pi kt}}e^{-\frac{x^2}{4kt}}=\lim_{n\to\infty}\frac{n}{\sqrt{4\pi k}}e^{-\frac{x^2n^2}{4k}}; \tag{6.60}$$

and

$$\lim_{\epsilon\to 0}\delta_\epsilon(x)=\lim_{n\to\infty}\begin{cases}n, x\in[-\frac{2}{n},\frac{2}{n}];\\ 0, otherwise.\end{cases} \tag{6.61}$$

Consider two sequences, $a_n=\frac{n}{\sqrt{4\pi k}}e^{-\frac{x^2n^2}{4k}}$ and $b_n=\begin{cases}n, x\in[-\frac{2}{n},\frac{2}{n}];\\ 0, otherwise\end{cases}$. Then, under the metric $d_1(f,g)=\int_{-\infty}^{\infty}|f(x)-g(x)|dx$, it follows that

$$d_1(a_n,b_n)=\int_{-\infty}^{-\frac{2}{n}}\frac{n}{\sqrt{4\pi k}}e^{-\frac{x^2n^2}{4k}}dx+\int_{-\frac{2}{n}}^{\frac{2}{n}}\left|\frac{n}{\sqrt{4\pi k}}e^{-\frac{x^2n^2}{4k}}-n\right|dx+\int_{\frac{2}{n}}^{\infty}\frac{n}{\sqrt{4\pi k}}e^{-\frac{x^2n^2}{4k}}dx. \tag{6.62}$$

Thus, $\forall \epsilon > 0$, there $\exists N =$, s.t. $\forall n \geqslant N$, we have

$$\begin{aligned}\int_{-\infty}^{-\frac{2}{n}} \frac{n}{\sqrt{4\pi k}} e^{-\frac{x^2 n^2}{4k}} dx &= \int_{\frac{2}{n}}^{\infty} \frac{n}{\sqrt{4\pi k}} e^{-\frac{x^2 n^2}{4k}} dx \\ &\leqslant \int_{\frac{2}{N}}^{\infty} \frac{N}{\sqrt{4\pi k}} e^{-\frac{x^2 n^2}{4k}} dx \\ &= \int_{\frac{2n}{2N\sqrt{k}}}^{\infty} \frac{N}{n\sqrt{\pi}} e^{-y^2} dy \\ &\leqslant \frac{N}{n\sqrt{\pi}} \frac{N\sqrt{k}}{2n} \int_{\frac{2n}{2N\sqrt{k}}}^{\infty} 2ye^{-y^2} dy\end{aligned} \tag{6.63}$$

Firstly, consider the case when $x = 0$,

$$\phi(0,0) = \lim_{t\to 0} \frac{1}{\sqrt{4\pi kt}} \to \infty. \tag{6.64}$$

Secondly, when $x \neq 0$, it gives that

$$\begin{aligned}\phi(x,0) &= \lim_{t\to 0} \frac{1}{\sqrt{4\pi kt}} e^{-\frac{x^2}{4kt}} \\ &\overset{\lambda=\frac{1}{\sqrt{t}}}{=} \lim_{\lambda\to\infty} \frac{1}{\sqrt{4\pi k}} \frac{\lambda}{e^{x^2\lambda^2/4k}} \\ &= \lim_{\lambda\to\infty} \frac{1}{\sqrt{4\pi k}} \frac{\lambda}{1 + x^2\lambda^2/4k} \\ &\leqslant \lim_{\lambda\to\infty} \frac{1}{\sqrt{4\pi k}} \frac{4k}{x^2} \frac{1}{\lambda} = 0 = \delta(x).\end{aligned} \tag{6.65}$$

Lastly, we need to calculate the integral of $\phi(x,t)$, it follows that

$$\begin{aligned}\int_{x=-\infty}^{x=\infty} \phi(x,t)dx &= \int_{-\infty}^{\infty} \phi(x,0)dx \\ &= \frac{1}{\sqrt{4\pi kt}} \int_{-\infty}^{\infty} e^{-\frac{x^2}{4kt}} dx \\ &\overset{z=\frac{x}{\sqrt{2kt}}}{=} \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}} dz \\ &= 1.\end{aligned} \tag{6.66}$$

Here, $N(y)$ is the cumulative standard normal distribution. And by taking $t = 0$, we have

$$\int_{-\infty}^{\infty} \phi(x,0)dx = 1. \tag{6.67}$$

From equations (6.64), (6.65) and (6.67), we can conclude that $\phi(x,0)$ is a delta function, which satisfies that

$$\phi(x,0) = \delta(x). \tag{6.68}$$

□

From the properties above, the link between $u(x,t)$ and $g(x)$ can be established.

Theorem 6.18.
We have the property that

$$u(x,0)=g(x). \tag{6.69}$$

Then we can rewrite the general solution for one dimensional heat equation,

$$u(x,t)=\int_{-\infty}^{\infty}\phi(x-y,t)g(y)dy=\int_{-\infty}^{\infty}\phi(x-y,t)u(y,0)dy. \tag{6.70}$$

Proof. Property 6.17 and 6.16 shows that

$$u(x,0)=\int_{-\infty}^{\infty}\phi(x-y,0)g(y)dy=\int_{-\infty}^{\infty}\delta(x-y)g(y)dy=g(x). \tag{6.71}$$

□

Next, we would like to present a few heat equation's solving problems with initial conditions to aid in your comprehension of heat equation.

Theorem 6.19 (quarter-plane problem I)**.**
We are willing to work with a system corresponding to the boundary conditions of heat equation in this problem, where

$$\begin{cases} u_t=ku_{xx} & ,t>0,x>0 \\ u(x,0)=g(x) & ,x>0, \\ u(0,t)=0 & ,t>0. \end{cases} \tag{6.72}$$

The solution of the system is that

$$u(x,t)=\int_{0}^{\infty}\left[\phi(x-y,t)-\phi(x+y,t)\right]g(y)dy. \tag{6.73}$$

Proof. Here, $g(x)$ is defined in the area that $x>0$, but, we would need to find a function defined throughout the x-axis in order to apply heat function's general solution. Additionally, we must ensure that the value of the function stays consistent with $g(x)$ when $x>0$.

Define the odd extension of $g(x)$, say

$$\widetilde{g}(x)=\begin{cases} g(x) & ,x>0, \\ -g(-x) & ,x<0, \\ 0 & ,x=0. \end{cases} \tag{6.74}$$

Then we can demonstrate that the solution for quarter-plane problem I is that

$$u(x,t)=\int_{-\infty}^{\infty}\phi(x-y,t)\widetilde{g}(y)dy. \tag{6.75}$$

Actually, the only thing we need to verify is that equation (6.75) satisfies the last boundary condition $u(0,t)=0$.

Note that, when $x = 0$, it follows that

$$u(0,t) = \int_{-\infty}^{\infty} \phi(-y,t)\widetilde{g}(y)dy. \tag{6.76}$$

Since $\phi(-y,t)$ is an even function, $\widetilde{g}(y)$ is an odd function, this suggests that $\phi(-y,t)\cdot\widetilde{g}(y)$ is an odd function. So we have $u(0,t) = 0$ and we are certain that equation (6.75) is the solution.

To simplify,

$$\begin{aligned}
u(x,t) &= \int_{-\infty}^{\infty} \phi(x-y,t)\widetilde{g}(y)dy \\
&= -\int_{-\infty}^{0} \phi(x-y,t)g(-y)dy + \int_{0}^{\infty} \phi(x-y,t)g(y)dy \\
&\overset{z=-y}{=} \int_{\infty}^{0} \phi(x+z,t)g(z)dz + \int_{0}^{\infty} \phi(x-y,t)g(y)dy \\
&= -\int_{0}^{\infty} \phi(x+y,t)g(y)dy + \int_{0}^{\infty} \phi(x-y,t)g(y)dy \\
&= \int_{0}^{\infty} \left[\phi(x-y,t) - \phi(x+y,t)\right] g(y)dy.
\end{aligned} \tag{6.77}$$

Then we have verified the theorem. □

Theorem 6.20 (quarter-plane problem II)**.**
Then we consider another case where the boundary condition of heat equation has changed, that is

$$\begin{cases} u_t = ku_{xx} & ,t > 0, x > 0 \\ u(x,0) = g(x) & ,x > 0, \\ u_x(0,t) = 0 & ,t > 0. \end{cases} \tag{6.78}$$

The solution is that

$$u(x,t) = \int_{0}^{\infty} \left[\phi(x-y,t) + \phi(x+y,t)\right] g(y)dy. \tag{6.79}$$

Proof. Unlike quarter plane problem I, here, we define the odd extension of $g(x)$, say

$$\widetilde{g}(x) = \begin{cases} g(x) & ,x \geqslant 0, \\ -g(-x) & ,x < 0. \end{cases} \tag{6.80}$$

Then we directly calculate the solution:

$$
\begin{aligned}
u(x,t) &= \int_{-\infty}^{\infty} \phi(x-y,t)\widetilde{g}(y)dy \\
&= \int_{-\infty}^{0} \phi(x-y,t)g(-y)dy + \int_{0}^{\infty} \phi(x-y,t)g(y)dy \\
&\overset{z=-y}{=} -\int_{\infty}^{0} \phi(x+z,t)g(z)dz + \int_{0}^{\infty} \phi(x-y,t)g(y)dy \\
&= \int_{0}^{\infty} \phi(x+y,t)g(y)dy + \int_{0}^{\infty} \phi(x-y,t)g(y)dy \\
&= \int_{0}^{\infty} \left[\phi(x-y,t) + \phi(x+y,t)\right] g(y)dy.
\end{aligned}
\tag{6.81}
$$

If we check whether $u(x,t)$ would satisfy the boundary condition, it gives that

$$
u_x(0,t) = \int_{0}^{\infty} \left[\phi_x(-y,t) + \phi_x(y,t)\right] g(y)dy. \tag{6.82}
$$

Note that ϕ_x is odd and g is even, we conclude that $u_x(0,t) = 0$, then we have proved that equation (6.79) is the solution for quarter plane problem II. □

6.3.2 Transferring Black-Scholes-Merton Equation into Heat Equation

Let us come back to Black-Scholes-Merton partial differential equation we have discussed before, that is

$$
c_t(t,x) + rxc_x(t,x) + \frac{1}{2}\sigma^2 x^2 c_{xx}(t,x) = rc(t,x). \tag{6.83}
$$

And the terminal value for the partial differential equation should be

$$
c(T,x) = (x-K)^+. \tag{6.84}
$$

Here, we would like to introduce a method to the equation into heat equation, please referring to Wilmott, Howison and Dewynne (1995).

The first step is permutation, which is the intentional modification to change the backward condition into a forward condition, and in the meanwhile, eliminating the annoying coefficients in front of c_{xx} term. We set new variable α, τ and $v(\tau, r)$ such that

$$
\alpha = \ln(\frac{x}{K}),\ \tau = \frac{1}{2}\sigma^2(T-t) \ \textit{and}\ v(\tau,\alpha) = \frac{1}{K}c(t,x). \tag{6.85}
$$

And then

$$
x = Ke^{\alpha},\ t = T - \frac{2\tau}{\sigma^2} \ \textit{and}\ c(t,x) = Kv(\tau,\alpha). \tag{6.86}
$$

It follows the chain rule that

$$
\begin{aligned}
c_x(t,x) &= Kv_x(\tau,\alpha) \\
&= K\left(\frac{\partial v}{\partial \tau}\frac{\partial \tau}{\partial x} + \frac{\partial v}{\partial \alpha}\frac{\partial \alpha}{\partial x}\right) \\
&= K\left(0 + \frac{1}{x}v_\alpha\right) = \frac{K}{x}v_\alpha.
\end{aligned} \tag{6.87}
$$

$$
\begin{aligned}
c_t(t,x) &= Kv_t(\tau,\alpha) \\
&= K\left(\frac{\partial v}{\partial \tau}\frac{\partial \tau}{\partial t} + \frac{\partial v}{\partial \alpha}\frac{\partial \alpha}{\partial t}\right) \\
&= K\left(-\frac{1}{2}\sigma^2 v_\tau + 0\right) = -\frac{1}{2}K\sigma^2 v_\tau.
\end{aligned} \tag{6.88}
$$

$$
\begin{aligned}
c_{xx}(t,x) &= Kv_{xx}(\tau,\alpha) \\
&= K\left(v_x(\tau,\alpha)\right)_x \\
&= K^2\left(\frac{1}{x}v_\alpha\right)_x \\
&= K^2\left(-\frac{1}{x^2}v_\alpha + \frac{1}{x}\left(v_\alpha\right)_x\right) \\
&= -\frac{K^2}{x^2}v_\alpha + \frac{K^2}{x}\left(\frac{\partial v_\alpha}{\partial \tau}\frac{\partial \tau}{\partial x} + \frac{\partial v_\alpha}{\partial \alpha}\frac{\partial \alpha}{\partial x}\right) \\
&= -\frac{K^2}{x^2}v_\alpha + \frac{K^2}{x}\left(0 + \frac{1}{x}v_{\alpha\alpha}\right) \\
&= \frac{K^2}{x^2}(v_{\alpha\alpha} - v_\alpha).
\end{aligned} \tag{6.89}
$$

Then we can substitute equation (6.86), (6.87), (6.88) and (6.89) into Black-Scholes-Merton equation with the terminal condition. This suggests that

$$
v_\tau = v_{\alpha\alpha} + \left(\frac{2r}{\sigma^2} - 1\right)v_\alpha - \frac{2r}{\sigma^2}v. \tag{6.90}
$$

We set $l = \frac{2r}{\sigma^2}$, then

$$
v_\tau = v_{\alpha\alpha} + (l-1)\,v_\alpha - lv. \tag{6.91}
$$

For the boundary condition, note that, when $(t,x) = (T,x)$, we have $(\tau,\alpha) = (0,\alpha)$. And $(x-K)^+ = (Ke^\alpha - K)^+ = K(e^\alpha - 1)^+$. By substituting above results into the terminal condition, we have

$$
\begin{aligned}
c(T,x) &= Kv(,\alpha) \\
&= (x-K)^+ = K(e^\alpha - 1)^+,
\end{aligned} \tag{6.92}
$$

which is equivalent to say that

$$v(0, \alpha) = (e^{\alpha} - 1)^{+}. \tag{6.93}$$

Equation (6.93) is the designed boundary condition for equation (6.91) after change of variable.

The next step is to convert equation (6.91) into the wave equation, and according to Wilmott, Howison and Dewynne (1995), this can be achieved by using a simple change of variable. We define a new function $u(\tau, \alpha)$ and variables p, q such that

$$v(\tau, \alpha) = e^{p\tau + q\alpha} u(\tau, \alpha). \tag{6.94}$$

The left hand side of equation (6.91) says that

$$v_{\tau} = pe^{p\tau + q\alpha} u + e^{p\tau + q\alpha} u_{\tau}; \tag{6.95}$$

while the right hand side of equation (6.91) says that

$$\begin{aligned} v_{\alpha\alpha} + (l-1)\, v_{\alpha} - lv = {} & \left(e^{p\tau+q\alpha} u_{\alpha\alpha} + 2qe^{p\tau+q\alpha} u_{\alpha} + q^2 e^{p\tau+q\alpha} u\right) \\ & + (l-1)\left(e^{p\tau+q\alpha} u_{\alpha} + qe^{p\tau+q\alpha} u\right) - le^{p\tau+q\alpha} u. \end{aligned} \tag{6.96}$$

By equating the left hand side and the right hand side, we can conclude that

$$pu + u_{\tau} = u_{\alpha\alpha} + (2q + l - 1)u_{\alpha} + (q^2 + (l-1)q - l)u. \tag{6.97}$$

For equation (6.97), we only need to ensure that the coefficients of u term and u_{α} term are the same. It follows that

$$\begin{cases} 2q + l - 1 = 0, \\ p = q^2 + (l-1)q - l. \end{cases} \tag{6.98}$$

We have

$$q = -\frac{1}{2}(l-1) \ and \ p = -\frac{1}{4}(l+1)^2. \tag{6.99}$$

Then with

$$v(\tau, \alpha) = exp\{-\frac{1}{4}(l+1)^2\tau - \frac{1}{2}(l-1)\alpha\}u(\tau, \alpha), \tag{6.100}$$

it gives that

$$u_{\tau} = u_{\alpha\alpha}, \tag{6.101}$$

which is the heat equation with initial value

$$u(0, \alpha) = \left(e^{\frac{1}{2}(l+1)\alpha} - e^{\frac{1}{2}(l-1)\alpha}\right)^{+}. \tag{6.102}$$

6.3.3 deriving the solution

In this subsection, we want to deduce the solution for Black-Scholes-Merton equation; however, obtaining the ultimate result is difficult, so we would proceed step by step. Let us starts from the heat equation that we demonstrated after change of variable. That is

$$\begin{aligned} u_{\tau} &= u_{\alpha\alpha}, \\ with \ u(0, \alpha) &= \left(e^{\frac{1}{2}(l+1)\alpha} - e^{\frac{1}{2}(l-1)\alpha}\right)^{+}. \end{aligned} \tag{6.103}$$

From equation (6.51) and Theorem 6.18, we find that

$$u(\tau,\alpha)=\frac{1}{\sqrt{4\pi\tau}}\int_{-\infty}^{\infty}e^{-\frac{(\alpha-y)^2}{4\tau}}g(y)dy, \tag{6.104}$$

where

$$g(\alpha)=u(0,\alpha)=\left(e^{\frac{1}{2}(l+1)\alpha}-e^{\frac{1}{2}(l-1)\alpha}\right)^+. \tag{6.105}$$

Be careful that here we need to modify the order of two variables τ and α to fit heat equation's initial value. To calculate the integral, we would apply the properties of standard normal distribution. Firstly, for permutation, we assume $y'=(y-\alpha)/\sqrt{2\tau}$, and correspondingly $dy=\sqrt{2\tau}dy'$, then we have

$$\begin{aligned}u(\tau,\alpha)&=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty}e^{-\frac{1}{2}y'^2}\left(e^{\frac{1}{2}(l+1)(\sqrt{2\tau}y'+\alpha)}-e^{\frac{1}{2}(l-1)(\sqrt{2\tau}y'+\alpha)}\right)^+dy'\\&=\frac{1}{\sqrt{2\pi}}\int_{-\alpha/\sqrt{2\tau}}^{\infty}e^{-\frac{1}{2}y'^2}e^{\frac{1}{2}(l+1)(\sqrt{2\tau}y'+\alpha)}dy\\&\quad-\frac{1}{\sqrt{2\pi}}\int_{-\alpha/\sqrt{2\tau}}^{\infty}e^{-\frac{1}{2}y'^2}e^{\frac{1}{2}(l-1)(\sqrt{2\tau}y'+\alpha)}dy.\end{aligned} \tag{6.106}$$

For the first term, it gives that

$$\begin{aligned}&\frac{1}{\sqrt{2\pi}}\int_{-\alpha/\sqrt{2\tau}}^{\infty}e^{-\frac{1}{2}y'^2}e^{\frac{1}{2}(l+1)(\sqrt{2\tau}y'+\alpha)}dy\\&=\frac{1}{\sqrt{2\pi}}\int_{-\alpha/\sqrt{2\tau}}^{\infty}e^{-\frac{1}{2}(y'-\frac{1}{2}(l+1)\sqrt{2\tau})^2}e^{\frac{1}{4}(l+1)^2\tau+\frac{1}{2}(l+1)\alpha}dy\\&=\frac{1}{\sqrt{2\pi}}e^{\frac{1}{4}(l+1)^2\tau+\frac{1}{2}(l+1)\alpha}\int_{-\alpha/\sqrt{2\tau}-\frac{1}{2}(l+1)\sqrt{2\tau}}^{\infty}e^{-\frac{1}{2}\rho^2}d\rho\\&=e^{\frac{1}{4}(l+1)^2\tau+\frac{1}{2}(l+1)\alpha}N(d_1)\end{aligned} \tag{6.107}$$

During the calculation, we set $\rho=y'-\frac{1}{2}(l+1)\sqrt{2\tau}$. And $N(d_1)$ is the cumulative probability for normal distribution that satisfies

$$N(d_1)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{d_1}e^{-\frac{1}{2}y^2}dy, \tag{6.108}$$

where

$$d_1=\alpha/\sqrt{2\tau}+\frac{1}{2}(l+1)\sqrt{2\tau}. \tag{6.109}$$

Similarly, the second term of equation (6.106) satisfies

$$\frac{1}{\sqrt{2\pi}}\int_{-\alpha/\sqrt{2\tau}}^{\infty}e^{-\frac{1}{2}y'^2}e^{\frac{1}{2}(l-1)(\sqrt{2\tau}y'+\alpha)}dy=e^{\frac{1}{4}(l-1)^2\tau+\frac{1}{2}(l-1)\alpha}N(d_2). \tag{6.110}$$

And

$$N(d_2)=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{d_2}e^{-\frac{1}{2}y^2}dy, \tag{6.111}$$

where

$$d_2 = \alpha/\sqrt{2\tau} + \frac{1}{2}(l-1)\sqrt{2\tau}. \tag{6.112}$$

Recall that we have

$$v(\tau, \alpha) = exp\{-\frac{1}{4}(l+1)^2 - \frac{1}{2}(l-1)\alpha\}u(\tau, \alpha); \tag{6.113}$$

and in the meanwhile,

$$\alpha = \ln(\frac{x}{k}), \tau = \frac{1}{2}\sigma^2(T-t) \ and \ c(t,x) = Kv(\tau, \alpha). \tag{6.114}$$

All that remains is to substitute two equations above into the general solution of $u(\tau, \alpha)$, and it follows that

$$c(t,x) = xN(d_1) - Ke^{-r(T-t)}N(d_2), \tag{6.115}$$

where

$$d_1 = \frac{log(\frac{x}{k}) + (r + \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}, \tag{6.116}$$

$$d_2 = \frac{log(\frac{x}{k}) + (r - \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}. \tag{6.117}$$

Up to this point, we have correctly deduced the Black-Scholes-Merton partial differential equation, and we can see that the entire journey are challenging and fantastic. What it implied during the journey is that there might be some minor connections between all of the equations applied to real life, just like Black-Scholes-Merton equation and heat equation. When we struggle to find out these connections, we can see the light that leads us through the darkness. Actually, the most crucial thing in mathematical learning is the ongoing pursuit of enjoyment and beauty during the process. Remember that while these conclusions may help you find a job and achieve high incomes, life is more than that, and the experience is the most essential part of life.

For all of you who are reading now, I hope you can maintain your enthusiasm and faith in life and pursue every single thing that can make you delighted. Try to catch every chance that comes in your way, just as what we do in the early stage to derive the Black-Scholes-Merton equation, beginning with the potential connections with heat equation and trying to find a way to change the variables.

7 Sensitive Analysis for Stochastic Processes

7.1 Variance, Skewness and Kurtosis of Black-Scholes-Merton Equation

Variance, skewness and kurtosis are useful methods to determine the sensitivity of Black-Scholes-Merton equation. Here, we would give an introduction how these methods are calculated.

Definition 7.1 (variance). *Consider a stochastic process $X(t)$, the variance is defined as*

$$Var[X(t)] = \mathbb{E}[(X(t) - \mathbb{E}[X(t)])^2]. \tag{7.1}$$

Definition 7.2 (skewness). *Consider a stochastic process $X(t)$, the skewness is defined as*

$$Skew[X(t)] = \mathbb{E}[(X(t) - \mathbb{E}[X(t)])^3]. \tag{7.2}$$

Definition 7.3 (kurtosis). *Consider a stochastic process $X(t)$, the kurtosis is defined as*

$$Kur[X(t)] = \mathbb{E}[(X(t) - \mathbb{E}[X(t)])^4]. \tag{7.3}$$

Theorem 7.4. *For some $\alpha, \beta \in R$, it could be verified that*

$$Skew[\alpha X(t) + \beta] = Skew[X(t)],\ \ \alpha > 0; \tag{7.4}$$

while

$$Skew[\alpha X(t) + \beta] = -Skew[X(t)],\ \ \alpha < 0. \tag{7.5}$$

Theorem 7.5. *For some $\alpha, \beta \in R$, it could be verified that*

$$Kur[\alpha X(t) + \beta] = Kur[X(t)],\ \ \forall \alpha \in R. \tag{7.6}$$

7.2 Lévy measure

7.2.1 Lévy's Process and Infinite Divisible Random Variables

Definition 7.6 (Lévy's process). *Suppose the stochastic process $X(t)$ satisfies the following properties, then it is called Lévy's process:*
i) At time 0, we have $X(0) = 0$;
ii) any increment $X(t) - X(s)$ are independent of other increments with misalignment and time t;
iii) the process $X(t)$ is right continuous.

Definition 7.7 (infinite divisible). *For a random variable Γ, presume Γ is infinite divisible, then we could find a sequence $\{\Gamma_k\}_{k=1}^{k=n}$ such that Γ has the same distribution with the sum of the sequence, that is*

$$\Gamma \equiv \sum_{k=1}^{n} \Gamma_k. \tag{7.7}$$

Theorem 7.8 (properties of characteristic functions). *Suppose that Γ is infinite divisible, then its characteristic function satisfies the property that*

$$\Phi(w) = [\Phi_n(w)]^n, \tag{7.8}$$

where $\Phi_n(w)$ is the characteristic function of some random variable Γ_n.

Definition 7.9 (space of infinite divisible random variables)**.** *We use the notation $\mathcal{I}$ to denote the space of infinite divisible processes, this suggests that*

$$\mathcal{I} = \left\{ \Gamma : \Gamma \equiv \sum_{k=1}^{n} \Gamma_k, \ \exists \{\Gamma_k\}_{k=1}^{k=n} \right\} \tag{7.9}$$

Theorem 7.10. *The space of infinite divisible random variables $\mathcal{I}$ is a vector space.*

Proof. Take $\Gamma^1, \Gamma^2 \in \mathcal{I}$ and $\alpha \in R$, such that

$$\Gamma^i \equiv \sum_{k=1}^{n_i} \Gamma_k^i, \ i = 1, 2. \tag{7.10}$$

Suppose that $n = \max\{n_1, n_2\}$, we can write

$$\Gamma^i \equiv \sum_{k=1}^{n} \Gamma_k^i, \ i = 1, 2. \tag{7.11}$$

Therefore,

$$\alpha\Gamma^1 + \Gamma^2 \equiv \sum_{k=1}^{n} \alpha\Gamma_k^1 + \Gamma_k^2, \tag{7.12}$$

which implies that $\alpha\Gamma_1 + \Gamma_2 \in \mathcal{I}$. Hence, $\mathcal{I}$ is a vector space. □

Theorem 7.11. *$\mathcal{I}$ is a closed space.*

Sketch of proof. Recall that a closed space is equivalent to say that any convergent sequence converges in the space. We suppose that $\Gamma^n \to \Gamma$ and $(\Gamma^n)_n \in \mathcal{I}$, then we need to prove that $\Gamma \in \mathcal{I}$. □

7.2.2 Lévy Measure and Lévy-Khintchine Formula

Definition 7.12 (Lévy measure)**.** *We say $\mathcal{V}$ is a Lévy measure when*

$$\int_R \max\{x^2, 1\} \mathcal{V}(dx) < \infty. \tag{7.13}$$

Theorem 7.13 (Lévy-Khintchine Theorem)**.** *Consider a triple $(b, \sigma, \mathcal{V})$ with $b \in R$ and $\sigma \in R$, the random variable Γ is infinite divisible if and only if $\mathcal{V}$ is the Lévy's measure and the following equation holds true that*

$$\mathbb{E}[e^{iu\Gamma}] = \exp\left\{ ibu - \frac{1}{2}\sigma^2 u^2 + \int_{-\infty}^{\infty} (e^{iux} - 1 - iux1_{x\in(-1,1)})\mathcal{V}(dx) \right\} \tag{7.14}$$

Properties 7.14 (characteristic exponent)**.** *Suppose $X(t)$ is a Lévy process, then there exists $\Psi(w)$ such that*

$$\mathbb{E}[e^{iwX(t)}] = e^{-t\Psi(w)}, \tag{7.15}$$

where $\Psi(w)$ is the characteristic exponent of $X(t)$.

Theorem 7.15 (Lévy process, Lévy-Khintchine formula).
Consider a triple $(b, \sigma^2, \mathcal{V})$ with $b \in R$ and $\sigma \in R^+$, suppose that the random variable $X(t)$ is a Lévy process and $\mathcal{V}$ is the Lévy's measure, then the characteristic exponent of $X(t)$ satisfies the formula that

$$\Psi(w) = ibw - \frac{1}{2}\sigma^2 w^2 + \int_{-\infty}^{\infty} (e^{iwx} - 1 - iwx 1_{x \in (-1,1)})\mathcal{V}(dx). \tag{7.16}$$

7.2.3 Cumulants

Before the introduction to cumulants, we would firstly interpret the definition of cumulant generating function.

Definition 7.16 (cumulant generating function). *For a stochastic process $X(t)$, the cumulant generating function is defined as*

$$K(t) = \log(\mathbb{E}[e^{tX(t)}]), \tag{7.17}$$

where $\mathbb{E}[e^{tX(t)}]$ is the moment generating function.

Definition 7.17 (cumulants). *Cumulants of a stochastic process could be derived by applying Taylor's expansion on cumulant generating function, it follows that*

$$K(t) = \sum_{n=0}^{\infty} k_n \frac{t^n}{n!}. \tag{7.18}$$

Then $(k_n)_n$ are the sequence of cumulants.

Theorem 7.18. *Suppose the characteristic function of a random variable $X(t)$ is $\Phi(w)$, then it follows that*

$$\log(\Phi(w)) = \int_{n=0}^{\infty} k_n \frac{t^n}{n!}. \tag{7.19}$$

Hence, the k-th cumulant satisfies the formula that

$$k_n = i^{-n} \left[\frac{d^n}{dw^n} \Phi(w) \right] \Bigg|_{w=0}. \tag{7.20}$$

Theorem 7.19 (cumulants, sensitive analysis methods on unit time).
The relationship between unit time variance, skewness, kurtosis and cumulants is shown below.

$$Var[X(t)] = k_2; \tag{7.21}$$

$$Skew[X(t)] = \frac{k_3}{k_2^{\frac{3}{2}}}; \tag{7.22}$$

$$Kur[X(t)] = \frac{k_4}{k_2^2}. \tag{7.23}$$

7.3 Sensitive Analysis for Black-Scholes-Merton Equation

Definition 7.20 (Black-Scholes-Merton log-return density function).
Note that the Black-Scholes-Merton equation is not a Lévy's process, then we would analyze the sensitivity using Black-Scholes-Merton log-return density function. Define

$$\Gamma(t) = \log\left(\frac{S(t)}{S(0)}\right) = \int_0^t \sigma dW(t) + \int_0^t \left(r - \frac{1}{2}\sigma^2\right) dt. \tag{7.24}$$

Theorem 7.21 (distribution of $\Gamma(t)$). *The Black-Schole-Merton log-return density function $\Gamma(t)$ follows a linear normal distribution with*

$$\Gamma(t) \sim N((\alpha - \frac{1}{2}\sigma^2)t, \sigma^2 t). \tag{7.25}$$

Proof. Note that the formula for $\Gamma(t)$ should be

$$\Gamma(t) = \log\left(\frac{S(t)}{S(0)}\right) = \int_0^t \sigma dW(t) + \int_0^t \left(r - \frac{1}{2}\sigma^2\right) dt. \tag{7.26}$$

Consider time period $\tau \to 0$, we have

$$\begin{aligned}\Gamma(t+\tau) - \Gamma(t) &= \int_t^{t+\tau} \sigma dW(u) + \int_t^{t+\tau} \left(\alpha - \frac{1}{2}\sigma^2\right) du \\ &= \sigma(W(t+\tau) - W(t)) + \left(\alpha - \frac{1}{2}\sigma^2\right)\tau.\end{aligned} \tag{7.27}$$

Hence, for a time interval $[0, T]$ and separation $\Pi = \{t_0, t_1, \cdots, t_n\}$, it follows that

$$\Gamma(T) - \Gamma(0) = \sum_{i=1}^{n} \Gamma((i+1)\tau) - \Gamma(i\tau) = \sigma(W(T - W(0))) + \left(\alpha - \frac{1}{2}\sigma^2\right) T. \tag{7.28}$$

Therefore,

$$\Gamma(t) = \sigma W(t) + \left(\alpha - \frac{1}{2}\sigma^2\right) t. \tag{7.29}$$

It emerges that $\Gamma(t)$ follows a Brownian motion with $B(\alpha - \frac{1}{2}\sigma^2, \sigma)$, and then

$$\Gamma(t) \sim N((\alpha - \frac{1}{2}\sigma^2)t, \sigma^2 t). \tag{7.30}$$

□

Theorem 7.22 (characteristic component for $\Gamma(t)$).
Consider Lévy's process $\Gamma(t)$, the characteristic component satisfies the formula that

$$\Psi(w) = i\left(\alpha - \frac{1}{2}\sigma^2\right) w - \frac{1}{2}\sigma^2 w^2. \tag{7.31}$$

Proof. Since Black-Scholes-Merton log-return density function is a Lévy's process, there exists a triple $(b, \sigma^2, \mathcal{V})$ such that the characteristic component is defined as

$$\Psi(w) = ibw - \frac{1}{2}\sigma^2 w^2 + \int_{-\infty}^{\infty} (e^{iwx} - 1 - iwx1_{x\in(-1,1)})\mathcal{V}(dx) \tag{7.32}$$

Note that the Black-Scholes-Merton log-return density function is independent of jump processes, while the Lévy measure $\mathcal{V}(dx)$ denotes the arrival rate of jumps, so we have

$$\mathcal{V}(dx) = 0. \tag{7.33}$$

Since $\Gamma(t)$ follows the linear normal distribution with $N((\alpha - \frac{1}{2}\sigma^2)t, \sigma^2 t)$, then

$$b = \alpha - \frac{1}{2}\sigma^2. \tag{7.34}$$

Therefore, the characteristic component could be written as

$$\Psi(w) = i\left(\alpha - \frac{1}{2}\sigma^2\right)w - \frac{1}{2}\sigma^2 w^2. \tag{7.35}$$

□

Theorem 7.23 (first four cumulants of Black-Scholes-Merton equation)**.**
The first four cumulants of Black-Scholes-Merton equation, or the Black-Scholes-Merton log-return density function, can be computed as

$$k_1 = \alpha - \frac{1}{2}\sigma^2; \tag{7.36}$$

$$k_2 = \sigma^2; \tag{7.37}$$

$$k_3 = k_4 = 0. \tag{7.38}$$

Proof. To calculate the first four cumulants, we apply the equation for the k-th cumulant, that is

$$k_n = i^{-n}\left[\frac{d^n}{dw^n}\Psi(w)\right]\bigg|_{w=0}. \tag{7.39}$$

The first order derivative of the characteristic component satisfies the formula that

$$\frac{d}{dw}\Psi(w) = i\left(\alpha - \frac{1}{2}\sigma^2\right) - \sigma^2 w; \tag{7.40}$$

the second order derivative of the characteristic component satisfies the formula that

$$\frac{d^2}{dw^2}\Psi(w) = -\sigma^2; \tag{7.41}$$

the third and fourth order derivatives of the characteristic component are equal to zero. Hence,

$$k_1 = \alpha - \frac{1}{2}\sigma^2; \tag{7.42}$$

$$k_2 = \sigma^2; \tag{7.43}$$

$$k_3 = k_4 = 0. \tag{7.44}$$

□

Theorem 7.24 (unit time variance, skewness and kurtosis for Black-Scholes-Merton equation)**.**

The unit time variance, skewness and kurtosis could be calculated as follow:

$$Var[\Gamma(t)] = k_2 = \sigma^2; \tag{7.45}$$

$$Skew[\Gamma(t)] = \frac{k_3}{k_2^{\frac{3}{3}}} = 0; \tag{7.46}$$

$$Kur[\Gamma(t)] = \frac{k_4}{k_3^2} = 0. \tag{7.47}$$

For the Black-Scholes-Merton equation, we only need to consider the variance in the sensitive analysis.

8 Risk-Neutral Pricing

8.1 Girsanov's Theorem

In previous sections, random variables are defined under a probability measure. It is uncertain whether this point has been noticed, but if not, that is fine. We do not emphasize the probability space because random variable that follows some sort of distribution must belong to well-defined probability measures. So we would roughly review some properties of probability measure.

Properties 8.1 (a revision for probability space)**.**
For the probability space $(\Omega, \mathscr{F}, \mathbb{P})$*, suppose* $\Omega = \{a_1, a_2, \ldots, a_n\}$*, which denotes the sample space. Then, for* $i, j \in \{1, 2, \ldots, n\}$*, we say*

$$\emptyset \in \mathscr{F},\ \Omega \in \mathscr{F},\ a_i^c \in \mathscr{F} \ and\ a_i \cup a_j \in \mathscr{F}. \tag{8.1}$$

For $\forall A \in \mathscr{F}$*, it has a probability* $\mathbb{P}(A) \geqslant 0$*. Suppose every event* $a_i \in \Omega$ *contains a value, with the probability, we can then calculate the expected value and other properties.*

With the basic idea of probability space, we would like to introduce a method to vary the probability measure $\mathbb{P}$ into $\widetilde{\mathbb{P}}$.

Firstly, consider a probability space for discrete variable $(\Omega, \mathscr{F}, \mathbb{P})$, according to Properties 8.1, we have

$$\mathbb{P}(\Omega) = 1,\ \mathbb{P}(A) = \sum_{X_i \in A} P(X_i)\ for\ A \in \mathscr{F}, \tag{8.2}$$

where A is a discrete set. In order to reset the probability measure, the only thing we need to do is to consider the probability of each element in set A as a whole, and assign precise weight. For new probability measure $\widetilde{\mathbb{P}}$ and weights $Z(\omega)$, this suggests that,

$$\widetilde{\mathbb{P}}(A) = \sum_{X_i \in A} Z(X_i)\mathbb{P}(X_i)\ for\ A \in \mathscr{F}. \tag{8.3}$$

Then for continuous variable, the probability under measure $\mathbb{P}$ satisfies

$$\mathbb{P}(A) = \int_A dP(\omega) \ for \ A \in \mathscr{F}. \tag{8.4}$$

Consider a non-negative random variable Z such that the expected value $\mathbb{E}[Z] = 1$, and we can define the new probability measure that

$$\widetilde{\mathbb{P}}(A) = \int_A Z(\omega) d\mathbb{P}(\omega) \ for \ A \in \mathscr{F}. \tag{8.5}$$

Then we would demonstrate the relationship between probability measure $\mathbb{P}$ and $\widetilde{\mathbb{P}}$.

Theorem 8.2 (expected value under $\widetilde{\mathbb{P}}$)**.**
Under new probability measure $\widetilde{\mathbb{P}}$, consider the expected value of random variable Y, we have the following property that

$$\widetilde{\mathbb{E}}[Y] = \mathbb{E}[YZ] \ for \ Y \in \mathscr{F}. \tag{8.6}$$

And when $P(Z = 0) = 0$, it follows that

$$\mathbb{E}[Y] = \widetilde{\mathbb{E}}\left[\frac{Y}{Z}\right]. \tag{8.7}$$

Proof. We assume the corresponding value for event ω is $x(\omega)$, it follows that

$$\mathbb{E}[Y] = \sum_{\omega \in Y} \mathbb{E}[x(\omega)] = \sum_{\omega \in Y} x(\omega)\mathbb{P}(\omega); \tag{8.8}$$

and then,

$$\widetilde{\mathbb{E}}[Y] = \sum_{\omega \in Y} x(\omega)Z(\omega)\mathbb{P}(\omega) = \sum_{\omega \in Y} \mathbb{E}[x(\omega)Z(\omega)] = \mathbb{E}[YZ]. \tag{8.9}$$

□

In previous interpretations, Z is defined to be a random variable; while since we would like to analyze the efforts of change of measure on stochastic process, we specialize $Z(t)$ to be a stochastic process and $\mathscr{F}(t)$ to be a filtration. And, in the meantime, $t \in [0, t)$ and $\mathbb{E}[Z(t)] = 1$.

Definition 8.3 (Radon-Nikodým derivative process)**.**
We say that a stochastic process $Z(t)$ is the Radon-Nikodým derivative process if and only if

$$Z(t) = \mathbb{E}[Z|\mathscr{F}(t)]. \tag{8.10}$$

Properties 8.4. *The Radon-Nikodým derivative process is a martingale.*

Proof. Suppose $0 \leqslant s \leqslant t \leqslant T$, then we have

$$\mathbb{E}[Z(t)|\mathscr{F}(t)] = \mathbb{E}[\mathbb{E}[Z|\mathscr{F}(t)]|\mathscr{F}(s)] = \mathbb{E}[Z|\mathscr{F}(s)] = Z(s). \tag{8.11}$$

Note that the second equality is due to the iterated conditioning rule for conditional expectations. □

Actually, with the definition of Radon-Nikodým derivative process, we can then rewrite Theorem 8.2 for Randon-Nikodým derivative process $Z(t)$.

Theorem 8.5. *Under probability space $(\Omega, \mathscr{F}, \mathbb{P})$ and $(\Omega, \mathscr{F}, \widetilde{\mathbb{P}})$, for Randon-Nikodým derivative process $Z(t)$ and random variable Y which is $\mathscr{F}(t)$-measurable, it gives that*

$$\widetilde{\mathbb{E}}[Y] = \mathbb{E}[YZ(t)]. \tag{8.12}$$

Proof. According to Properties 8.3, we can say

$$\widetilde{\mathbb{E}}[Y] = \mathbb{E}[YZ] = \mathbb{E}[\mathbb{E}[YZ|\mathscr{F}(t)]] = \mathbb{E}[Y\mathbb{E}[Z|\mathscr{F}(t)]] = \mathbb{E}[YZ(t)]. \tag{8.13}$$

In the equation, the third and forth equality is due to independence and taking out what is known for conditional expectations. □

And then we can calculate the conditional expectation under new probability measure, denoted by original probability space.

Theorem 8.6. *We suppose $0 \leqslant s \leqslant t \leqslant T$. Still for Randon-Nikodým derivative process $Z(t)$ and $\mathscr{F}(t)$-measurable random variable Y, we have*

$$\widetilde{\mathbb{E}}[Y|\mathscr{F}(s)] = \frac{1}{Z(t)}\mathbb{E}[YZ(t)|\mathscr{F}(s)]. \tag{8.14}$$

Proof. To prove the theorem, we start by leveraging the partial-averaging property on the left hand side of equation (8.14), that is to say,

$$\int_A \frac{1}{Z(t)}\mathbb{E}[YZ(t)|\mathscr{F}(s)]d\widetilde{\mathbb{P}} = \int_A Y d\widetilde{\mathbb{P}} \quad for \quad \forall A \in \mathscr{F}(s). \tag{8.15}$$

As for the right hand side of equation (8.14), we can also utilize partial-averaging property under probability space $\widetilde{\mathbb{P}}$, then

$$\int_A \frac{1}{Z(t)}\widetilde{\mathbb{E}}[YZ(t)|\mathscr{F}(s)]d\widetilde{\mathbb{P}} = \int_A Y d\widetilde{\mathbb{P}} \quad for \quad \forall A \in \mathscr{F}(s), \tag{8.16}$$

and this equation implies that the expected value of $\frac{1}{Z(t)}\widetilde{\mathbb{E}}[YZ(t)|\mathscr{F}(s)]$ should be equal to that of Y in set A, by using an indicator 1_A, we have

$$\widetilde{\mathbb{E}}\left[1_A \frac{1}{Z(t)}\widetilde{\mathbb{E}}[YZ(t)|\mathscr{F}(s)]\right] = \widetilde{\mathbb{E}}[1_A Y]. \tag{8.17}$$

Therefore,

$$
\begin{aligned}
\widetilde{\mathbb{E}}[Y|\mathscr{F}(s)] &= \widetilde{\mathbb{E}}\left[\widetilde{\mathbb{E}}[Y|Y\in A]\middle|\mathscr{F}(s)\right] \\
&= \widetilde{\mathbb{E}}\left[\widetilde{\mathbb{E}}[1_A Y]\middle|\mathscr{F}(s)\right] \\
&= \widetilde{\mathbb{E}}\left[1_A\frac{1}{Z(t)}\widetilde{\mathbb{E}}[YZ(t)|\mathscr{F}(s)]\right] \\
&= \mathbb{E}\left[1_A\mathbb{E}[YZ(t)|\mathscr{F}(s)]\right] \\
&= \mathbb{E}[1_A YZ(t)] \\
&= \widetilde{\mathbb{E}}[1_A Y] \\
&= \int_A Y d\widetilde{\mathbb{P}}
\end{aligned}
\tag{8.18}
$$

Then, equation (8.15) and equation (8.18) verifies the theorem. □

Then we would discuss Girsanov's Theorem for one dimensional Brownian motion, it demonstrates the form of Brownian motion under new probability measure $\widetilde{\mathbb{P}}$.

Lemma 8.7. *(Novikov)*
For adapted process $\theta(t)$, define

$$
Z(t) = \exp\left\{-\int_0^t \theta(s)dW(s) - \frac{1}{2}\int_0^t \theta^2(s)ds\right\}. \tag{8.19}
$$

And we suppose that

$$
\mathbb{E}\ \exp\left\{\int_0^t \theta^2(s)ds\right\} < \infty, \tag{8.20}
$$

then we can verify that $Z(t)$ is a martingale and $\mathbb{E}[Z(t)] = 1$.

Sketch of proof. We would leverage Itô-Deoblin formula to prove the lemma, and suppose $Z(t) = f(X(t)) = e^{X(t)}$, $X(t) = -\int_0^t \theta(s)dW(s) - \frac{1}{2}\int_0^t \theta^2(s)ds$, then we have

$$
dZ(t) = f'(X(t))dX(t) + \frac{1}{2}f''(X(t))dX(t)dX(t). \tag{8.21}
$$

We apply the property that

$$
dX(t)dX(t) = [X,X](t) = \theta^2 dt, \tag{8.22}
$$

and this suggests that

$$
dZ(t) = -\theta(t)Z(t)dW(t). \tag{8.23}
$$

Then, equation (8.23) implies that $Z(t)$ is a martingale since $\theta(t)Z(t)$ is left-continuous and $W(t)$ is a martingale. Therefore, $\mathbb{E}[Z(t)] = Z(0) = 1$. □

Lemma 8.8 (Lévy's Theorem).
A $\mathscr{F}(t)$-measurable stochastic process $M(t)$ is a Brownian motion under probability space $(\Omega, \mathscr{F}, \mathbb{P})$ with $t \in [0,T]$ if $M(t)$ satisfies following conditions:
i) $M(t)$ starts at zero at time zero, namely,

$$P(M(0) = 0) = 1; \tag{8.24}$$

ii) $M(t)$ is a continuous martingale with filtration $\mathscr{F}(t)$;
iii) the quadratic variation $[M,M](t) = t$ for all $t \in [0,T]$.

Theorem 8.9 (Girsanov's Theorem for one dimensional Brownian motion).
Suppose that $W(t)$ is the Brownian motion under probability space $(\Omega, \mathscr{F}, \mathbb{P})$, and $\theta(t)$ is an adapted process, $t \in [0,T]$. Then we define

$$Z(t) = \exp\left\{-\int_0^t \theta(s)dW(s) - \frac{1}{2}\int_0^t \theta^2(s)ds\right\}, \quad Z = Z(T), \tag{8.25}$$

$$\widetilde{W}(t) = W(t) + \int_0^t \theta(u)du. \tag{8.26}$$

If we have

$$\mathbb{E}\int_0^T \theta^2(u)Z^2(u)du < \infty, \tag{8.27}$$

then it is noteworthy that $\mathbb{E}[Z] = 1$ and $\widetilde{W}(t)$ is a Brownian motion under probability measure $\widetilde{\mathbb{P}}$.

Proof. According to Lemma 8.7, $Z(t)$ is a martingale, and certainly, $\mathbb{E}[Z] = \mathbb{E}[Z(T)] = 1$. So the key point is to prove that $\widetilde{W}(t)$ is a Brownian motion.

To verify that $\widetilde{W}(t)$ is the Brownian motion under $\widetilde{\mathbb{P}}$, from Theorem 8.8, Lévy's Theorem, it is prone to see that $\widetilde{M}(t)$ starts at zero, satisfies continuity property and $[\widetilde{W},\widetilde{W}](t) = [W,W](t) = t$, so it is equivalent to prove that $\widetilde{W}(t)$ is a martingale.

We firstly verify that $\widetilde{W}(t)Z(t)$ is a martingale under P. According to Itô's product rule, we have

$$\begin{aligned} d(\widetilde{W}(t)Z(t)) &= \widetilde{W}(t)dZ(t) + Z(t)d\widetilde{W}(t) + d\widetilde{W}(t)dZ(t) \\ &= -\widetilde{W}(t)\theta(t)Z(t)dW(t) + Z(t)dW(t) + Z(t)\theta(t)dt \\ &\quad +(dW(t) + \theta(t)dt)(-\theta(t)Z(t)dW(t)) \\ &= (-\widetilde{W}(t)\theta(t) + 1)Z(t)dW(t). \end{aligned} \tag{8.28}$$

Since $(-\widetilde{W}(t)\theta(t) + 1)Z(t)$ is left continuous and $W(t)$ is a martingale, then we say $\widetilde{W}(t)Z(t)$ is a martingale under probability measure P.

Note that $Z(t)$ is a martingale and $Z = Z(t)$ as defined, then it gives that

$$Z(t) = \mathbb{E}[Z(T)|\mathscr{F}(t)] = \mathbb{E}[Z|\mathscr{F}(t)] \quad for\ 0 \leqslant t \leqslant T, \tag{8.29}$$

which implies that $Z(t)$ is the Radon-Nikodým derivative process.

From Theorem 8.6, for $0 \leqslant s \leqslant t \leqslant T$, we have

$$\widetilde{\mathbb{E}}[\widetilde{W}(t)|\mathscr{F}(s)] = \frac{1}{Z(s)}\mathbb{E}[\widetilde{W}(t)Z(t)|\mathscr{F}(s)] = \frac{1}{Z(t)}\widetilde{W}(s)Z(s) = \widetilde{W}(s). \tag{8.30}$$

Thus, $\widetilde{W}(t)$ is a martingale under P, and we proved the Girsanov's Theorem for one dimensional Brownian motion. □

Under different probability measure, the expected value of stochastic processes changes, just like one dimensional Brownian motion $W(t)$ and $\widetilde{W}(t)$, but what remains unchanged is the probabilities. So, we would like to test whether the computations may be simplifies after changing the probability measure, which is the aim of introducing Girsanov's Theorem in the section.

8.2 Risk-Neutral Measure for Stock

Here, we would introduce the performance of stochastic model for stock price after applying Risk-Neutral measure.

Recall that the stock price is defined to be a generalized Brownian motion, that is

$$S(t) = S(0)\ \exp\{\int_0^t \sigma(s)dW(s) + \int_0^t \left(\alpha(s) - \frac{1}{2}\sigma^2(s)\right) ds\} \quad for\ 0 \leqslant t \leqslant T, \tag{8.31}$$

where $W(t)$ is the Brownian motion under probability space $(\Omega, \mathscr{F}, \mathbb{P})$ and $\alpha(t), \sigma(t)$ are adapted processes. And corresponding, the differential form of $S(t)$ satisfies

$$dS(t) = \alpha(t)S(t)dt + \sigma(t)S(t)dW(t). \tag{8.32}$$

By rewriting the above equation as

$$\frac{dS(t)}{S(t)}\alpha(t)dt + \sigma(t)dW(t). \tag{8.33}$$

Then we can see that the mean rate of return for stock price $S(t)$ and volatility should be $\alpha(t)$ and $\sigma(t)$ respectively.

In order to leverage risk-neutral measure on $S(t)$, we would define the discount process.

Definition 8.10 (continuous discount rate)**.**
Suppose an trader invests in the money market account, and we take no account of the market risk in the account. Now, assume the account price is $x(0)$ at time 0 and $R(t)$ denotes the interest rate, and then the rate of return at time t satisfies the property that

$$\frac{x(t+\Delta(t))-x(t)}{x(t)}=R(t)\Delta(t), \tag{8.34}$$

which suggests that

$$x(t)=x(0)\exp\left\{\int_0^t R(s)ds\right\}. \tag{8.35}$$

Let $D(t)$ denote the discount rate, then we have

$$D(t)x(t)=x(0), \tag{8.36}$$

or, after substituting equation (8.35),

$$D(t)=\exp\left\{-\int_0^t R(s)ds\right\}, \tag{8.37}$$

which is the formula for continuous discount rate.

Definition 8.11 (discount process)**.**
Since we have discussed the non-random discount rate in Definition 8.10, for an adapted interest rate process $R(t)$, we can hypothesis that the discount process $D(t)$ also satisfies equation (8.37) that

$$D(t)=\exp\left\{-\int_0^t R(s)ds\right\}, \tag{8.38}$$

and then

$$dD(t)=-R(t)D(t)dt. \tag{8.39}$$

Proof. We suppose that $I(t)=-\int_0^t R(s)ds$. Then it follows that $dI(t)=R(t)dt$ and $dI(t)dI(t)=R^2(t)dtdt=0$. Define $f(x)=e^{-x}$, and then $f'(x)=-e^{-x}=-f(x)$ and $f''(x)=e^{-x}=f(x)$. By utilizing Itô-Deoblin formula, we have

$$\begin{aligned}
dD(t)&=df(I(t))\\
&=f'(I(t))dI(t)+\frac{1}{2}f''(I(t))dI(t)dI(t)\\
&=-f(I(t))R(t)dt\\
&=-R(t)D(t)dt,
\end{aligned} \tag{8.40}$$

which implies the adapted interest rate should be $R(t)$ as defined. □

The reason why we introduce random discount process is that we want to calculate discounted stock price, and note that stock price $S(t)$ is a stochastic process with nonzero quadratic variation while discount rate $D(t)$ has zero quadratic variation.

Theorem 8.12 (discounted stock price process).
The discounted stock price is calculated to be

$$D(t)S(t) = S(0)\exp\left\{\int_0^t \sigma(s)dW(s) + \int_0^t \left(\alpha(s) - R(s) - \frac{1}{2}\sigma^2(s)\right)ds\right\}. \tag{8.41}$$

And the differential form satisfies

$$\begin{aligned} d(D(t)S(t)) &= (\alpha(t) - R(t))D(t)S(t)dt + \sigma(t)D(t)S(t)dW(t) \\ &= \sigma(t)D(t)S(t)[\theta(t)dt + dW(t)], \end{aligned} \tag{8.42}$$

where

$$\theta(t) = \frac{\alpha(t) - R(t)}{\sigma(t)}. \tag{8.43}$$

Proof. To deduce the differential form, we utilize Itô-Deoblin formula, and it gives that

$$\begin{aligned} d(D(t)S(t)) &= D(t)dS(t) + S(t)dD(t) + dD(t)dS(t) \\ &= D(t)\left(\alpha(t)S(t)dt + \sigma(t)S(t)dW(t)\right) + S(t)\left(-R(t)D(t)dt\right) \\ &= \left(\alpha(t) - R(t)\right)D(t)S(t)dt + \sigma(t)D(t)S(t)dW(t) \\ &= \sigma(t)D(t)S(t)[\theta(t)dt + dW(t)]. \end{aligned} \tag{8.44}$$

□

Recall that the stock price without discount process is that

$$dS(t) = \alpha(t)S(t)dt + \sigma(t)S(t)dW(t). \tag{8.45}$$

Compared to undiscounted stock price, the mean rate of return for discounted stock price varies from $\alpha(t)$ to $\alpha(t) - R(t)$, while the volatility remains the same, which is $\sigma(t)$.

In equation (8.42), $\theta(t)$ can be regarded as the market price of risk, then according to Girsanov's theorem, we define

$$\widetilde{W}(t) = W(t) + \int_0^t \theta(u)du, \text{ or } d\widetilde{W}(t) = dW(t) + \theta(t)dt, \tag{8.46}$$

where $\widetilde{W}(t)$ is the Brownian motion under probability measure $\widetilde{\mathbb{P}}$. Then the differential form of discounted stock price satisfies

$$d(D(t)S(t)) = \sigma(t)D(t)S(t)d\widetilde{W}(t). \tag{8.47}$$

Note that $\sigma(t)D(t)S(t)d\widetilde{W}(t)$ is left continuous and $\widetilde{W}(t)$ is a martingale, which implies that $D(t)S(t)$ is also a martingale.

And we can also rewrite the differential form for undiscounted stock price $S(t)$, that is

$$\begin{aligned} dS(t) &= \alpha(t)S(t)dt + \sigma(t)S(t)\left(d\widetilde{W}(t) - \theta(t)dt\right) \\ &= R(t)S(t)dt + \sigma(t)S(t)d\widetilde{W}(t). \end{aligned} \tag{8.48}$$

Then we can use the properties of stochastic differential equation to find the formula for $S(t)$, and it gives that

$$S(t) = S(0)\exp\left\{\int_0^t \sigma(s)d\widetilde{W}(s) + \int_0^t \left(R(s) - \frac{1}{2}\sigma^2(s)ds\right)\right\}. \tag{8.49}$$

From the formula above, we can see that what has varied after change of measure is the mean rate of return instead of the volatility.

8.3 Risk-Neutral Measure for Portfolio

Suppose an agent holds $X(t)$ amount of asset at time t, for $0 \leqslant t \leqslant T$. If the agent try to adopt some risk management strategies, then he need to create hedges. So part of money is used to purchase shares of stocks, while others is invested in money market account. Say the agent holds $\Delta(t)$ shares of stocks at time t, and then the amount of money in the account should be $(X(t) - \Delta(t)S(t))$ with interest rate $R(t)$. The stock price is defined in equation (8.31). It follows that

$$\begin{aligned} dX(t) &= \Delta(t)dS(t) + R(t)\left(X(t) - \Delta(t)S(t)\right)dt \\ &= \Delta(t)d\left(\alpha(t)S(t)dt + \sigma(t)S(t)dW(t)\right) + R(t)\left(X(t) - \Delta(t)S(t)\right)dt \\ &= R(t)X(t)dt + \Delta(t)\sigma(t)S(t)\left[\frac{\alpha(t) - R(t)}{\sigma(t)}dt + dW(t)\right] \\ &= R(t)X(t)dt + \Delta(t)\sigma(t)S(t)\left[\theta(t)dt + dW(t)\right]. \end{aligned} \tag{8.50}$$

And for the discounted portfolio price, using the property that $dW(t)dt = dtdt = 0$, we have

$$\begin{aligned} d(D(t)X(t)) &= D(t)dX(t) + X(t)dD(t) + dD(t)dX(t) \\ &= D(t)\Big\{R(t)X(t)dt + \Delta(t)\sigma(t)S(t)\left[\theta(t)dt + dW(t)\right]\Big\} - X(t)R(t)D(t)dt \\ &= \Delta(t)\sigma(t)D(t)S(t)\left[\theta(t)dt + dW(t)\right] \\ &= \Delta(t)d(D(t)S(t)). \end{aligned} \tag{8.51}$$

Substituting $d(D(t)S(t)) = \sigma(t)D(t)S(t)d\widetilde{W}(t)$ into equation (8.51), it follows that, under probability measure $\widetilde{\mathbb{P}}$,

$$d(D(t)X(t)) = \Delta(t)\sigma(t)D(t)S(t)d\widetilde{W}(t). \tag{8.52}$$

And similarly,

$$dX(t) = R(t)X(t)dt + \Delta(t)\sigma(t)S(t)d\widetilde{W}(t). \tag{8.53}$$

8.4 Deriving the Black-Scholes-Merton Equation Under Risk-Neutral Measure

Let us come back to the problem that an agent want to hedge a short position in the European call. $V(t)$ denotes the underlying payoff of a derivative security and $S(t)$ denotes the stock price where $0 \leqslant t \leqslant T$, then according to the definition of call price, at time T,

$$V(T) = (S(T) - K)^+ . \tag{8.54}$$

In order to evaluate the value of $V(t)$, namely, the payoff of a derivative security, recall that the basic idea is no-arbitrage pricing using a hedging portfolio. And the meaning of hedging portfolio is to reproduce reproduce the payoff of the derivative security $V(t)$ at any time t. So we define the hedging portfolio with value $X(t)$. What we are certain is that

$$X(T) = V(T) = (S(T) - K)^+ . \tag{8.55}$$

Theorem 8.13. *The $\mathscr{F}(t)$-measurable discounted stock price satisfies the property that*

$$D(t)X(t) = \widetilde{\mathbb{E}}[D(T)V(T)|\mathscr{F}(t)]. \tag{8.56}$$

Proof. Since $D(t)X(t)$ is a martingale under probability space tP, we have

$$D(t)X(t) = \widetilde{\mathbb{E}}[D(T)X(t)|\mathscr{F}(s)] \quad for\ 0 \leqslant s \leqslant t \leqslant T. \tag{8.57}$$

And we use the fact that $X(T) = V(T)$, then

$$D(t)X(t) = \widetilde{\mathbb{E}}[D(T)V(T)|\mathscr{F}(t)]. \tag{8.58}$$

□

Theorem 8.14. *Assume $R(t)$ is the interest rate of the payoff in the derivative security, then this suggests that*

$$V(t) = \widetilde{\mathbb{E}}\left[\exp\{-\int_t^T R(u)du\}V(T)|\mathscr{F}(t)\right] \quad for\ 0 \leqslant s \leqslant t \leqslant T. \tag{8.59}$$

Proof. Note that $X(t)$ is suggested to be in consistence with $V(t)$. By rewriting Theorem 8.13, we have

$$D(t)V(t) = \widetilde{\mathbb{E}}[D(T)V(T)|\mathscr{F}(t)]. \tag{8.60}$$

Since $D(t)$ satisfies

$$D(t) = \exp\{-\int_t^T R(u)du\}, \tag{8.61}$$

equation (8.60) then becomes

$$\exp\{-\int_0^t R(u)du\}V(T) = \widetilde{\mathbb{E}}\left[\exp\{-\int_0^T R(u)du\}V(T)|\mathscr{F}(t)\right] \tag{8.62}$$

which is equivalent to say that

$$V(t) = \widetilde{\mathbb{E}}\left[\exp\{-\int_t^T R(u)du\}V(T)|\mathscr{F}(t)\right]. \tag{8.63}$$

That is because $D(t)$ is $\mathscr{F}(t)$-measurable and thus we can move $\frac{1}{D(t)}$ into the conditional expectation. $\square$

Then we would like to employ the theorems for hedging process to derive the Black-Scholes-Merton equation of a European call, and we suppose the volatility and interest rate of payoff are constant, especially, $\sigma(t) = \sigma$ and $\mathbb{R}(t) = r$. And since the payoff in the derivative security depends on the stock price $S(t)$ and time t, we can rewrite $V(t)$ as $c(t, x)$ where $x = S(t)$. Theorem 8.14 is equivalent to

$$c(t, S(t)) = \widetilde{\mathbb{E}}\left[e^{-r(T-t)}(S(T) - K)^+ | \mathscr{F}(t)\right]. \tag{8.64}$$

And the equation for $S(t)$ then becomes

$$S(t) = S(0) \exp\left\{\sigma \widetilde{W}(t) + \left(r - \frac{1}{2}\sigma^2\right) t\right\} \tag{8.65}$$

Theorem 8.15 (Black-Scholes-Merton equation)**.**
We can derive the Black-Scholes-Merton equation using change of measure, that is

$$c(t, x) = xN(d_+(\tau, x)) - e^{-r\tau} KN(d_-(\tau, x)), \tag{8.66}$$

where $\tau = T - t$ *and*

$$d_+(\tau, x) = \frac{1}{\sigma\sqrt{\tau}}\left[\log\frac{x}{K} + \left(r + \frac{1}{2}\sigma^2\right)\tau\right]; \tag{8.67}$$

$$d_-(\tau, x) = \frac{1}{\sigma\sqrt{\tau}}\left[\log\frac{x}{K} + \left(r - \frac{1}{2}\sigma^2\right)\tau\right]. \tag{8.68}$$

Proof.
From equation (8.65), we can derive that

$$\begin{aligned} S(T) &= S(t) \exp\left\{\sigma(\widetilde{W}(T) - \widetilde{W}(t)) + \left(r - \frac{1}{2}\sigma^2\right) t\right\} \\ &= S(t) \exp\left\{\sigma\sqrt{\tau} Y + \left(r - \frac{1}{2}\sigma^2\right) t\right\}, \end{aligned} \tag{8.69}$$

where

$$Y = -\frac{\widetilde{W}(T) - \widetilde{W}(t)}{\sqrt{T-t}}, \ \tau = T - t. \tag{8.70}$$

Note that random variable Y is standard normal distributed the formula of $S(T)$ is $\mathscr{F}(t)$-measurable, then it gives that

$$\begin{aligned} c(t, x) &= \widetilde{\mathbb{E}}\left[e^{-r\tau}\left(x \exp\left\{-\sigma\sqrt{\tau}Y + \left(r - \frac{1}{2}\sigma^2\right)\tau\right\} - K\right)^+\right] \\ &= \int_{-\infty}^{\infty} e^{-r\tau}\left(x \exp\left\{-\sigma\sqrt{\tau}y + \left(r - \frac{1}{2}\sigma^2\right)\tau\right\} - K\right)^+ N(y)dy \\ &= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} e^{-r\tau}\left(x \exp\left\{-\sigma\sqrt{\tau}y + \left(r - \frac{1}{2}\sigma^2\right)\tau\right\} - K\right)^+ e^{-\frac{1}{2}y^2}dy. \end{aligned} \tag{8.71}$$

Note that

$$\left(x\exp\left\{-\sigma\sqrt{\tau}y+\left(r-\frac{1}{2}\sigma^2\right)\tau\right\}-K\right)^+>0, \tag{8.72}$$

when

$$y<\frac{1}{\sigma\sqrt{\tau}}\left[\log\frac{x}{K}+\left(r-\frac{1}{2}\sigma^2\right)\tau\right]=d_-(\tau,x); \tag{8.73}$$

otherwise,

$$\left(x\exp\left\{-\sigma\sqrt{\tau}y+\left(r-\frac{1}{2}\sigma^2\right)\tau\right\}-K\right)^+=0. \tag{8.74}$$

Consequently,

$$\begin{aligned}
c(t,x)&=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{d_-(\tau,x)}e^{-r\tau}\left(x\exp\left\{-\sigma\sqrt{\tau}y+\left(r-\frac{1}{2}\sigma^2\right)\tau\right\}-K\right)e^{-\frac{1}{2}y^2}dy\\
&=\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{d_-(\tau,x)}x\exp\left\{-\frac{1}{2}y^2-\sigma\sqrt{\tau}y-\frac{1}{2}\sigma^2\tau\right\}dy\\
&\quad-\frac{1}{\sqrt{2\pi}}\int_{-\infty}^{d_-(\tau,x)}e^{-r\tau}Ke^{-\frac{1}{2}y^2}dy\\
&=\frac{x}{\sqrt{2\pi}}\int_{-\infty}^{d_-(\tau,x)}\exp\left\{-\frac{1}{2}(y+\sigma\sqrt{\tau})^2\right\}dy-e^{-r\tau}KN(d_-(\tau,x))\\
&\overset{z=y+\sigma\sqrt{\tau}}{=}\frac{x}{\sqrt{2\pi}}\int_{-\infty}^{d_-(\tau,x)+\sigma\sqrt{\tau}}\exp\left\{-\frac{z^2}{2}\right\}dz-e^{-r\tau}KN(d_-(\tau,x))\\
&=xN(d_+(\tau,x))-e^{-r\tau}KN(d_-(\tau,x)).
\end{aligned} \tag{8.75}$$

Then we have verified the theorem. □

9 Jump Processes

9.1 Poisson Process

9.1.1 Definition of Poisson Process

Definition 9.1 (Poisson Distribution)**.** For the time t, we say t follows a Poisson distribution if its probability density function has the form

$$P(t)=\begin{cases}\lambda e^{-\lambda t}, & t\geqslant 0,\\ 0, & t<0.\end{cases} \tag{9.1}$$

Then the expected value of t

$$E[t]=\frac{1}{\lambda}. \tag{9.2}$$

And the cumulative distribution function

$$F(t)=P(t\leqslant t)=\int_0^t uf(u)du=1-e^{-\lambda t}. \tag{9.3}$$

Then we can calculate the conditional probability.

$$P(t > \tau + s | t > s) = \frac{P(t > \tau + s \quad and \quad t > s)}{P(t > s)} = e^{-\lambda\tau}. \tag{9.4}$$

The formula above gives us the probability that we need to wait extra τ time units or more if we had waited s time units.

9.1.2 Construction of Poisson Process

Our aim is to utilize the properties of Poison distribution to describe the random occur of jumps. Let τ_1 be the duration from time 0 to the first jump and τ_i be the duration from $(i-1)$th jump to ith jump, where $i = 1, 2, \ldots$. Each duration has expected value $E[\tau_i] = \frac{1}{\lambda}$, or we say that each jump would occur at the rate of λ unit times.

Then we notice that the nth jump would occur at time S_n, where

$$S_n = \sum_{i=1}^{n} \tau_i. \tag{9.5}$$

Let $N(t)$ be the number of jumps in [0,t]. Then

$$N(t) = \begin{cases} 0, & 0 \leqslant t < \tau_1, \\ 1, & \tau_1 \leqslant t < \tau_2, \\ \ldots \end{cases} \tag{9.6}$$

Properties 9.2. *We use the gamma density function $g_n(s)$ to describe the probability density of the value of total time S_n when there are n jumps. Function $g_n(s)$ has the form*

$$g_n(s) = \frac{(\lambda s)^{n-1}}{(n-1)!}\lambda e^{-\lambda s}. \tag{9.7}$$

Proof. We would prove the property by induction, when $n = 1$, exactly one jump would occur at time $S_1 = \tau_1$, then it follows that

$$g_1(s) = \lambda e^{-\lambda s}. \tag{9.8}$$

Suppose the formula holds for $n = k$, namely, we can assume the following formula is true.

$$g_k(s) = \frac{(\lambda s)^{k-1}}{(k-1)!}\lambda e^{-\lambda s}. \tag{9.9}$$

Then we consider $n = k + 1$, we have $S_{k+1} = S_k + \tau_{k+1}$, and the density of S_{n+1} can be computed as

$$
\begin{aligned}
g_{n+1}(s) &= \int_0^s g_n(s)f(s-v)dv \\
&= \int_0^s \frac{(\lambda v)^{n-1}}{(n-1)!}\lambda e^{-\lambda v}\cdot \lambda e^{-\lambda(s-v)}dv \\
&= \frac{\lambda^{n+1}e^{-\lambda s}}{(n-1)!}\int_0^s v^{n-1}ds \\
&= \frac{(\lambda s)^n}{n!}\lambda e^{-\lambda s}
\end{aligned}
\tag{9.10}
$$

Then by induction, we can verify that S_n follows a gamma distribution as we have introduced.
□

Properties 9.3. *Then we would describe the possibility when there has been n jumps at time t. We have*

$$
P(N(t)=k) = \frac{(\lambda t)^k}{k!}e^{-\lambda t} \tag{9.11}
$$

Here, we can use the difference of cumulative distribution function of S_k to calculate $P(N(t) = k)$.

Proof. We would firstly calculate the probability that $N(t) \geqslant k$ and $N(t) \geqslant k+1$. Hence,

$$
\mathbb{P}(N(t)\geqslant k) = \mathbb{P}(S_k \leqslant t) = \int_0^t \frac{(\lambda s)^{k-1}}{(k-1)!}\lambda e^{-\lambda s}ds. \tag{9.12}
$$

And on the other hand,

$$
\begin{aligned}
\mathbb{P}(N(t)\geqslant k+1) &= \mathbb{P}(S_{k+1}\leqslant t) \\
&= \int_0^t \frac{(\lambda s)^k}{k!}\lambda e^{-\lambda s}ds \\
&= -\frac{(\lambda s)^k}{k!}e^{-\lambda s}\Big|_{s=0}^{s=t} + \int_0^t \frac{(\lambda s)^{k-1}}{(k-1)!}\lambda e^{-\lambda s}ds \\
&= -\frac{(\lambda s)^k}{k!}e^{-\lambda s}\Big|_{s=0}^{s=t} + \mathbb{P}(N(t)\geqslant k).
\end{aligned}
\tag{9.13}
$$

Then by calculating the difference, we have

$$
\mathbb{P}(N(t)=k) = \mathbb{P}(N(t)\geqslant k) - \mathbb{P}(N(t)\geqslant k+1) = \frac{(\lambda t)^k}{k!}e^{-\lambda t}\ for\ k\geqslant 1. \tag{9.14}
$$

When $k=0$,

$$
\mathbb{P}(N(t)=0) = \mathbb{P}(S_1 > t) = e^{-\lambda t} \tag{9.15}
$$

□

Definition 9.4 (Poisson Increments).
The increment $N(t+s) - N(s)$ represents the quantity of jumps in $(s, t+s]$. Notice that the expected value of $N(t+s) - N(s)$ does not depends on the information up to time s, or $\mathscr{F}(s)$. However, the Poisson increment is not a martingale since the expected number of jumps is not zero, giving jumps happen with mean $\frac{1}{\lambda}$ unit time. Then we would see some properties of Poisson Increment.

Properties 9.5. *Assume that N(t) is a Poisson Process with intensity λ and $0 = t_0 < t_1 < \cdots < t_n = T$ is a partition of $[0, T]$. For each increments, we have*

$$\begin{aligned} P(N(t_{i+1}) - N(t_i) = k) = \frac{[\lambda(t_{i+1} - t_i)]^k}{k!} e^{-\lambda(t_{i+1} - t_i)} \\ for \quad i = 0, 1, \ldots, n-1 \end{aligned} \tag{9.16}$$

Properties 9.6 (mean of Poisson increment). *We can calculate the mean value of Poisson increment as*

$$\begin{aligned} \mathbb{E}[N(t) - N(s)] &= \sum_{k=0}^{\infty} k \frac{\lambda^k (t-s)^k}{k!} e^{-\lambda(t-s)} \\ &= \lambda(t-s) e^{-\lambda(t-s)} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}(t-s)^{k-1}}{(k-1)!} \\ &= \lambda(t-s) e^{-\lambda(t-s)} e^{\lambda(t-s)} \\ &= \lambda(t-s). \end{aligned} \tag{9.17}$$

Properties 9.7 (variance of Poisson increment). *The variance of the Poisson increment should be*

$$Var[N(t) - N(s)] = \lambda(t-s). \tag{9.18}$$

Definition 9.8 (Compensated Poisson Process).
Since Poisson process is not a martingale, we can define a new process which has no tendency to rise or drop, that is

$$M(t) = N(t) - \lambda t. \tag{9.19}$$

9.1.3 Compound Poisson Process

In ordinary Poisson process, the jump size is always equal to one unit. So we want to construct a process with random jump size. Then we define the formula known as compound Poisson process.

$$Q(t) = \sum_{i=0}^{N(t)} Y_i. \tag{9.20}$$

Here, $N(t)$ describes Poisson process. Y_i are independent random variables and $E[Y_i] = \beta$. Notice that $Q(t)$ depends only on time t. Analyze the formula at time $t = \tau$, we can see that the number of jumps would be $N(\tau)$, say $N(\tau) = N$. And then $Q(t)$ is equal to the sum of first N random variables Y_i.

As for the graph of $Q(t)$, we can then state that the jump in $Q(t)$ occurs if and only if the jump in $N(t)$ occurs.

Properties 9.9. *The increments $Q(t) - Q(s)$ are independent of $\mathscr{F}(s)$.*

Proof. Note that

$$Q(t) - Q(s) = \sum_{i=N(s)+1}^{N(t)} Y_i, \tag{9.21}$$

and each jump size are independent random variables, so we have the property. □

Properties 9.10 (Expected value of compound Poisson Process)**.**

$$E[Q(t)] = \beta\lambda t \tag{9.22}$$

Proof. To calculate the mean, we would better regard Q(t) as a function of N(t), namely, $Q(t) = F(N(t))$. Then by taking $N(t) = 0, 1, \dots, k, \dots$, we can calculate the expected value using conditional expected value, say

$$\begin{aligned}
\mathbb{E}[Q(t)] &= \sum_{k=0}^{\infty} \mathbb{E}\left[Q(t)|N(t) = k\right] \mathbb{P}\{N(t) = k\} \\
&= \sum_{k=0}^{\infty} \mathbb{E}\left[\sum_{i=1}^{k} Y_i \middle| N(t) = k\right] \mathbb{P}\{N(t) = k\} \\
&= \sum_{k=0}^{\infty} \beta k \frac{(\lambda k)^k}{k!} e^{-\lambda t} \\
&= \beta\lambda t e^{-\lambda t} \sum_{k=1}^{\infty} \frac{(\lambda k)^{k-1}}{(k-1)!} \\
&= \beta\lambda t e^{-\lambda t} e^{\lambda t} = \beta\lambda t
\end{aligned} \tag{9.23}$$

□

Definition 9.11 (compensated compound Poisson process)**.**
Similar to compensated Brownian motion, we define new process $Q(t) - \beta\lambda s$ to form a martingale.

Proof. We apply the property that $Q(t) - Q(s)$ is independent of $Q(s)$, then we have

$$\begin{aligned}
\mathbb{E}[Q(t) - \beta\lambda t|\mathscr{F}(s)] &= \mathbb{E}[Q(t) - Q(s) + Q(s) - \beta\lambda t|\mathscr{F}(s)] \\
&= \mathbb{E}[Q(t) - Q(s)] + Q(s) - \beta\lambda t \\
&= \beta\lambda t - \beta\lambda s + Q(s) - \beta\lambda t = Q(s) - \beta\lambda s,
\end{aligned} \tag{9.24}$$

which implies that the compensated Poisson process is a martingale. □

9.2 Stochastic Integral for Poisson Process

Recall that in previous section, we introduce Itô integral based on Brownian motion, Brownian motion is a martingale since we find that the increment $W(t) - W(s)$ is independent of filtration $\mathscr{F}(t)$. And later we verify that Itô integral is also a martingale.

As for Poisson process, we are also willing to search for the integral form. That is the purpose of this section. However, Poisson process $J(t)$ may not be a martingale, namely, $E[J(t) - J(s)|\mathscr{F}(s)] \neq 0$.

9.2.1 Definition and Properties

Definition 9.12 (right-continuous function).
Function $f : A \to R$ where $A \in R$ is defined to be right-continuous if and only if at every jump of f, $f(t)$ takes its value after the jump. In analysis, we can say

$$\lim_{t\to\tau^+} f(t) = f(\tau) \quad for \quad \forall\tau \in A. \tag{9.25}$$

Or it is equivalent to say that

$$\forall\epsilon > 0, \exists\delta > 0 \quad s.t. \quad \tau < t < \tau + \delta \Rightarrow |f(x) - f(\tau)| < \epsilon. \tag{9.26}$$

Definition 9.13 (stochastic integral for jump processes).
The stochastic integral has the form

$$\int_0^t \Phi(s)dX(s). \tag{9.27}$$

Here, $X(t)$ is defined to be a right continuous function. And then

$$X(t) = X(0) + I(t) + R(t) + J(t). \tag{9.28}$$

In this formula, $I(t) = \int_0^t \Gamma(s)dW(s)$ is the Itô integral; $R(t) = \int_0^t \Theta(s)ds$ is the Riemann integral. Then we let

$$X^c(t) = X(0) + I(t) + R(t) = X(0) + \int_0^t \Gamma(s)dW(s) + \int_0^t \Theta(s)ds. \tag{9.29}$$

Note that $X^c(t)$ is the Itô process which we have defined before. Here, we can also say that $X^c(t)$ is the continuous part of $X(t)$.

Properties 9.14. Recall that $X^c(t)$ satisfies following properties:
i) $X^c(t)$ has quadratic variation is

$$[X^c, X^c](t) = \int_0^t \Gamma^2(s)ds. \tag{9.30}$$

ii) the differentiation of $X^c(t)$ is

$$dX^c(t) = \Gamma(t)dW(t) + \Theta(t)dt. \tag{9.31}$$

iii) $X^c(t)$ satisfies

$$\int_0^t f(s)dX^c(s) = \int_0^t f(s)\Gamma(s)dW(s) + \int_0^t f(s)\Theta(s)ds. \tag{9.32}$$

for some function $f(t)$.

Then we consider the jump part $J(s)$. Since $X(t)$ is defined to be right-continuous, then $J(s)$ is also right-continuous. Generally, we said $J(t)$ is pure jump process if $J(t)$ is piecewise constant, namely, $J(t)$ remains a constant value between two consecutive jumps.

Properties 9.15.
Compensated Poisson process $M(t) = N(t) - \lambda t$ is not a pure jump process since $M(t)$ drops with rate $-\lambda$ after the jump time.

But we can still define the stochastic integral based on compensated Poisson process even if it is not a pure jump process.

Definition 9.16 (jump size)**.**
Now, we have seen that a process $X(t)$ consists of two parts: continuous part and jump part. Then we would like to figure out the change of value right before the jump and right after the jump, which is called the jump size. Recall that $X(t)$ is right continuous, which implies that the value of X at jump time t is exact the value right after the jump, so we only need to find the value right before the jump.

Define $X(t-)$ to be the value right before the jump:

$$X(t-) = \lim_{\tau \to t-} X(\tau). \tag{9.33}$$

Then the jump size is the difference between $X(t)$ and $X(t-)$ at jump time t, say

$$\Delta X(t) = X(t) - X(t-). \tag{9.34}$$

Since $X(t) = X^c(t) + J(t)$ and $X^c(t)$ is a continuous function without jumps. So for jump time t

$$\Delta X(t) = X(t) - X(t-) = \Delta J(t) = J(t) - J(t-). \tag{9.35}$$

We want the jump size to be 0 when jump does not happens, and the function above also satisfies this property. Thus, for any t defined on the domain, we also have

$$\Delta X(t) = X(t) - X(t-) = \Delta J(t) = J(t) - J(t-). \tag{9.36}$$

Theorem 9.17.
The stochastic integral has the property that

$$\int_0^t \Phi(s)dX(s) = \int_0^t \Phi(s)\Gamma(s)dW(s) + \int_0^t \Phi(s)\Theta(s)ds + \sum_{0<s\leqslant t} \Phi(s)\Delta J(s). \tag{9.37}$$

Sketch of proof.
To prove the property, we share some similar idea in the section 5.2.2 '*Proof of function convergence for integrant* $\Delta(t)$' that dividing the integral into two parts: continuous part and jump part.

For continuous part, according to Property 9.14 iii), we have

$$\int_0^t \Phi(s)dX^c(s) = \int_0^t \Phi(s)\Gamma(s)dW(s) + \int_0^t \Phi(s)\Theta(s)ds. \tag{9.38}$$

While for jump part, let $n \in R \cup \{\infty\}$

$$\int_0^t \Phi(s)dJ(s) = \sum_{i=0}^{n} \Phi(t_i)\left(J(t_i) - J(t_i-)\right) = \sum_{0<s\leqslant t} \Phi(s)\Delta J(s). \tag{9.39}$$

By adding them together,

$$\begin{aligned}\int_0^t \Phi(s)dX(s) &= \int_0^t \Phi(s)dX^c(s) + \int_0^t \Phi(s)dJ(s)\\ &= \int_0^t \Phi(s)\Gamma(s)dW(s) + \int_0^t \Phi(s)\Theta(s)ds + \sum_{0<s\leqslant t} \Phi(s)\Delta J(s).\end{aligned} \tag{9.40}$$

Then we have proved the property. □

Actually, we can rewrite the integral into differentiation form,

$$\begin{aligned}\Phi(t)dX(t) &= \Phi(t)dX^c(t) + \Phi(t)dJ(t)\\ &= \Phi(t)dI(t) + \Phi(t)dR(t) + \Phi(t)dJ(t)\\ &= \Phi(t)\Gamma(t)dW(t) + \Phi(t)\Theta(t)dW(t) + \Phi(t)dJ(t).\end{aligned} \tag{9.41}$$

Usually, the integrand $\Phi(t)$ is defined to be left -continuous and adapted. In contrast to right-continuous functions, left-continuous functions has unique properties.

Theorem 9.18.
Assume $\Phi(t)$ is a left-continuous function,then $\Delta\Phi(t)$ is always equal to 0.

Sketch of proof.
Let Π be the set of jumps, then we consider two cases.
i) For $p \in \Pi$, $\Phi(p)$ is the value of Φ just before the jump, which is equal to $\Phi(p-)$. Then $\Delta\Phi(p) = \Phi(p) - \Phi(p-) = 0$
ii) For $p \notin \Pi$, assume the nearest jump before time p is at time q. Then what we need to prove is that there is at least a point between time q and p. In fact, we can simply let $\epsilon = p - q$, and choose $r = p - \frac{\epsilon}{n}$ where $n \geqslant 2$. Then Φ is continuous in the interval $[r, p]$. It gives that

$$\Delta\Phi(p) = \Phi(p) - \Phi(p-) = \Phi(p) - \lim_{n\to\infty} \Phi(p - \frac{\epsilon}{n}) = 0. \tag{9.42}$$

□

To explain why we require $\Phi(t)$ to be left-continuous and $X(t)$ to be right-continuous, we can analyze the behavior of stochastic integral of jump process. Since $\Delta\Phi(t) = 0$, then whenever $X(s)$ jumps, the integral jumps. On the other hand, the jump size should be contained when calculating the integral. So $\Delta X(t) \neq 0$ at jump time t. Thus, $X(t)$ must be right-continuous.

Then we would introduce a extremely significant theorem.

Theorem 9.19.
For left-continuous and adapted $\Phi(t)$, assume a process $X(s)$ is a martingale, then $\int_0^t \Phi(s)dX(s)$ is a martingale if

$$\mathbb{E}\int_0^t \Gamma^2(s)\Phi^2(s)ds < \infty. \tag{9.43}$$

An example would be utilized to help us understand the theorem.

Example 9.20.
For stochastic integral $\int_0^t \Phi(s)dM(s)$, assume $M(t) = N(t) - \lambda t$ is compensated Poisson process and $\Phi(t) = 1_{[0,S_1]}(s)$. Note that $\Phi(t)$ is left-continuous and $S_1 = \tau_1$ is the time when first jump happens. Since $\Phi(t)$ is defined to be an indicator, we need to consider two cases.

Case 1) $t \in [0, S_1)$, $N(t) = 0$ because the first jump does not happen. $M(t) = 0 - \lambda t = -\lambda t$. Then

$$\int_0^t \Phi(s)dM(s) = \int_0^t dM(s) = M(t) - M(0) = -\lambda t. \tag{9.44}$$

Case 2) $t \in [S_1, \infty)$, $M(s_1) = 1 - \lambda S_1$ since the first jump happens at time S_1. $\Phi(t) = 0$ for $t > S_1$. Then

$$\int_0^t \Phi(s)dM(s) = \int_0^{S_1} \Phi(s)dM(s) = M(S_1) - M(0) = 1 - \lambda S_1. \tag{9.45}$$

To sum up, we have

$$\begin{aligned}\int_0^t \Phi(s)dM(s) &= \begin{cases} -\lambda t, & 0 \leqslant t < S_1, \\ 1 - \lambda S_1, & t \geqslant S_1. \end{cases} \\ &= 1_{[S_1,\infty]}(t) - \lambda \cdot \min\{S_1, t\}.\end{aligned} \tag{9.46}$$

In order to check whether the stochastic integral is a martingale, we would compute the value of $\mathbb{E}\left[1_{[S_1,\infty]}(t) - \lambda \cdot \min\{S_1, t\}|\mathscr{F}(s)\right]$ for $t > s > 0$. Since

$$\mathbb{E}\left[1_{[S_1,\infty]}(t)|\mathscr{F}(s)\right] = 1 \cdot \mathbb{P}[t \geqslant S_1|F(s)] + 0 \cdot \mathbb{P}[t < S_1|\mathscr{F}(s)], \tag{9.47}$$

It follows,

$$\mathbb{E}\left[1_{[S_1,\infty]}(t) - \lambda \cdot \min\{S_1, t\}|\mathscr{F}(s)\right] = \mathbb{P}[t \geqslant S_1|\mathscr{F}(s)] - \lambda\mathbb{E}[\min\{t, S_1\}|\mathscr{F}(s)]. \tag{9.48}$$

To calculate $\mathbb{E}\left[1_{[S_1,\infty]}(t) - \lambda \cdot \min\{S_1, t\}|\mathscr{F}(s)\right]$, we still consider two cases:

Case 1) $S_1 \leqslant s < t$, then we already have $\mathbb{E}\left[1_{[S_1,\infty]}(t) - \lambda \cdot \min\{S_1, t\}|\mathscr{F}(s)\right] = 1 - \lambda S_1$;
Case 2) $S_1 > s$, then

$$\begin{aligned} \mathbb{P}[t \geqslant S_1|\mathscr{F}(s)] &= \mathbb{P}[t \geqslant S_1|S_1 > s] \\ &= 1 - \mathbb{P}[S_1 > t|S_1 > s] \\ &= 1 - \mathbb{P}[S_1 > t - s] \\ &= 1 - e^{-\lambda(t-s)}. \end{aligned} \tag{9.49}$$

In order to calculate $\lambda\mathbb{E}\left[\lambda \cdot \min\{S_1, t\}|\mathscr{F}(s)\right]$, we have to use the probability density of S_1, which is

$$-\frac{\partial}{\partial u}\mathbb{P}[S_1 > t|S_1 > s] = -\frac{\partial}{\partial u}e^{-\lambda(u-s)} = \lambda e^{-\lambda(u-s)}. \tag{9.50}$$

And then

$$\begin{aligned} \lambda\mathbb{E}\left[\min\{S_1, t\}|\mathscr{F}(s)\right] &= \lambda\mathbb{E}\left[\min\{S_1, t\}|S_1 > s\right] \\ &= \lambda^2 \int_s^\infty \min\{t, u\} e^{-\lambda(u-s)} du \\ &= \lambda^2 \int_s^t u e^{-\lambda(u-s)} du + \lambda^2 \int_t^\infty t e^{-\lambda(u-s)} du \\ &= -\lambda \int_s^t u d e^{-\lambda(u-s)} - \lambda t e^{-\lambda(u-s)} \Big|_{u=t}^{u=\infty} \\ &= -\lambda u e^{-\lambda(u-s)} \Big|_{u=s}^{u=t} + \lambda \int_s^t e^{-\lambda(u-s)} du - \lambda t e^{-\lambda(u-s)} \Big|_{u=t}^{u=\infty} \\ &= \lambda s - e^{-\lambda(t-s)} + 1. \end{aligned} \tag{9.51}$$

Thus,

$$\mathbb{E}\left[1_{[S_1,\infty]}(t) - \lambda \cdot \min\{S_1, t\}|\mathscr{F}(s)\right] = -\lambda s. \tag{9.52}$$

In both two cases, we have

$$\mathbb{E}\left[1_{[S_1,\infty)}(t) - \lambda \cdot \min\{S_1, t\}|\mathscr{F}(s)\right] = 1_{[S_1,\infty)}(s) - \lambda \cdot Min\{S_1, s\}. \tag{9.53}$$

We have finished our calculation and finally observe that $\int_0^t \Phi(s)dX(s)$ is a martingale. Here, we let $\Phi(t)$ be a left-continuous function. However, if we let $\Phi(t)$ be a right-continuous function, i.e. $\Phi(s) = 1_{[0,S_1)}(t)$, then the integral would not be a martingale.

Example 9.20 is one particular case that corresponds to the description in Theorem 9.19. Although it can not verify that Theorem 9.19 is true, it illustrates that Theorem 9.19 has some meaning even if the proof would not be included in this report. For instance, famous Physicist De Broglie put forward a hypothesis based on Einstein's conclusion: 'wave is a kind of particle'. De Broglie thought that 'since wave is a kind of particle, what if particle is also a kind of wave'. And he believed that electron also satisfies the equation for Quanta purposed by Einstein. He wrote down the equation and found that it satisfies light quantum equation purposed in 1905 by Einstein. But actually, his hypothesis does not say anything and it may be just a particular case. No one believed it to be true at that time and even his supervisor Langevin has to write a letter to Einstein so that De Broglie's hypothesis can be published. Later we all know his hypothesis does tell something, and Schrodinger's equation is established based on De Broglie's work. So in mathematics study, we should not ignore the importance of particular cases.

Definition 9.21 (quadratic variation and cross variation)**.**
For a jump process $X(t)$ defined on $[0,T]$, suppose $\Pi=\{t_0,t_1,\dots,t_n\}$ is a partition of [0,T] and $\|\Pi\|=\max_j(t_{j+1}-t_j)$. Then the quadratic variation satisfies

$$\begin{aligned}[X,X](t) &= \lim_{\|\Pi\|\to 0}\sum_{j=0}^{n-1}\left(X(t_{j+1})-X(t_j)\right)^2 \\ &= \lim_{\|\Pi\|\to 0} Q_\Pi(X).\end{aligned} \tag{9.54}$$

And we can also define cross variation between $X_1(t)$ and $X_2(t)$.

$$\begin{aligned}[X_1,X_2](T) &= \lim_{\|\Pi\|\to 0} C_\Pi(X_1,X_2) \\ &= \lim_{\|\Pi\|\to 0}\sum_{j=0}^{n-1}\left(X_1(t_{j+1})-X_1(t_j)\right)\left(X_2(t_{j+1})-X_2(t_j)\right).\end{aligned} \tag{9.55}$$

Theorem 9.22.
Suppose $X_1(t)=X_1^c(t)+J_1(t)$ with integrand $\Gamma_1(t)$ and $X_2(t)=X_2^c(t)+J_2(t)$ with integrand $\Gamma_2(t)$. Then the cross variation between X_1 and X_2 satisfies

$$[X_1,X_2](T)=\int_0^T\Gamma_1(s)\Gamma_2(s)ds+\sum_{0<s\leqslant T}\Delta J_1(s)\Delta J_2(s). \tag{9.56}$$

Proof. We would first compute the quadratic variation for X_1.

$$[X_1,X_1](T)=[X_1^c,X_1^c](T)+[J_1,J_1](t) \tag{9.57}$$

Since $[X_1^c,X_1^c](T)=\int_0^T\Gamma_1^2(s)ds$ and quadratic variation for J_1 is the sum of square of jump sizes, i.e. $[J_1,J_1](t)=\sum_{0<s\leqslant T}(\Delta J_1(s))^2$. Then

$$[X_1,X_1](T)=\int_0^T\Gamma_1^2(s)ds+\sum_{0<s\leqslant T}(\Delta J_1(s))^2. \tag{9.58}$$

Similarly, X_2 satisfies

$$[X_2,X_2](T)=\int_0^T\Gamma_2^2(s)ds+\sum_{0<s\leqslant T}(\Delta J_2(s))^2. \tag{9.59}$$

From Properties 3.125, we have

$$[X_1, X_2](T) = [X_1^c, X_2^c](T) + [J_1, J_2](t). \tag{9.60}$$

By applying bracket process in Theorem 3.126, we verify that

$$\begin{aligned}
[X_1^c, X_2^c](t) &= \frac{1}{4}\{[X_1^c + X_2^c, X_1^c + X_2^c](t) - [X_1^c - X_2^c, X_1^c - X_2^c](t)\} \\
&= \frac{1}{4}\left\{\int_0^T (\Gamma_1 + \Gamma_2)^2 ds - (\Gamma_1 - \Gamma_2)^2 ds\right\} \\
&= \int_0^T \Gamma_2^2(s) ds.
\end{aligned} \tag{9.61}$$

And

$$\begin{aligned}
[J_1, J_2](T) &= \frac{1}{4}\{[J_1 + J_2](t) - [J_1 - J_2](t)\} \\
&= \frac{1}{4}\left\{\sum_{0<s\leqslant T} (\Delta J_1(s) + \Delta J_2(s))^2 - \sum_{0<s\leqslant T} (\Delta J_1(s) - \Delta J_2(s))^2\right\} \\
&= \sum_{0<s\leqslant T} \Delta J_1(s) \Delta J_2(s).
\end{aligned} \tag{9.62}$$

Adding the two equations together, we have proved the theorem that

$$[X_1, X_2](T) = \int_0^T \Gamma_1(s)\Gamma_2(s) ds + \sum_{0<s\leqslant T} \Delta J_1(s) \Delta J_2(s). \tag{9.63}$$

□

Actually, we can rewrite the theorem into differentiation form

$$dX_1(t)dX_2(t) = dX_1^c(t)dX_2^c(t) + dJ_1(t)dJ_2(t). \tag{9.64}$$

9.2.2 Itô-Doeblin Formula for Jump Processes

In this subsection, we would demonstrate the integral form and differential form of $f(t, X_1(t), X_2(t), \dots)$ where $X_i(t)$ is the jump process. Both the integral form and differential form are called Itô-Doeblin formula for jump process, which would be frequently utilized in later calculations.

In the beginning, we would discuss Itô-Doeblin formula for a single jump process.

Theorem 9.23 (Itô-Doeblin formula for one jump)**.**
For jump process $X(t) = X^c(t) + J(t)$, and function $f \in C^2[0, t]$, we have

$$\begin{aligned}
f(X(t)) =& f(X(0)) + \int_0^t f'(X(s)) dX^c(s) + \frac{1}{2}\int_0^t f''(X(s)) dX^c(s) dX^c(s) \\
&+ \sum_{0<s\leqslant t} [f(X(s)) - f(X(s-))].
\end{aligned} \tag{9.65}$$

Proof. *Consider the jump process*

$$X(t) = X^c(t) + J(t). \tag{9.66}$$

For the continuous part

$$X^c(t) = X^c(0) + \int_0^t \Gamma(s)dW(s) + \int_0^t \Theta(s)ds. \tag{9.67}$$

According to Itô-Doeblin formula for continuous part, we have

$$\begin{aligned} f(X^c(t)) =& f(X^c(0)) + \int_0^t f'(X^c(s))\Gamma(s)dW(s) + \int_0^t f'(X^c(s))\Theta(s)ds \\ &+ \frac{1}{2}f''(X^c(s))\Gamma^2(s)ds. \end{aligned} \tag{9.68}$$

Suppose the process $X(t)$ jumps at time $0 < t_1 < t_2 < \cdots < t_{n-1}$, and $t_0 = 0$, $t_n = t$, then $X(t)$ is continuous in the interval $[t_i, t_{i+1})$,implying that we can use Itô-Doeblin formula in each interval. For any $u < v \in [t_i, t_{i+1})$, it follows that

$$\begin{aligned} dX(t) &= \Delta X(t) = X(t) - X(t-) = X^c(t) - X^c(t-) + J(t) - J(t-) \\ &= X^c(t) - X^c(t-) = \Delta X^c(t) = dX(t). \end{aligned} \tag{9.69}$$

And

$$[X, X](t) = [X^c, X^c](t) + \Delta J(t)\Delta J(t) = [X^c, X^c](t). \tag{9.70}$$

Then we have

$$\begin{aligned} f(X(v)) - f(X(u)) &= \int_v^u f'(X(s))dX(s) + \frac{1}{2}\int_v^u f''(X(s))d[X, X](s) \\ &= \int_v^u f'(X(s))dX^c(s) + \frac{1}{2}\int_v^u f''(X(s))dX^c(s)dX^c(s). \end{aligned} \tag{9.71}$$

Set $u = t_i$ and $v = t_{i+1}$, then equation (9.71) can be rewritten as

$$f(X(t_{i+1}-)) - f(X(t_i)) = \int_{t_i}^{t_{i+1}} f'(X(s))dX^c(s) + \frac{1}{2}\int_{t_i}^{t_{i+1}} f''(X(s))dX^c(s)dX^c(s). \tag{9.72}$$

Note that when $X(t)$ jumps at time t_{i+1}, $f(X(t))$ also jumps. This suggests that

$$\begin{aligned} f(X(t_{i+1})) - f(X(t_i)) =& \int_{t_i}^{t_{i+1}} f'(X(s))dX^c(s) + \frac{1}{2}\int_{t_i}^{t_{i+1}} f''(X(s))dX^c(s)dX^c(s) \\ &+ f(X(t_{i+1})) - f(X(t_{i+1}-)). \end{aligned} \tag{9.73}$$

By adding up the increments over each interval, we obtain that

$$\begin{aligned} f(X(t)) - f(X(0)) &= \sum_{i=0}^{n} [f(X(t_{i+1})) - f(X(t_i))] \\ &= \int_0^t f'(X(s))dX^c(s) + \frac{1}{2}\int_0^t f''(X(s))dX^c(s)dX^c(s) \\ &\quad + \sum_{i=0}^{n-1} f(X(t_{i+1})) - f(X(t_{i+1}-)). \end{aligned} \tag{9.74}$$

This is equivalent to what we need to verify, because when jump does not happen, $f(X(t)) - f(X(t-)) = 0$. □

9.2.3 Geometric Poisson Process

Then we would like to introduce a famous example based on jump process and Itô-Doeblin formula.

Similar to geometric Brownian motion, we can also define geometric Poisson process. Geometric Poisson process can be applied to describe non-arbitrage stock price with jumps. And the prominent advantage is that geometric Poisson process is non-negative, compared to pure jump processes.

Before we introduce the definition of geometric Poisson process, we would firstly propose a significant lemma about how to deal to exponential function of Poisson process.

Lemma 9.24 (Expected value for an exponential function of Poisson process)**.**
for a Poisson process $N(t)$ with intensity λ, and constant α, we have

$$E\left[(\alpha+1)^{N(t)}\right] = e^{\alpha\lambda t}. \tag{9.75}$$

Proof. Note that Poisson distribution has the property that $E[N(t)] = \lambda t$,and then

$$\sum_{i=0}^{\infty} \frac{(\lambda t)^i}{i!} e^{-\lambda t} = 1. \tag{9.76}$$

We would use the property and definition of expected value to prove the property.

$$\begin{aligned} E\left[(\alpha+1)^{N(t)}\right] &= \sum_{i=0}^{\infty} (\alpha+1)^i \frac{(\lambda t)^i}{i!} e^{-\lambda t} \\ &= e^{\alpha\lambda t} \sum_{i=0}^{\infty} \frac{((\alpha+1)\lambda t)^i}{i!} e^{-(\alpha+1)\lambda t} \\ &= e^{\alpha\lambda t}. \end{aligned} \tag{9.77}$$

For more general exponential form, we can use similar idea to calculate and we would not demonstrate them in this section. □

With the lemma, then we would talk about geometric Poisson process.

Definition 9.25 (geometric Poisson process)**.**
Geometric Poisson process is defined to be

$$S(t) = S(0)e^{(\alpha-\lambda\sigma)t}(\sigma+1)^{N(t)}, \tag{9.78}$$

where $N(t)$ is the Poisson process and $\sigma > -1$.

Properties 9.26 (graphs of geometric Poisson process)**.**
The behavior of geometric Poisson process is greatly determined by σ. Note that, when $\sigma > 0$, the value for $S(t)$ would increase suddenly at jump times while decrease between two consecutive jumps, or we say, the process $S(t)$ jumps up and moves down between jumps. On the contrary, when $-1 < \sigma < 0$, the process jumps down and moves up between jumps.

Properties 9.27 (mean rate of return for geometric Poisson process)**.**
Geometric Poisson process $S(t) = S(0)e^{(\alpha-\lambda\sigma)t}(\sigma+1)^{N(t)}$ *has mean rate of return* α*.*

Proof. Actually, we notice that geometric Poisson process share the same mean rate of return with geometric Brownian motion.
As usual, we calculate the expected value of $S(t)$,

$$\begin{aligned} E[S(t)] &= E\left[S(0)e^{(\alpha-\lambda\sigma)t}(\sigma+1)^{N(t)}\right] \\ &= S(0)e^{(\alpha-\lambda\sigma)t}E\left[(\sigma+1)^{N(t)}\right]. \end{aligned} \tag{9.79}$$

To find the value for $E\left[(\sigma+1)^{N(t)}\right]$, we utilize Lemma 9.24, then

$$E\left[(\sigma+1)^{N(t)}\right] = e^{\sigma\lambda t}. \tag{9.80}$$

Namely,

$$E[S(t)] = S(0)e^{(\alpha-\lambda\sigma)t}e^{\sigma\lambda t} = S(0)e^{\alpha t}. \tag{9.81}$$

Recall that we have seen the formula $E[S(t)] = E[S(0)]e^{\alpha t}$ when calculating the mean rate for geometric Brownian motion. So from Lemma 6.2, geometric Poisson process has mean rate of return α. ☐

Like geometric Poisson process, which is not a martingale but a submartingale, it is noteworthy that geometric Poisson process is also a submartingale. In later part, we would verify that geometric Poisson process with $\alpha = 0$ is a martingale and then analyze the principle behind this.

Theorem 9.28.
Geometric Poisson process with $\alpha = 0$ *is a martingale.*

Proof. The basic idea for the proof is to figure out the differential form of geometric Poisson process and then use theorem 9.19 to verify that the process is a martingale
Assume $f(x) = e^x$ and $X(t) = N(t)\log(\sigma+1) - \lambda\sigma t$. Then $X(t)$ has continuous part $X^c(t) = -\lambda\sigma t$ and pure jump part $J(t) = N(t)\log(\sigma+1)$. Applying Itô-Doeblin formula for

jump process, it follows that

$$
\begin{aligned}
S(t) &= S(0)f(X(t)) \\
&= S(0)f(X(0)) + S(0)\int_0^t f'(X(u))dX^c(u) + S(0)\frac{1}{2}\int_0^t f''(X(u))d[X^c,X^c](u) \\
&\quad + S(0)\sum_{0<s\leqslant t}[f(X(u)) - f(X(u-))] \\
&= S(0)f(X(0)) + S(0)\int_0^t f'(X(u))dX^c(u) + S(0)\frac{1}{2}\int_0^t f''(X(u))d[X^c,X^c](u) \\
&\quad + S(0)\sum_{0<s\leqslant t} f(X(u-))\sigma\Delta N(u) \\
&= S(0) - \lambda\sigma\int_0^t S(u)du + \sigma\int_0^t S(u-)dN(u) \\
&= S(0) + \sigma\int_0^t S(u-)d(N(u) - \lambda t).
\end{aligned}
\tag{9.82}
$$

To derive the third equality, it is noteworthy that, when jump happens,

$$
\begin{aligned}
f(X(u)) - f(X(u-)) &= e^{N(u)\log(\sigma+1)-\lambda\sigma t} - e^{N(u-)\log(\sigma+1)-\lambda\sigma t} \\
&= e^{(N(u-)+1)\log(\sigma+1)-\lambda\sigma t} - e^{N(u-)\log(\sigma+1)-\lambda\sigma t} \\
&= (\sigma+1-1)e^{N(u-)\log(\sigma+1)-\lambda\sigma t} \\
&= \sigma f(X(u-)) = \sigma f(X(u-))dN(t)
\end{aligned}
\tag{9.83}
$$

And the formula above also holds when jump does not happen, hence, the third equality could be derived. From equation 9.82, the integral is a martingale, and consequently, $S(t)$ is a martingale when $\alpha = 0$. □

9.2.4 Itô's Product Rule and Doleans-Dade Exponential

Theorem 9.29 (Itô product rule for jump process)**.**
For jump process $X_1(t)$ and $X_2(t)$, we can verify that

$$
\begin{aligned}
X_1(t)X_2(t) = X_1(0)X_2(0) &+ \int_0^t X_2(s)dX_1^c(s) + \int_0^t X_1(s)dX_2^c(s) + \\
&[X_1^c, X_2^c](t) + \sum_{0<s\leqslant t}[X_1(s)X_2(s) - X_1(s-)X_2(s-)] \\
&= X_1(0)X_2(0) + \int_0^t X_2(s-)dX_1(s) + \int_0^t X_1(s-)dX_2(s) + [X_1, X_2](t).
\end{aligned}
\tag{9.84}
$$

Proof. By applying Itô-Doeblin formula for jump processes, we have

$$
\begin{aligned}
X_1(t)X_2(t) = X_1(0)X_2(0) &+ \int_0^t X_2(s)dX_1^c(s) + \int_0^t X_1(s)dX_2^c(s) \\
&+ [X_1^c, X_2^c](t) + \sum_{0<s\leqslant t}[X_1(s)X_2(s) - X_1(s-)X_2(s-)].
\end{aligned}
\tag{9.85}
$$

Then we could verify that the second equality is equivalent to the first equality. Recall that $X_i(t) = X_i^c(t) + J_i(t)$ for $i = 1, 2$. It follows that

$$\begin{aligned} & X_1(0)X_2(0) + \int_0^t X_2(s-)dX_1(s) + \int_0^t X_1(s-)dX_2(s) + [X_1, X_2](t) \\ & = X_1(0)X_2(0) + \int_0^t X_2(s-)dX_1^c(s) + \int_0^t X_2(s-)dJ_1(s) \\ & + \int_0^t X_1(s-)dX_2^c(s) + \int_0^t X_1(s-)dJ_2(s) \\ & + [X_1^c, X_2^c](t) + \sum_{0<s\leqslant t} \Delta J_1(s)J_2(s). \end{aligned} \tag{9.86}$$

Since $X_i^c(t)$ for $i = 1, 2$ are martingales and $\Delta X_i(t) = \Delta J_i(t)$, then it can be derived that

$$\begin{aligned} & X_1(0)X_2(0) + \int_0^t X_2(s-)dX_1(s) + \int_0^t X_1(s-)dX_2(s) + [X_1, X_2](t) \\ & = X_1(0)X_2(0) + \int_0^t X_2(s)dX_1^c(s) + \int_0^t X_1(s)dX_2^c(s) + [X_1^c, X_2^c](t) \\ & + \sum_{0<s\leqslant t} \left[X_2(s-)\Delta X_1(s) + X_1(s-)\Delta X_2(s) + \Delta X_1(s)\Delta X_2(s)\right]. \end{aligned} \tag{9.87}$$

And we find that

$$\begin{aligned} & X_1(s)X_2(s) - X_1(s-)X_2(s-) \\ & = \left(X_1(s-) + \Delta X_1(s)\right)\left(X_2(s-) + \Delta X_2(s)\right) - X_1(s-)X_2(s-) \\ & = X_2(s-)\Delta X_1(s) + X_1(s-)\Delta X_2(s) + \Delta X_1(s)\Delta X_2(s). \end{aligned} \tag{9.88}$$

Then we have verified the second equality. □

Theorem 9.30 (Property of Doleans-Dade exponential)**.**
For a jump process $X(t)$, the Doleans-Dade exponential of X is defined as the following formula

$$Z^X = \exp\{X^c(t) - \frac{1}{2}[X^c, X^c](t)\} \prod_{0<s\leqslant t} (1 + \Delta X(s)). \tag{9.89}$$

Suppose $Z^X(0) = 1$, then $Z^X(t)$ can be rewritten into integral form that

$$Z^X(t) = 1 + \int_0^t Z^X(s-)dX(s). \tag{9.90}$$

Proof. We would use the definition of Doleans-Dade exponential to deduce the integral form. Suppose that $Y(t) = exp\{X^c(t) - \frac{1}{2}[X^c, X^c](t)\}$ and

$$K(t) = \begin{cases} \prod_{0<s\leqslant t}(1 + \Delta X(s)), & t > 0, \\ 1, & t = 0. \end{cases} \tag{9.91}$$

According to Itô's product rule for jump processes,

$$Z^X(t) = Y(t)K(t) = Y(0)\cdot 1 + \int_0^t K(s-)dY(s) + \int_0^t Y(s-)dK(s). \tag{9.92}$$

Then we need substitute $dY(s)$ and $dK(s)$ in another way to further compute $Z^X(t)$.

Since $Y(t) = exp\{X^c(t) - \frac{1}{2}[X^c, X^c](t)\}$, we assume $A(t) = X^c(t) - \frac{1}{2}[X^c, X^c](t)$ and $f(x) = e^x$. Then $Y(t) = f(A(t))$.

From Itô-Doeblin formula, we have

$$dY(t) = df(A(t)) = f'(A(t))dA(t) + \frac{1}{2}f''(A(t))d[A, A](t). \tag{9.93}$$

Notice that $f''(x) = f'(x) = f(x) = e^x$, which implies that $f'(A(t)) = f''(A(t)) = f(A(t))$. In order to calculate $[A, A](t)$, we use the fact that $[X^c, X^c](t) = dX^c dX^c$ does not have quadratic variation, i.e. $[A, A](t) = [X^c, X^c](t)$.

$$\begin{aligned} dY(t) &= f'(A(t))dA(t) + \frac{1}{2}f''(A(t))d[A, A](t) \\ &= f(A(t))d\left(X^c(t) - \frac{1}{2}[X^c, X^c](t)\right) + f(A(t))d\left(\frac{1}{2}[X^c, X^c](t)\right) \\ &= F(A(t))dX^c(t) \\ &= Y(t)dX^c(t). \end{aligned} \tag{9.94}$$

Since the jump process is independent of Itô process $X^c(t)$, then we can write $Y(t)dX^c(t) = Y(t-)dX^c(t)$. Or

$$dY(t) = Y(t-)dX^c(t). \tag{9.95}$$

For $K(t) = \prod_{0<s\leqslant t}(1 + \Delta X(s))$ for $t > 0$, it is noteworthy that $K(t) = 1$ between time 0 and the time of the first jump; $K(t) = 1 + \Delta X(t_1)$ between time of first jump and time of second jump, where t_1 is the time for the first jump; $K(t) = (1+\Delta X(t_1))(1+\Delta X(t_2))\cdots(1+\Delta X(t_n))$ between time of nth jump and $(n+1)$th jump, where $t_1, t_2, \cdots, t_n$ are the time for jumps.

Then we can see that at jump time t, $K(t) = \lim_{\tau\to 0^+} K(t-\tau)(1+\Delta X(t)) = K(t-)(1+\Delta X(t))$. And then

$$dK(t) = \Delta K(t) = K(t) - K(t-) = K(t-)\Delta X(t). \tag{9.96}$$

This formula also satisfies for t at non-jump time because $\Delta K(t) = \Delta X(t) = 0$. Then we just need to replace $dY(t)$ and $dK(t)$ into the formula for $Z^X(t)$. Note that $X(t) = X^c(t) + J(t) =$

$X^c(t)+\sum_{0<s\leqslant t}\Delta J(t)=X^c(t)+\sum_{0<s\leqslant t}\Delta X(t)$. Then we have

$$\begin{aligned}Z^X(t)&=Y(0)\cdot 1+\int_0^t K(s-)dY(s)+\int_0^t Y(s-)dK(s)\\&=1+\int_0^t Y(s-)K(s-)dX^c(s)+\sum_{0<s\leqslant t}Y(s-)K(s-)\Delta X(s)\\&=1+\int_0^t Y(s-)K(s-)dX(s)\\&=1+\int_0^t Z^X(s-)dX(s).\end{aligned}\tag{9.97}$$

Thus, we have obtained the integral form using the definition of Doleans-Dade exponential. □

9.3 Change of Measure

Again, $N(t)$ is a Poisson process with intensity λ. For a positive number $\widetilde{\lambda}$, we can define

$$Z(t)=e^{(\lambda-\widetilde{\lambda})t}\left(\frac{\widetilde{\lambda}}{\lambda}\right)^{N(t)}\tag{9.98}$$

Since we would like to demonstrate that $Z(t)$ can be applied to change the measure, we need to show that $N(t)$ have intensity $\widetilde{\lambda}$ after change of measure.

9.3.1 Adapted Process Z(t) and Probability Measure $\widetilde{P}$

Lemma 9.31. *Suppose $M(t)=N(t)-\lambda t$ and $M(t)$ is a compensated Poisson process. Then we have*

$$dZ(t)=\frac{\widetilde{\lambda}-\lambda}{\lambda}Z(t-)dM(t).\tag{9.99}$$

Proof. To prove the lemma, we would use a cunning construction to transform $Z(t)$ into Doleans-Dade exponential.

Let $X(t)=\frac{\widetilde{\lambda}-\lambda}{\lambda}M(t)$. $X(t)$ is a jump process with continuous part $X^c(t)=(\lambda-\widetilde{\lambda})t$ and pure jump part $J(t)=\frac{\widetilde{\lambda}-\lambda}{\lambda}N(t)$. For continuous part,

$$[X^c,X^c](t)=(\lambda-\widetilde{\lambda})^2dtdt=0\tag{9.100}$$

While for jump part,

$$\Delta X(t)=\Delta J(t)=\frac{\widetilde{\lambda}}{\lambda}-1\tag{9.101}$$

Then we can rewrite $Z(t)$

$$\begin{aligned}Z(t)&=e^{(\lambda-\widetilde{\lambda})t}(\frac{\widetilde{\lambda}}{\lambda})^{N(t)}\\&=\exp\{X^c(t)-\frac{1}{2}[X^c,X^c](t)\}\prod_{0<s\leqslant t}(1+\Delta X(s)).\end{aligned}\tag{9.102}$$

According to Doleans-Dade exponential, $Z(t) = 1 + \int_0^t Z(s-)dX(s)$. Then we use the property that $Z(t-) = \lim_{\tau \to 0} Z(t-\tau)$ to obtain that

$$\begin{aligned} dZ(t) = Z(t) - Z(t-) &= \int_{t-\tau}^{t} Z(s-)dX(s) \\ &= \frac{\widetilde{\lambda} - \lambda}{\lambda} Z(t-) \int_{t-\tau}^{t} dM(t) \\ &= \frac{\widetilde{\lambda} - \lambda}{\lambda} Z(t-)[M(t) - M(t-\tau)] \\ &= \frac{\widetilde{\lambda} - \lambda}{\lambda} Z(t-)dM(t). \end{aligned} \tag{9.103}$$

□

Lemma 9.32 (a second state of the extremely significant theorem).
Consider the formula

$$\int_0^t f(s-)dX(s) \tag{9.104}$$

Here, $f(s-)$ satisfies $\Delta f(s-) = 0$ and $X(s)$ is a martingale with some limiting conditions. Then we have $\int_0^t f(s-)dX(s)$ is a martingale. (see Theorem 9.19)

Properties 9.33. *$Z(t)$ is a martingale and $\mathbb{E}[Z(t)] = 1$.*

Proof. From Lemma 6.3.1, we notice that X is a martingale and $\Delta Z(t-) = 0$. Then $Z(t) = 1 + \int_0^t Z(s-)dX(s)$ is a martingale (from Lemma 6.3.2). So we have the property that $\mathbb{E}[Z(t)] = Z(0) = 1$. □

We define the new probability measure to be

$$\widetilde{\mathbb{P}} = \int_A Z(t)d\mathbb{P} \quad for \quad A \in \mathscr{F}. \tag{9.105}$$

Theorem 9.34. *For the Poisson process $N(t)$ defined on the new probability measure $\widetilde{P}$, it would have intensity $\widetilde{\lambda}$.*

Proof. It is equivalent to prove that the moment-generating function of $N(t)$ has the property $\mathbb{E}[e^{uN(t)}Z(t)] = \exp\{\widetilde{\lambda}t(e^u - 1)\}$.

$$\begin{aligned} \mathbb{E}[e^{uN(t)}Z(t)] &= e^{(\lambda-\widetilde{\lambda})}\mathbb{E}[e^{uN(t)}\left(\frac{\widetilde{\lambda}}{\lambda}\right)^{N(t)}] \\ &= e^{(\lambda-\widetilde{\lambda})}\mathbb{E}[exp\{(u + log\left(\frac{\widetilde{\lambda}}{\lambda}\right))N(t)\}]. \end{aligned} \tag{9.106}$$

For a random variable x following Poisson distribution with intensity λ, we have $\mathbb{E}[x] = \lambda$. Then for Poisson process $N(t)$ with intensity λ, $\mathbb{E}[N(t)] = Var(N(t)) = \lambda t$. And then

$$\mathbb{E}[e^{uN(t)}] = \exp\{\lambda t(e^u - 1)\}. \tag{9.107}$$

By replacing u with $u + log(\frac{\widetilde{\lambda}}{\lambda})$,

$$\mathbb{E}[\exp\{(u + log(\frac{\widetilde{\lambda}}{\lambda}))N(t)\}] = \exp\{\lambda t(e^{u+log(\frac{\widetilde{\lambda}}{\lambda})} - 1)\}. \tag{9.108}$$

It follows

$$\begin{aligned}\mathbb{E}[e^{uN(t)}Z(t)] &= e^{(\lambda-\widetilde{\lambda})t}\exp\{\lambda t(e^{u+log(\frac{\widetilde{\lambda}}{\lambda})} - 1)\} \\ &= \exp\{\widetilde{\lambda}t(e^{u} - 1)\}.\end{aligned} \tag{9.109}$$

According to the property of moment generating function, it is explicit that $N(t)$ has intensity $\widetilde{\lambda}$ under the new measure. □

Example 9.35. Recall that for geometric Poisson process $S(t)$, we have shown that it has mean rate of return α and $dS(t) = \alpha S(t)dt + \sigma S(t-)dM(t)$, where $M(t) = N(t) - \lambda t$.
Then we need to define $S(t)$ under new probability measure $\widetilde{P}$. That is

$$dS(t) = rS(t)dt + \sigma S(t-)d\widetilde{M}(t) \tag{9.110}$$

Here, $\widetilde{M}(t) = N(t) - \widetilde{\lambda}t$.
Since we want to maintain $dS(t)$ after changing the measure, then

$$(\alpha - \lambda\sigma)S(t)dt = (r - \widetilde{\lambda}\sigma)S(t)dt. \tag{9.111}$$

Denote

$$\widetilde{\lambda} = \lambda - \frac{\alpha - r}{\sigma}. \tag{9.112}$$

As we must let $\widetilde{\lambda} > 0$, then

$$\lambda > \frac{\alpha - r}{\sigma}. \tag{9.113}$$

9.3.2 Change of Measure on Jump Processes

Then we interpret the effects of change of measure on jump processes During these examples, we would interpret the original processes, define the adapted process $Z(t)$, new probability measure $\widetilde{\mathbb{P}}$ and examine the validity of the adapted process. And the last but most significant step is showing that the intensity $\widetilde{\lambda}$ of the jump processes would vary.

Example 9.36 (revision of discrete compound Poisson process)**.** *Recall that the compound Poisson process is a combination of pure Poisson processes:*

$$Q(t) = \sum_{i=1}^{N(t)} Y_i = \sum_{m=1}^{M} y_m N_m(t). \tag{9.114}$$

Here, Y_i denotes the jump size when jump happens, so at jump time t, $\Delta Q(t) = Y_{N(t)}$. Also, $Q(t)$ can be denoted by calculating the number of existence of each jump sizes before time t

and summing them up.

Suppose there are finite possible jump sizes, i.e. $Y_i \in \{y_1, y_2, \ldots, y_M\}$*. We define* $p(y_m) = P\{Y_i = y_m\}$*. Then it is obvious that*

$$\sum_{m=1}^{M} p(y_m) = 1. \tag{9.115}$$

For each y_m*, we assume that there are* $N_m(t)$ *jumps with jump size* y_m*. Then total number of jumps satisfies*

$$N(t) = \sum_{m=1}^{M} N_m(t). \tag{9.116}$$

$N_m(t)$ *should be pure Poisson processes with fixed jump sizes. And the intensity of* $N_m(t)$ *satisfies* $\lambda_m = \lambda p(y_m)$*.*

Definition 9.37 (new probability measure for discrete compound Poisson process)**.** *Since for each Poisson process, Radon-Nikodým derivative* $Z_m(t)$ *is defined to be*

$$Z_m(t) = e^{(\lambda_m - \widetilde{\lambda}_m)t}\left(\frac{\widetilde{\lambda}_m}{\lambda_m}\right)^{N_m(t)}. \tag{9.117}$$

For compound Poisson process, we have the adapted process $Z(t)$ *satisfies the property that*

$$Z(t) = \prod_{m=1}^{M} Z_m(t). \tag{9.118}$$

We would illustrate the reasonability for constructing $Z(t)$ *as the formula above. With the process* $Z(t)$*, we can find the probability measure as defined before*

$$\widetilde{\mathbb{P}} = \int_A Z(t) d\mathbb{P} \quad for \quad A \in \mathscr{F}. \tag{9.119}$$

Properties 9.38 (validity of $Z(t)$ for discrete compound Poisson process)**.**
It emerges that $Z(t)$ *is a martingale and* $\mathbb{E}[Z(t)] = 1$*.*

Proof. According to Lemma 9.31, we have $Z_m(t)$ is a martingale and $\mathbb{E}[Z_m(t)] = 1$.
Also,

$$dZ_m(t) = \frac{\widetilde{\lambda}_m - \lambda_m}{\lambda_m} Z_m(t-) dM_m(t) \tag{9.120}$$

Here, $M_m(t) = N_m(t) - \lambda_m t$.

Note that N_m and N_n do not simultaneous jumps as designed, which means the two Poisson processes N_m and N_n are independent of each other. We can write $Z_m(t) = f(N_m(t))$ and $Z_n(t) = f(N_n(t))$, implying that Z_m and Z_n are two independent processes. From Theorem

3.128, we verify that the cross variation $[Z_m, Z_n](t) = 0$, or $dZ_m(t)dZ_n(t) = 0$. According to Theorem 9.29, we have

$$\begin{aligned} Z_1(t)Z_2(t) &= Z_1(0)Z_2(0) + \int_0^t Z_2(t)dZ_1(t) + \int_0^t Z_1(t)dZ_2(t) + [Z_1, Z_2](t) \\ &= \int_0^t Z_2(t)dZ_1(t) + \int_0^t Z_1(t)dZ_2(t). \end{aligned} \tag{9.121}$$

Then,

$$\begin{aligned} d\left(Z_1(t)Z_2(t)\right) &= \lim_{\tau\to 0^+} \int_{t-\tau}^t Z_2(t)dZ_1(t) + \int_{t-\tau}^t Z_1(t)dZ_2(t) \\ &= \lim_{\tau\to 0^+} \int_{t-\tau}^t \left(Z_2(t-) + \Delta Z_2(t)\right) dZ_1(t) + \int_{t-\tau}^t \left(Z_1(t-) + \Delta Z_1(t)\right) dZ_2(t) \\ &= Z_2(t-)[Z_1(t) - Z_1(t-)] + \lim_{\tau\to 0^+} \int_{t-\tau}^t dZ_2(t)dZ_1(t) \\ &\quad + Z_1(t-)[Z_2(t) - Z_2(t-)] + \lim_{\tau\to 0^+} \int_{t-\tau}^t dZ_1(t)dZ_2(t) \\ &= Z_2(t-)\Delta Z_1(t) + 0 + Z_1(t-)\Delta Z_2(t) + 0 \\ &= Z_2(t-)dZ_1(t) + Z_1(t-)dZ_2(t). \end{aligned} \tag{9.122}$$

Since Z_1 and Z_2 are martingales and $\Delta Z_1(t-) = \Delta Z_2(t-) = 0$, from Lemma 9.32, we conclude that Z_1Z_2 is a martingale.

Now we define a new process which is left-continuous

$$S_n(t) := Z_1(t) \cdot Z_2(t) \dots Z_n(t) \ for\ n = 1, 2, \dots, M. \tag{9.123}$$

From the definition, we can see that $Z(t) = S_M(t)$. Suppose that, for $n = k$, $S_k(t)$ is a martingale, then consider $n = k + 1$,

$$\begin{aligned} d(S_{k+1}(t)) &= d(S_k(t) \cdot Z_{k+1}(t)) \\ &= S_k(t-)dZ_{k+1}(t) + Z_{k+1}(t-)dS_k(t). \end{aligned} \tag{9.124}$$

Since Z_{k+1}, S_k are martingales and $\Delta Z_{k+1}(t-) = \Delta S_k(t-) = 0$, then $S_{k+1}(t)$ is also a martingale.

By induction, we verify that $Z(t) = S_M(t)$ is a martingale.

Since Z_m and Z_n are two independent processes, then the covariance $Cov(Z_m, Z_n) = 0$. Since $\mathbb{E}[Z_m] = 1$, then

$$\mathbb{E}[Z(t)] = \mathbb{E}[\prod_{m=1}^M Z_m(t)] = \prod_{m=1}^M \mathbb{E}[Z_m(t)] = 1.$$

We have proved that $Z(t)$ is a martingale and $Z(t)$ has mean 1. □

Theorem 9.39 (well-defined change of measure for discrete compound Poisson process). *Under new probability measure $\widetilde{\mathbb{P}}$, the compound Poisson process has intensity $\widetilde{\lambda} = \sum_{m=1}^{M} \widetilde{\lambda}_m$, the jump size Y_i has probability*

$$\widetilde{\mathbb{P}}\{Y_i = y_m\} = \widetilde{p}(y_m) = \frac{\widetilde{\lambda}_m}{\widetilde{\lambda}}. \tag{9.125}$$

Sketch of proof.
We calculate the moment generating function of $Q(t)$:

$$\begin{aligned}
\widetilde{\mathbb{E}}\left[e^{uQ(t)}\right] &= \mathbb{E}\left[e^{uQ(t)}Z(t)\right] \\
&= \mathbb{E}\left[\exp\{u\sum_{m=1}^{M} y_m N_m(t)\} \cdot \prod_{m=1}^{M} e^{(\lambda_m - \widetilde{\lambda}_m)t} \frac{\widetilde{\lambda}_m}{\lambda_m}^{N_m(t)}\right] \\
&= \prod_{m=1}^{M} \exp\left\{(\lambda_m - \widetilde{\lambda}_m)t\right\} \exp\left\{\lambda_m t(e^{uy_m + log(\widetilde{\lambda}_m/\lambda_m)} - 1)\right\} \\
&= \prod_{m=1}^{M} \exp\left\{\widetilde{\lambda}_m t(e^{uy_m -1})\right\}.
\end{aligned} \tag{9.126}$$

After replacing $\widetilde{\lambda}_m$ by $\widetilde{\lambda}\widetilde{p}(y_m)$, we have

$$\begin{aligned}
\widetilde{\mathbb{E}}\left[e^{uQ(t)}\right] &= \exp\left\{\widetilde{\lambda}t(\sum_{m=1}^{M} \widetilde{p}(y_m)e^{uy_m}) - (\sum_{m=1}^{M} \widetilde{\lambda}_m)t\right\} \\
&= \exp\left\{\widetilde{\lambda}t(\sum_{m=1}^{M} \widetilde{p}(y_m)e^{uy_m} - 1)\right\}.
\end{aligned} \tag{9.127}$$

From the definition of moment generating function, we verify that the intensity of compound Poisson process under measure $\widetilde{\mathbb{P}}$ should be $\widetilde{\lambda}$. □

Then we can rewrite $Z(t)$ as

$$\begin{aligned}
Z(t) &= \exp\left\{\sum_{m=1}^{M}(\lambda_m - \widetilde{\lambda}_m)t\right\} \cdot \prod_{m=1}^{M}\left(\frac{\widetilde{\lambda}\widetilde{p}(y_m)}{\lambda p(y_m)}\right)^{N_m(t)} \\
&= e^{(\lambda - \widetilde{\lambda})t} \prod_{i=1}^{N(t)} \frac{\widetilde{\lambda}\widetilde{p}(Y_i)}{\lambda p(Y_i)}
\end{aligned} \tag{9.128}$$

Two expressions of $Z(t)$ means two ways of grouping jumps up to time t. The first equality represents sorting jumps according to their jump sizes; while the second equality represents sorting jumps according to their order of occurrence.

By applying Radon-Nikodým derivative process $Z(t)$, we can both change the intensity of $Q(t)$ from λ to $\widetilde{\lambda}$ and alter the density of $\{Y_i\}$. Actually, in a generalized case, if we assume

Y_i is not discrete, then the discrete probability $p(Y_i)$ becomes the density distribution $f(y)$ and similarly $\widetilde{p}(Y_i)$ becomes $\widetilde{f}(y)$.

$$Z(t) = e^{(\lambda-\widetilde{\lambda})t} \prod_{i=1}^{N(t)} \frac{\widetilde{\lambda}\widetilde{f}(Y_i)}{\lambda f(Y_i)}. \tag{9.129}$$

Based on how we construct $Z(t)$, we must ensure that $\widetilde{f}(y) = 0$ if and only if $f(y) = 0$, i.e. jumps sizes would not appear suddenly after change of measure, but what varies merely is the density.

Anyway, since we are considering the compound Poisson process under discrete cases, we still need to prove that change of measure for non-discrete compound Poisson process is well-defined.

Example 9.40 (revision of non-discrete compound Poisson process). *For the non-discrete compound Poisson process, we write*

$$Q(t) = \sum_{i=1}^{N(t)} Y_i = \sum_{m=1}^{\infty} y_m N_m(t). \tag{9.130}$$

Here, compared to discrete compound Poisson process, the only difference is that Y_i is a continuous random variable, which has the probability density $f(Y_i)$.

Definition 9.41 (new probability measure for non-discrete compound Poisson process). *For compound Poisson process, the adapted process $Z(t)$ is almost the same with that in discrete compound Poisson process, it follows that*

$$\begin{aligned} Z(t) = \prod_{m=1}^{\infty} Z_m(t) &= \prod_{m=1}^{\infty} e^{(\lambda_m-\widetilde{\lambda}_m)t} \left(\frac{\widetilde{\lambda}_m}{\lambda_m}\right)^{N_m(t)} \\ &= \prod_{m=1}^{\infty} e^{(\lambda f(y_m)-\widetilde{\lambda}\widetilde{f}(y_m))t} \left(\frac{\widetilde{\lambda}\widetilde{f}(y_m)}{\lambda f(y_m)}\right)^{N_m(t)} \\ &= e^{(\lambda-\widetilde{\lambda})t} \prod_{i=1}^{N(t)} \frac{\widetilde{\lambda}\widetilde{f}(Y_i)}{\lambda f(Y_i)}. \end{aligned} \tag{9.131}$$

With the process $Z(t)$, we can find the probability measure as defined before

$$\widetilde{\mathbb{P}} = \int_A Z(t) d\mathbb{P} \quad for \quad A \in \mathscr{F}. \tag{9.132}$$

Properties 9.42 (validity of $Z(t)$ for non-discrete compound Poisson process). *Process $Z(t)$ in non-discrete cases is a martingale and $\mathbb{E}[Z(t)] = 1$.*

Proof. We observe that

$$Z(t) = e^{(\lambda-\widetilde{\lambda})t} \prod_{i=1}^{N(t)} \frac{\widetilde{\lambda}\widetilde{f}(Y_i)}{\lambda f(Y_i)}. \tag{9.133}$$

The process is the product of the continuous part and jump part, say continuous part $Z^c(t) = e^{(\lambda-\widetilde{\lambda})t}$ and jump part

$$J(t) = \prod_{i=1}^{N(t)} \frac{\widetilde{\lambda}\widetilde{f}(Y_i)}{\lambda f(Y_i)}. \tag{9.134}$$

Then we have

$$Z(t) = Z^c(t)J(t) \tag{9.135}$$

We notice that $e^{(\lambda-\widetilde{\lambda})t}$ and $J(t)$ are independent, then $[e^{(\lambda-\widetilde{\lambda})t}, J(t)] = 0$. According to Theorem 9.29 on Itô's product rule, it follows that

$$Z(t) = Z(0) + \int_0^t J(s)(\lambda - \widetilde{\lambda})e^{(\lambda-\widetilde{\lambda})s}ds + \int_0^t e^{(\lambda-\widetilde{\lambda})s}dJ(s). \tag{9.136}$$

For further calculation, we need to determine the term $dJ(t)$, thus, at jump time,

$$\begin{aligned} dJ(t) &= \Delta J(t) \\ &= J(t) - J(t-) \\ &= J(t-)\frac{\widetilde{\lambda}\widetilde{f}(Y_{N(t)})}{\lambda f(Y_{N(t)})} - J(t-) \\ &= \left[\frac{\widetilde{\lambda}\widetilde{f}(Y_{N(t)})}{\lambda f(Y_{N(t)})} - 1\right] J(t-). \end{aligned} \tag{9.137}$$

It is noteworthy the following equation holds true at both jump and non-jump time due to the property of $N(t)$, it says

$$dJ(t) = \left[\frac{\widetilde{\lambda}\widetilde{f}(Y_{N(t)})}{\lambda f(Y_{N(t)})} - 1\right] J(t-)dN(t). \tag{9.138}$$

Then we would like to define

$$H(t) = \sum_{i=1}^{N(t)} \frac{\widetilde{\lambda}\widetilde{f}(Y_i)}{\lambda f(Y_i)}. \tag{9.139}$$

At jump times t, we have

$$dH(t) = \Delta H(t) = \frac{\widetilde{\lambda}\widetilde{f}(Y_{N(t)})}{\lambda f(Y_{N(t)})}. \tag{9.140}$$

We can also derive the differential form of $H(t)$ that holds no matter it jumps or not, and this suggests that

$$dH(t) = \frac{\widetilde{\lambda}\widetilde{f}(Y_{N(t)})}{\lambda f(Y_{N(t)})}dN(t). \tag{9.141}$$

From equation 9.137 and 9.141, we can write

$$\begin{aligned} dJ(t) &= J(t-)\frac{\widetilde{\lambda}\widetilde{f}(Y_{N(t)})}{\lambda f(Y_{N(t)})}dN(t) - J(t-)dN(t) \\ &= J(t-)dH(t) - J(t-)dN(t) \end{aligned} \tag{9.142}$$

Note that this formula also holds when jump does not happen. By substituting the formula for $dJ(t)$ into the Itô product rule, we have

$$\begin{aligned} Z(t) &= Z(0) + \int_0^t J(s)(\lambda - \widetilde{\lambda})e^{(\lambda-\widetilde{\lambda})s}ds + \int_0^t e^{(\lambda-\widetilde{\lambda})s}dJ(s) \\ &= 1 + \int_0^t J(s)(\lambda - \widetilde{\lambda})e^{(\lambda-\widetilde{\lambda})s}ds + \int_0^t e^{(\lambda-\widetilde{\lambda})s}J(s-)dH(s) - \int_0^t e^{(\lambda-\widetilde{\lambda})s}J(s-)dN(s) \\ &= 1 + \int_0^t e^{(\lambda-\widetilde{\lambda})s}J(s-)d(H(s) - \widetilde{\lambda}s) - \int_0^t e^{(\lambda-\widetilde{\lambda})s}J(s-)d(N(s) - \lambda s) \\ &= 1 + \int_0^t Z(s-)d(H(s) - \widetilde{\lambda}s) - \int_0^t Z(s-)d(N(s) - \lambda s). \end{aligned} \tag{9.143}$$

And $H(t) - \widetilde{\lambda}t$ is a martingale because

$$\begin{aligned} \mathbb{E}\left[H(t) - \widetilde{\lambda}t|\mathscr{F}(s)\right] &= \mathbb{E}\left[H(t) - H(s)\right] + H(s) - \widetilde{\lambda}t \\ &= \mathbb{E}[(N(t) - N(s))] \cdot \mathbb{E}\left[\frac{\widetilde{\lambda}\widetilde{f}(Y_i)}{\lambda f(Y_i)}\right] + H(s) - \widetilde{\lambda}t \\ &= \mathbb{E}[(N(t) - N(s))] \cdot \int_{-\infty}^{\infty} \frac{\widetilde{\lambda}\widetilde{f}(y)}{\lambda f(y)} f(y)dy + H(s) - \widetilde{\lambda}t \\ &= \mathbb{E}[(N(t) - N(s))] \cdot \frac{\widetilde{\lambda}}{\lambda} \int_{-\infty}^{\infty} \widetilde{f}(y)dy + H(s) - \widetilde{\lambda}t \\ &= \lambda(t-s) \cdot \frac{\widetilde{\lambda}}{\lambda} + H(s) - \widetilde{\lambda}t \\ &= H(s) - \widetilde{\lambda}s. \end{aligned} \tag{9.144}$$

Since we already know that $N(t) - \lambda t$ is a martingale and $\Delta Z(t-) = 0$, then according to Lemma 9.32, we verify that $Z(t)$ is a martingale and $\mathbb{E}[Z(t)] = Z(0) = 1$. □

In differential form, we have

$$dZ(t) = Z(t-)d(H(t) - \widetilde{\lambda}t) - Z(t-)d(N(t) - \lambda t). \tag{9.145}$$

Theorem 9.43 (well-defined change of measure for non-discrete compound Poisson process)**.** *Under new probability measure $\widetilde{\mathbb{P}}$, $Q(t)$ has intensity $\widetilde{\lambda}$ and the jump sizes has density $\widetilde{f}(y)$. The proof of the theorem would not be shown in this section, for more details, please refer to Shreve (2004).*

9.3.3 Change of Measure for Brownian motion

In order to perceive the effects of change of measure, we would like to introduce some examples about Brownian motion.

Definition 9.44 (linear Brownian motion). *The linear Brownian motion is an extension of Brownian motion, which is denoted as*

$$W_l(t) = \mu t + \sigma W(t), \tag{9.146}$$

where μ represents the drift and σ denotes the volatility.

The linear Brownian motion provides a framework that combines the linear trends with stochastic processes. In this example, we would demonstrate that the linear Brownian motion could not be transferred into standard Brownian motion due to the linear trends μt.

Theorem 9.45 (undefined change of measure for linear Brownian motion).
The linear Brownian motion could not be transferred into standard Brownian motion using change of measure.

Proof. Suppose the adapted process for change of measure could be defined, then for a stochastic process $\theta_l(t) = \alpha W(t) + \beta t + \gamma$. Since the adapted process must be a martingale, the adapted process for linear Brownian motion satisfies the formula that

$$\begin{aligned} Z_l(t) &= \exp\left\{-\theta_l(t) - \frac{1}{2}[\theta_l, \theta_l](t)\right\} \\ &= \exp\left\{-\alpha W(t) - \left(\beta + \frac{1}{2}\alpha^2\right) t - \gamma\right\}. \end{aligned} \tag{9.147}$$

And the corresponding probability measure should be

$$\widetilde{\mathbb{P}} = \int_A Z_l(T) d\mathbb{P}, \forall A \in \mathscr{F}. \tag{9.148}$$

Then assume that, under new probability measure $\widetilde{\mathbb{P}}$, the linear Brownian motion satisfies

$$\widetilde{W_l}(t) = \frac{1}{\sigma}(W_l(t) - \mu t) = W(t). \tag{9.149}$$

It is equivalent to say that the moment-generating function under $\widetilde{\mathbb{P}}$ satisfies the following formula that

$$\widetilde{\mathbb{E}}[e^{u\widetilde{W}_l(t)}] = \exp\left\{\frac{1}{2}u^2 t\right\}. \tag{9.150}$$

Then it follows that

$$\begin{aligned} \widetilde{\mathbb{E}}[e^{u\widetilde{W}_l(t)}] &= \mathbb{E}[e^{u\widetilde{W}_l(t)} Z_l(t)] \\ &= \mathbb{E}\left[\exp\left\{\left(\frac{u}{\sigma} - \alpha\right) W_l(t) - \left(\frac{u\mu}{\sigma} + \beta + \frac{1}{2}\alpha\right) t - \gamma\right\}\right] \\ &= \exp\left\{\left(\frac{u}{\sigma} - \alpha\right)\left(\mu t + \frac{1}{2}\sigma^2\right) - \left(\frac{u\mu}{\sigma} + \beta + \frac{1}{2}\alpha^2\right) t - \gamma\right\} \\ &= \exp\left\{\frac{1}{2}\sigma^2\left(\frac{u}{\sigma} - \alpha\right) - \left(\beta + \frac{1}{2}\alpha^2 + \frac{1}{2}\alpha\sigma^2\right) t - \gamma\right\}. \end{aligned} \tag{9.151}$$

Since u in moment-generating function is a random variable, it could not be eliminated by choosing the parameters α and σ. Hence, the adapted process that transferring linear Brownian motion into standard Brownian motion could not be found. We verify the theorem. □

Then we would give an example referring to Shreve (2004).

Definition 9.46 (adapted process and new probability measure for Brownian motion). *We define Radon-Nikodým derivative processes for Brownian motion and Compound Poisson process:*

$$Z_1(t) = \exp\left\{-\int_0^t \Theta(u)dw(u) - \frac{1}{2}\int_0^t \Theta^2(u)du\right\}, \tag{9.152}$$

$$Z_2(t) = e^{(\lambda-\widetilde{\lambda})t}\prod_{i=1}^{N(t)} \frac{\widetilde{\lambda}\widetilde{f}(Y_i)}{\lambda f(Y_i)}. \tag{9.153}$$

And set

$$Z(t) = Z_1(t)Z_2(t). \tag{9.154}$$

As usual, we define $\widetilde{\mathbb{P}}(A) = \int_A Z(T)d\mathbb{P}$ for $A \in \mathscr{F}$.

Lemma 9.47. *We have shown that $Z_2(t)$ is a martingale in last section. Now we would like to point out that $Z_1(t)$ is also a martingale.*

Proof. *To prove the lemma, we need to verify that $\mathbb{E}[Z_1(t)|\mathscr{F}(s)] = Z_1(s)$. And since $\mathbb{E}[Z_1(t)|\mathscr{F}(s)] = \mathbb{E}[Z_1(t) - Z_1(s) + Z_1(s)|\mathscr{F}(s)] = \mathbb{E}[Z_1(t) - Z_1(s)] + Z_1(s)$, it is equivalent to prove that $\mathbb{E}[Z_1(t) - Z_1(s)] = 0$. If we can prove that $\mathbb{E}[Z_1(t)]$ is always a constant, then the lemma always holds true.*
Assume $X(t) = -\int_0^t \Theta(u)dW(u) - \frac{1}{2}\int_0^t \Theta^2(u)du$ and $f(x) = e^x$, then $Z_1(t) = f(X(t))$. Let $I(t) = \int_0^t \Theta(u)dw(u)$. Note that

$$X(t) = -\int_0^t \Theta(u)dw(u) - \frac{1}{2}[I, I](t). \tag{9.155}$$

And since

$$[X, X](t) = [-I, -I](t) + 0 = [I, I](t), \tag{9.156}$$

we get

$$X(t) = -\int_0^t \Theta(u)dW(u) - \frac{1}{2}[X, X](t) = -I(t) - \frac{1}{2}[X, X](t). \tag{9.157}$$

We apply Itô-Doeblin formula,

$$\begin{aligned}Z_1(t) &= f(X(t)) \\ &= f(X(0)) + \int_0^t f'(X(s))dX(s) + \frac{1}{2}\int_0^t f''(X(s))d[X,X](s) \\ &= 1 + \int_0^t e^{X(s)}d\left(-I(t) - [X,X](s)\right) + \frac{1}{2}\int_0^t e^{X(s)}d[X,X](s) \\ &= 1 - \int_0^t e^{X(s)}dI(s) \\ &= 1 - \int_0^t Z_1(s)dI(s) \\ &= 1 - \int_0^t Z_1(s-)dI(s).\end{aligned} \tag{9.158}$$

Since $Z_1(t-)$ is left-continuous and $I(t)$, regarded as Itô integral, is a martingale. According to Lemma 9.32, $\mathbb{E}[Z_1(t)] = 1$. Then $Z_1(t)$ is a martingale. □

Properties 9.48. *Process $Z(t)$ is a martingale and $\mathbb{E}[Z(t)] = 1$.*

Proof. From Lemma 9.47, we already illustrate that Z_1 and Z_2 are martingales. And Z_2 not containing Itô integral part implies that Z_1 and Z_2 are independent. From Itô's product rule (Theorem 9.29), it follows that

$$Z_1(t)Z_2(t) = Z_1(0)Z_2(0) + \int_0^t Z_1(s-)dZ_2(s) + \int_0^t Z_2(s-)dZ_1(s). \tag{9.159}$$

According to Lemma 9.32, $Z(t)$ is a martingale and $E[Z(t)] = Z(0) = 1$. □

Theorem 9.49. *Under probability measure $\widetilde{\mathbb{P}}$, we define*

$$\widetilde{W}(t) = W(t) + \int_0^t \Theta(s)ds. \tag{9.160}$$

$\widetilde{W}(t)$ is a Brownian motion, likewise, $Q(t)$ is a compound Poisson process with intensity $\widetilde{\lambda}$ and density of jump sizes $\widetilde{f}(y)$. If $\Theta(t)$ is independent of $Q(t)$, then $\widetilde{W}(t)$ is independent of $Q(t)$.

Sketch of proof.
To prove $\widetilde{W}(t)$ and $Q(t)$ are independent, it is equivalent to clarify that the moment-generating function under $\widetilde{\mathbb{P}}$ satisfies

$$\widetilde{\mathbb{E}}[e^{u_1\widetilde{W}(t)+u_2Q(t)}] = \exp\{\frac{1}{2}u_1^2 t\} \cdot \exp\{\widetilde{\lambda}t(\widetilde{\varphi}_Y(u) - 1)\}, \tag{9.161}$$

where

$$\widetilde{\varphi}_Y(u) = \int_{-\infty}^{\infty} e^{uy}\widetilde{f}(y)dy. \tag{9.162}$$

Consider the case that $\Theta(t)$ is independent of $Q(t)$, then Z_1 is independent of Q and Z_2 is independent of $\widetilde{W}$. This suggests that

$$\begin{aligned}\widetilde{\mathbb{E}}[e^{u_1\widetilde{W}(t)+u_2Q(t)}] &= \mathbb{E}[e^{u_1\widetilde{W}(t)}Z_1(t)\cdot e^{u_2Q(t)}Z_2(t)] \\ &= \mathbb{E}[e^{u_1\widetilde{W}(t)}Z_1(t)]\cdot\mathbb{E}[e^{u_2Q(t)}Z_2(t)]\end{aligned} \tag{9.163}$$

It has been demonstrated that

$$\mathbb{E}[e^{u_1\widetilde{W}(t)}Z_1(t)] = \exp\{\frac{1}{2}u_1^2t\}. \tag{9.164}$$

And

$$\mathbb{E}[e^{u_2Q(t)}Z_2(t)] = \exp\{\widetilde{\lambda}t(\widetilde{\varphi}_Y(u)-1)\}. \tag{9.165}$$

As described, we have proved equation 9.161 ,and correspondingly, proved the theorem. □

Theorem 9.50 (A generalized condition of Theorem 9.49).
It is interesting to note that, even if $\Theta(t)$ might depend on $Q(t)$, $\widetilde{W}(t)$ is still independent of $Q(t)$.

Proof. Suppose

$$X_1(t) = \exp\{u_1\widetilde{W}(t) - \frac{1}{2}u_1^2t\}, \tag{9.166}$$

$$X_2(t) = \exp\{u_2Q(t) - \widetilde{\lambda}t(\widetilde{\varphi}_Y(u)-1)\}. \tag{9.167}$$

Assume $A(t) = u_1\widetilde{W}(t) - \frac{1}{2}u_1^2t$, $f(x) = e^x$, it follows that $X_1(t) = f(A(t))$. By applying Itô-Doeblin formula,

$$\begin{aligned}dX_1(t) = df(A(t)) &= f'(A(t))dA(t) + \frac{1}{2}f''(A(t))d[A,A](t) \\ &= f(A(t))d(u_1W(t) - \frac{1}{2}u_1^2t) + \frac{1}{2}f(A(t))d(u_1^2t) \\ &= u_1X_1(t)d\widetilde{W}(t) \\ &= u_1X_1(t)dW(t) + u_1\Theta(t)X_1(t)dt.\end{aligned} \tag{9.168}$$

During the calculation, we apply the fact that

$$[A,A](t) = [u_1\widetilde{W}, u_1\widetilde{W}](t) = u_1^2[W,W](t) = u_1^2t.$$

On the other hand, in lemma 9.47, we have illustrated that

$$Z_1(t) = 1 - \int_0^t Z_1(s)dI(s), \tag{9.169}$$

where

$$I(s) = \int_0^t \Theta(s)dW(s).$$

Consequentially,

$$dZ_1(t) = -\Theta(t)Z_1(t)dW(t). \tag{9.170}$$

By using Itô's product rule, we have

$$\begin{aligned} d(X_1(t)Z_1(t)) &= X_1(t)dZ_1(t) + Z_1(t)dX_1(t) + dX_1(t)dZ_1(t) \\ &= (u_1 - \Theta(t))X_1(t)Z_1(t)dW(t). \end{aligned} \tag{9.171}$$

This suggests that $X_1(t)Z_1(t)$ is a martingale (Lemma 9.32). Likewise, $X_2(t)Z_2(t)$ is also a martingale.

Ultimately, $X_2(t)Z_2(t)$, due to not containing Itô integral part, should be independent of $X_1(t)Z_1(t)$, implying that $[X_2(t)Z_2(t), X_1(t)Z_1(t)] = 0$. Utilizing Itô's product rule,

$$\begin{aligned} X_1(t)Z_1(t)X_2(t)Z_2(t) = 1 &+ \int_0^t X_1(s-)Z_1(s-)d(X_2(s)Z_2(s)) \\ &+ \int_0^t X_2(s-)Z_2(s-)d(X_1(s)Z_1(s)). \end{aligned} \tag{9.172}$$

It Gives that $X_1(t)Z_1(t)X_2(t)Z_2(t)$ is a martingale and

$$\mathbb{E}[X_1(t)Z_1(t)X_2(t)Z_2(t)] = 1. \tag{9.173}$$

Then we clarify the theorem. □

9.3.4 Pricing a European Call

Recall that we can leverage geometric Poisson process to denote the stock price. Suppose the stock price $S(t)$ has constant rate of return α and constant volatility σ, say

$$\begin{aligned} S(t) &= S(0)\exp\{\alpha t + N(t)\log(\sigma+1) - \lambda\sigma t\} \\ &= S(0)e^{(\alpha-\lambda\sigma)t}(\sigma+1)^{N(t)}, \end{aligned} \tag{9.174}$$

and its differential form satisfies

$$dS(t) = \alpha S(t)dt + \sigma S(t-)dM(t), \tag{9.175}$$

where $N(t)$ is a Poison process with intensity λ and $M(t) = N(t) - \lambda t$ is the compensated Poisson process.

Theorem 9.51 (stock price under risk-neutral measure).
In risk-neutral world, suppose the interest rate in finance market is constant, denoted by r. The stock price under new probability space $(\Omega, \mathscr{F}, \widetilde{\mathbb{P}})$ can be well-defined when we have

$$\lambda > \frac{\alpha - r}{\sigma}. \tag{9.176}$$

Suppose $\widetilde{\lambda} = \lambda - \frac{\alpha - r}{\sigma}$, as we have discussed, by setting $Z(t) = e^{(\lambda - \widetilde{\lambda})t}\left(\frac{\widetilde{\lambda}}{\lambda}\right)^{N(t)}$, the risk-neutral measure satisfies

$$\widetilde{\mathbb{P}}(A) = \int_A Z(T)dP \; for \; \forall A \in \mathscr{F}. \tag{9.177}$$

Under risk-neutral measure, the differential form of stock price then becomes

$$dS(t) = rS(t)dt + \sigma S(t-)d\widetilde{M}(t). \tag{9.178}$$

and in the meanwhile,

$$S(t) = S(0)e^{(r-\widetilde{\lambda}\sigma)t}(\sigma+1)^{N(t)}. \tag{9.179}$$

For the discounted stock price, note that the interest rate is a constant, which suggests that

$$d(e^{-rt}S(t)) = \sigma e^{-rt}S(t-)d\widetilde{M}(t). \tag{9.180}$$

Let us say that an agent would like to invest in a European call, and the call price is denoted by $V(t)$, then the underlying payoff of the derivative security satisfies $V(T) = (S(T) - K)^+$. But we still need to figure out the formula for $V(t)$.

Theorem 9.52. *The call price $V(t)$ satisfies the property that*

$$V(t) = \widetilde{\mathbb{E}}\left[e^{-r(T-t)}\left(S(t)e^{(r-\widetilde{\lambda}\sigma)(T-t)}(\sigma+1)^{N(T)-N(t)} - K\right)^+ |\mathscr{F}(t)\right]. \tag{9.181}$$

Proof. It is different to find the equality directly, however, we can deduce the formula from the discounted call price $e^{-rt}V(t)$. That is because the stock price is suggested to have no arbitrage, and then the discounted stock price must be a martingale. It follows that

$$e^{-rt}V(t) = \widetilde{\mathbb{E}}[e^{-rT}V(T)|\mathscr{F}(t)] = \widetilde{\mathbb{E}}\left[e^{-rT}(S(T)-K)^+|\mathscr{F}(t)\right]. \tag{9.182}$$

According to equation (9.179), we have

$$S(T) = S(0)e^{(r-\widetilde{\lambda}\sigma)t}(\sigma+1)^{N(t)}\dot{e}^{(r-\widetilde{\lambda}\sigma)t}(\sigma+1)^{N(T)-N(t)}. \tag{9.183}$$

And then

$$\begin{aligned} V(t) &= \widetilde{\mathbb{E}}\left[e^{-rT}(S(T)-K)^+|\mathscr{F}(t)\right] \\ &= \widetilde{\mathbb{E}}\left[e^{-rT}(S(t)e^{(r-\widetilde{\lambda}\sigma)(T-t)}(\sigma+1)^{N(T)-N(t)} - K)^+|\mathscr{F}(t)\right], \end{aligned} \tag{9.184}$$

this suggests that $V(t)$ is $\mathscr{F}(t)$-measurable. □

The call price $V(t)$ depends on time t and the stock price $S(t)$, so we set $V(t) = c(t, x)$ where $x = S(t)$. From Theorem 9.52, we have

$$\begin{aligned} c(t,x) &= \widetilde{\mathbb{E}}\left[e^{-rT}(S(t)e^{(r-\widetilde{\lambda}\sigma)(T-t)}(\sigma+1)^{N(T)-N(t)} - K)^{+}|\mathscr{F}(t)\right] \\ &= \widetilde{\mathbb{E}}\left[e^{-rT}(S(t)e^{(r-\widetilde{\lambda}\sigma)(T-t)}(\sigma+1)^{N(T)-N(t)} - K)^{+}\right] \end{aligned} \tag{9.185}$$

In order to calculate the conditional expectation in equation (9.185), in the first place, we need to decide which variable can be treated as research object, i.e., to figure out, for which variable, we want to investigate its distribution and use it to compute the expectation. Actually, we are not going to choose t or $S(t)$ as our research object, because $c(t, x)$ depends on these two variables. What we would choose is $N(T) - N(t)$ as a whole, and it should follow a Poisson distribution. And this is our starting point to solve this problem.

Set $j = N(T) - N(t)$, then it follows that

$$P(N(T) - N(t)) = \frac{\widetilde{\lambda}^j (T-t)^j}{j!} e^{-\widetilde{\lambda}(T-t)}. \tag{9.186}$$

Equation (9.185) then becomes

$$\begin{aligned} c(t,x) &= \widetilde{\mathbb{E}}\left[e^{-rT}(S(t)e^{(r-\widetilde{\lambda}\sigma)(T-t)}(\sigma+1)^{N(T)-N(t)} - K)^{+}\right] \\ &= \sum_{j=0}^{\infty} e^{-rT}(xe^{(r-\widetilde{\lambda}\sigma)(T-t)}(\sigma+1)^{N(T)-N(t)} - K)^{+} P(N(T)-N(t)) \\ &= \sum_{j=0}^{\infty} e^{-rT}(xe^{(r-\widetilde{\lambda}\sigma)(T-t)}(\sigma+1)^{j} - K)^{+} \frac{\widetilde{\lambda}^j (T-t)^j}{j!} e^{-\widetilde{\lambda}(T-t)} \\ &= \sum_{j=0}^{\infty} (xe^{-\widetilde{\lambda}\sigma(T-t)}(\sigma+1)^{j} - K)^{+} \frac{\widetilde{\lambda}^j (T-t)^j}{j!} e^{-\widetilde{\lambda}(T-t)}. \end{aligned} \tag{9.187}$$

Then we would like to calculate $d(e^{-rt}c(t, S(t)))$. From equation (9.182), we write

$$e^{-rt}c(t, S(t)) = \widetilde{\mathbb{E}}[e^{-rT}(S(T) - K)^{+}|\mathscr{F}(t)]; \tag{9.188}$$

and equation (9.178) can be written as

$$dS(t) = (r - \widetilde{\lambda}\sigma)S(t)dt + \sigma S(t-)dN(t). \tag{9.189}$$

We can notice that equation (9.189) contains contains continuous part $S^c(u) = (r - \widetilde{\lambda}\sigma)S(t)dt$ and jump part $\sigma S(t-)dN(t)$. For the jump part, we define the jump size:

$$\Delta S(t) = S(t) - S(t-) = \sigma S(t-),\ S(t) = (\sigma + 1)S(t-) \tag{9.190}$$

Properties 9.53 (deriving the formula for discounted call price)**.**
We would utilize the lemma for jump processes, that is

$$\int_0^t f(x-)dg(x) = \int_0^t f(x)dg(x) \text{ when } g(x) \text{ is continuous.} \tag{9.191}$$

And this lemma may not be true if $g(x)$ is not a continuous function. Also, we would apply the fact that $dS(t)dS(t) = 0$. Then according to Itô-Doeblin formula, it follows that

$$\begin{aligned} e^{-rt}c(t, S(t)) &= c(0, S(0)) + \int_0^t \frac{\partial(e^{-rt}c(t, S(t)))}{\partial t} dt + \int_0^t \frac{\partial(e^{-rt}c(t, S(t)))}{\partial x} dS^c(t) \\ &\quad + \sum_{0<u\leqslant t} e^{-ru}[c(u, S(u)) - c(u, S(u-))] \\ &= c(0, S(0)) + \int_0^t e^{-ru}[-rc(u, S(u))du + c_t(u, S(u))du] \\ &\qquad + c_x(u, S(u))dS^c(u)] \\ &\quad + \sum_{0<u\leqslant t} e^{-ru}[c(u, S(u)) - c(u, S(u-))] \end{aligned} \tag{9.192}$$

Since $S(t) = (\sigma + 1)S(t-)$, then

$$\begin{aligned} e^{-rt}c(t, S(t)) &= c(0, S(0)) + \int_0^t e^{-ru}[-rc(u, S(u)) + c_t(u, S(u)) \\ &\qquad + (r - \widetilde{\lambda}\sigma)S(u)c_x(u, S(u))]du \\ &\quad + \int_0^t e^{-ru}[c(u, (\sigma + 1)S(u-)) - c(u, S(u-))]dN(u) \\ &= c(0, S(0)) + \int_0^t e^{-ru}[-rc(u, S(u)) + c_t(u, S(u)) \\ &\qquad + (r - \widetilde{\lambda}\sigma)S(u)c_x(u, S(u))]du \\ &\quad + \int_0^t e^{-ru}[c(u, (\sigma + 1)S(u-)) - c(u, S(u-))]\widetilde{\lambda}du \\ &\quad + \int_0^t e^{-ru}[c(u, (\sigma + 1)S(u-)) - c(u, S(u-))]d\widetilde{M}(t) \\ &= c(0, S(0)) + \int_0^t e^{-ru}[-rc(u, S(u)) + c_t(u, S(u)) \\ &\qquad + (r - \widetilde{\lambda}\sigma)S(u)c_x(u, S(u))]du \\ &\quad + \int_0^t e^{-ru}[\widetilde{\lambda}c(u, (\sigma + 1)S(u)) - \widetilde{\lambda}c(u, S(u))]du \\ &\quad + \int_0^t e^{-ru}[c(u, (\sigma + 1)S(u-)) - c(u, S(u-))]d\widetilde{M}(t). \end{aligned} \tag{9.193}$$

Till now, we have derived the formula for discounted call price.

In equation (9.193), the last term in last equality is a martingale because the integrand is left-continuous and $\widetilde{M}(t)$ is a martingale; and we have verified that $e^{-rt}c(t,S(t))$ is a martingale, then the remaining term in last equality must be a martingale, say

$$\begin{aligned} c(0,S(0)) + \int_0^t e^{-ru}[&-rc(u,S(u)) + c_t(u,S(u)) + (r-\widetilde{\lambda}\sigma)S(u)c_x(u,S(u)) \\ &+ \widetilde{\lambda}(c(u,(\sigma+1)S(u)) - c(u,S(u)))]du. \end{aligned} \tag{9.194}$$

Lemma 9.54.
For function $f(t,S(t)) \in C^2$, $S(t)$ is a continuous stochastic process, if the term

$$\int_0^t f(u,S(u))du \tag{9.195}$$

is a martingale, then $f(u,S(u)) = 0$

Proof. Assume $g(t) = \int_0^t f(u,S(u))du$, since $\int_0^t f(u,S(u))du$ is a martingale, then for $\tau \to 0$, we have

$$\mathbb{E}[g(t+\tau)|\mathscr{F}(t)] = g(t). \tag{9.196}$$

And then

$$\mathbb{E}[g(t+\tau)|\mathscr{F}(t)] = \mathbb{E}\int_t^{t+\tau} f(u,S(u))du = \int_t^{t+\tau} f(u,S(u))du. \tag{9.197}$$

Since $\tau \to 0$, $f(u,S(u))$ can be regarded as a constant in the interval $[t,t+\tau]$, which suggests that

$$\int_t^{t+\tau} f(u,S(u))du = f(u,S(u))\tau \tag{9.198}$$

Equations (9.196), (9.197), (9.198) implies that

$$g(t) = \lim_{\tau\to 0} f(u,S(u))\tau = 0. \tag{9.199}$$

And note that

$$\int_0^t f(u,S(u))du = 0 \ for \ t > 0, \tag{9.200}$$

which implies that $f(u,S(u)) = 0$. □

From Lemma 9.54, equation (9.194) is a martingale is equivalent to say that

$$\begin{aligned} e^{-ru}[&-rc(u,S(u)) + c_t(u,S(u)) + (r-\widetilde{\lambda}\sigma)S(u)c_x(u,S(u)) \\ &+ \widetilde{\lambda}(c(u,(\sigma+1)S(u)) - c(u,S(u)))] \\ = &-rc(t,x) + c_t(t,x) + (r-\widetilde{\lambda}\sigma)xc_x(t,x) + \widetilde{\lambda}(c(t,(\sigma+1)x) - c(t,x)) = 0. \end{aligned} \tag{9.201}$$

Equation (9.201) is called the differential-difference equation and it holds true when $0 \leqslant t < T$ and $x \geqslant 0$.

From equation (9.194) and equation (9.201), we can rewrite the formula foe discounted call price, that is

$$e^{-rt}c(t,S(t)) = c(0,S(0)) + \int_0^t e^{-ru}[c(u,(\sigma+1)S(u-)) - c(u,S(u-))]d\widetilde{M}(u). \tag{9.202}$$

And at time T, we have

$$\begin{aligned} e^{-rT}(S(T)) &= e^{-rT}c(T,S(T)) \\ &= c(0,S(0)) + \int_0^T e^{-ru}[c(u,(\sigma+1)S(u-)) - c(u,S(u-))]d\widetilde{M}(u). \end{aligned} \tag{9.203}$$

Definition 9.55 (hedging portfolio).
In order to hedge a short position in the European call, we use the hedging portfolio with capital $X(t)$ to reproduce the payoff, and then $X(t) = c(t,S(t))$.

The value of the portfolio consists of $\Gamma(t)$ shares of stocks and $X(t) - \Gamma(t)S(t)$ in money market account. Then the differential form of $X(t)$ satisfies that

$$dX(t) = \Gamma(t-)dS(t) + r[X(t) - \Gamma(t)S(t)]dt. \tag{9.204}$$

Thus, for the discounted hedging portfolio, we have

$$\begin{aligned} d(e^{-rt}X(t)) &= e^{-rt}[-rX(t)dt + dX(t)] \\ &= e^{-rt}[\Gamma(t-)dS(t) - r\Gamma(t)S(t)dt] \\ &= e^{-rt}\sigma\Gamma(t-)S(t-)d\widetilde{M}(t). \end{aligned} \tag{9.205}$$

Since $X(t) = c(t,S(t))$, then $d(e^{-rt}X(t)) = d(e^{-rt}c(t,S(t)))$, namely,

$$e^{-rt}\sigma\Gamma(t-)S(t-)d\widetilde{M}(t) = e^{-rt}[c(t,(\sigma+1)S(t-)) - c(t,S(t-))]d\widetilde{M}(t). \tag{9.206}$$

We deduce the formula above, and we can find that

$$\Gamma(t-) = \frac{c(t,(\sigma+1)S(t-)) - c(t,S(t-))}{\sigma S(t-)}, \tag{9.207}$$

or we write

$$\Gamma(t) = \frac{c(t,(\sigma+1)S(t)) - c(t,S(t))}{\sigma S(t)}. \tag{9.208}$$

With $\Gamma(t)$ defined in equation (9.208), the hedging portfolio $X(t)$ can be well-defined.

9.4 Asset defined by a Brownian motion and a Compound Poisson

Under probability space $(\Omega, \mathscr{F}, \mathbb{P})$, let $W(t)$ be a Brownian motion and $N_1(t), N_2(t), \cdots, N_M(t)$ be M independent Poisson processes, where $0 \leqslant t \leqslant T$. And each $N_m(t)$ contains intensity λ_m. Define

$$N(t) = \sum_{m=1}^M N_m(t), \ \ Q(t) = \sum_{m=1}^M y_m N_m(t), \tag{9.209}$$

where $y_m > -1$, $y_m \neq 0$ and $y_1 < y_2 < \cdots < y_M$. We notice that $N(t)$ is a Poisson process with intensity $\lambda = \sum_{m=1}^{M} Y_i$ and $Q(t)$ is a compound Poisson process.

Since we do not when $N_m(t)$ would jump and we merely know the jump size of each $N_m(t)$, then for $Q(t)$, the jump size for each jump is random, range from y_1 to y_M. To better understand this, let us consider the first time when jump happens in $Q(t)$, and this depends on the first jump time time for Poisson processes $N_1(t), N_2(t), \cdots, N_M(t)$, say $\tau_1, \tau_2, \cdots, \tau_M$. And suppose $\tau_k = \min \tau_1, \tau_2, \cdots, \tau_M$, then the first jump size of $Q(t)$ should be equal to the first jump size of $N_k(t)$, that is τ_k.

Now use Y_i to denote the i-th jump size of the compound Poisson process $Q(t)$, we can see the set of Y_i contains i random variables and $Y_i \in y_1, y_2, \cdots, y_M$. Then we can rewrite the formula for $Q(t)$, namely,

$$Q(t) = \sum_{i=1}^{N(t)} Y_i. \tag{9.210}$$

Definition 9.56 (the probability of jump size for $Q(t)$)**.**
For compound Poisson process $Q(t)$ defined in equations (9.209) and (9.210), the jump size $Y_i \in y_1, y_2, \cdots, y_M$ are independent with each other. However, which value Y_i would take range from y_1 to y_M is not equal likely, and it depends on the intensity of Poisson processes $N_1(t), \cdots, N_M(t)$. So we can hypothesis the probability of jump size,

$$\mathbb{P}\{Y_i = y_m\} = p(y_m) = \frac{\lambda_m}{\lambda}. \tag{9.211}$$

With the probability distribution of each jump size for $Q(t)$, we can calculate the expected value of Y_i.

Properties 9.57 (Expected value of Y_i)**.**
For $Q(t)$ as defined, at each jump time, the mean jump size should be

$$\beta = \mathbb{E}[Y_i] = \sum_{m=1}^{M} y_m p(y_m) = \frac{1}{\lambda} \sum_{m=1}^{M} \lambda_m y_m \tag{9.212}$$

And as we have discussed before, the formula

$$Q(t) - \beta\lambda t = Q(t) - t \sum_{m=1}^{M} \lambda_m y_m \tag{9.213}$$

is a martingale.

Here, we would like to leverage the stochastic differential equation to describe the stock price $S(t)$, then

$$\begin{aligned} dS(t) &= \alpha S(t)dt + \sigma S(t)dW(t) + S(t-)d(Q(t) - \beta\lambda t) \\ &= (\alpha - \beta\lambda)S(t)dt + \sigma S(t)dW(t) + S(t-)dQ(t). \end{aligned} \tag{9.214}$$

From equation (9.214), the mean rate of return of the stock price is α and the volatility of the stock price is σ. Then we would talk about how to find the general solution of equation (9.214).

Theorem 9.58.
The general solution of equation (9.214) *is*

$$S(t) = S(0)\exp\{\sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right) t\} \prod_{i=1}^{N(t)} (Y_i + 1). \tag{9.215}$$

Proof. In the first place, we define the continuous stochastic process $X(t)$ such that

$$X(t) = S(0)\exp\{\sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right) t\}, \tag{9.216}$$

The differential form could be computed by applying Itô-Doeblin formula, and it follows that

$$dX(t) = (\alpha - \beta\lambda)X(t)dt + \sigma X(t)dW(t). \tag{9.217}$$

We can also define the pure jump process

$$J(t) = \prod_{i=1}^{N(t)} (Y_i + 1). \tag{9.218}$$

At the time when jump happens, we have

$$J(t) = J(t-)(Y_i + 1), \tag{9.219}$$

and the jump size satisfies

$$\Delta J(t) = J(t) - J(t-) = J(t-)Y_i = J(t-)\Delta Q(t). \tag{9.220}$$

We can check the formula above also holds when jump does not happen, so we can rewrite the formula into differential form, that is

$$dJ(t) = J(t-)dQ(t). \tag{9.221}$$

Note that $S(t) = X(t)J(t)$, according to Itô's product rule for jump processes, we can verify that

$$\begin{aligned} S(t) = X(t)J(t) &= S(0) + \int_0^t X(s-)dJ(s) + \int_0^t J(s)dX(s) + [X, J](t) \\ &= S(0) + \int_0^t X(s-)dJ(s) + \int_0^t J(s)dX(s). \end{aligned} \tag{9.222}$$

The second equality of equation (9.222) holds because continuous stochastic process $X(t)$ and pure jump process $J(t)$ are independent, which implies that $[X, J](t) = 0$.

Then we can replace $dX(t)$ and $dJ(t)$ into equation (9.222), and it gives that

$$\begin{aligned} S(t) &= X(t)J(t) \\ &= S(0) + \int_0^t X(s-)J(s-)dQ(s) + (\alpha - \beta\lambda)\int_0^t J(s)X(s)ds \\ &\quad \sigma \int_0^t J(s)X(s)dW(s). \end{aligned} \tag{9.223}$$

By taking the differential form, we have

$$\begin{aligned} d(S(t)) &= d(X(t)J(t)) \\ &= X(t-)J(t-)dQ(t) + (\alpha - \beta\lambda)J(t)X(t)dt + \sigma J(t)X(t)dW(t) \\ &= S(t-)dQ(t) + (\alpha - \beta\lambda)S(t)dt + \sigma S(t)dW(t). \end{aligned} \tag{9.224}$$

Then we have verified the theorem. □

10 Introduction to Merton Jump Diffusion Model

10.1 Backgrounds

There are numerical stochastic models to describe the stock price, and in this section, we would utilize a Brownian motion and a compound Poisson process under probability space $(\Omega, \mathscr{F}, \mathbb{P})$ to define the stock price $S(t)$, say

$$S(t) = S(0)\exp\left\{\sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right)t\right\}\prod_{i=1}^{N(t)}(Y_i + 1), \tag{10.1}$$

with the differential form

$$dS(t) = (\alpha - \beta\lambda)S(t)dt + \sigma S(t)dW(t) + S(t-)dQ(t). \tag{10.2}$$

In the equations above, $W(t)$ denotes the Brownian motion and $Q(t) = \sum_{i=1}^{N(t)} Y_i$ denotes the compound Poisson process with intensity λ.

Definition 10.1 (log jump size for stock price $S(t)$).
The log jump size at i-th jump time for $S(t)$ is defined to be

$$F_i = \log\left(Y_{N(t)} + 1\right) \tag{10.3}$$

Proof. Here, it is important to understand the reason why we define log stock price jump size instead of the normal one. Consequently, we would try calculating the stock price jump size without logarithm to better interpret the underlying factors. Suppose one jump happens at time t, then we have

$$\begin{aligned}\Delta(t) &= S(t) - S(t-)\\ &= S(0)\exp\left\{\sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right)t\right\}\left\{\prod_{i=1}^{N(t)}(Y_i+1) - \prod_{i=1}^{N(t)-1}(Y_i+1)\right\} \\ &= S(0)\exp\left\{\sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right)t\right\}(Y_{N(t)}+1).\end{aligned} \tag{10.4}$$

From the formula above, the jump size depends on the behavior of Brownian motion and time t, which expands the difficulty to analyze its behavior. So instead, we would leverage log jump size.

Note that the logarithm of $S(t)$ satisfies

$$\log(S(t)) = \log(S(0))\left(\sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right)t\right)\left(\log(Y_1+1) + \cdots + \log(Y_{N(t)}+1)\right). \tag{10.5}$$

And we can see from equation (10.1) that

$$S(t) = S(t-)(Y_{N(t)}+1), \tag{10.6}$$

which gives that

$$\log(S(t)) - \log(S(t-)) = \log\left(\frac{S(t)}{S(t-)}\right) = \log(Y_{N(t)}+1). \tag{10.7}$$

□

Actually, Definition 10.1 can be verified using the properties of the asset price under a Brownian motion and a compound Poisson process, but it is hard to figure out the magnitude for Y_i. So Merton made an hypothesis on the distribution of log jump size many years ago, and he suggested that the log jump size follows a normal distribution. We would discuss about it further while introducing the following theorem.

Theorem 10.2 (Merton's hypothesis on log stock price jump size)**.**
For the jump size at time t

$$F_i = \log(S(t)) - \log(S(t-)) = \log(Y_{N(t)}+1), \tag{10.8}$$

We say F_i follows a normal distribution with mean μ and standard deviation δ, or $F_i \sim N(\mu, \delta)$. Precisely, the probability distribution $f(F_i)$ satisfies the formula that

$$f(F_i) = \frac{1}{\sqrt{2\pi\delta^2}}\exp\left\{-\frac{(F_i-\mu)^2}{2\delta^2}\right\}. \tag{10.9}$$

Definition 10.3 (rewriting the stock price with log jump size).
We can rewrite the stock price in equation (10.1) *with the log jump size* F_i*, and it follows that*

$$S(t) = S(0)\exp\left\{\sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right)t + \sum_{i=1}^{N(t)} F_i\right\}. \tag{10.10}$$

Then we would introduce the Levy's process, which is strictly deduced from the stock price defined by a Brownian motion and a compound Poisson process.

Definition 10.4 (Levy's process).
According to Definition 10.3, the Levy's process has the form that

$$L(t) = \log\left(\frac{S(t)}{S(0)}\right) = \sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right)t + \sum_{i=1}^{N(t)} F_i. \tag{10.11}$$

10.2 Probability Distribution

With all of these definitions and theorems, especially theorem 10.2, which causes the uniqueness of Merton's jump diffusion model, we are going to deduce the general form of Merton's model.

Firstly, we would discuss the behavior of stock price on a tiny interval, say $\Delta t \to 0$. So whether there would be one jump or more could be measured by the probability.

Theorem 10.5 (appearance of jumps).
On the time interval interval $\Delta t \to 0$*, to consider the probability of jumps occur in stock price, it is equivalent to analyze that in the compound Poisson process. And recall that the compound Poisson process* $\prod_{i=1}^{N(t)} Y_i$ *has intensity* λ*, so the following properties hold true:*
i) The probability that the stock price jumps once in Δt *is that*

$$P\{\Delta N(t) = 1\} \approx \lambda\Delta t; \tag{10.12}$$

ii) The probability that the stock price jumps more than once in Δt *is that*

$$P\{\Delta N(t) > 1\} \approx 0; \tag{10.13}$$

iii) The probability that the stock price does not jump in Δt *is that*

$$P\{\Delta N(t) = 0\} \approx \lambda\Delta t. \tag{10.14}$$

Proof. Here, we would merely prove property i), by analyzing the definition of Poisson process, we say a Poisson process has intensity λ, it means that λ jumps are supposed to occur in one unit time. To determine how many jumps are supposed to occur in time interval Δt, by using Taylor's expansion, we simply write $\lambda\Delta t$. $\square$

Properties 10.6.
Remember Y_i is used to describe the jump size of the compound Poisson process, and we have the property that the stock price jumps from $S(t)$ to $(Y_i + 1)S(t)$ if it is the i-th jump at time t.

Sketch of proof. We suppose the stock price $S(t)$ jumps to $\alpha_i S(t)$ at time t, then we calculate the term $\frac{S(t)}{S(t-)}$, and it follows that

$$\frac{S(t)}{S(t-)} = \alpha_i. \tag{10.15}$$

According to Theorem 10.2, we have

$$\frac{S(t)}{S(t-)} = Y_i + 1. \tag{10.16}$$

Consequently, $\alpha_i = Y_i + 1$ as defined. □

Note that $F_i = \log(Y_i + 1)$, and the log jump size F_i following a normal distribution implies that $Y_i + 1$ follows a lognormal distribution, say $\log(Y_i + 1) \sim N(\mu, \delta^2)$

Theorem 10.7 (lognormal distribution of $Y_i + 1$)**.**
Since we have $\log(Y_i + 1) \sim N(\mu, \delta^2)$*, we can then calculate the expected value and variance,*

$$\mathbb{E}[Y_i + 1] = e^{\mu + \frac{1}{2}\sigma^2}, \tag{10.17}$$

and

$$\sigma^2 = \mathbb{E}[(Y_i + 1 - \mathbb{E}[Y_i + 1])^2] = e^{2\mu+\sigma^2}(e^{\sigma^2} - 1). \tag{10.18}$$

So, it follows that

$$Y_i + 1 \sim LN(e^{\mu+\frac{1}{2}\sigma^2}, e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)). \tag{10.19}$$

And in the meantime, we can also write

$$Y_i \sim LN(e^{\mu+\frac{1}{2}\sigma^2} - 1, e^{2\mu+\sigma^2}(e^{\sigma^2} - 1)). \tag{10.20}$$

Here, we use the lemma that, for any random variable x, $Var[x - 1] = Var[x]$.

To derive Merton's model, we mainly work on Levy's process, the prominent step is to analyze the distribution of Levy's process at time t. And recall that Levy's process satisfies the property that

$$L(t) = \log\left(\frac{S(t)}{S(0)}\right) = \sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right)t + \sum_{i=1}^{N(t)} F_i. \tag{10.21}$$

From the equation above, we notice that at time t, the term $\sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right)t$ is normal distributed, say

$$\sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right)t \sim N(\alpha - \beta\lambda - \frac{1}{2}\sigma^2, \sigma^2 t). \tag{10.22}$$

And the term $\sum_{i=1}^{N(t)} F_i$ is also normal distributed. Normally, the Lévy's process is called Merton's log-return density function.

10.3 Deriving the Equation Containing Random Jump sizes

The Merton's jump diffusion model is applied to describe the call price in the European call, and we would use $c(t, x)$ where $x = S(t)$ to denote the call price.

Theorem 10.8 (rate of return for call price $c(t, S(t))$).
Let $c(t, S(t))$ denote the call price, $S(t)$ is the stock price defined under a Brownian motion $W(t)$ and a compound Poisson process $Q(t)$ with jump sizes Y_i ranging from y_i to y_M and intensity λ. The stock price has interest rate α and volatility σ. Then the rate of return for the call price satisfies the property that

$$\begin{aligned}\frac{dc(t,S(t))}{c(t,S(t))} = &\left(\frac{1}{2}\sigma^2 S^2(t) c_{xx}(t,S(t)) + (\alpha-\beta\lambda)S(t)c_x(t,S(t)) + c_t(t,S(t))\right)/c(t,S(t))\ dt\\ &+ \sigma S(t) c_x(t,S(t))/c(t,S(t))\ dW(t)\\ &+ \sum_{m=1}^{M}[c(t,(y_m+1)S(t-)) - c(t,S(t-))]/c(t,S(t))\ dN_m(t).\end{aligned} \tag{10.23}$$

Proof. Suppose

$$Q(t) = \sum_{m=1}^{M} y_m N_m(t) = \sum_{i=1}^{N(t)} Y_i. \tag{10.24}$$

From equation (10.1), the differential form of the stock price satisfies the property that

$$dS(t) = (\alpha-\beta\lambda)S(t)dt + \sigma S(t)dW(t) + S(t-)dQ(t), \tag{10.25}$$

with the continuous part

$$dS^c(t) = (\alpha-\beta\lambda)S(t)dt + \sigma S(t)dW(t). \tag{10.26}$$

Then we can apply Itô-Deoblin formula on jump processes to express $c(t, S(t))$, and it gives that

$$\begin{aligned}c(t,S(t)) = &\int_0^t c_t(u,S(u))du + \int_0^t c_x(u,S(u))dS^c(u)\\ &+ \frac{1}{2}c_{xx}(u,S(u))d[S^c,S^c](u) + \sum_{0<u\leqslant t}[c(u,S(u)) - c(u,S(u-))].\end{aligned} \tag{10.27}$$

To calculate the quadratic variation $[S^c, S^c](t)$, we utilize the property that $dW(t)dW(t) = dt$, then we have

$$\begin{aligned}[S^c,S^c](t) &= dS^c(t)dS^c(t)\\ &= (\alpha-\beta\lambda)^2 S^2(t)dtdt + \sigma^2 S^2(t)dW(t)dW(t)\\ &= \sigma^2 S^2(t)dt.\end{aligned} \tag{10.28}$$

Recall that the compound Poisson process $Q(t)$ represents a combination of several branch Poisson processes, each of which happens independently and inconsistently, so every jump size of $Q(t)$ should be equal to that of exactly one branch Poisson process. When the jump happens for the stock price $S(t)$, we can compute the jump size that

$$\Delta S(t) = S(t) - S(t-) = S(t-)dQ(t) = y_m S(t-),\ m \in \{1, 2, \cdots, M\}, \tag{10.29}$$

which implies that

$$S(t) = (y_m + 1)S(t-). \tag{10.30}$$

The call price $c(t, S(t))$ jumps as $S(t)$ jumps, so at the jump time, the jump size should be equal to exactly one branch Poisson process with $\Delta N_m(t) = 1$. Meanwhile, for other branch Poisson processes, $\Delta N_j(t) = 0$ with $j \neq m$. Then we can rewrite the term $[c(u, S(u)) - c(u, S(u-))]$ as

$$c(u, S(u)) - c(u, S(u-)) = \sum_{m=1}^{M} c(u, (y_m + 1)S(u-)) - c(u, S(u-))\Delta N_m(u), \tag{10.31}$$

which suggests that

$$\begin{aligned}\sum_{0<u\leqslant t} [c(u, S(u)) - c(u, S(u-))] &= \sum_{m=1}^{M} \sum_{0<u\leqslant t} [c(u, (y_m + 1)S(u-)) - c(u, S(u-))]\Delta N_m(u) \\ &= \sum_{m=1}^{M} \int_0^t [c(u, (y_m + 1)S(u-)) - c(u, S(u-))]dN_m(u).\end{aligned} \tag{10.32}$$

Then we can substitute equation (10.28) and equation (10.32) into equation (10.27) to simplify the form of $c(t, S(t))$ and transform it into a format that is suitable to differentiate, and we have

$$\begin{aligned}c(t, S(t)) = &\int_0^t c_t(u, S(u))du + \int_0^t c_x(u, S(u))dS^c(u) \\ &+ \frac{1}{2}c_{xx}(u, S(u))\sigma^2 S^2(t)dt + \sum_{m=1}^{M} \int_0^t [c(u, (y_m + 1)S(u-)) - c(u, S(u-))]dN_m(u).\end{aligned} \tag{10.33}$$

Therefore, we can calculate $dc(t, S(t))$ that,

$$\begin{aligned}dc(t, S(t)) = &c_t(t, S(t))dt + (\alpha - \beta\lambda)S(t)c_x(t, S(t))dt + \sigma S(t)c_x(t, S(t))dW(t) \\ &+ \frac{1}{2}\sigma^2 S^2(t)c_{xx}(t, S(t))dt \\ &+ \sum_{m=1}^{M}[c(t, (y_m + 1)S(t-)) - c(t, S(t-))]dN_m(t).\end{aligned} \tag{10.34}$$

We may write the dt term and the $dW(t)$ term on the right hand side of equation (10.34), respectively,

$$\left(\frac{1}{2}\sigma^2 S^2(t)c_{xx}(t, S(t)) + (\alpha - \beta\lambda)S(t)c_x(t, S(t)) + c_t(t, S(t))\right)dt; \tag{10.35}$$

$$\sigma S(t)c_x(t,S(t))dW(t). \tag{10.36}$$

Then, consider the rate of return of call price $c(t,S(t))$, we have

$$\begin{aligned}\frac{dc(t,S(t))}{c(t,S(t))} &= \left(\frac{1}{2}\sigma^2 S^2(t)c_{xx}(t,S(t)) + (\alpha-\beta\lambda)S(t)c_x(t,S(t)) + c_t(t,S(t))\right)/c(t,S(t))\ dt \\ &+ \sigma S(t)c_x(t,S(t))/c(t,S(t))\ dW(t) \\ &+ \sum_{m=1}^{M}[c(t,(y_m+1)S(t-)) - c(t,S(t-))]/c(t,S(t))\ dN_m(t),\end{aligned} \tag{10.37}$$

which proves the theorem. □

Recall the rate of return for stock price is that

$$\frac{dS(t)}{S(t)} = (\alpha-\beta\lambda)dt + \sigma dW(t) + \frac{S(t-)}{S(t)}dQ(t). \tag{10.38}$$

We observe that the rate of return for call price is analogue to the rate of return for stock price, which consists of three components, dt term, $dW(t)$ term and a Poisson process, thus we would like to utilize certain notations to further emphasize the similarity of two formulas, namely, we intend to rewrite the rate of return of call price $c(t,S(t))$ in equation (10.37) as

$$\frac{dc(t,S(t))}{c(t,S(t))} = (\alpha_w - \beta_w\lambda)dt + \sigma_w dW(t) + \frac{c(t,S(t-))}{c(t,S(t))}dQ_w(t), \tag{10.39}$$

where

$$\alpha_w - \beta_w\lambda = \left(\frac{1}{2}\sigma^2 S^2(t)c_{xx}(t,S(t)) + (\alpha-\beta\lambda)S(t)c_x(t,S(t)) + c_t(t,S(t))\right)/c(t,S(t)), \tag{10.40}$$

$$\sigma_w = \sigma S(t)c_x(t,S(t))/c(t,S(t)), \tag{10.41}$$

$$\frac{c(t,S(t-))}{c(t,S(t))}dQ_w(t) = \sum_{m=1}^{M}[c(t,(y_m+1)S(t-)) - c(t,S(t-))]/c(t,S(t))\ dN_m(t). \tag{10.42}$$

Firstly, in equation (10.38), we notice that the compound Poisson process $Q(t)$ has jump sizes Y_i and β is the mean jump size, say $\beta = \mathbb{E}[Y_i]$. So we can hypothesis that the Poisson process for call price, say $Q_w(t)$, would have the mean jump size β_w.

Therefore, in order to verify that equation (10.39) do have some specific meanings, we need to show that $Q_w(t)$ is a compound Poisson process with intensity λ and mean jump size β_w, and α_w is a martingale. It would be educed in next theorem to compute the mean jump size of $Q_w(t)$.

Theorem 10.9 (mean jump size for $Q_w(t)$)**.**
The mean jump size of $Q_w(t)$ satisfies the following formula that

$$\beta_w = \mathbb{E}[Y_j^w] = \mathbb{E}\frac{c(t,(y_m+1)S(t)) - c(t,S(t))}{c(t,S(t))}. \tag{10.43}$$

Proof. Consider the term $\sum_{m=1}^{M}[c(t,(Y_m+1)S(t-)) - c(t,S(t-))]/c(t,S(t))\ dN_m(t)$ in equation (10.37) and according to our assumption, we can derive that

$$\begin{aligned} dQ_w(t) &= \frac{c(t,S(t))}{c(t,S(t-))}\sum_{m=1}^{M}\frac{[c(t,(y_m+1)S(t-)) - c(t,S(t-))]}{c(t,S(t))}\ dN_m(t) \\ &= \sum_{m=1}^{M}\frac{[c(t,(y_m+1)S(t-)) - c(t,S(t-))]}{c(t,S(t-))}\ dN_m(t). \end{aligned} \tag{10.44}$$

For the jump size Y_j^w, it should be equal to the jump size of exactly one branch Poisson process of $Q_w(t)$, say the m-th branch Poisson process without loss of generality, so $dN_m(t)=1$ and $dN_k(t)=0$ with $\forall k \neq m$. Then we have

$$\begin{aligned} Y_j^w &= Q_w(t) - Q_w(t-) \\ &= \frac{c(t,(y_m+1)S(t-)) - c(t,S(t-))}{c(t,S(t-))}, \end{aligned} \tag{10.45}$$

and the mean jump size for $Q_w(t)$ should be

$$\beta_w = \mathbb{E}[Y_j^w] = \mathbb{E}\frac{c(t,(y_m+1)S(t)) - c(t,S(t))}{c(t,S(t))}. \tag{10.46}$$

□

Theorem 10.10. *We can verify that the jump process $Q_w(t)$ is a compound Poisson process.*

Proof. In fact, by integrating equation (10.44), we can rewrite $Q_w(t)$ as a combination of all the jump sizes before time t, that is

$$Q_w(t) = \sum_{m=1}^{M} y_m^w N_m(t) = \sum_{j=1}^{N_w(t)} Y_j^w, \tag{10.47}$$

where

$$y_m^w = \frac{c(t,(y_m+1)S(t-)) - c(t,S(t-))}{c(t,S(t-))}. \tag{10.48}$$

And Y_j^w here should be a random variation taken from the set $\{y_1^w, y_2^w, \cdots, y_M^w\}$. Note that equation (10.47) shows that $Q_w(t)$ is defined to be a compound Poisson process consisting of branch Poisson processes with jump sizes $\{y_1^w, y_2^w, \cdots, y_M^w\}$. □

In the above two theorems, we have interpreted that $Q_w(t)$ is a compound Poisson process with mean jump size β_w. Here, we still need to show that $Q_w(t)$ has intensity λ, or $Q_w(t)$ has the same intensity with $Q(t)$ as defined before.

Theorem 10.11 (intensity of $Q_w(t)$).
Recall the compound Poisson process $Q_w(t)$ satisfies that

$$Q_w(t) = \sum_{m=1}^{M}\frac{c(t,(y_m+1)S(t-)) - c(t,S(t-))}{c(t,S(t-))}N_m(t). \tag{10.49}$$

We have the property that $Q_w(t)$ has intensity λ.

Sketch of proof. Note that the compound Poisson process $Q(t)$ defined before has the intensity λ with

$$Q(t) = \sum_{m=1}^{M} y_m N_m(t). \tag{10.50}$$

The definition of intensity shows that the intensity is independent of jump sizes but corresponding to pure Poisson process $N_m(t)$. Comparing the formula for $Q_w(t)$ and $Q(t)$, we figure out that the only thing that varies is the jump size, so the intensity of $Q_w(t)$ should also be equal to λ. □

Properties 10.12. *The increments $Q(t) - Q(s)$ with $t \geqslant s > 0$ is independent of $\mathscr{F}(s)$, since jump sizes Y_j^w are independent random variables.*

Properties 10.13 (expected value of compound Poisson process $Q_w(t)$)**.**

$$\mathbb{E}[Q_w(t)] = \beta_w \lambda t. \tag{10.51}$$

Proof. We can firstly calculate the conditional expected value corresponding to $N_m(t) = k$ respectively and then sum them up. Therefore, it follows that

$$\begin{aligned}\mathbb{E}[Q_w(t)] &= \sum_{k=0}^{\infty} \mathbb{E}[Q_w(t)|N(t) = k]\mathbb{P}\{N(t) = k\} \\ &= \mathbb{E}\left[\sum_{j=0}^{k} Y_j^w \middle| N_m(t) = k\right] \mathbb{P}\{N_m(t) = k\}.\end{aligned} \tag{10.52}$$

Note that the expected value of jump size satisfies the property that

$$\mathbb{E}[Y_j^w] = \beta_w, \tag{10.53}$$

then it gives that

$$\begin{aligned}\mathbb{E}[Q_w(t)] &= \mathbb{E}\left[\sum_{j=0}^{k} Y_j^w \middle| N_m(t) = k\right] \mathbb{P}\{N_m(t) = k\} \\ &= \sum_{k=0}^{\infty} \beta_w k \frac{(\lambda k)^k}{k!} e^{-\lambda t} \\ &= \beta_w \lambda t e^{-\lambda t} \sum_{k=1}^{\infty} \frac{(\lambda k)^{k-1}}{(k-1)!} \\ &= \beta_w \lambda t e^{-\lambda t} e^{\lambda t} = \beta_w \lambda t\end{aligned} \tag{10.54}$$

Hence, we have $\mathbb{E}[Q_w(t)] = \beta_w \lambda t$. □

Properties 10.14. *The formula $Q_w(t) - \beta_w \lambda t$ is a martingale.*

Proof. By calculating the expected value of $Q_w(t) - \beta_w \lambda t$ with respect to $\mathscr{F}(s)$, we derive the formula that

$$\begin{aligned}\mathbb{E}[Q_w(t) - \beta_w \lambda t|\mathscr{F}(s)] &= \mathbb{E}[Q_w(t) - Q_w(s) + Q_w(s) - \beta_w \lambda t|\mathscr{F}(s)] \\ &= \mathbb{E}[Q_w(t) - Q_w(s)] + Q_w(s) - \beta_w \lambda t \\ &= \beta_w \lambda(t-s) + Q_w(s) - \beta_w \lambda t \\ &= Q_w(s) - \beta_w \lambda s.\end{aligned} \tag{10.55}$$

According the definition of martingale, we show that $Q_w(t) - \beta_w \lambda$ is a martingale. □

Theorem 10.15 (rewriting rate of return for call price).
Define

$$\begin{aligned}\alpha_w &= \alpha_w - \beta_w \lambda + \beta_w \lambda \\ &= \bigg(\frac{1}{2}\sigma^2 S^2(t) c_{xx}(t, S(t)) + (\alpha - \beta\lambda) S(t) c_x(t, S(t)) + c_t(t, S(t)) \\ &\quad + \lambda \mathbb{E}[c(t, (Y_i + 1)S(t)) - c(t, S(t))]\bigg)/c(t, S(t));\end{aligned} \tag{10.56}$$

and

$$\sigma_w = \sigma S(t) c_x(t, S(t))/c(t, S(t)). \tag{10.57}$$

Then, with $Q_w(t)$ and β_w in Theorem (10.9), we can rewrite the differential form of call price $dc(t, S(t))$ as

$$\begin{aligned}dc(t, S(t)) &= (\alpha_w - \beta_w \lambda) c(t, S(t)) dt + \sigma_w dW(t) + c(t, S(t-)) dQ_w(t) \\ &= \alpha_w c(t, S(t)) dt + \sigma_w dW(t) + c(t, S(t-)) d(Q_w(t) - \beta_w \lambda t).\end{aligned} \tag{10.58}$$

We have already verified that $Q_w(t)$ is a compound Poisson process with intensity λ and mean jump size β_w. The compensated compound Poisson process $Q_w(t) - \beta_w \lambda t$ is a martingale. α_w and σ_w are adapted processes. Since $Q_w(t)$, α_w and σ_w are well-defined processes, then the formula for call price after rewriting has specific financial meanings. And the interest rate for call price should be α_w, the volatility should be σ_w and jumps would be expected to occur every $\frac{1}{\lambda}$ unit times with expected jump size β_w.

Consequently, we can write the mean rate of return for call price in the following formula, and it meets our assumption.

$$\frac{dc(t, S(t))}{c(t, S(t))} = (\alpha_w - \beta_w \lambda) dt + \sigma_w dW(t) + \frac{c(t, S(t-))}{c(t, S(t))} dQ_w(t). \tag{10.59}$$

With equation (10.59), we can then consider the return of a portfolio as an example.

Example 10.16 (return of a portfolio).
Let assume that an agent holds a portfolio with value P. And w_1 percentage of the portfolio is invested in the stock, w_2 percent in the call option and w_3 in the riskless asset respectively.

We define the risk-free rate to be r.

We embezzle the notations we have defined before to avoid any misapprehension. So the interest rate of the stock should be α and volatility should be σ. Thus, The return of the stock price satisfies the formula that

$$\frac{dS(t)}{S(t)} = (\alpha - \beta\lambda)dt + \sigma dW(t) + \frac{S(t-)}{S(t)}dQ(t), \tag{10.60}$$

where

$$Q(t) = \sum_{i=1}^{N(t)} Y_i. \tag{10.61}$$

Similarly, the interest rate and volatility of the call option should be α_w and σ_w respectively, and the return should be

$$\frac{dc(t,S(t))}{c(t,S(t))} = (\alpha_w - \beta_w\lambda)dt + \sigma_w dW(t) + \frac{c(t,S(t-))}{c(t,S(t))}dQ_w(t), \tag{10.62}$$

where

$$Q_w(t) = \sum_{m=1}^{M} \frac{[c(t,(y_m+1)S(t-)) - c(t,S(t-))]}{c(t,S(t-))}\, dN_m(t). \tag{10.63}$$

Furthermore, the riskless asset is easier to describe since it does not have volatility or jumps, and we can define the asset price to be

$$\frac{dq(t)}{q(t)} = rdt. \tag{10.64}$$

Since the portfolio consists of three components: stock, call option and riskless asset, we observe that the portfolio value P should be a formula of stock price $S(t)$, call price $c(t,S(t))$ and asset price $q(t)$, together with their weights w_1, w_2, w_3. Then we have

$$P(t) = w_1 S(t) + w_2 c(t,S(t)) + w_3 q(t). \tag{10.65}$$

Therefore, the return of portfolio should be

$$\begin{aligned}
\frac{dP(t)}{P(t)} &= w_1\frac{dS(t)}{S(t)} + w_2\frac{dc(t,S(t))}{c(t,S(t))} + w_3\frac{dq(t)}{q(t)} \\
&= \left(w_1(\alpha-\beta\lambda) + w_2(\alpha_w - \beta_w\lambda) + w_3 r\right)dt + (w_1\sigma + w_2\sigma_w)dW(t) \\
&\quad + \left(\frac{S(t-)}{S(t)}dQ(t) + \frac{c(t,S(t-))}{c(t,S(t))}dQ_w(t)\right) \\
&= \left((w_1\alpha + w_2\alpha_w + w_3 r) - (w_1\beta + w_2\beta_w)\lambda\right)dt + (w_1\sigma + w_2\sigma_w)dW(t) \\
&\quad + \left(w_1\frac{S(t-)}{S(t)}dQ(t) + w_2\frac{c(t,S(t-))}{c(t,S(t))}dQ_w(t)\right).
\end{aligned} \tag{10.66}$$

We set

$$\begin{aligned}
\alpha_p &= w_1\alpha + w_2\alpha_w + w_3 r; \\
\beta_p &= w_1\beta + w_2\beta_w; \\
\sigma_p &= w_1\sigma + w_2\sigma_w; \\
\frac{P(t-)}{P(t)}dQ_p(t) &= \left(w_1\frac{S(t-)}{S(t)}dQ(t) + w_2\frac{c(t,S(t-))}{c(t,S(t))}dQ_w(t)\right).
\end{aligned} \tag{10.67}$$

Then we have

$$\frac{dP(t)}{P(t)} = (\alpha_p - \beta_p\lambda)dt + \sigma_p dW(t) + \frac{P(t-)}{P(t)}dQ_p(t). \tag{10.68}$$

Properties 10.17.
Continue our analysis in Example 10.16. Note that, in equations (10.67)*, the Poisson process* $Q_p(t)$ *can be rewritten as a sum of random variable* Y_k^p*, where* Y_k^p *denotes the jump sizes of* $Q_p(t)$ *and it depends on* Y_i *and* Y_j^w*. Here,* Y_i *is the jump sizes of* $Q(t)$ *and* Y_j^w *is the jump sizes of* $Q_p(t)$ *as defined before.*

The equality holds that

$$\begin{aligned}
Y^p &= w_1 Y + w_2 Y^w \\
&= w_1 Y + w_2\frac{c(t,(Y+1)S(t)) - c(t,S(t))}{c(t,S(t))}.
\end{aligned} \tag{10.69}$$

With the equality above, we can write

$$Q_p(t) = \sum_{k=1}^{N_p(t)} Y_k^p. \tag{10.70}$$

To further analyze equation (10.68), we can make an assumption that the portfolio is risk-free, and then observe its behavior.

Example 10.18 (Black-Scholes analysis of the portfolio)**.**
We support that the portfolio is risk-free and the intensity for compound Poisson process $Q(t)$ is zero. Then this suggests that the volatility should be equal to zero, i.e.,

$$\sigma_p = w_1\sigma + w_2\sigma_w = 0 \tag{10.71}$$

And in risk-neutral world, if a portfolio is risk-free, then it must have no arbitrage. Note that equation (10.68) could be rewritten as

$$\frac{dP(t)}{P(t)} = \alpha_p dt. \tag{10.72}$$

Then it follows that

$$\alpha_p = w_1\alpha + w_2\alpha_w + w_3 r = w_1\alpha + w_2\alpha_w + (1 - w_1 - w_2)r = r \tag{10.73}$$

It is equivalent to say that

$$w_1(\alpha - r) + w_2(\alpha_w - r) = 0. \tag{10.74}$$

From equation (10.71) and equation (10.74), we could derive that

$$\frac{\alpha - r}{\sigma} = \frac{\alpha_w - r}{\sigma_w}. \tag{10.75}$$

And the difficulty here is to figure out the relationship between w_1 and w_2. We can deduce that

$$w_1 = -w_2 S(t)\frac{c_x(t, S(t))}{c(t, S(t))}, \tag{10.76}$$

then we have

$$Y^p = w_2 \frac{1}{c(t, S(t))}[c(t, (Y+1)S(t)) - c(t, S(t)) - YS(t)c_x(t, S(t))]. \tag{10.77}$$

10.4 An Option Pricing Formula

Actually, the risk-free portfolio is an idealistic assumption that could not be achieved in real life. However, this assumption gives us some hints in deriving more complicated formulas, such as the option pricing formula which we would introduce in this section.

Therefore, in this subsection, we would continue our assumption that the intensity for compound Poisson process λ is equal to zero.

Recall that we have calculated the mean rate of return for call price in Theorem 10.15, and interest rate should be

$$\begin{aligned}\alpha_w = \bigg(&\frac{1}{2}\sigma^2 S^2(t)c_{xx}(t, S(t)) + (\alpha - \beta\lambda)S(t)c_x(t, S(t)) + c_t(t, S(t)) \\ &+ \lambda\mathbb{E}[c(t, (Y_i+1)S(t)) - c(t, S(t))]\bigg)/c(t, S(t));\end{aligned} \tag{10.78}$$

Now we introduce a function $g(t, S(t))$ to denote the expected rate of return for the call option. It follows that

$$\begin{aligned}g(t, S(t)) = \bigg(&\frac{1}{2}\sigma^2 S^2(t)c_{xx}(t, S(t)) + (\alpha - \beta\lambda)S(t)c_x(t, S(t)) + c_t(t, S(t)) \\ &+ \lambda\mathbb{E}[c(t, (Y_i+1)S(t)) - c(t, S(t))]\bigg)/c(t, S(t)),\end{aligned} \tag{10.79}$$

or we say

$$\begin{aligned}&\frac{1}{2}\sigma^2 S^2(t)c_{xx}(t, S(t)) + (\alpha - \beta\lambda)S(t)c_x(t, S(t)) + c_t(t, S(t)) - g(t, S(t))c(t, S(t)) \\ &\quad + \lambda\mathbb{E}[c(t, (Y_i+1)S(t)) - c(t, S(t))] = 0.\end{aligned} \tag{10.80}$$

Here, we make a change-in-variables by setting $\tau = T - t$ and $c(\tau, S(T-\tau)) = c(t, S(t))$. So τ denotes the time before expiration of option. If we set $x = S(T-\tau) = S(t)$, then we can calculate the partial derivative of $c(\tau, x)$, and it gives that

$$\begin{aligned} c_x(\tau, S(t)) &= c_x(t, S(t)); \\ c_{xx}(\tau, S(t)) &= c_{xx}(t, S(t)); \\ c_\tau(\tau, S(t)) &= \frac{\partial c}{\partial t}\frac{\partial t}{\partial \tau} = -c_t(t, S(t)). \end{aligned} \tag{10.81}$$

So we can then substitute the derivatives in equation (10.81) into equation (10.80), then we have

$$\begin{aligned} &\frac{1}{2}\sigma^2 S^2(t) c_{xx}(\tau, S(t)) + (\alpha - \beta\lambda) S(t) c_x(\tau, S(t)) \\ &\quad - c_\tau(\tau, S(t)) - g(\tau, S(t)) c(\tau, S(t)) \\ &\quad + \lambda \mathbb{E}[c(\tau, (Y_i + 1) S(t)) - c(\tau, S(t))] = 0. \end{aligned} \tag{10.82}$$

Since we are using $c(\tau, S(t))$ to denote the call price, then it must satisfies the boundary conditions that

$$c(\tau, 0) = 0, \ \ c(0, S(t)) = (S(t) - K)^+. \tag{10.83}$$

The first equation holds true because the call price is invalid when there is no stock price; the second equation is the terminal condition.

In risk-neutral world, the stock and the call option should both be non-arbitrage, so we can set the rate of return to be the interest rate t, and it follows that

$$\alpha = r, \ \ g(\tau, S(t)) = r. \tag{10.84}$$

Theorem 10.19 (the stochastic differential equation for call price)**.**
According to equation (10.82) *and* (10.83)*, we can then rewrite it to be*

$$\begin{aligned} &\frac{1}{2}\sigma^2 S^2(t) c_{xx}(\tau, S(t)) + (r - \beta\lambda) S(t) c_x(\tau, S(t)) \\ &\quad - c_\tau(\tau, S(t)) - r c(\tau, S(t)) \\ &\quad + \lambda \mathbb{E}[c(\tau, (Y_i + 1) S(t)) - c(\tau, S(t))] = 0, \end{aligned} \tag{10.85}$$

with the boundary conditions

$$c(\tau, 0) = 0, \ \ c(0, S(t)) = (S(t) - K)^+. \tag{10.86}$$

Then we may recall the Black-Scholes-Merton equation for call price without jump processes, denoted by $BS(S(t), \tau, K, r, \sigma^2)$.

$$BS(S(t), \tau, K, r, \sigma^2) = S(t) N(d_+(\tau, x)) - e^{-r\tau} K N(d_-(\tau, x)), \tag{10.87}$$

where $\tau = T - t$ and

$$d_+(\tau, x) = \frac{1}{\sigma\sqrt{\tau}}\left[\log\frac{S(t)}{K} + \left(r + \frac{1}{2}\sigma^2\right)\tau\right]; \tag{10.88}$$

$$d_-(\tau, x) = \frac{1}{\sigma\sqrt{\tau}}\left[\log\frac{S(t)}{K} + \left(r - \frac{1}{2}\sigma^2\right)\tau\right]. \tag{10.89}$$

With the definition of Black-Scholes-Merton equation, we are willing to rewrite the formula for stock price $c(\tau, S(t))$ with respect to Black-Scholes-Merton equation. And before that, we would compute the partial differential form of $BS(S(t), \tau, K, r, \sigma^2)$, which is crucial for understanding latter calculations.

Lemma 10.20 (Leibniz rule)**.** *Suppose that $I(t)$ is a integral that*

$$I(t) = \int_{a(t)}^{b(t)} f(x, t)dx, \tag{10.90}$$

then we have the property that

$$\frac{dI(t)}{dt} = \int_{a(t)}^{b(t)} \frac{\partial f}{\partial t}(x, t)dx + f(b(t), t)b'(t) - f(a(t), t)a'(t). \tag{10.91}$$

Properties 10.21 (partial derivative of Black-Scholes-Merton equation over $S(t)$)**.** *Note that we have introduced the Black-Scholes-Merton equation in* (10.87)*, that is*

$$BS(S(t), \tau, K, r, \sigma^2) = S(t)N(d_+(\tau, S(t))) - e^{-r\tau}KN(d_-(\tau, S(t))) \tag{10.92}$$

The we have the property that

$$\begin{aligned} BS_x(S(t), \tau, K, r, \sigma^2) &= \frac{\partial}{\partial S(t)}BS \\ &= N(d_+(\tau, S(t))) + \frac{1}{\sigma\sqrt{2\pi\tau}}e^{-\frac{1}{2}d_+^2(\tau, S(t))} \\ &\quad - e^{-r\tau}K\frac{1}{\sigma S(t)\sqrt{2\pi\tau}}e^{-\frac{1}{2}d_-^2(\tau, S(t))}. \end{aligned} \tag{10.93}$$

Proof. In order to calculate $\frac{\partial}{\partial S(t)}BS$, it is essential to figure out the formula for $\frac{\partial N(d_\pm(\tau, S(t)))}{\partial S(t)}$. Note that $N(y)$ is the cumulative normal distribution, and $N(y) = \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{y} e^{-\frac{1}{2}z^2}dz$. Then it follows that

$$\begin{aligned} \frac{\partial N(d_+(\tau, S(t)))}{\partial S(t)} &= \frac{dN(d_+(\tau, S(t)))}{d\ d_+(\tau, S(t))}\frac{\partial d_+(\tau, S(t))}{\partial S(t)} \\ &= \frac{1}{\sqrt{2\pi}}\left(e^{-\frac{1}{2}y^2}\right)\Big|_{y=d_+} \cdot \frac{1}{K\sigma\sqrt{\tau}}\frac{K}{S(t)} \\ &= \frac{1}{\sigma S(t)\sqrt{2\pi\tau}}e^{-\frac{1}{2}d_+^2(\tau, S(t))}; \end{aligned} \tag{10.94}$$

and similarly,

$$
\begin{aligned}
\frac{\partial N(d_-(\tau, S(t)))}{\partial S(t)} &= \frac{dN(d_-(\tau, S(t)))}{d\ d_-(\tau, S(t))} \frac{\partial d_-(\tau, S(t))}{\partial S(t)} \\
&= \frac{1}{\sqrt{2\pi}} \left(e^{-\frac{1}{2}y^2}\right)\Bigg|_{y=d_-} \cdot \frac{1}{K\sigma\sqrt{\tau}} \frac{K}{S(t)} \\
&= \frac{1}{\sigma S(t)\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_-^2(\tau, S(t))}.
\end{aligned}
\tag{10.95}
$$

Then we compute $\frac{\partial}{\partial S(t)} BS$, and it gives that

$$
\begin{aligned}
\frac{\partial}{\partial S(t)} BS &= \frac{\partial}{\partial S(t)} \big(S(t)N(d_+(\tau, S(t))) - e^{-r\tau} K N(d_-(\tau, S(t)))\big) \\
&= N(d_+(\tau, S(t))) + S(t) \frac{1}{\sigma S(t)\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_+^2(\tau, S(t))} \\
&\quad - e^{-r\tau} K \frac{1}{\sigma S(t)\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_-^2(\tau, S(t))} \\
&= N(d_+(\tau, S(t))) + \frac{1}{\sigma\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_+^2(\tau, S(t))} \\
&\quad - e^{-r\tau} K \frac{1}{\sigma S(t)\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_-^2(\tau, S(t))}.
\end{aligned}
\tag{10.96}
$$

□

Properties 10.22 (second partial derivative of Black-Scholes-Merton equation over $S(t)$)**.**

The second partial derivative of Black-Scholes-Merton equation satisfies the formula that

$$
\begin{aligned}
BS_{xx} = &\frac{1}{\sigma S(t)\sqrt{2\pi\tau}} \left(e^{-\frac{1}{2}d_+^2(\tau, S(t))} + \frac{K}{S(t)} e^{-r\tau - \frac{1}{2}d_-^2(\tau, S(t))}\right) \\
&- \frac{1}{\sigma^2 S^2(t)\tau\sqrt{2\pi}} \left(e^{-\frac{1}{2}d_+^2(\tau, S(t))} d_+(\tau, S(t)) - K e^{-r\tau - \frac{1}{2}d_-^2(\tau, S(t))} d_-(\tau, S(t))\right).
\end{aligned}
\tag{10.97}
$$

Proof. We continue Properties 10.21, and then we can derive that

$$
\begin{aligned}
BS_{xx} &= \frac{\partial}{\partial S(t)} BS_x \\
&= \frac{\partial}{\partial S(t)}\left(N(d_+(\tau, S(t))) + \frac{1}{\sigma\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_+^2(\tau,S(t))} - e^{-r\tau} K \frac{1}{\sigma S(t)\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_-^2(\tau,S(t))} \right) \\
&= \frac{1}{\sigma S(t)\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_+^2(\tau,S(t))} - \frac{1}{\sigma\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_+^2(\tau,S(t))} d_+(\tau, S(t)) \frac{1}{\sigma S(t)\sqrt{\tau}} \\
&\quad - e^{-r\tau} K \left(- \frac{1}{\sigma S^2(t)\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_-^2(\tau,S(t))} - \frac{1}{\sigma S(t)\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_-^2(\tau,S(t))} d_-(\tau, S(t)) \frac{1}{\sigma S(t)\sqrt{\tau}} \right) \\
&= \frac{1}{\sigma S(t)\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_+^2(\tau,S(t))} - \frac{1}{\sigma^2 S(t)\tau\sqrt{2\pi}} e^{-\frac{1}{2}d_+^2(\tau,S(t))} d_+(\tau, S(t)) \\
&\quad - e^{-r\tau} K \left(- \frac{1}{\sigma S^2(t)\sqrt{2\pi\tau}} e^{-\frac{1}{2}d_-^2(\tau,S(t))} \right. \\
&\quad \left. - \frac{1}{\sigma^2 S^2(t)\tau\sqrt{2\pi}} e^{-\frac{1}{2}d_-^2(\tau,S(t))} d_-(\tau, S(t)) \right) \\
&= \frac{1}{\sigma S(t)\sqrt{2\pi\tau}} \left(e^{-\frac{1}{2}d_+^2(\tau,S(t))} + \frac{K}{S(t)} e^{-r\tau-\frac{1}{2}d_-^2(\tau,S(t))} \right) \\
&\quad - \frac{1}{\sigma^2 S^2(t)\tau\sqrt{2\pi}} \left(e^{-\frac{1}{2}d_+^2(\tau,S(t))} d_+(\tau, S(t)) - K e^{-r\tau-\frac{1}{2}d_-^2(\tau,S(t))} d_-(\tau, S(t)) \right).
\end{aligned}
\tag{10.98}
$$

Then we verify the property. □

Properties 10.23 (partial derivative of Black-Schole-Merton equation over τ)**.**
The following formula holds true that

$$
\begin{aligned}
BS_\tau = &\frac{1}{\sigma\sqrt{2\pi\tau}}(r + \frac{1}{2}\sigma^2) S(t) e^{-\frac{1}{2}d_+^2} - \frac{1}{2\tau\sqrt{2\pi}} S(t) d_+ e^{-\frac{1}{2}d_+^2} + r e^{-r\tau} K N(d_-) \\
&- \frac{K}{\sigma\sqrt{2\pi\tau}}(r - \frac{1}{2}\sigma^2) e^{-r\tau-\frac{1}{2}d_-^2} - \frac{K}{2\tau\sqrt{2\pi}} d_- e^{-r\tau-\frac{1}{2}d_-^2}.
\end{aligned}
\tag{10.99}
$$

Proof. By observing the structure of $BS(S(t), \tau, K, r, \sigma^2)$, we notice that we may begin calculating the terms $\frac{\partial N(d_\pm(\tau,S(t)))}{\partial\tau}$, and it gives that

$$
\begin{aligned}
\frac{\partial N(d_+(\tau, S(t)))}{\partial\tau} &= \frac{d\ N(d_+(\tau, S(t)))}{d\ d_+(\tau, S(t))} \frac{\partial d_+(\tau, S(t))}{\partial\tau} \\
&= \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}d_+^2} \cdot \left(-\frac{1}{2}\frac{1}{\tau} d_+ + \frac{1}{\sigma\sqrt{\tau}}(r + \frac{1}{2}\sigma^2) \right) \\
&= \frac{1}{\sigma\sqrt{2\pi\tau}}(r + \frac{1}{2}\sigma^2) e^{-\frac{1}{2}d_+^2} - \frac{1}{2\tau\sqrt{2\pi}} d_+ e^{-\frac{1}{2}d_+^2}.
\end{aligned}
\tag{10.100}
$$

And similarly,

$$\begin{aligned}\frac{\partial N(d_-(\tau,S(t)))}{\partial\tau} &= \frac{d\ N(d_-(\tau,S(t)))}{d\ d_-(\tau,S(t))}\frac{\partial d_-(\tau,S(t))}{\partial\tau}\\ &= \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}d_-^2}\cdot\left(-\frac{1}{2}\frac{1}{\tau}d_- + \frac{1}{\sigma\sqrt{\tau}}(r-\frac{1}{2}\sigma^2)\right)\\ &= \frac{1}{\sigma\sqrt{2\pi\tau}}(r-\frac{1}{2}\sigma^2)e^{-\frac{1}{2}d_-^2} - \frac{1}{2\tau\sqrt{2\pi}}d_-e^{-\frac{1}{2}d_-^2}.\end{aligned} \tag{10.101}$$

Then we utilize the chain rule to calculate the partial derivative of $BS(S(t),\tau,K,r,\sigma^2)$ over τ,

$$\begin{aligned}BS_\tau &= S(t)\frac{\partial N(d_+)}{\partial\tau} + re^{-r\tau}KN(d_-) - e^{-r\tau}K\frac{\partial N(d_-(\tau,S(t)))}{\partial\tau}\\ &= \frac{1}{\sigma\sqrt{2\pi\tau}}(r+\frac{1}{2}\sigma^2)S(t)e^{-\frac{1}{2}d_+^2} - \frac{1}{2\tau\sqrt{2\pi}}S(t)d_+e^{-\frac{1}{2}d_+^2} + re^{-r\tau}KN(d_-)\\ &- \frac{K}{\sigma\sqrt{2\pi\tau}}(r-\frac{1}{2}\sigma^2)e^{-r\tau-\frac{1}{2}d_-^2} + \frac{K}{2\tau\sqrt{2\pi}}d_-e^{-r\tau-\frac{1}{2}d_-^2}.\end{aligned} \tag{10.102}$$

□

Since we have calculated the partial derivative of Black-Scholes-Merton equation in Properties 10.21, 10.22 and 10.23, then we would discuss the important formula for the call price.

Theorem 10.24 (One solution of the partial differential equation)**.** *Recall that we would like to figure out the general solution for the partial differential problem that*

$$\begin{aligned}&\frac{1}{2}\sigma^2S^2(t)c_{xx}(\tau,S(t)) + (r-\beta\lambda)S(t)c_x(\tau,S(t))\\ &\quad - c_\tau(\tau,S(t)) - rc(\tau,S(t))\\ &\quad + \lambda\mathbb{E}[c(\tau,(Y_i+1)S(t)) - c(\tau,S(t))] = 0,\end{aligned} \tag{10.103}$$

with the boundary conditions

$$c(\tau,0)=0,\ \ c(0,S(t)) = (S(t)-K)^+. \tag{10.104}$$

We can verify that the following formula satisfies equation (10.103) *that*

$$c(\tau,S(t)) = \sum_{n=0}^{\infty}\frac{e^{-\lambda t}(\lambda t)^n}{n!}\mathbb{E}_n\left[BS(S(t)(X_n+1)e^{-\beta\lambda\tau},\tau,K,r,\sigma^2)\right], \tag{10.105}$$

where X_n+1 is defined as the product of n independent random variables Y_i+1 with $\mathbb{E}[X_n+1]=1$ and $\mathbb{E}_n$ is defined to be the expected value with respect to X_n+1. More specified, we have the property that

$$X_n+1 \equiv \prod_{i=0}^{n}(Y_i+1),\ \ X_0+1\equiv 1. \tag{10.106}$$

Sketch of proof. Equation (10.105) is difficult to derive directly, but we would demonstrate that it satisfies Theorem (10.19).

Suppose that equation (10.105) is correct, which gives that

$$c(\tau, S(t)) = \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[BS(V_n, \tau, K, r, \sigma^2)\right], \tag{10.107}$$

where $P_n(\tau) = \frac{e^{-\lambda\tau}(\lambda\tau)^n}{n!}$ and $V_n = S(t)(X_n + 1)e^{-\beta\lambda\tau}$. Then by differentiating $c(\tau, S(t))$ with respect to $x = S(t)$, we have

$$\begin{aligned} c_x(\tau, S(t)) &= \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[BS_x(V_n, \tau, K, r, \sigma^2)\frac{\partial V_n}{\partial S(t)}\right] \\ &= \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[BS_x(V_n, \tau, K, r, \sigma^2)(X_n + 1)e^{-\beta\lambda\tau}\right]. \end{aligned} \tag{10.108}$$

In the calculation above, we leverage the chain rule to calculate the partial differential $BS_x(V_n, \tau, K, r, \sigma^2)$. It is equivalent to say

$$\begin{aligned} S(t)c_x(\tau, S(t)) &= \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[BS_x(V_n, \tau, K, r, \sigma^2)S(t)(X_n + 1)e^{-\beta\lambda\tau}\right] \\ &= \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[V_n BS_x(V_n, \tau, K, r, \sigma^2)\right]. \end{aligned} \tag{10.109}$$

Applying similar process for the second order derivative $c_{xx}(\tau, S(t))$, it gives that

$$c_{xx} = \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[BS_{xx}(V_n, \tau, K, r, \sigma^2)(X_n + 1)^2 e^{-2\beta\lambda\tau}\right]; \tag{10.110}$$

or we say

$$S^2(t)c_{xx} = \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[V_n^2 BS_{xx}(V_n, \tau, K, r, \sigma^2)\right]; \tag{10.111}$$

Then we would determine the partial derivative of call price over τ, and it follows that

$$\begin{aligned} c_\tau(\tau, S(t)) &= \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[\frac{\partial BS}{\partial V_n}\frac{\partial V_n}{\partial \tau} + \frac{\partial BS}{\partial \tau}\right] + \sum_{n=0}^{\infty} \frac{\partial P_n(\tau)}{\partial \tau}\mathbb{E}_n[BS] \\ &= \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[\frac{\partial BS}{\partial V_n}\frac{\partial V_n}{\partial \tau}\right] + \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[\frac{\partial BS}{\partial \tau}\right] + \sum_{n=0}^{\infty} \frac{\partial P_n(\tau)}{\partial \tau}\mathbb{E}_n[BS] \end{aligned} \tag{10.112}$$

So, we would calculate the three terms in second equality of equation (10.112). For the first term,

$$\begin{aligned}\sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[\frac{\partial BS}{\partial V_n}\frac{\partial V_n}{\partial \tau}\right] &= \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[BS_x(V_n,\tau,K,r,\sigma^2)(-\beta\lambda)S(t)(X_n+1)e^{-\beta\lambda\tau}\right] \\ &= -\beta\lambda\sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[V_nBS_x(V_n,\tau,K,r,\sigma^2)\right].\end{aligned} \tag{10.113}$$

For the second term,

$$\sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[\frac{\partial BS}{\partial \tau}\right] = \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[BS_\tau\right]. \tag{10.114}$$

For the third term,

$$\sum_{n=0}^{\infty} \frac{\partial P_n(\tau)}{\partial \tau}\mathbb{E}_n[BS] = -\lambda c(\tau,S(t)) + \lambda\sum_{n=1}^{\infty}\frac{(\lambda\tau)^{n-1}e^{-\lambda t}}{(n-1)!}\mathbb{E}_n[BS]. \tag{10.115}$$

By adding up equations (10.113), (10.114) and (10.115), equation (10.112) then becomes

$$\begin{aligned}c_\tau(\tau,S(t)) &= -\lambda k\sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[V_nBS_x\right] + \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[BS_\tau\right] \\ &\quad - \lambda c(\tau,S(t)) + \lambda\sum_{n=1}^{\infty}\frac{(\lambda\tau)^{n-1}e^{-\lambda t}}{(n-1)!}\mathbb{E}_n[BS] \\ &= -\lambda c(\tau,S(t)) - \lambda kS(t)c_x(\tau,S(t)) + \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[BS_\tau\right] \\ &\quad + \lambda\sum_{n=1}^{\infty}\frac{(\lambda\tau)^{n-1}e^{-\lambda t}}{(n-1)!}\mathbb{E}_n[BS].\end{aligned} \tag{10.116}$$

And in particular, note that the Black-Scholes-Merton equation here should be $BS = BS(V_n,\tau,K,r,\sigma^2)$, so we may write

$$\begin{aligned}c_\tau(\tau,S(t)) &= -\lambda c(\tau,S(t)) - \beta\lambda S(t)c_x(\tau,S(t)) \\ &\quad + \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[BS_\tau(V_n,\tau,K,r,\sigma^2)\right] \\ &\quad + \lambda\sum_{n=1}^{\infty}\frac{(\lambda\tau)^{n-1}e^{-\lambda t}}{(n-1)!}\mathbb{E}_n[BS(V_n,\tau,K,r,\sigma^2)] \\ &= -\lambda c(\tau,S(t)) - \beta\lambda S(t)c_x(\tau,S(t)) \\ &\quad + \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[BS_\tau(V_n,\tau,K,r,\sigma^2)\right] \\ &\quad + \lambda\sum_{m=0}^{\infty} P_m(\tau)\mathbb{E}_{m+1}[BS(V_{m+1},\tau,K,r,\sigma^2)].\end{aligned} \tag{10.117}$$

And it is noteworthy that

$$\mathbb{E}_Y[c(\tau,(Y_i+1)S(t))] = \mathbb{E}_Y\left[\sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n[BS((Y_i+1)V_n,\tau,K,r,\sigma^2)]\right]. \tag{10.118}$$

From the definition,

$$X_{n+1}+1 \equiv \prod_{i=0}^{n+1}(Y_i+1) \equiv (Y_i+1)\prod_{i=0}^{n}(Y_i+1) \equiv (Y_i+1)(X_i+1), \tag{10.119}$$

so $(Y_i+1)(X_i+1)$ has the same probability distribution with $(X_{i+1}+1)$, and then $(Y_i+1)V_n$ is identically distributed with V_{n+1}. Therefore,

$$\begin{aligned}\mathbb{E}_Y[c(\tau,(Y_i+1)S(t))] &= \mathbb{E}_Y\left[\sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n[BS((Y_i+1)V_n,\tau,K,r,\sigma^2)]\right]\\ &= \sum_{m=0}^{\infty} P_m(\tau)\mathbb{E}_{m+1}[BS(V_{m+1},\tau,K,r,\sigma^2)].\end{aligned} \tag{10.120}$$

By substituting equation (10.120) into equation (10.117), we obtain that

$$\begin{aligned}c_\tau(\tau,S(t)) = {}& -\lambda c(\tau,S(t)) - \beta\lambda S(t)c_x(\tau,S(t))\\ &+ \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n\left[BS_\tau(V_n,\tau,K,r,\sigma^2)\right]\\ &+ \lambda\mathbb{E}_Y[c(\tau,(Y_i+1)S(t))].\end{aligned} \tag{10.121}$$

According to equations (10.109), (10.111) and (10.121), we have

$$\begin{aligned}&\frac{1}{2}\sigma^2S^2(t)c_{xx} + (r-\beta\lambda)S(t)c_x - c_\tau - rc\\ &= \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n[\frac{1}{2}\sigma^2V_n^2BS_{xx} + rV_nBS_x - BS_\tau - rBS]\\ &\quad - \beta\lambda S(t)c_x + \lambda c + \beta\lambda S(t)c_x - \lambda\mathbb{E}_Y[c(\tau,(Y_i+1)S(t))].\end{aligned} \tag{10.122}$$

By substituting the partial derivative of Black-Scholes-Merton equation, we have

$$
\begin{aligned}
&\frac{1}{2}\sigma^2 V_n^2 BS_{xx} + rV_n BS_x - BS_\tau - rBS \\
&= \frac{1}{\sigma S(t)\sqrt{2\pi\tau}}\left(e^{-\frac{1}{2}d_+^2} + Ke^{-r\tau-\frac{1}{2}d_-^2}\right)\frac{1}{2}\sigma^2 S^2(t)(X_n+1)^2 e^{-2\beta\lambda\tau} \\
&- \frac{1}{\sigma^2 S^2(t)\tau\sqrt{2\pi}}\left(e^{-\frac{1}{2}d_+^2}d_+ - Ke^{-r\tau-\frac{1}{2}d_-^2}d_-\right)\frac{1}{2}\sigma^2 S^2(t)(X_n+1)^2 e^{-2\beta\lambda\tau} \\
&+ N(d_+)rS(t)(X_n+1)e^{-\beta\lambda\tau} + \frac{1}{\sigma\sqrt{2\pi\tau}}e^{-\frac{1}{2}d_+^2}rS(t)(X_n+1)e^{-\beta\lambda\tau} \\
&- e^{-r\tau}K\frac{1}{\sigma S(t)\sqrt{2\pi\tau}}e^{-\frac{1}{2}d_-^2}rS(t)(X_n+1)e^{-\beta\lambda\tau} \\
&- \frac{1}{\sigma\sqrt{2\pi\tau}}(r+\frac{1}{2}\sigma^2)S(t)e^{-\frac{1}{2}d_+^2} + \frac{1}{2\tau\sqrt{2\pi}}S(t)d_+e^{-\frac{1}{2}d_+^2} - re^{-r\tau}KN(d_-) \\
&+ \frac{K}{\sigma\sqrt{2\pi\tau}}(r-\frac{1}{2}\sigma^2)e^{-r\tau-\frac{1}{2}d_-^2} - \frac{K}{2\tau\sqrt{2\pi}}d_-e^{-r\tau-\frac{1}{2}d_-^2} \\
&- rS(t)N(d_+) + re^{-r\tau}KN(d_-).
\end{aligned}
\tag{10.123}
$$

When $n = 0$, note that $\mathbb{E}_0[X_0+1] = \mathbb{E}_0[(X_0+1)^2] = 1$ and $\lambda \equiv 0$, which implies that

$$
\begin{aligned}
&\frac{1}{2}\sigma^2 V_n^2 BS_{xx} + rV_n BS_x - BS_\tau - rBS \\
&= \frac{1}{2\sqrt{2\pi\tau}}\left(e^{-\frac{1}{2}d_+^2} + \frac{K}{S(t)}e^{-r\tau-\frac{1}{2}d_-^2}\right)\sigma S(t) \\
&- \frac{1}{2\tau\sqrt{2\pi}}\left(e^{-\frac{1}{2}d_+^2}d_+ - Ke^{-r\tau-\frac{1}{2}d_-^2}d_-\right) \\
&+ N(d_+)rS(t) + \frac{1}{\sigma\sqrt{2\pi\tau}}e^{-\frac{1}{2}d_+^2}rS(t) \\
&- \frac{rK}{\sigma S(t)\sqrt{2\pi\tau}}e^{-r\tau-\frac{1}{2}d_-^2}S(t) \\
&- \frac{1}{\sigma\sqrt{2\pi\tau}}(r+\frac{1}{2}\sigma^2)S(t)e^{-\frac{1}{2}d_+^2} + \frac{1}{2\tau\sqrt{2\pi}}S(t)d_+e^{-\frac{1}{2}d_+^2} - re^{-r\tau}KN(d_-) \\
&+ \frac{K}{\sigma\sqrt{2\pi\tau}}(r-\frac{1}{2}\sigma^2)e^{-r\tau-\frac{1}{2}d_-^2} - \frac{K}{2\tau\sqrt{2\pi}}d_-e^{-r\tau-\frac{1}{2}d_-^2} \\
&- rS(t)N(d_+) + re^{-r\tau}KN(d_-) \\
&= e^{-\frac{1}{2}d_+^2}\left(\frac{\sigma S(t)}{2\sqrt{2\pi\tau}} + \frac{rS(t)}{\sigma\sqrt{2\pi\tau}} - \frac{rS(t)}{\sigma\sqrt{2\pi\tau}} - \frac{\sigma S(t)}{2\sqrt{2\pi\tau}}\right) + d_+e^{-\frac{1}{2}d_+^2}\left(-\frac{1}{2\tau\sqrt{2\pi}} + \frac{1}{2\tau\sqrt{2\pi}}\right) \\
&+ e^{-r\tau-\frac{1}{2}d_-^2}\left(\frac{K\sigma}{2\sqrt{2\pi\tau}} - \frac{rK}{\sigma\sqrt{2\pi\tau}} + \frac{rK}{\sigma\sqrt{2\pi\tau}} - \frac{K\sigma}{2\sqrt{2\pi\tau}}\right) + d_-e^{-r\tau-\frac{1}{2}d_-^2}\left(\frac{K}{2\tau\sqrt{2\pi}} - \frac{K}{2\tau\sqrt{2\pi}}\right) \\
&+ rS(t)N(d_+) - rS(t)N(d_+) - re^{-r\tau}KN(d_-) + re^{-r\tau}KN(d_-) \\
&= 0.
\end{aligned}
\tag{10.124}
$$

When $n \neq 0$, note that $\mathbb{E}_n[X_n+1] = \mathbb{E}_n[(X_n+1)^2] = 0$, thus,

$$
\begin{aligned}
&\frac{1}{2}\sigma^2 V_n^2 BS_{xx} + rV_n BS_x - BS_\tau - rBS \\
&= -\frac{rK}{\sigma S(t)\sqrt{2\pi\tau}} e^{-r\tau-\frac{1}{2}d_-^2} S(t) \\
&\quad - \frac{1}{\sigma\sqrt{2\pi\tau}}(r+\frac{1}{2}\sigma^2)S(t)e^{-\frac{1}{2}d_+^2} + \frac{1}{2\tau\sqrt{2\pi}}S(t)d_+e^{-\frac{1}{2}d_+^2} - re^{-r\tau}KN(d_-) \\
&\quad + \frac{K}{\sigma\sqrt{2\pi\tau}}(r-\frac{1}{2}\sigma^2)e^{-r\tau-\frac{1}{2}d_-^2} - \frac{K}{2\tau\sqrt{2\pi}}d_-e^{-r\tau-\frac{1}{2}d_-^2} \\
&\quad - rS(t)N(d_+) + re^{-r\tau}KN(d_-) \\
&= -\frac{rK}{\sigma S(t)\sqrt{2\pi\tau}} e^{-r\tau-\frac{1}{2}d_-^2} S(t) \\
&\quad - \frac{1}{\sigma\sqrt{2\pi\tau}}(r+\frac{1}{2}\sigma^2)S(t)e^{-\frac{1}{2}d_+^2} + \frac{1}{2\tau\sqrt{2\pi}}S(t)d_+e^{-\frac{1}{2}d_+^2} \\
&\quad + \frac{K}{\sigma\sqrt{2\pi\tau}}(r-\frac{1}{2}\sigma^2)e^{-r\tau-\frac{1}{2}d_-^2} - \frac{K}{2\tau\sqrt{2\pi}}d_-e^{-r\tau-\frac{1}{2}d_-^2} - rS(t)N(d_+) \\
&\neq 0.
\end{aligned} \tag{10.125}
$$

However, the formula $\frac{1}{2}\sigma^2 V_n^2 BS_{xx} + rV_n BS_x - BS_\tau - rBS$ is a martingale since X_n+1 is a martingale, so we have

$$
\begin{aligned}
&\mathbb{E}_n[\frac{1}{2}\sigma^2 V_n^2 BS_{xx} + rV_n BS_x - BS_\tau - rBS] \\
&= \mathbb{E}_0[\frac{1}{2}\sigma^2 V_0^2 BS_{xx} + rV_n BS_x - BS_\tau - rBS] = 0.
\end{aligned} \tag{10.126}
$$

Therefore,

$$
\begin{aligned}
&\frac{1}{2}\sigma^2 S^2(t)c_{xx} + (r-\beta\lambda)S(t)c_x - c_\tau - rc \\
&= \sum_{n=0}^{\infty} P_n(\tau)\mathbb{E}_n[\frac{1}{2}\sigma^2 V_n^2 BS_{xx} + rV_n BS_x - BS_\tau - rBS] \\
&\quad - \beta\lambda S(t)c_x + \lambda c + \beta\lambda S(t)c_x - \lambda\mathbb{E}_Y[c(\tau,(Y_i+1)S(t))] \\
&= \lambda c - \lambda\mathbb{E}_Y[c(\tau,(Y_i+1)S(t))] \\
&= -\lambda\mathbb{E}_Y[c(\tau,(Y_i+1)S(t)) - c(\tau,S(t))].
\end{aligned} \tag{10.127}
$$

Then we have

$$
\begin{aligned}
&\frac{1}{2}\sigma^2 S^2 c_{xx}(\tau,S(t)) + (r-\beta\lambda)S(t)c_x(\tau,S(t)) - c_\tau(\tau,S(t)) - rc(\tau,S(t)) \\
&\quad = -\lambda\mathbb{E}_Y[c(\tau,(Y_i+1)S(t)) - c(\tau,S(t))].
\end{aligned} \tag{10.128}
$$

We have verified the theorem that equation (10.105) is a solution of partial differential equation (10.103). □

Theorem 10.25 (satisfying the boundary conditions)**.** *We need to show that the solution of equation* (10.103) *with*

$$c(\tau, S(t)) = \sum_{n=0}^{\infty} \frac{e^{-\lambda t}(\lambda t)^n}{n!} \mathbb{E}_n \left[BS(S(t)(X_n+1)e^{-\beta\lambda\tau}, \tau, K, r, \sigma^2) \right], \tag{10.129}$$

would satisfy the boundary conditions that

$$c(\tau, 0) = 0, \;\; c(0, S(t)) = (S(t) - K)^+. \tag{10.130}$$

Proof. When $S(t) = 0$, the equation $c(\tau, 0)$ can be derived as

$$c(\tau, 0) = \sum_{n=0}^{\infty} \frac{e^{-\lambda t}(\lambda t)^n}{n!} \mathbb{E}_n \left[0, \tau, K, r, \sigma^2) \right] = 0, \tag{10.131}$$

since $BS(0, \tau, k, r, \sigma^2) = 0$. And consider $\tau \to 0$, it follows that

$$\begin{aligned}
&\lim_{\tau\to 0} \mathbb{E}_n[BS(S(t)(X_n+1)e^{-\beta\lambda\tau}, \tau, K, r, \sigma^2)] \\
&\quad = \lim_{\tau\to 0} \mathbb{E}_n[(V_n - K)^+] \\
&\quad \leqslant \lim_{\tau\to 0} \mathbb{E}_n[S(t)(X_n+1)e^{-\beta\lambda\tau}] \\
&\quad = \lim_{\tau\to 0} S(t)e^{-\beta\lambda\tau} \\
&\quad \leqslant \lim_{\tau\to 0} S(t)(1 - \beta\lambda\tau + \frac{1}{2}\beta^2\lambda^2\tau^2) \leqslant S(t).
\end{aligned} \tag{10.132}$$

Then we have

$$\begin{aligned}
&\lim_{\tau\to 0} c(\tau, S(t)) \sum_{n=1}^{\infty} P_n(\tau)\mathbb{E}_n[BS(S(t)(X_n+1)e^{-\beta\lambda\tau}, \tau, K, r, \sigma^2)] \\
&\leqslant \lim_{\tau\to 0} \sum_{n=1}^{\infty} \frac{S(t)e^{-\lambda\tau(\lambda\tau)^n}}{n!} \\
&= \lim_{\tau\to 0} S(t)e^{-\lambda\tau}[e^{\lambda\tau} - 1] = 0.
\end{aligned} \tag{10.133}$$

So the formula for $c(\tau, S(t))$ satisfies the boundary conditions. □

Then we would like to show that the partial differential equation (10.103) has exactly one solution, and before that, we would introduce an important lemma corresponding to the property.

Lemma 10.26 (the energy method)**.**
Suppose that $u(\tau, x)$ *where* $x = S(t)$ *satisfies the formula that*

$$\begin{aligned}
&\frac{1}{2}\sigma^2 x^2 u_{xx}(\tau, x) + (r - \beta\lambda)x u_x(\tau, x) \\
&\quad - u_\tau(\tau, x) - r u(\tau, x) \\
&\quad + \lambda\mathbb{E}[u(\tau, (Y_i+1)x) - u(\tau, x)] = 0,
\end{aligned} \tag{10.134}$$

and the boundary condition says that

$$\lim_{x\to\pm\infty} u(\tau, x) = 0,\ \ u(0, x) = 0. \tag{10.135}$$

Then the energy is defined as

$$E(\tau) = \int_{-\infty}^{\infty} u^2(\tau, x)dx. \tag{10.136}$$

It gives that

$$E'(\tau) = \int_{-\infty}^{\infty} 2uu_\tau dx, \tag{10.137}$$

and for further calculation, we have

$$\begin{aligned} E'(\tau) &= \int_{-\infty}^{\infty} 2uu_\tau dx \\ &= \sigma^2 \int_{-\infty}^{\infty} x^2 u u_{xx} dx + 2(r - \beta\lambda) \int_{-\infty}^{\infty} x u u_x dx - 2r \int_{-\infty}^{\infty} u^2 dx \\ &\quad + 2\lambda \int_{-\infty}^{\infty} u \mathbb{E}[u(\tau, (Y_i + 1)S(t)) - u(\tau, S(t))]dx. \end{aligned} \tag{10.138}$$

In the equation above, the first term is equivalent to say that

$$\begin{aligned} \sigma^2 \int_{-\infty}^{\infty} x^2 u u_{xx} dx &= \sigma^2 x^2 u u_x \Big|_{-\infty}^{\infty} - \sigma^2 \int_{-\infty}^{\infty} x^2 u_x^2 dx \\ &= 0 - \sigma^2 \int_{-\infty}^{\infty} x^2 u_x^2 dx \\ &- \sigma^2 \int_{-\infty}^{\infty} x^2 u_x^2 dx; \end{aligned} \tag{10.139}$$

for the second term, the integral can be calculated as

$$\begin{aligned} \int_{-\infty}^{\infty} x u u_x dx &= xuu \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} u d(xu) \\ &= 0 - \int_{-\infty}^{\infty} u(du + udx) \\ &= -\int_{-\infty}^{\infty} u u_x dx - \int_{-\infty}^{\infty} u^2 dx, \end{aligned} \tag{10.140}$$

and using the trick that

$$\begin{aligned} \int_{-\infty}^{\infty} u u_x dx &= u^2 \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} u u_x dx \\ &= 0 - \int_{-\infty}^{\infty} u u_x dx \\ &= -\int_{-\infty}^{\infty} u u_x dx, \end{aligned} \tag{10.141}$$

which implies that

$$\int_{-\infty}^{\infty} uu_x dx = 0. \tag{10.142}$$

Therefore,

$$2(r-\beta\lambda)\int_{-\infty}^{\infty} xuu_x dx = -2(r-\beta\lambda)\int_{-\infty}^{\infty} u^2 dx \tag{10.143}$$

Then it follows that

$$\begin{aligned}
E'(\tau) &= \int_{-\infty}^{\infty} 2uu_\tau dx \\
&= \sigma^2 \int_{-\infty}^{\infty} x^2 uu_{xx} dx + 2(r-\beta\lambda)\int_{-\infty}^{\infty} xuu_x dx - 2r\int_{-\infty}^{\infty} u^2 dx \\
&\quad + 2\lambda \int_{-\infty}^{\infty} u\mathbb{E}[u(\tau,(Y_i+1)S(t)) - u(\tau,S(t))]dx \\
&= -\sigma^2 \int_{-\infty}^{\infty} x^2 u_x^2 dx - 4r\int_{-\infty}^{\infty} u^2 dx + 2\beta\lambda \int_{-\infty}^{\infty} u^2 dx \\
&\quad + 2\lambda \int_{-\infty}^{\infty} u\mathbb{E}[u(\tau,(Y_i+1)S(t)) - u(\tau,S(t))]dx
\end{aligned} \tag{10.144}$$

Then we consider a special case that $\beta < 0$ *and*

$$\begin{aligned}
E'(\tau) &= -\sigma^2 \int_{-\infty}^{\infty} x^2 u_x^2 dx - 4r\int_{-\infty}^{\infty} u^2 dx + 2\beta\lambda \int_{-\infty}^{\infty} u^2 dx \\
&\quad + 2\lambda \int_{-\infty}^{\infty} u\mathbb{E}[u(\tau,(Y_i+1)S(t)) - u(\tau,S(t))]dx \\
&\leqslant -\sigma^2 \int_{-\infty}^{\infty} x^2 u_x^2 dx - 4r\int_{-\infty}^{\infty} u^2 dx + 2\beta\lambda \int_{-\infty}^{\infty} u^2 dx \\
&\quad + 2\beta\lambda \int_{-\infty}^{\infty} u^2 dx \\
&= -\sigma^2 \int_{-\infty}^{\infty} x^2 u_x^2 dx - 4(r-\beta\lambda)\int_{-\infty}^{\infty} u^2 dx \leqslant 0.
\end{aligned} \tag{10.145}$$

Then the derivative of $E(\tau)$ *is always negative, and using the definition that* $u(0,x) = 0$*, we have the property that*

$$\int_{-\infty}^{\infty} u(\tau,x)dx \leqslant \int_{-\infty}^{\infty} u(\tau,0)dx = 0. \tag{10.146}$$

Then $u(\tau,x) = 0$ *for all* x *in the domain.*

Properties 10.27. *In Lemma 10.26, the domain of* x *can be changed into* $[a,b]$ *and the boundary condition then becomes* $\lim_{x\to a} u(\tau,x) = 0$, $\lim_{x\to b} u(\tau,x) = 0$ *and* $u(0,x) = 0$*, while the result would be the same.*

If we reduce one of the boundary conditions, say

$$\lim_{x\to a} u(\tau,x) = 0,\ u(0,x) = 0, \tag{10.147}$$

the result would also be the same.

Theorem 10.28. *The partial differential equation system*

$$\begin{aligned}\frac{1}{2}\sigma^2 S^2(t)c_{xx}(\tau, S(t)) &+ (r-\beta\lambda)S(t)c_x(\tau, S(t))\\ &- c_\tau(\tau, S(t)) - rc(\tau, S(t))\\ &+ \lambda\mathbb{E}[c(\tau,(Y_i+1)S(t)) - c(\tau, S(t))] = 0,\end{aligned} \tag{10.148}$$

with the boundary conditions

$$c(\tau, 0) = 0,\ \ c(0, S(t)) = (S(t)-K)^+, \tag{10.149}$$

and the special case that $\beta < 0$,

$$\int_{-\infty}^{\infty} u\mathbb{E}[u(\tau,(Y_i+1)S(t)) - u(\tau, S(t))]dx \leqslant 2\int_{-\infty}^{\infty} u^2 dx. \tag{10.150}$$

has exactly one solution.

Proof. Suppose that c_1 and c_2 are two solutions of the partial differential equation, then we set $u = c_1 - c_2$, u would also satisfies the partial differential equation with boundary conditions that

$$c(\tau, 0) = 0,\ \ c(0, S(t)) = 0. \tag{10.151}$$

Using the energy method, it follows that

$$u(\tau, S(t)) = 0. \tag{10.152}$$

Hence,

$$c_1 = c_2. \tag{10.153}$$

□

It is noteworthy that we have not verified the partial differential equation system has exactly one solution in all the cases, and the property is not prone to prove.

10.5 Deriving Merton's Jump Diffusion Model

According to theorem 10.24, 10.25 and 10.28, the formula

$$c(\tau, S(t)) = \sum_{n=0}^{\infty}\frac{e^{-\lambda t}(\lambda t)^n}{n!}\mathbb{E}_n\left[BS(S(t)(X_n+1)e^{-\beta\lambda\tau}, \tau, K, r, \sigma^2)\right] \tag{10.154}$$

is the only solution to partial differential system

$$\begin{aligned}\frac{1}{2}\sigma^2 x^2 u_{xx}(\tau, x) &+ (r-\beta\lambda)xu_x(\tau, x)\\ &- u_\tau(\tau, x) - ru(\tau, x)\\ &+ \lambda\mathbb{E}[u(\tau,(Y_i+1)x) - u(\tau, x)] = 0,\end{aligned} \tag{10.155}$$

with the boundary conditions

$$c(\tau, 0) = 0,\ c(0, S(t)) = (S(t) - K)^+. \tag{10.156}$$

We suppose that the random variable $Y_i + 1$ is log-normal distributed, say $\log(Y_i + 1) \sim N(\mu, \delta^2)$. Since the random variable $X_n + 1$ is defined as

$$X_n + 1 = \prod_{i=0}^{n} (Y_i + 1). \tag{10.157}$$

So $X_n + 1$ would also follow a log-normal distribution with

$$\mathbb{E}_n[X_n + 1] = e^{n\mu}, \tag{10.158}$$

and

$$Var[\log(X_n + 1)] = n\delta^2 \tag{10.159}$$

We can rewrite the formula for call option by letting

$$c_n(\tau, S(t)) = BS(S(t), \tau, K, \frac{r - \beta\lambda + n\mu}{\tau}, \frac{\sigma^2 + n\delta^2}{\tau}), \tag{10.160}$$

then we can derive that

$$c(\tau, S(t)) = \sum_{n=0}^{\infty} \frac{e^{-(1+\beta)\lambda\tau}((1+\beta)\lambda\tau)^n}{n!} c_n(\tau, S(t)). \tag{10.161}$$

The equation above is known as Merton's jump diffusion model.

10.6 Principles of Sensitive Analysis for Merton's Jump Diffusion Model

10.6.1 Deriving the cumulants

Theorem 10.29 (Lévy measure for Merton's log-return density function)**.** *The Lévy measure for Merton's log-return density function is denote as*

$$\mathcal{V}(dx) = \frac{\lambda}{\sqrt{2\pi\delta^2}} \exp\left\{ -\frac{(dx - \mu)^2}{2\delta^2} \right\}. \tag{10.162}$$

Proof. The Merton's log-return density function is a Lévy's process relative to Merton's jump diffusion model, which has the form that

$$L(t) = \log\left(\frac{S(t)}{S(0)}\right) = \sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right) t + \sum_{i=1}^{N(t)} \log(Y_i + 1). \tag{10.163}$$

Then probability distribution for the term $\log(Y_i + 1)$ should be

$$f(\log(Y_i + 1)) = \frac{1}{\sqrt{2\pi\delta^2}} \exp\left\{ -\frac{(\log(Y_i + 1) - \mu)^2}{2\delta^2} \right\}. \tag{10.164}$$

For the Merton's log-return density function, the Lévy measure $\mathcal{V}$ denotes the jumps' arrival rate, which satisfies the following formula that

$$\int_{-\infty}^{\infty} \mathcal{V}(dx) = \lambda. \tag{10.165}$$

Therefore,

$$\mathcal{V}(dx) = \lambda f(dx) = \frac{\lambda}{\sqrt{2\pi\delta^2}} \exp\left\{-\frac{(dx-\mu)^2}{2\delta^2}\right\}. \tag{10.166}$$

□

Theorem 10.30 (characteristic component of Merton's log-return density function)**.** *Recall that Merton's jump diffusion model is not a Lévy's process, but the corresponding Merton's log-return density function constitutes a Lévy's process, denoted as*

$$L(t) = \log\left(\frac{S(t)}{S(0)}\right) = \sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right) t + \sum_{i=1}^{N(t)} \log(Y_i + 1), \tag{10.167}$$

where $\log(Y_i + 1) \sim N(\mu, \delta^2)$*. For a Lévy's process, its characteristic component can be computed as*

$$\Psi(w) = i(\alpha - \beta\lambda - \frac{1}{2}\sigma^2)w - \frac{1}{2}\sigma^2 w^2 + \lambda\left\{\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\} - 1\right\}. \tag{10.168}$$

Proof. According to Lévy-Khintchine formula, for a triple $(b, \sigma^2, \mathcal{V})$ where $b, \sigma \in R$ and $\mathcal{V}$ is a Lévy measure such that

$$\int_{-\infty}^{\infty} \max\{x^2, 1\}\mathcal{V}(dx) < \infty, \tag{10.169}$$

the characteristic component of $L(t)$ satisfies the formula that

$$\Psi(w) = ibw - \frac{1}{2}\sigma^2 w^2 + \int_{-\infty}^{\infty} \Big(\exp\{iwx\} - 1 - iwx1_{x\in\{-1,1\}}\Big)\mathcal{V}(dx). \tag{10.170}$$

Since the continuous part follows the distribution of $N(\alpha - \beta\lambda - \frac{1}{2}\sigma^2, \sigma^2 t)$ and the jump size $\log(Y_i + 1)$ follows the distribution of $N(\mu, \sigma^2)$, we can derive that

$$b = \alpha - \beta\lambda - \frac{1}{2}\sigma^2, \tag{10.171}$$

and then

$$\mathcal{V}(dx) = \frac{\lambda}{\sqrt{2\pi\delta^2}} \exp\left\{-\frac{(dx-\mu)^2}{2\delta^2}\right\}. \tag{10.172}$$

By substituting equations 10.171 and 10.172 into equation 10.170, it follows that

$$\begin{aligned}\Psi(w) &= ibw - \frac{1}{2}\sigma^2 w^2 + \int_{-\infty}^{\infty}\Big(\exp\{iwx\} - 1\Big)\mathcal{V}(dx)\\ &= ibw - \frac{1}{2}\sigma^2 w^2 + \lambda\int_{-\infty}^{\infty}\Big(\exp\{iwx\} - 1\Big)f(dx)\\ &= ibw - \frac{1}{2}\sigma^2 w^2 - \lambda + \lambda\int_{-\infty}^{\infty}\exp\{iwx\}\, f(dx)\\ &= i(\alpha - \beta\lambda - \frac{1}{2}\sigma^2)w - \frac{1}{2}\sigma^2 w^2 + \lambda\left\{\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\} - 1\right\}.\end{aligned} \tag{10.173}$$

□

Theorem 10.31 (first four cumulants for Merton's jump diffusion model)**.**
The first four cumulants satisfies the formula that

$$k_1 = \alpha - \beta\lambda + \lambda\mu - \frac{1}{2}\sigma^2; \tag{10.174}$$

$$k_2 = \lambda\delta^2 + \lambda\mu^2 + \sigma^2; \tag{10.175}$$

$$k_3 = 3\lambda\delta^2\mu + \lambda\mu^3; \tag{10.176}$$

$$k_4 = 3\lambda\delta^4 + 6\lambda\delta^2\mu^2 + \lambda\mu^4. \tag{10.177}$$

Proof. Since we have derived the characteristic component for Merton's log-return density function, which is given by

$$\Psi(w) = i(\alpha - \beta\lambda - \frac{1}{2}\sigma^2)w - \frac{1}{2}\sigma^2 w^2 + \lambda\left\{\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\} - 1\right\}. \tag{10.178}$$

Then recall that the k-th cumulants can be calculated by

$$k_n = i^{-n}\left[\frac{d^n}{dw^n}\Psi(w)\right]\Bigg|_{w=0}. \tag{10.179}$$

The first order derivative of the characteristic component can be computed as

$$\frac{d}{dw}\Psi(w) = i(\alpha - \beta\lambda - \frac{1}{2}\sigma^2) - \sigma^2 w + \lambda(i\mu - \delta^2 w)\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\}; \tag{10.180}$$

the second order derivative of the characteristic component can be computed as

$$\frac{d^2}{dw^2}\Psi(w) = -\sigma^2 + \lambda(-\mu^2 - 2i\delta^2\mu w + \delta^4 w^2 - \delta^2)\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\}; \tag{10.181}$$

the third order derivative of the characteristic component can be computed as

$$\begin{aligned}\frac{d^3}{dw^3}\Psi(w) &= \lambda(-\mu^2 - 2i\delta^2\mu w + \delta^4 w^2 - \delta^2)(i\mu - \delta^2 w)\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\}\\ &\quad + \lambda(-2i\delta^2\mu + 2\delta^4 w)\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\}\\ &= \lambda(-i\mu^3 + 3\delta^2\mu^2 w + 3i\delta^4\mu w^2 - \delta^6 w^3 - i\delta^2\mu + \delta^4 w)\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\}\\ &\quad + \lambda(-2i\delta^2\mu + 2\delta^4 w)\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\}\\ &= \lambda(-i\mu^3 + 3\delta^2\mu^2 w + 3i\delta^4\mu w^2 - \delta^6 w^3 - 3i\delta^2\mu + 3\delta^4 w)\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\};\end{aligned} \tag{10.182}$$

and the fourth order derivative of the characteristic component can be computed as

$$\begin{aligned}\frac{d^4}{dw^4}\Psi(w) &= \lambda(-i\mu^3 - 3i\delta^2\mu + f_1(w))(i\mu - \delta^2 w)\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\}\\ &\quad + \lambda(3\delta^2\mu^2 + 3\delta^4 + f_2(w))\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\}\\ &= \lambda(\mu^4 + 3\delta^2\mu^2 + f_2(w))\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\}\\ &\quad + \lambda(3\delta^2\mu^2 + 3\delta^4 + f_2(w))\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\}\\ &= \lambda(3\delta^4 + 6\delta^2\mu^2 + \mu^4 + f_4(w))\exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\},\end{aligned} \tag{10.183}$$

where $f_i(w)$ denotes the functions in terms of w with $f_i(0) = 0$. Therefore,

$$k_1 = \alpha - \beta\lambda + \lambda\mu - \frac{1}{2}\sigma^2; \tag{10.184}$$

$$k_2 = \lambda\delta^2 + \lambda\mu^2 + \sigma^2; \tag{10.185}$$

$$k_3 = 3\lambda\delta^2\mu + \lambda\mu^3; \tag{10.186}$$

$$k_4 = 3\lambda\delta^4 + 6\lambda\delta^2\mu^2 + \lambda\mu^4. \tag{10.187}$$

Then we derive the formula of first four cumulants. □

10.6.2 Variance, Skewness and Kurtosis for Merton's Jump Diffusion Model

In this section, we would interpret the variance, skewness and kurtosis corresponding to Merton's jump diffusion model. An revision of these methods would be discussed.

Definition 10.32 (revision of variance)**.** *Consider a stochastic process $X(t)$, the variance is defined as*

$$Var[X(t)] = \mathbb{E}[(X(t) - \mathbb{E}[X(t)])^2]. \tag{10.188}$$

Definition 10.33 (revision of skewness). *Consider a stochastic process $X(t)$, the skewness is defined as*

$$Skew[X(t)] = \mathbb{E}[(X(t) - \mathbb{E}[X(t)])^3]. \tag{10.189}$$

Definition 10.34 (revision of kurtosis). *Consider a stochastic process $X(t)$, the kurtosis is defined as*

$$Kur[X(t)] = \mathbb{E}[(X(t) - \mathbb{E}[X(t)])^4]. \tag{10.190}$$

Theorem 10.35. *Therefore, we can derive the variance, skewness and kurtosis of Merton's jump diffusion model, or Merton's log-return density function using the cumulants. It follows that*

$$Var[L(t)] = k_2 = \lambda\delta^2 + \lambda\mu^2 + \sigma^2; \tag{10.191}$$

$$Skew[L(t)] = \frac{k_3}{(k_2)^{\frac{3}{3}}} = \frac{3\lambda\delta^2\mu + \mu^3}{(\sigma^2 + \lambda\delta^2 + \lambda\mu^2)^{\frac{3}{2}}}; \tag{10.192}$$

$$Kur[L(t)] = \frac{k_4}{k_2^2} = \frac{3\lambda\delta^4 + 6\lambda\mu^2\delta^2 + \lambda\mu^4}{(\lambda\delta^2 + \lambda\mu^2 + \sigma^2)^2}. \tag{10.193}$$

The numerical results of these methods are essential to analyze the sensitivity of the model.

11 Option Pricing with Python

In this section, we propose the problems we intend to tackle, as well as associated algorithms employed in data analysis. Multiple perspectives has been provided for understanding the stochastic models in valuation of options. We would estimate option prices with Black-Scholes-Merton equation, GARCH model and Merton's jump diffusion model and utilize error analysis to evaluate their performance under various circumstances.

11.1 Model Construction

11.1.1 Black-Scholes-Merton Equation

The most general way to simulate call options is guaranteeing non-arbitrage pricing and constructing self-financing dynamic hedging. Regardless of jump processes, the formula can be derived as

$$c_t(t,x) + rxc_x(t,x) + \frac{1}{2}\sigma^2 x^2 c_{xx}(t,x) = rc(t,x),\ c(T,x) = (x-K)^+. \tag{11.1}$$

By utilizing either heat equations or risk-neutral measure, the Black-Scholes-Merton equation is computed as

$$BS(S(t), \tau, K, r, \sigma^2) = S(t)N(d_+(\tau, S(t))) - e^{-r\tau}KN(d_-(\tau, S(t))), \tag{11.2}$$

where $\tau = T - t$, $N(y)$ denotes the cumulative probability density and

$$d_+(\tau, S(t)) = \frac{1}{\sigma\sqrt{\tau}}\left[\log\frac{S(t)}{K} + \left(r + \frac{1}{2}\sigma^2\right)\tau\right]; \tag{11.3}$$

$$d_{-}(\tau, S(t)) = \frac{1}{\sigma\sqrt{\tau}}\left[\log\frac{S(t)}{K} + \left(r - \frac{1}{2}\sigma^2\right)\tau\right]. \tag{11.4}$$

And the stock price $S(t)$ is simulated by geometric Brownian motion,

$$S(t) = S(0)\exp\left\{\int_0^t \sigma dW(t) + \int_0^t \left(r - \frac{1}{2}\sigma^2\right)dt\right\}. \tag{11.5}$$

11.1.2 Merton's Jump Diffusion Model

Despite the efficacy of Black-Scholes-Merton equation in valuation of options, Merton's jump diffusion equation is another model that takes into account jump processes. The Merton's jump diffusion model can be derived by deriving the equation that

$$\begin{aligned}\frac{1}{2}\sigma^2 S^2(t)c_{xx}(\tau, S(t)) &+ (r - \beta\lambda)S(t)c_x(\tau, S(t)) \\ &- c_\tau(\tau, S(t)) - c(\tau, S(t)) \\ &+ \lambda\mathbb{E}[c(\tau, (Y_i + 1)S(t)) - c(\tau, S(t))] = 0.\end{aligned} \tag{11.6}$$

Suppose that Y_i follows a log-normal distribution such that $\log(Y_i + 1) \sim N(\mu, \delta^2)$, then the Merton's jump diffusion model is calculated as

$$W(S(t), \tau, K, r, \beta, \lambda, \mu, \delta^2, \sigma^2) = \sum_{n=0}^{\infty}\frac{e^{-(1+\beta)\lambda\tau}((1+\beta)\lambda\tau)^n}{n!}c_n(\tau, S(t)), \tag{11.7}$$

where

$$c_n(\tau, S(t)) = BS(S(t), \tau, K, \frac{r - \beta\lambda + n\mu}{\tau}, \frac{\sigma^2 + n\delta^2}{\tau}). \tag{11.8}$$

The stock price $S(t)$ in Merton's jump diffusion model is defined as a geometric Poisson process,

$$S(t) = S(0)\exp\left\{\sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right)t\right\}\prod_{i=1}^{N(t)}(Y_i + 1). \tag{11.9}$$

11.2 Error Analysis

Error in a model refers to the inevitable uncertainty that permeates every prediction. Generally, the error is unavoidable, so we can estimate its unconstrained optima. An optimization problem is aimed to figure out the global minimum of the error by identifying parameters that control the accuracy and adjusting parameters to raise precision. Then we can examine the model's performance by checking whether it can attain desired accuracy. If there are more than one models, we compare these models to determine which one is superior.

Consider the function $f : X \subset R^n \to R$ and $y' = f(x)$, say y'_i is an approximation to y_i, the least square error is defined to be

$$e(x) = \sum_{i=1}^{n}\frac{|y'_i - y_i|^2}{|y_i|^2}; \tag{11.10}$$

and by contrast, the mean absolute percentage error is defined as

$$e(x) = \sum_{i=N+1}^{N+M} \frac{|y_i' - y_i|}{M|y_i|}. \tag{11.11}$$

11.3 Optimization Problem

11.3.1 Optimization Problem for Black-Scholes-Merton Equation

For the Black-Scholes-Merton equation, our target function is denoted as

$$BS(S(t), \tau, K, r, \sigma^2) = S(t)N(d_+(\tau, S(t))) - e^{-r\tau}KN(d_-(\tau, S(t))), \tag{11.12}$$

where σ is a parameter and $S(t), \tau, K, r$ are variables. However, in the optimization problem, we aim to determine the optimal value of σ. Thus, given $S(t), \tau, K, r$, the call price can be estimated using σ and the Black-Scholes-Merton equation. So the Black-Scholes-Merton equation can be represented by $y' = f(x)$ where $x = \sigma \in R$. Then the corresponding optimization problem is defined as

$$\begin{aligned} &\min \sum_{i=1}^{n} \frac{|y_i' - y_i|^2}{|y_i|^2} \\ &subject\ to: \ y_i' = S(t)N(d_+(\tau, S(t))) - e^{-r\tau}KN(d_-(\tau, S(t))). \end{aligned} \tag{11.13}$$

11.3.2 Optimization Problem for Merton's Jump Diffusion Model

For the Merton's jump diffusion model, we tackle the equation that

$$\begin{aligned} &W(S(t), \tau, K, r, \beta, \lambda, \mu, \delta^2, \sigma^2) \\ &= \sum_{n=0}^{\infty} \frac{e^{-(1+\beta)\lambda\tau}((1+\beta)\lambda\tau)^n}{n!} BS(S(t), \tau, K, \frac{r - \beta\lambda + n\mu}{\tau}, \frac{\sigma^2 + n\delta^2}{\tau}). \end{aligned} \tag{11.14}$$

In the equation above, β, λ, μ, δ, σ are parameters and $S(t)$, τ, K, r are variables. Similar to optimization problem for Black-Scholes-Merton equation, we would like to figure out the optimal value of $x = (\beta, \lambda, \mu, \delta, \sigma)$. And the optimization problem can be written as

$$\begin{aligned} &\min \sum_{i=1}^{n} \frac{|y_i' - y_i|^2}{|y_i|^2} \\ &subject\ to: \ y_i' = \sum_{n=0}^{\infty} \frac{e^{-(1+\beta)\lambda\tau}((1+\beta)\lambda\tau)^n}{n!} c_n(\tau, S(t)). \end{aligned} \tag{11.15}$$

11.4 Numerical Analysis Methods

Several methods has been proposed for solving the optimization problem, such as Newton's method, Cauchy-Newton method and Levenberg-Marquardt algorithm. An introduction of these methods is provided in this section.

11.4.1 Pure Newton's method

Pure Newton's method can be employed to figure out the minimum of a multivariate function in the unconstrained minimization problem. The adaption encompasses the iterative refinement on the random variable. The sequence of iterates form a convergent sequence $(x_k)_k$ where $x_k \to x$. At the point x, the target function would attain its minimum. To seek the next iterate, we apply Taylor's expansion on the error $e(x)$, and it gives that

$$e(x_{k+1}) = e(x_k) + J(x_k)(x_{k+1} - x_k) + \frac{1}{2}(x_{k+1} - x_k)^T H(x_k)(x_{k+1} - x_k). \tag{11.16}$$

In the equation above, $J(x) = \nabla f(x)^T$ denotes the Jacobian matrix and $H(x) = \nabla^2 f(x)$ represents the Hessian matrix. Then at the stationary point,

$$J(x_k) + H(x_k)(x_{k+1} - x_k) = 0. \tag{11.17}$$

Therefore, the iterative adaption satisfies the formula that

$$x_{k+1} = x_k - (H(x_k))^{-1} J(x_k). \tag{11.18}$$

The iteration is designed leveraging the update rule above to approximate the critical point. And the update rule contains the information of the first and second derivative of the target function. Since the iteration refinement is in descend directions, the pure Newton's method is classified as a scaled gradient method.

11.4.2 DFP Algorithm

In pure Newton's method, calculating the inverse of Hessian matrix is laborious. To improve the algorithm, DFP algorithm, one of the quasi-Newton methods, would be introduced and interpreted. And the basic idea of DFP method is to leverage matrix $G(x_k)$ as a substitution of $(H(x_k))^{-1}$. Note that any point in pure Newton's method satisfies the formula that

$$J(x_{k-1}) = J(x_k) + H(x_k)(x_{k-1} - x_k). \tag{11.19}$$

If we let $g_k = J(x_k) - J(x_{k-1})$ and $\delta_k = x_k - x_{k-1}$, then the secant equation can be derived as

$$(H(x_{k+1}))^{-1} g_k = \delta_k. \tag{11.20}$$

Since the matrix $G(x_k)$ is a substitution of $(H(x_{k+1}))^{-1}$, it should also satisfies the secant equation, we say

$$G(x_{k+1}) g_k = \delta_k. \tag{11.21}$$

In DFP algorithm, we assume that $G(x_k)$ has the iteration formula that $G(x_{k+1}) = G(x_k) + P(x_k) + Q(x_k)$. By multiplying g_k on both sides of the equation, we can derive that

$$G(x_{k+1}) g_k = G(x_k) g_k + P(x_k) g_k + Q(x_k) g_k = \delta_k. \tag{11.22}$$

We set $P(x_k) y_k = \delta_k$ and $Q(x_k) g_k = G(x_k) g_k$. To meet the conditions, we can assume

$$P(x_k) = \frac{\delta_k \delta_k^T}{\delta_k^T y_k},\ Q(x_k) = -\frac{G(x_k) y_k y_k^T G(x_k)}{y_k^T G(x_k) y_k}. \tag{11.23}$$

Therefore, the iteration of $G(x_k)$ can be calculated as

$$G(x_{k+1}) = G_k + \frac{\delta_k \delta_k^T}{\delta_k^T y_k} - \frac{G(x_k) y_k y_k^T G(x_k)}{y_k^T G(x_k) y_k}. \tag{11.24}$$

11.4.3 BFGS Algorithm

The BFGS algorithm is another sort of quasi-Newton methods. Contradictory to DFP algorithm, the BFGS method use $B(x_k)$ to approximate the Hessian matrix $H(x_k)$, then the corresponding equation is calculated as

$$B(x_k)\delta_k = g_k, \tag{11.25}$$

where $g_k = J(x_k) - J(x_{k-1})$ and $\delta_k = x_k - x_{k-1}$ as defined. We assume that $B(x_k)$ has the iteration that $B(x_{k+1}) = B(x_k) + P(x_k) + Q(x_k)$, and it gives that

$$B(x_{k+1})\delta_k = B(x_k)\delta_k + P(x_k)\delta_k + Q(x_k)\delta_k = g_k. \tag{11.26}$$

To ensure each iterate of $B(x_k)$ would satisfies the corresponding equation, we can define $P(x_k)\delta_k = g_k$ and $Q(x_k)\delta_k = -B(x_k)\delta_k$. Then we take

$$P(x_k) = \frac{g_k g_k^T}{g_k^T \delta_k},\ Q(x_k) = -\frac{B(x_k)\delta_k \delta_k^T B(x_k)}{\delta_k^T B(x_k)\delta_k}. \tag{11.27}$$

Hence, the iteration of $B(x_k)$ should be

$$B(x_{k+1}) = B(x_k) + \frac{g_k g_k^T}{g_k^T \delta_k} - \frac{B(x_k)\delta_k \delta_k^T B(x_k)}{\delta_k^T B(x_k)\delta_k}. \tag{11.28}$$

We would briefly describe the operation of the algorithm. In the first step, the initial value x_0 and B_0 is chosen, where B_0 is positive definite. In the second step, we use the equation $B(x_k)p_k = J(x_k)$ to figure out the value p_k. In the third step, we assign the value p_k to the next iterate such that $x_{k+1} = x_k - p_k$. In the last step, determine whether $|J(x_{k+1})| < \epsilon$ for some ϵ we define, and if so, the minimum attains when we take the value x_{k+1}. We repeat the steps above until the minimum solution is found.

11.5 Option Pricing with a small example

11.5.1 Option Pricing with Black-Scholes-Merton Equation

An example is given on option pricing utilizing Black-Scholes-Merton equation. The Black-Scholes-Merton equation has provided a breakthrough on valuation of options, which definitely revolutionized the finance market. Giving a credible method in option pricing, the Black-Scholes-Merton equation entitles us to understanding the dynamics of finance markets. As we have introduced, the Black-Scholes-Merton equation is based on non-arbitrage pricing and self-financing dynamic hedging. The stock price is assumed to follow a geometric Brownian motion, and the Black-Scholes-Merton equation is an continuous multivariate function depends on stock price $S(t)$, time to expiration τ, strike price K, interest rate r. However, it is significant to figure out the optimal value of the parameter σ before we apply the equation to predict call price.

Here, we choose an example from Dow Jones Industrial Average from Sep. 28, 2022 to

Feb. 28, 2023 to test the efficacy of Balck-Scholes-Merton equation on valuation of options. The index contains broad members across numerous corporation, so it is a suitable choice to examine the performance of Black-Scholes-Merton equation.

When dealing with data, it's crucial to guarantee that the data is valuable and pertinent. We need to figure out the data necessary to train the model. Then the explicit interpretation about the data would be given and discussed.

Time to Expiration τ: τ represents the remaining time before the expiration of call option. In the data frame, it is noteworthy that the release date and expire date of the stock is provided. When dealing with these data, we need to calculate the time interval between the release date and expire date. It is essential to understand that the time interval has two meanings, the first meaning is time to expiration τ and the second meaning is time to maturity T. The two meanings overlap since the strike price during the option trading cannot be determined.

Call Price $c(\tau, S(t))$: The call price $c(\tau, S(t))$ can be calculated as the average of the best bid and best offer prices which are provided in the data frame.

Strike Price K: The strike price denotes the price at expiration time. The option holder would have the right to purchase the derivative with strike price K.

Stock Price $S(t)$: The stock price is a fundamental input in Black-Scholes-Merton equation. Although the stock price could be simulated using stochastic processes, in valuation of options, we would consider stock price as a variable. Hence, the call price can be predicted when inputting the stock price and other relevant variables. Actually, in the data frame, the time to expiration τ should be equal to the time to maturity T, so the stock price should the initial stock price $S(0)$ at time 0 when tackling the data.

Interest Rate r: We would use the zero coupon yield curve to represent the interest rate. In the data frame, we are given the interest rate in different periods, say R. Thus, we have to solve the formula

$$e^{rt} = 1 + R \tag{11.29}$$

to figure out the interest rate in one unit time.

In valuation of options, we do not delete outliers. The accuracy and consistence of these outliers can be guaranteed since these data is revealed in the market. After cleaning the data, we can apply Black-Scholes-Merton equation to determine the parameter. Here, the BFGS algorithm is leveraged to reach the minimal value of the error, namely, the distance between the approximation value of call price and actual call price is minimized.

In table 1, the information of call options in Dow Jones Industrial Average is listed and each value in the data frame is calculated as mentioned. Before leveraging the algorithm, we would separate the data frame into training set and testing set in a proportion of 4 : 1. The training set is shown in table 2 and the testing set is shown in table 3. In the table, note that the stock price is provided, so we do not need to estimate it when dealing with the data. Thus, we simply requires to consider the non-random partial differential equation.

index	tau	call price	strike price	stock price	rate
0	1.0	5.050	311.0	314.55	0.042906
1	1.0	2.720	314.0	314.55	0.042906
2	1.0	1.935	315.0	314.55	0.042906
3	1.0	1.295	316.0	314.55	0.042906
4	1.0	0.735	317.0	314.55	0.042906
...	...	...	...	...	...
3323	52.0	0.830	350.0	328.73	0.054651
3324	52.0	0.425	355.0	328.73	0.054651
3325	52.0	0.215	360.0	328.73	0.054651
3326	52.0	0.080	370.0	328.73	0.054651
3327	479.0	25.000	375.0	328.73	0.054651

Table 1: Data frame of Dow Jones call options from Sep. 28, 2022 to Feb. 28, 2023

index	tau	call price	strike price	stock price	rate
0	1.0	5.050	311.0	314.55	0.042906
1	1.0	2.720	314.0	314.55	0.042906
2	1.0	1.935	315.0	314.55	0.042906
3	1.0	1.295	316.0	314.55	0.042906
4	1.0	0.735	317.0	314.55	0.042906
...	...	...	...	...	...
2657	3.0	54.075	285.0	342.22	0.054465
2658	3.0	19.275	320.0	342.22	0.054465
2659	3.0	9.325	330.0	342.22	0.054465
2660	3.0	4.675	335.0	342.22	0.054465
2661	3.0	1.250	340.0	342.22	0.054465

Table 2: Training set of Dow Jones call options from Sep. 28, 2022 to Feb. 28, 2023

From the algorithm we discussed, the initial value of σ can be defined as $\sigma_0 = 0.2$, then we can calculate the approximation of the call price, denoted by y_i' using the Black-Scholes-Merton equation. Each y_i' can be compared with the real value, and we can compute the single error as $|y' - y|/y$. Since there are 2662 data elements in the training set table 2, we sum up 2662 single errors to obtain total error of the model when $\sigma = \sigma_0 = 0.2$. Then as proposed in the BFGS algorithm, we would adjust the value of τ using the iteration refinement in descend directions, and the aim of the iteration is to minimize the total error. After a couple of minutes, the strict minimum is about to be attained. In our model, the minimum is attained when

index	tau	call price	strike price	stock price	rate
0	3.0	0.555	342.0	342.22	0.054465
1	3.0	0.345	343.0	342.22	0.054465
2	3.0	0.125	345.0	342.22	0.054465
3	3.0	0.025	355.0	342.22	0.054465
4	3.0	0.025	360.0	342.22	0.054465
...	...	...	...	...	...
661	52.0	0.830	350.0	328.73	0.054651
662	52.0	0.425	355.0	328.73	0.054651
663	52.0	0.215	360.0	328.73	0.054651
664	52.0	0.080	370.0	328.73	0.054651
665	479.0	25.000	375.0	328.73	0.054651

Table 3: Testing set of Dow Jones call options from Sep. 28, 2022 to Feb. 28, 2023

$\sigma \approx 0.0834$ and the mean error is $1797.7253/2662 \approx 0.6753$. Then we substitute $\sigma = 0.0834$ into the Black-Scholes-Merton equation, which is one model for valuation of call options in Dow Jones Investment Average. Later, we can examine the efficacy of the model using the testing set. Substituting the value of τ, strike price, stock price, rate and $\sigma = 0.0834$ into the Black-Scholes-Merton equation, we can compare the estimated and real value of call prices by calculating the accuracy score. Here, the accuracy score is defined as R^2, where

$$R^2 = 1 - \frac{\sum_{i=1}^{n} |y_i - y_i'|^2}{\sum_{i=1}^{n} |y_i - \mathbb{E}[y_i]|^2}. \tag{11.30}$$

The closer R^2 is to 1, the more accurate is the model in valuation of options. In our model, R^2 can be calculated as $R^2 \approx 0.979$, we say the Black-Scholes-Merton equation is effective in pricing the call options of Dow Jones Industrial Average over the six-month period. And a graph that displays the effect of Black-Scholes-Merton equation in the testing set is shown in 1.

Figure 1: Black-Scholes-Merton equation in predicting Dow Jones call prices over six-month period

11.5.2 Option Pricing with Merton's Jump Diffusion Model

Then the efficacy of Merton's jump diffusion model would be tested. We would still leverage data from Dow Jones Industrial Average from Sep. 28, 2022 to Feb. 28, 2023 as an example. The training set is shown in table 2 and testing set in table 3. We set the initial value as $x_0 = (\beta_0, \lambda_0, \mu_0, \delta_0, \sigma_0) = (0, 1, 0, 0.1, 0.15)$. After the iteration refinement in BFGS, the minimized mean total error is calculated as $1252.4537/2662 \approx 0.4705$ and the minimum is attained when $x = (\beta, \lambda, \mu, \delta, \sigma) = (-0.0720, 3.0182, -0.0102, 0.1093, 0.0578)$. It is noteworthy that the sum of infinity terms below cannot be calculated directly, so a numerical method is employed during the calculation. The sum is defined as

$$\sum_{n=0}^{\infty} a_n = \sum_{n=0}^{\infty} \frac{e^{-(1+\beta)\lambda\tau}((1+\beta)\lambda\tau)^n}{n!} BS(S(t), \tau, K, \frac{r-\beta\lambda+n\mu}{\tau}, \frac{\sigma^2+n\delta^2}{\tau}). \tag{11.31}$$

The sequence $(a_n)_n$ satisfied the property that $(a_n)_n \in C_0 = \{(x_n)_n \subset R : \lim_n a_n = 0\}$. Since C_0 is a Banach space and $\sum_{n=0}^{\infty} |a_n| < \infty$, then $\sum_{n=0}^{\infty} a_n$ converges in C_0. There exists some M such that $|\sum_{n=0}^{M} a_n - \sum_{n=0}^{\infty} a_n| \to 0$. In our calculation, we set

$$\begin{aligned}\sum_{n=0}^{\infty} a_n &\approx \sum_{n=0}^{40} a_n \\ &= \sum_{n=0}^{\infty} \frac{e^{-(1+\beta)\lambda\tau}((1+\beta)\lambda\tau)^n}{n!} BS(S(t), \tau, K, \frac{r-\beta\lambda+n\mu}{\tau}, \frac{\sigma^2+n\delta^2}{\tau}).\end{aligned} \tag{11.32}$$

Figure 2: Merton's jump diffusion model in predicting Dow Jones call prices over six-month period

After calculating the parameters of Merton's jump diffusion model, we substitute the data in testing set (see table 3) to determine the accuracy of the model. Then the accuracy score R^2 is computed as $R^2 \approx 0.987$, which implies the efficacy of Merton's jump diffusion model in valuation of option in Dow Jones for six months. We leverage a graph in figure 2 to demonstrate the predict call price, compared with real value.

11.5.3 Comparison of Two Models

The calculations above reveals that both Black-Scholes-Merton equation and Merton's jump diffusion model exhibit high accuracy, where Merton's jump diffusion model performs slightly better than Black-Scholes-Merton equation in call pricing of Dow Jones Industial Average over the six-month period. We say Merton's jump diffusion model is more accurate than Black-Scholes-Merton equation in this example. Furthermore, by examining the speed of convergence of these two models, we can then evaluate the recursion rate of these two models.

The graph of least square error with respect to number of iterations is shown in figure 3. Overall, compared to Merton's jump diffusion model, Black-Scholes-Merton equation converges more quickly and it requires less iterations to attain the optimal solution. However, Merton's jump diffusion model entitles us to getting a more precise result. Actually, the total error of Merton's jump diffusion model is less than 2000 merely after one iteration, while using Black-Scholes-Merton equation can never allow us to obtain a result with this accuracy.

To compare the complexity of two models, we introduce Big-O notation and Big-theta

(a) Convergent rate of Black-Scholes-Merton equation

(b) Convergent rate of Merton jump diffusion

Figure 3: Comparing the convergent rate

notation. Big-O notation denotes the upper bound for the running time. For instance, we say the running time of the algorithm satisfies that $f(n) = O(g(n))$ if there $\exists c \in R$ such that $f(n^2) \leqslant cg(n)$ for n large enough. And Big-theta notation denotes the precise asymptotic growth of the running time. It is noteworthy that both Big-O notation and Big-theta notation are not exact value, but they represents the asymptotic behavior of an algorithm in terms of time.

On one hand, consider the Black-Scholes-Merton equation, the complexity of BFGS algorithm is $\theta(n)$ since the Hessian matrix is replaced by iterations. And the running time of the equation has asymptotic growth $\theta(n)$. So the complexity of Black-Scholes-Merton equation is $\theta(n) \cdot \theta(n) = \theta(n^2)$, implying that the running time of Black-Scholes-Merton equation is quadratic.

On the other hand, consider Merton's jump diffusion model, the complexity of BFGS algorithm is also $\theta(n)$ as defined. While as for the equation, since the partial sum corresponding to the Black-Scholes-Merton equation should be computed, then the running time is $\theta(n^2)$. The complexity of Merton's jump diffusion model is $\theta(n) \cdot \theta(n^2) = \theta(n^3)$, implying that the running time of Merton's jump diffusion model is cubic.

In summary, compared to Black-Scholes-Merton equation, Merton's jump diffusion model performs better in convergence and has more accuracy, but it also contains worse time complexity. When the data frame is extremely large, Merton's jump diffusion model requires additional time to operate, which increases workloads.

11.6 Different Time to Maturity in Valuation of Options

In the example we have discussed, we take into account all possible time to maturity in the data frame when we train the models. And in terms of the arbitrary time to maturity, or arbitrary τ, the parameters for Black-Scholes-Merton equation and Merton's jump diffusion model are calculated before the efficacy of two models are examined. However, in reality, it is suggested that the parameters of models in valuation of options may alter for different time to maturity. Therefore, we would like to train our models using data with same time to maturity in order to determine whether the parameters would alter for various time to maturity and assess the accuracy. In training the models, it is crucial to note that the time to expiration τ is exactly the time to maturity T in the data frame.

We would choose the data from MKS Instruments as an example. MKS Instruments is a global company that provides systems, instruments and process control solutions for manufacturing and packaging. The beginning date we select should be Feb. 28, 2013 and the end date is Feb. 28, 2023.

The information of call prices in MKS Instruments is attached in table 4. These data would be grouped by time to expiration τ, and we separate out the data where $\tau = 1$, $\tau = 31$, $\tau = 92$, $\tau = 183$ respectively. These data frames are shown in table 5, table 6, table 7 and table 8. Each data frame is then split into training set and testing set with the proportion $4:1$.

index	tau	call price	strike price	stock price	rate
0	16.0	12.250	100.0	112.75	0.040162
1	16.0	8.450	105.0	112.75	0.040162
2	16.0	4.400	110.0	112.75	0.040162
3	16.0	2.100	115.0	112.75	0.040162
4	16.0	0.875	120.0	112.75	0.040162
...	...	...	...	...	...
12716	143.0	3.425	120.0	95.53	0.061019
12717	234.0	8.300	110.0	95.53	0.061019
12718	234.0	2.350	140.0	95.53	0.061019
12719	234.0	14.950	95.0	95.53	0.061019
12720	290.0	34.050	70.0	95.53	0.061019

Table 4: information in MKS Instruments from Feb. 28, 2013 to Feb. 28, 2023

index	tau	call price	strike price	stock price	rate
0	1.0	8.150	115.0	124.05	0.040407
1	1.0	3.275	120.0	124.05	0.040407
2	1.0	0.275	125.0	124.05	0.040407
3	1.0	1.575	105.0	115.80	0.041599
4	1.0	0.175	110.0	115.80	0.041599
...	...	...	...	...	...
147	1.0	1.725	95.0	97.45	0.060256
148	1.0	0.575	100.0	98.70	0.060335
149	1.0	0.175	105.0	98.70	0.060335
150	1.0	9.050	90.0	98.70	0.060335
151	1.0	3.325	95.0	98.70	0.060335

Table 5: Data frame when $\tau = 1$ in MKS Instruments
from Feb. 28, 2013 to Feb. 28, 2023

index	tau	call price	strike price	stock price	rate
0	31.0	19.600	105.0	123.650	0.040872
1	31.0	14.850	110.0	123.650	0.040872
2	31.0	10.700	115.0	123.650	0.040872
3	31.0	7.200	120.0	123.650	0.040872
4	31.0	4.300	125.0	123.650	0.040872
...	...	...	...	...	...
253	31.0	1.150	95.0	81.125	0.060072
254	31.0	5.800	100.0	100.030	0.060276
255	31.0	2.175	110.0	100.030	0.060276
256	31.0	3.350	105.0	97.940	0.060186
257	31.0	1.475	115.0	97.940	0.060186

Table 6: Data frame when $\tau = 30$ in MKS Instruments
from Feb. 28, 2013 to Feb. 28, 2023

11.6.1 Training with Black-Scholes-Merton equation

We would firstly leverage the Black-Scholes-Merton equation to test the accuracy under different time to maturity. We substitute the information in data frames with $\tau = 1, 31, 91, 183$ into the equation and estimate the value of σ using BFGS algorithm. Then the accuracy score R^2 is calculated utilizing the testing set. The value of σ and the accuracy of the equation corresponding to different time to maturity τ is displayed in table 9. And we use graphs to compare the predicted call prices and real call prices. From the table and graphs, we notice that the computing result of optimal σ would have a different value when time to maturity varies. For valuation of options at different time to maturity, we are required to apply different parameter in Black-Scholes-Merton equation to get a desired prediction of call options.

Our conclusion can be further discussed when training the equation with the whole data frame regardless of fixed time to maturity. The optimal solution is attained when $\sigma \approx 0.3057$, the accuracy score $R^2 \approx 0.7645$ and the graph for prediction of call options is shown in figure 5. It is noteworthy that the accuracy of the equation when neglecting fixed time to maturity is less than that of any model in terms of specific time to maturity as discussed. In conclusion, by using MKS Industrial call prices as an example, we should leverage alternative parameter of Black-Scholes-Merton equation to tackle various time to maturity in order to guarantee the efficacy of the equation in option pricing.

11.6.2 Training with Merton's Jump Diffusion Model

Then we would apply the training set in MKS Instruments when $\tau = 1, 31, 91, 183$ to determine the efficacy of Merton's jump diffusion model. The parameters of Merton's jump diffusion model are $x = (\beta, \lambda, \mu, \delta, \sigma)$. By using BFGS algorithm, the result of the parameters and accuracy score is attached in table 10. The graphs in figure 6 implies the error between predicted call prices and real call prices corresponding to different τ. The accuracy scores when $\tau = 1$, $\tau = 31$ and $\tau = 92$ are above 0.75, but the value is lower than 0.6 when $\tau = 183$.

index	tau	call price	strike price	stock price	rate
0	92.0	11.350	100.0	115.80	0.041599
1	92.0	8.650	105.0	115.80	0.041599
2	92.0	4.700	115.0	115.80	0.041599
3	92.0	3.125	120.0	115.80	0.041599
4	92.0	2.150	125.0	115.80	0.041599
...	...	...	...	...	...
42	92.0	3.375	130.0	112.16	0.047760
43	92.0	33.700	80.0	112.16	0.047760
44	92.0	1.400	100.0	78.64	0.056512
45	92.0	4.750	85.0	78.64	0.056512
46	92.0	2.025	115.0	97.45	0.060256

Table 7: Data frame when $\tau = 92$ in MKS Instruments from Feb. 28, 2013 to Feb. 28, 2023

Regardless of arbitrary time to maturity, the parameters of Merton's jump diffusion model is calculated as $x = (\beta, \lambda, \mu, \delta, \sigma) = (-0.0694, 4.0688, -0.0713, 0.1376, 0.1847)$. And then the accuracy score for any τ satisfies $R^2 = 0.7575$. In summary, when leveraging Merton's jump diffusion model to pricing call options in MKS Instruments, in majority, estimating the parameters of Merton's jump diffusion model separately for different τ yields more accuracy than evaluating the parameters for all possible τs. But the accuracy degrades when $\tau = 183$, and the reason for this is that Merton's jump diffusion model is more complicated and contains numerous parameters. When time to maturity is large, the model may overfit and perform badly.

11.6.3 Comparing Two Models for Option Pricing

In the above example, the observation is that the accuracy score of Black-Scholes-Merton equation is greater than Merton's jump diffusion model in valuation of options when we choose different time to expiration, for instance, the accuracy score of Black-Scholes-Merton equation when $\tau = 30$ is 0.8233 while that of Merton's jump diffusion model is 0.7811. Actually, there are several factors that may contribute to this phenomenon.

Market conditions is one of the factors that affects the accuracy score of the models. The Black-Scholes-Merton equation do not consider the jump processes while the Merton's jump diffusion model would record the jumps by calculating the optimal expected value and volatility of jump sizes. Suppose there are few jumps in stock prices aligned with the call options we would like to predict, then the Black-Scholes-Merton equation would provide more accurate results.

The accuracy of the models is also affected by the duration in the data frame. Black-Scholes-Merton equation demonstrates generalized performance of the call options, thus, it is adept in tackling long duration; while Merton's jump diffusion model is better at capturing

index	tau	call price	strike price	stock price	rate
0	183.0	15.300	100.0	115.80	0.041599
1	183.0	13.400	105.0	115.80	0.041599
2	183.0	8.050	115.0	115.80	0.041599
3	183.0	6.450	120.0	115.80	0.041599
4	183.0	4.950	125.0	115.80	0.041599
...	...	...	...	...	...
23	183.0	13.600	185.0	179.66	0.018226
24	183.0	2.425	190.0	123.33	0.035077
25	183.0	10.350	120.0	112.16	0.047760
26	183.0	3.175	125.0	97.45	0.060256
27	183.0	12.950	95.0	97.45	0.060256

Table 8: Data frame when $\tau = 183$ in MKS Instruments
from Feb. 28, 2013 to Feb. 28, 2023

jumps, allowing better performance when handling short-term data.

The MKS instruments is a tech company, the release of new products may have an impact on tech companies. Many times, the stock prices might not rise immediately after a new product is launched. Due to the defects of new products, their value would not be demonstrated until improvements of the quality of the products. This may impede the tracking of jumps, thus, Merton's jump diffusion model is less accurate than Black-Scholes-Merton equation.

11.7 Sensitive Analysis

Utilizing the call options in MKS instruments during the five-year period, we would examine the sensitivity of Black-Scholes-Merton equation and Merton's jump diffusion model corresponding to time to maturity $\tau = 31$. In the experiment, the variance, skewness and kurtosis would be computed to determine the sensitivity of the parameters corresponding to the models.

The variance, skewness and kurtosis for stochastic processes could be derived from cumulants. Firstly, we need to figure out the corresponding Lévy's process of Black-Scholes-Merton equation and Merton's jump diffusion model, and consequently, the Lévy-Khintchine formula could be applied to derive the characteristic component of these processes. Then we leverage Taylor's expansion on the cumulants generating function to calculate the cumulants. For a triple $(b, \sigma^2, \mathcal{V})$, where Lévy measure $\mathcal{V}$ satisfies the formula that $\int_{-\infty}^{\infty} \max\{x^2, 1\}\mathcal{V}(dx)$, the characteristic component has the form

$$\Psi(w) = ibw - \frac{1}{2}\sigma^2 w^2 + \int_{-\infty}^{\infty} (e^{iwx} - 1 - iwx1_{x\in(-1,1)})\mathcal{V}(dx), \tag{11.33}$$

index	$\tau = 1$	$\tau = 31$	$\tau = 92$	$\tau = 183$
σ	0.0435	0.3484	0.2482	0.3098
R^2	0.9458	0.8233	0.9483	0.8753

Table 9: σ the accuracy in terms of different time to maturity in MKS with Black-Scholes-Merton equation

(a) Black-Scholes-Merton equation in predicting call prices in MKS when $\tau = 1$

(b) Black-Scholes-Merton equation in predicting call prices in MKS when $\tau = 31$

(c) Black-Scholes-Merton equation in predicting call prices in MKS when $\tau = 92$

(d) Black-Scholes-Merton equation in predicting call prices in MKS when $\tau = 183$

Figure 4: Black-Scholes-Merton equation for different time to maturity in MKS

Figure 5: Black-Scholes-Merton for any time to maturity in MKS

index	$\tau = 1$	$\tau = 31$	$\tau = 92$	$\tau = 183$
β	0.5731	-0.0253	-0.1023	-0.1464
λ	34.2019	46.6381	3.0325	3.4686
μ	-0.8238	-0.0240	-0.6063	-0.2489
δ	0.4111	0.0015	0.4777	0
σ	0.0219	0.296	0.0323	0.2334
R^2	0.7612	0.7811	0.8271	0.5983

Table 10: σ and the accuracy in terms of different time to maturity in MKS with Merton's jump diffusion model

(a) Merton's jump diffusion model in predicting call prices in MKS when $\tau = 1$

(b) Merton's jump diffusion model in predicting call prices in MKS when $\tau = 31$

(c) Merton's jump diffusion model in predicting call prices in MKS when $\tau = 92$

(d) Merton's jump diffusion model in predicting call prices in MKS when $\tau = 183$

Figure 6: Merton's jump diffusion model for different time to maturity in MKS

Figure 7: Merton's jump diffusion model for any time to maturity in MKS

and then the n-th cumulant can be computed as

$$k_n = i^{-n}\left[\frac{d^n}{dw^n}\Psi(w)\right]\Bigg|_{w=0}. \tag{11.34}$$

The sensitive analysis method can be derived utilizing the cumulants, and it follows that

$$Var[X] = \mathbb{E}[(X-\mathbb{E}[X])^2] = k_2; \tag{11.35}$$

$$Skew[X] = \mathbb{E}[(X-\mathbb{E}[X])^3] = \frac{k_3}{k_2^{\frac{3}{3}}}; \tag{11.36}$$

$$Kur[X] = \mathbb{E}[(X-\mathbb{E}[X])^4] = \frac{k_4}{k_2^2}. \tag{11.37}$$

11.7.1 Sensitive Analysis: Black-Scholes-Merton equation

For the Black-Scholes-Merton equation, the Black-Scholes-Merton log-return density function $\Gamma(t) = \sigma W(t) + \left(\alpha - \frac{1}{2}\sigma^2\right)t$ is the corresponding Lévy's process and the characteristic component is denoted as

$$\Psi(w) = i\left(\alpha - \frac{1}{2}\sigma^2\right)w - \frac{1}{2}\sigma^2 w^2. \tag{11.38}$$

Since the skewness and kurtosis are zero, we apply the unit time variance to represent the sensitivity, and $Var[\Gamma(t)] = \sigma^2$. The Black-Scholes-Merton equation merely has one parameter σ, so we would analysis the sensitivity of the equation by changing the value of σ.

We utilize the information of call options in MKS instruments when time to maturity $\tau = 31$ as an example. In the optimal solution, $\sigma = 0.3484$, we would choose five different levels of σ: $0.1484, 0.2482, 0.3484, 0.4484, 0.5484$ to examine the results of variance and the graph for the predicted call prices in the test set. The variance is shown in table 11 and the graph is shown in table 8.

index	$\sigma = 0.1484$	$\sigma = 0.2484$	$\sigma = 0.3484$	$\sigma = 0.4484$	$\sigma = 0.5484$
variance	0.0220	0.0617	0.1214	0.2021	0.3007
R^2	0.6380	0.7408	0.8233	0.8537	0.8083

Table 11: variance and R^2 for sensitive analysis in Black-Scholes-Merton equation

Figure 8: Sensitivity in Black-Scholes-Merton equation

From the graph, we notice that the call prices is positive relative to σ. And the accuracy score drops sharply when the parameter σ decreases; while the accuracy score remains stable when σ increases.

11.7.2 Sensitive Analysis: Merton's Jump Diffusion Model in terms of β

Consider the Merton's jump diffusion model, the corresponding Lévy's process is denoted as

$$L(t) = \sigma W(t) + \left(\alpha - \beta\lambda - \frac{1}{2}\sigma^2\right) t + \sum_{i=1}^{N(t)} \log(Y_i + 1). \tag{11.39}$$

The jump size $\log(Y_i + 1)$ follows a normal distribution as $N(\mu, \delta^2)$. For $(b, \sigma^2, \mathcal{V})$, the characteristic component should be

$$\Psi(w) = i(\alpha - \beta\lambda - \frac{1}{2}\sigma^2)w - \frac{1}{2}\sigma^2 w^2 + \lambda \left\{exp\left\{i\mu w - \frac{1}{2}\delta^2 w^2\right\} - 1\right\}. \tag{11.40}$$

Thus, the unit time variance, skewness, kurtosis could be computed as

$$Var[L(t)] = \lambda\delta^2 + \lambda\mu^2 + \sigma^2; \tag{11.41}$$

$$Skew[L(t)] = \frac{3\lambda\delta^2\mu + \mu^3}{(\sigma^2 + \lambda\delta^2 + \lambda\mu^2)^{\frac{3}{2}}}; \tag{11.42}$$

$$Kur[L(t)] = \frac{3\lambda\delta^4 + 6\lambda\mu^2\delta^2 + \lambda\mu^4}{(\lambda\delta^2 + \lambda\mu^2 + \sigma^2)^2}. \tag{11.43}$$

We would firstly study the sensitivity of Merton's jump diffusion model in terms of β using the call options in MKS instruments when $\tau = 31$. The parameters leveraged in the sensitive analysis is shown below:

- $\tau = 31$
- $\lambda = 46.6381$
- $\mu = -0.0240$
- $\delta = 0.0015$
- $\sigma = 0.296$

The optimal value when pricing the options is $\beta = -0.0253$, so we would choose the following value of β as comparison: $-0.2253, -0.1253, -0.0253, 0.0747, 0.1747$. By substituting the data in test set to examine the sensitivity, we demonstrate the variance, skewness, kurtosis in table 12 and the graph in figure 9.

Figure 9: Sensitivity in terms of β

From the graph, it is noteworthy that the change with 0.1 in the value of β would greatly affect the accuracy of the model, while β is independent of the sensitive analysis methods: variance, skewness and kurtosis. Actually, if the value of β altered with 0.1 or 0.2, the predicted option prices would show significant variation, making the accuracy score of the model extremely low. This indicates that β contains a high level of sensitivity in Merton's jump diffusion model.

11.7.3 Sensitive Analysis: Merton's Jump Diffusion model in terms of λ

Then the sensitivity of λ in Merton's jump diffusion model would be introduced in this subsection and we would use the same example in MKS instruments. The parameters leveraged in the sensitive analysis is shown below:

- $\tau = 31$
- $\beta = -0.0253$
- $\mu = -0.0240$
- $\delta = 0.0015$
- $\sigma = 0.296$

When $\lambda = 46.6381$, the optimal solution is attained for option pricing. Then we select the following level of λ: $46.4381, 46.5381, 46.6381, 46.7381, 46.8381$. Using these value of λ, the variance, skewness, kurtosis is shown in table 13 and the graph is shown in figure 10.

Figure 10: Sensitivity in terms of λ

From the table and graph, we notice that the change in λ with 0.1 or 0.2 would cause tiny changes in sensitive analysis methods, variance, skewness and kurtosis. While a tiny variation in the value of λ would increase the accuracy score R^2 in this example.

11.7.4 Sensitive Analysis: Merton's Jump Diffusion model in terms of μ

Then we would determine the sensitivity of μ in Merton's jump diffusion model from the call options in MKS instrument. The parameters leveraged in the sensitive analysis is shown below:

- $\tau = 31$
- $\beta = -0.0253$
- $\lambda = 46.6381$
- $\delta = 0.0015$
- $\sigma = 0.296$

The optimal solution is attained when $\mu = -0.0240$. Then we choose the value of μ as: $-0.2240, -0.1240, -0.0240, 0.0760, 0.1760$. These different values of μ and the parameters defined would be substituted into the Merton's jump diffusion model. To test the sensitivity of Merton's jump diffusion model, the variance, skewness, kurtosis is shown in table 14 and the graph is shown in figure 11.

We demonstrate that μ is positive relative to the call prices in Merton's jump diffusion model. Both the sensitive analysis methods and R^2 shows that μ is greatly sensitive in the model, and a variation in μ would ruin the prediction of call options though other parameters remain unchanged.

index	$\beta = -0.2253$	$\beta = -0.1253$	$\beta = -0.0253$	$\beta = 0.0747$	$\beta = 0.1747$
variance	0.1145	0.1145	0.1145	0.1145	0.1145
skewness	-0.0006	-0.0006	-0.0006	-0.0006	-0.0006
kurtosis	0.0012	0.0012	0.0012	0.0012	0.0012
R^2	-0.0229	0.0033	0.7811	-0.6820	-0.6820

Table 12: variance, skewness, kurtosis and R^2 for sensitive analysis in terms of β in Merton's jump diffusion model

index	$\lambda = 46.4381$	$\lambda = 46.5381$	$\lambda = 46.6381$	$\lambda = 46.7381$	$\lambda = 46.8381$
variance	0.1144	0.1144	0.1145	0.1146	0.1146
skewness	-0.0006	-0.0006	-0.0006	-0.0006	-0.0006
kurtosis	0.0012	0.0012	0.0012	0.0012	0.0012
R^2	0.9993	1.0	0.7811	1.0	1,0

Table 13: variance, skewness, kurtosis and R^2 for sensitive analysis in terms of λ in Merton's jump diffusion model

index	$\mu = -0.2240$	$\mu = -0.1240$	$\mu = -0.0240$	$\mu = 0.0760$	$\mu = 0.1760$
variance	2.4285	0.8051	0.1145	0.3566	1.5315
skewness	-0.0030	-0.0027	-0.0006	0.0022	0.0030
kurtosis	0.0199	0.0170	0.0012	0.0122	0.0191
R^2	-0.6286	-0.4749	0.7811	0.0077	-0.0226

Table 14: variance, skewness, kurtosis and R^2 for sensitive analysis in terms of μ in Merton's jump diffusion model

Figure 11: Sensitivity in terms of μ

11.7.5 Sensitive Analysis: Merton's Jump Diffusion model in terms of δ

Then we study the sensitivity of Merton's jump diffusion model with respect to δ. The parameters leveraged in the sensitive analysis is shown below:

- $\tau = 31$
- $\beta = -0.0253$
- $\lambda = 46.6381$
- $\mu = -0.0240$
- $\sigma = 0.296$

In the optimal solution that we have computed, $\delta = 0.0015$. We choose the following value of δ by varying ± 0.1 or ± 0.2, which is given by $\delta = -0.1985, -0.0985, 0.0015, 0.1015, 0.2015$. Using these value of δ, the variance, skewness, kurtosis is shown in table 15 and the graph is shown in figure 12.

index	$\delta = -0.1985$	$\delta = -0.0985$	$\delta = 0.0015$	$\delta = 0.1015$	$\delta = 0.2015$
variance	2.0082	0.5950	0.1145	0.5668	1.9519
skewness	-0.0480	-0.0756	-0.0006	-0.0765	-0.0486
kurtosis	0.0588	0.0467	0.0012	0.0459	0.0587
R^2	0.0375	0.5128	0.7811	0.5517	0.0371

Table 15: variance, skewness, kurtosis and R^2 for sensitive analysis in terms of δ in Merton's jump diffusion model

Figure 12: Sensitivity in terms of δ

According to the accuracy score and the value of δ, the accuracy seems to follow a quasi normal distribution in terms of δ. The optimal value $\delta = 0.0015$ where the error attains the minimum should be the center of the quasi normal distribution. When δ deviates some distance from the center, the accuracy will descend by a certain magnitude regardless of whether the direction is positive or negative. The larger the deviation, the greater the drop in the accuracy score. From the variance, skewness and kurtosis, although δ shows less sensitivity than β and μ in Merton's jump diffusion model, the variation in δ would greatly affect the result of option pricing.

11.7.6 Sensitive Analysis: Merton's Jump Diffusion model in terms of σ

Then the sensitivity of σ in Merton's jump diffusion model would be discussed in this subsection from the data set in MKS instrument. The parameters leveraged in the sensitive analysis is shown below:

- $\tau = 31$
- $\beta = -0.0253$
- $\lambda = 46.6381$
- $\mu = -0.0240$
- $\delta = 0.0015$

The optimal value of σ when the error is minimized should be $\sigma = 0.296$. Then we select the following level of σ: $0.096, 0.196, 0.296, 0.396, 0.496$. By substituting these values into Merton's jump diffusion model,we can display the variance, skewness, kurtosis in table 16 and the graph in figure 13.

index	$\sigma = 0.096$	$\sigma = 0.196$	$\sigma = 0.296$	$\sigma = 0.396$	$\sigma = 0.496$
variance	0.0362	0.0654	0.1145	0.1836	0.2728
skewness	-0.0031	-0.0013	-0.0006	-0.0003	-0.0002
kurtosis	0.0122	0.0037	0.0012	0.0005	0.0002
R^2	0.9398	0.9767	0.7811	0.9697	0.8743

Table 16: variance, skewness, kurtosis and R^2 for sensitive analysis in terms of σ in Merton's jump diffusion model

Figure 13: Sensitivity in terms of σ

By observing the graph, accuracy score R^2 and sensitivity analysis methods: variance, skewness and kurtosis, we find that slight changes in σ may cause increase in the accuracy score. However, the change in accuracy is not significant. When the value of σ is adjusted by 0.1 or 0.2, the variation in variance, skewness and kurtosis is unnoticeable. We conclude that changing the σ does not significantly affect the model and σ is a less sensitive parameter.

12 Conclusion

In this report, we give an introduction of stochastic calculus in valuation of options. We focus on the derivation and employment of several stochastic processes. We start from Benoulli random walk to derive Brownian motion, then the stochastic integral corresponding to

Brownian motion is interpreted. Then we talk about Black-Scholes-Merton equation utilizing the geometric Brownian motion, which represents the formula for stock prices. To derive the Black-Scholes-Merton equation, we would use two methods. The first method should be partial differential equation method. The basic idea is to convert the Black-Scholes-Merton equation into one dimensional heat equation. The second method is the risk-neutral measure. We interpret the Radon-Nikodým derivative where the adapted process is defined on. The idea of risk-neutral measure on Brownian motion and the non-jump stochastic processes depends on Brownian motion is to change its drift. Then we leverage risk-neutral measure to ensure the non-arbitrage of the hedging portfolio in order to derive the Black-Scholes-Merton equation. The sensitive analysis is proposed separately. We give an introduction to Lévy measure and Lévy-Khintchine formula, thus, the characteristic component and cumulants would be computed for Lévy processes. And then we could calculate the variance, skewness and kurtosis from the cumulants. Then we present an innovative method to derive Merton's jump diffusion. During the calculation, we focus on the inclusive compound Poisson process, along with its mean jump size. This calculation provides a completely different understanding compared to Merton (1976). In the example with Python, we show that Merton's jump diffusion model is more accurate than Black-Scholes-Merton equation in the short term, while Black-Scholes-Merton equation is better in the long term. Different time to maturity affect the accuracy of the models and the optimal parameters of the models. If we consider the time to maturity in the data frame separately, the accuracy of both model is higher than the accuracy for any time to maturity. Then we conduct sensitive analysis. For the Black-Scholes-Merton equation, the parameter σ is more sensitive when it decreases and less sensitive when it increases. For Merton's jump diffusion model, parameters β, μ, σ is more sensitive, while λ and δ is less sensitive. Then we provide the details of each section.

In the preliminaries, we focus on normed space, functional analysis, optimization theory and probability theorem. In the functional analysis, we introduce Banach space, Dual space and inner product space. The inner product space is quite important in the report because it is applied in the introduction of cross variation. We define the space

$$\mathcal{N} = \{f | f \ is\ nowhere\ differentiable\ or\ f = 0\}. \tag{12.1}$$

The space is a vector space and then we verified that the cross product defined under $\mathcal{N}$ is an inner product space. Then we could verify the bracket process using the properties of inner product space. The cross variation is significant because quadratic variation is a specific case of cross variation. And the bracket process is leveraged to prove the Itó's product rule. The optimization theory is employed when dealing with the data.

Then the Brownian motion, the starting point of stochastic process, is introduced. We have interpreted the symmetric random walk to construct Brownian motion by taking the limit such that

$$W(t) = \lim_{n} \frac{1}{\sqrt{n}} M_{nt}. \tag{12.2}$$

Then Brownian motion is nowhere differential. The difference between nowhere differential stochastic processes and differentiable functions can be noticed from the quadratic variation. Differentiable functions have zero quadratic variation, while for Brownian motion, the quadratic variation satisfies the formula that

$$[W, W](t) = t. \tag{12.3}$$

Then we have discussed the Itó integral and Itó-Doeblin formula. Itó integral is the integral with an adapted process and Brownian motion, defined as

$$I(t) = \int_0^t \Delta(u)du. \tag{12.4}$$

Itó integral is a martingale with

$$[I, I](t) = \int_0^t \Delta^2(u)du. \tag{12.5}$$

While the Itó-Doeblin formula is the differentiable form for a function of Brownian motion. As I have mentioned, Itô integral is defined under Lebesgue integral. So we have to consider the quadratic variation term when calculating the differential form. That is to say, for continuous stochastic process $X(t)$, we have

$$df(X(t)) = f'(X(t))dX(t) + \frac{1}{2}f''(X(t))dX(t)dX(t). \tag{12.6}$$

And that is different compared to a differentiable function. The tradition methods to prove that stochastic processes could be approximated by step functions is introduced in this report. We consider whether each step in the step functions, and then separate the domain into two parts. We leverage the ideas in metric spaces to show that the step functions converges to the stochastic function in L^p space. Using metric space to prove the theorem is not an easy thing, and I spent about half a month thinking about that problem. I realized that if $\Delta(t)$ is a continuous function, then the problem are easy to handle. Especially, if $\Delta(t)$ is monotone increasing function, then under L_1 metric, we can verify the step function converges to $\Delta(t)$ using the definition of Riemann integral. However, the tricky thing is that $\Delta(t)$ may have finitely many jumps and the jumps in $\Delta(t)$ is not synchronous with the jumps in the step functions. I figure out a way to divide the region into two parts: area before the consecutive jumps and after and then scaling the areas respectively. We can also finish the proof using Stone-Weierstress, Luzin Theorem, Bernstein polynomial, etc.

Then we introduce Black-Scholes-Merton equation, the Black-Scholes-Merton equation is highly related to the geometric Poisson process, which is defined as

$$S(t) = S(0)e^{X(t)}, \tag{12.7}$$

where

$$X(t) = X(0) + \int_0^t \Delta(u)dW(u) + \int_0^t \Theta(u)du. \tag{12.8}$$

We leverage the definition of integral to show that geometric Brownian motion is a not martingale and

$$X(t) \sim \sigma W(t) + (\alpha - \frac{1}{2}\sigma^2)t. \tag{12.9}$$

The formula above is applied to calculate the characteristic components of the equation, which derives the variance, skewness and kurtosis for further conducting sensitive analysis. We use innovative method to show that $S(t)$ has mean rate of return α by separating the time interval into infinite parts, say

$$S(t) = \lim_{n\to\infty} \left(1 + \alpha\frac{t}{n}\right)^n = S(0)e^{\alpha t}. \tag{12.10}$$

Actually, if we use geometric Brownian motion $S(t)$ to describe the asset price, then $S(t)$ has mean rate of return α. To verify why the mean rate of return should be α is not an easy thing. After we obtained $\mathbb{E}[S(t)] = S(0)e^{\alpha}t$, we have to rewrite it into discrete form to show that mean rate of return is α. In fact, by taking limit of the binomial tree model, we can obtain geometric Brownian motion, please refer to Shreve (2004). To understand why geometric Brownian motion can describe stock prices with no arbitrage, I look through the notes in Columbia university and gives a finance example. Then, after considering risk-free rate, geometric Brownian motion can utilized as a reasonable model for non-arbitrage world. Note that geometric Brownian motion and geometric Poisson process are submartingales, but, after set $\alpha = 0$ or apply change of measure, we can find that geometric Brownian motion and geometric Poisson process are all martingales.

Then we construct non-arbitrage pricing and self-financing dynamic hedging to derive Black-Scholes-Merton partial differential equation. We introduced two methods for solving Black-Scholes-Merton equation. The first one is partial differential equation method, referring to Wilmott, Howison and Dewynne (1995). We change the variables in Black-Scholes-Merton equation to transfer it into one dimensional heat equation. Then we solve the equation and replace the variables. The second one is risk-neutral measure.

Then we give a brief introduction to conduct sensitive analysis for stochastic processes. The variance, skewness and kurtosis is defined as sensitive analysis methods. The variance denotes the distribution of the random variable compared to the mean value; skewness denotes the extent of asymmetric; kurtosis denotes whether the distribution is flat or peaked. We introduce the basic ideas for variance, skewness and kurtosis since considering their values are significant in the sensitive analysis. The variance, skewness, kurtosis is easily calculated for the Lévy's processes. Thus, we interpret the definition of Lévy's process and infinite divisibility in order to conduct sensitivity analysis, we also verified the space of infinite divisible random variables is a closed vector space, denoted as

$$\mathcal{I} = \left\{ \Gamma : \Gamma \equiv \sum_{k=1}^{n} \Gamma_k,\ \exists \{\Gamma_k\}_{k=1}^{k=n} \right\}. \tag{12.11}$$

Then we introduce the principle of Lévy-Khintchine formula, which is utilized to calculate the characteristic component of Lévy processes. For a triple $(b, \sigma, \mathcal{V})$, the characteristic components is donated as

$$\Psi(w) = ibw - \frac{1}{2}\sigma^2 w^2 + \int_{-\infty}^{\infty} (e^{iwx} - 1 - iwx1_{x\in(-1,1)})\mathcal{V}(dx), \tag{12.12}$$

the value of b and σ can be determined from the distribution of Lévy's processes. The cumulants is generated from the characteristic components and we have found the relationship between variance, skewness, kurtosis and the cumulants. We conduct an example to calculate the sensitive analysis methods for Black-Scholes-Merton equation. Note that Black-Scholes-Merton equation is not Lévy process, we define the corresponding Black-Scholes-Merton log-return density function as a Lévy process, which is denoted as

$$\Gamma(t) = \log\left(\frac{S(t)}{S(0)}\right) = \int_0^t \sigma dW(t) + \int_0^t \left(r - \frac{1}{2}\sigma^2\right) dt, \tag{12.13}$$

and the characteristic components could be calculated as

$$\Psi(w) = i\left(\alpha - \frac{1}{2}\sigma^2\right) w - \frac{1}{2}\sigma^2 w^2. \tag{12.14}$$

This enable us to determine the sensitive analysis method: variance, skewness and kurtosis.

Then we talk about the properties and theorems of jump process. We start from the definition of Poisson process from Poisson distribution. Then we derive some significant jump processes, such as compensated Poisson process and compound Poisson process. The stochastic integral of jump process is introduced, which is given by

$$\int_0^t \Phi(s)dX(s) = \int_0^t \Phi(s)\Gamma(s)dW(s) + \int_0^t \Phi(s)\Theta(s)ds + \sum_{0<s\leqslant t} \Phi(s)\Delta J(s), \tag{12.15}$$

where

$$X(t) = X(0) + \int_0^t \Gamma(s)dW(s) + \int_0^t \Theta(s)ds + J(s). \tag{12.16}$$

We give an example referring to Shreve (2004) and compute the cross variation between two jump processes. Then two significant properties: Itô product rule and Doleans-Dade exponential is presented and verified, which are leveraged in change of measure. Then we introduce the effects of change of measure. We provide an example of change of measure on discrete compound Poisson process and non-discrete compound Poisson process. In the examples, we firstly define the adapted process and the corresponding probability measure. Then, we consider examine the validity of adapted process by determining whether it is a martingale and it has mean value 1. Next, we check whether the change of measure is well-defined, for jump processes, the intensity of the processes would change, while others remain unchanged. Change of measure is not a panacea, for instance, the linear Brownian motion could not become a Brownian motion under new probability measure.

Then Merton's jump diffusion model is introduced. This report presents a completely innovative method to explain Merton's jump diffusion model, which is an extension of the approaches from Shreve (2004) and the principles of partial differential equations. The probability distributions of log stock price jump size and log-return density function would be discussed in the first place. For Merton's jump diffusion model, we leverage the geometric Poisson process as the stock price, which contains the jumps. We deduced the formula for log return option prices $dc(t, S(t))/c(t, S(t))$. And we notice that the rate of return for stock prices is derived as

$$\frac{dS(t)}{S(t)} = (\alpha - \beta\lambda)dt + \sigma dW(t) + \frac{S(t-)}{S(t)}dQ(t), \tag{12.17}$$

which has the term $dS(t-)/dS(t)$, thus we make an assumption that the rate of return for call option can be written as

$$\frac{dc(t, S(t))}{c(t, S(t))} = (\alpha_w - \beta_w\lambda)dt + \sigma_w dW(t) + \frac{c(t, S(t-))}{c(t, S(t))}dQ_w(t). \tag{12.18}$$

The assumption is totally innovative and we have verified the validity. After substituting the variables α_w, β_w, σ_w and $Q_w(t)$, we have showed that $Q_w(t)$ is a compound Poisson process with differential form

$$dQ_w(t) = \sum_{m=1}^{M} \frac{[c(t, (y_m + 1))S(t-) - c(t, S(t-))]}{c(t, S(t-))} dN_m(t) \tag{12.19}$$

with the jump size

$$Y_j^w = \frac{c(t, (y_m + 1))S(t-) - c(t, S(t-))}{c(t, S(t-))} \tag{12.20}$$

Thus the mean jump size

$$\beta_w = \mathbb{E}\frac{c(t, (y_m + 1))S(t-) - c(t, S(t-))}{c(t, S(t-))}, \tag{12.21}$$

which shows the validity of our assumption. Then we derived the stochastic differential equation for call option with boundary conditions, or it is called Merton integral partial differential equation. To solve the equation, we compute the derivatives of Black-Scholes-Merton equation, then we could verify the solution for Merton integral partial differential equation. Our method provides a clearer understanding of the compensated Poisson process in the equation, implying its mean rate of return and why it is non-arbitrage.

Then we use two examples about option pricing with Python. And we compare the efficacy of Black-Scholes-Merton equation and Merton's jump diffusion model in valuation of options. The formulas for these models are demonstrated and the least square error is employed. Then we explain the optimization problem to find the parameters of Black-Scholes-Merton equation and Merton's jump diffusion model. The pure Newton's method, DFP algorithm and BFGS algorithm is discussed and BFGS algorithm is applied in the examples. Firstly, we choose the example from Dow Jones Industrial Average from Sep. 28, 2022 to Feb. 28, 2023. We leverage the train set to determine the parameters and the test set to examine the accuracy. For the short period, compared to Black-Scholes-Merton equation, Merton's jump diffusion model performs better in convergence and has more accuracy, though it has more complexity. Next, we choose an example from MKS instruments with a long duration to analyze the effects of different time to maturity on valuation of options. We find that time to maturity would affect the accuracy of both Black-Scholes-Merton equation and Merton's jump diffusion model. When the time to maturity in the data frame are considered separately, the accuracy of both model would higher than the accuracy for all time to maturity. Also, we note that, in the long term, Black-Sholes-Merton equation is more accurate than Merton's jump diffusion model. Lastly, we conduct sensitive analysis by calculating variance, skewness and kurtosis. For the Black-Scholes-Merton equation, the parameter σ is more sensitive when it decreases and less sensitive when it increases. For Merton's jump diffusion model, parameters β, μ, σ is more sensitive, while λ and δ is less sensitive.

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References

A. Einstein: *On the Movement of Small Particles Suspended in a Stationary Liquid Demanded by the Molecular-Kinetic Theory of Heat*, Annalen der Physik, 17(8), 549-560, (1905).

L. Bachelier: *Théorie de la Spéculation.* Paris: Gauthier-Villars, (1900); (English) The Theory of Speculation.

N. Wiener: The Homogeneous Chaos. *American Journal of Mathematics*, 45(3), 243-273, (1923).

N. Wiener: Differential-Space. *Journal of Mathematics and Physics*, 9(4), 137-153, (1930).

K. Itō: Stochastic Integral. *Proceedings of the Imperial Academy*, 20(8), 519-524, (1944).

K. Itō: On Stochastic Differential Equations. *Memoirs of the American Mathematical Society*, 4(22), 1-51, (1947).

P. A. Samuelson: Proof that Properly Anticipated Prices Fluctuate Randomly. *Industrial Management Review*, 6(2), 41-49, (1965).

F. Black and M. Scholes: The Pricing of Options and Corporate Liabilities. *Journal of Political Economy*, 81(3), 637-654, (1973).

R. C. Merton: Theory of Rational Option Pricing. *The Bell Journal of Economics and Management Science*, 4(1), 141-183, (1973).

R. C. Merton: Option Pricing and the Valuation of Corporate Debt. *The Journal of Finance*, 31(2), 449-470, (1976).

X. Kou: A Jump-Diffusion Model for Option Pricing. *Management Science*, 48(8), 1086-1101, (2002).

S. D. Poisson: *Recherches sur la Probabilité des Jugements en Matière Criminelle et en Matière Civile*, Bachelier, Paris, (1837).

J. F. C. Kingman: *The single server queue in heavy traffic*, Mathematical Proceedings of the Cambridge Philosophical Society, (1962).

W. Feller: *An Introduction to Probability Theory and Its Applications*, Volume 1, Third Edition, John Wiley and Sons, (1968).

L. Kleinrock: *Queueing Systems. Volume 1: Theory*, Wiley-Interscience, New York, (1975).

R. Cont and P. Tankov, *Financial Modelling with Jump Processes*, Chapman and Hall/CRC Financial Mathematics Series, (2003).

E. Renshaw: *Modelling biological populations in space and time*, Cambridge Studies in Mathematical Biology, (1993).

C. P. Robert and G. Casella: *Monte Carlo Statistical Methods*, Second Edition, Springer, New York, (2004).

J. Grandell: *Mixed Poisson Processes*, Chapman & Hall/CRC, (1997).

D. Karlis and E. Xekalaki: *A mixed Poisson distribution*, Statistics: A Journal of Theoretical and Applied Statistics, (2003).

A. K. Erlang: *The Theory of Probabilities and Telephone Conversations*, Nyt Tidsskrift for Matematik B, (1909).

E. Rutherford and H. Geiger: *The probability variations in the distribution of α particles*, Proceedings of the Royal Society of London. Series A, Containing Papers of a Mathematical and Physical Character, (1910).

S. E. Shreve: *Stochastic Calculus for Finance*, Springer, New York, (2004).

P. Levy: *Théorie de l'addition des variables aléatoires*, Gauthier-Villars, Paris, (1937).

P. Levy: *Processus Stochastiques et Mouvement Brownien*, Gauthier-Villars, Paris, (1948).

P. Wilmott, S. Howison and J. Dewynne: *The Mathematics of Financial Derivatives*, Cambridge University Press, New York, (1995).

R. Durrett: *Probability: theory and examples. Cambridge Series in Statistical and Probabilistic Mathematics*, Cambridge University Press, Cambridge, (2010).

A. N. Shiryaev: *Probability, volume 95 of Graduate Texts in Mathematics*, Springer-Verlag, New York, (1996).

K. Sato: *Lévy Processes and Infinitely Divisible Distributions*, Cambridge University Press, Japan, (1999).

A. Itkin: *Pricing Derivatives Under Lévy Models*, Springer, Brooklyn, (2010).

A. E. Kyprianou: *An Introduction to The Theory of Lévy Processes*, University of Bath, Bath, (2007).

J. Manafian and M. Paknezhad: *Analytical solutions for the Black-Scholes equation*, Applications and Applied Mathematics, Iran, (2017).

G.M. Georgiou and K. Voigt: *Stochastic computation of moments, mean, variance, skewness and kurtosis*, The Institution of Engineering and Technology, (2015).

K. Matsuda: *Introduction to Merton Jump Diffusion Model*, The City University of New York, New York, (2004).

M. Frittelli and W. Runggaldier: *Stochastic Methods in Finance*, Springer, Italy, (2003).